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EXTRATERRESTRIAL LANDER FOR DATA
COLLECTION AND RF TRANSMISSION

FLIGHT READINESS REVIEW

NASA STUDENT LAUNCH COMPETITION
2024 - 2025

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Table 1: Acronyms and acronym definitions

Acronym	Acronym Meaning
ABS	Acrylonitrile Butadiene Styrene
AGL	Above Ground Level
AIAA	American Institute of Aeronautics
AOA	Angle of Attack
BMS	Battery Management System
CAD	Computer-Aided Design
CFD	Computational Fluid Dynamics
CFR	Code of Federal Regulations
CG	Center of Gravity
CNC	Computer Numerical Control
CP	Center of Pressure
DRI	Dynamic Response Index
ESB	Engineering and Science Building
FAA	Federal Aviation Administration
FEA	Finite Element Analysis
FOV	Field of Vision
FPV	First Person View
FRR	Flight Readiness Review
GPS	Global Positioning System
IMU	Inertial Measurement Unit
KE	Kinetic Energy
LiPo	Lithium Polymer (Battery)
LRE	Launch Rail Exit
MECO	Main Engine Cut Off
MSDS	Material Safety Data Sheet
NAR	National Association of Rocketry
NFPA	National Fire Protection Association
PC	Personal Computer
LSM	Lander Separation Mechanism
PID	Proportional, Integration, Derivative (Control)
PLA	Polylactic Acid
PPE	Personal Protective Equipment
RF	Radio Frequency
RSO	Range Safety Officer
R/C	Remote Control
SDR	Software Defined Radio
SCM	STEMnaut Crew Module
SL	Student Launch
PDM	Parachute Detachment Mechanism
Continued on next page	

Table 1 – continued from previous page

Acronym	Acronym Meaning
LDM	Leg Deployment Mechanism
AEM	Antenna Extension Mechanism
SSM	Static Stability Margin
SSO	Student Safety Officer
STEM	Science, Technology, Engineering, and Math
T/W	Thrust to Weight Ratio
USLI	University Student Launch Initiative
UTM	Universal Testing Machine
VADL	Vanderbilt Aerospace Design Lab
VDF	Vehicle Demonstration Flight
VUSE	Vanderbilt University School of Engineering

1 Summary

1.1 Team Summary

Vanderbilt Aerospace Design Laboratory

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The team intends to travel to Huntsville, Alabama for the competition launch on May 3, 2025. Since the CDR milestone, the team has spent 1497 hours combined working on the project.

1.2 Launch Vehicle Summary

Parameter	Specification	Parameter	Specification
Target Apogee	4200 ft	Motor Selection	Primary: Cesaroni L1720, Secondary: N/A
Rocket Length	97 in	Burnout mass without ballast	35.8 lbm
Length/Mass (nose cone, payload bay, and lander separation bay)	45 in, 15.2 lbm	Recovery system	Dual deployment
Length/Mass (avionics and recovery)	15.5 in, 12.5 lbm	Drogue parachute	Fruity Chutes Elliptical 18 in, 1.5 CD
Length/Mass (tail section)	36 in, 14.4 lbm	Drogue deployment	Primary: Apogee, Backup: Apogee + 1s
Landing Mass (vehicle/lander)	29.7 lb, 8.0 lb	Main Parachute	Fruity Chutes Toroidal 96 in, CD 2.2
Dry mass without ballast	35.8 lbm	Main deployment	Primary: 750 ft, Backup: 700 ft
Dry mass with ballast	38.1 lb	Lander separation mechanism	Electromagnetic latches, 300 ft
Wet mass with ballast	42.0 lb	Rail size	144 in

1.3 Payload Summary

The Extraterrestrial Lander for Data Collection and RF Transmission is a parachute-arrested lander that acts as the nose cone of the launch vehicle and jettisons at 300 ft AGL during main parachute descent. Jettison is enabled by electromechanical latches which release the payload, allowing gravity to pull it from the foresection of the launch vehicle. Once jettisoned, the payload will deploy a parachute and landing legs that double as the ground plane for the antenna system. The payload will land horizontally (nose cone parallel to the ground) upon which the parachute will be detached and, 15 seconds later, the antenna will be raised to a height of 24 in above the ground allowing the payload to begin RF transmission of flight telemetry. The payload structure is constructed out of aluminum and 3D printed bulk plates attached to two aluminum rods to transfer load throughout the structure. The airframe and nose cone are made of fiberglass to reduce RF interference and are

fixed to the skeleton via two bulk plates. The fore bulk plate contains three radial bolts while the aft bulk plate slots into a retention ring epoxied to the airframe itself. The legs and antenna are deployed through slots cut into the airframe. The nose cone is attached to the airframe directly via four radial bolts.

2 Changes Since CDR

Table 2: Vehicle criteria changes.

Change	Justification
Overall rocket length decreased from 104 to 97 in	Due to a smaller packaging volume of the newly procured lander parachute, the lander parachute tube was shortened from 11.7 to 6 in. This adjustment was paired with the omission of the bulkplate between the parachute tube and the electronics housing, therefore reducing the length of the LSM bay airframe by 5.5 in. Additionally, the recovery coupler sections were both decreased in length by 0.75 in to optimize volume while maintaining compliance with NASA Requirement 2.4.1.
Drogue shock cords shortened from 40/20 ft to 30/15 ft; aft main shock cord lengthened from 10 to 15 ft	The fore shock cord lengths were originally selected to minimize the effect of swinging on the payload prior to jettison. However, numerous drop tests have proven the payload's jettison performance even in cases of extreme swinging. The shock cords were therefore shortened to reduce acceleration forces experienced at shock cord extension. The aft tail shock cord was increased to a more appropriate length given the size of the tail section to which it is connected.
Addition of a 2.3-lbm ballast	Due to design optimization and the manufacturing process, the as-built vehicle came out 2.3 lbm lighter than the CDR projection. To ensure that all flight objectives were successfully met, the team has implemented 2.3 lbm of ballast in the LSM bay, thus increasing the CG and more closely resembling the projected mass from CDR.

2.1 Payload Criteria Changes

The payload functionality and high-level structure in the FRR design cycle remains consistent with the CDR version. Changes made to the payload design were driven by testing results to improve the reliability and functionality of the payload.

Table 3: Payload Criteria Changes.

Change	Justification
Landing legs were extended from 1.7 in to 3.4 in below the lander belly Landing leg material was switched to a softer aluminum alloy that plastically deforms upon impact Breakable and easily replaceable 3D printed shoes were added to the fore landing legs to reduce impact loading	Landing instability from bouncing was observed in previous design iterations, and di not satisfy VADL requirement 4.1.7.
The number of internal rods were switched from 4 rods to 2 rods	The reduced number of aluminum rods optimizes mass and space within the payload, while not affecting payload performance
The PDM was modified to a simpler single attachment point instead of a double attachment	The reliability of parachute detachment observed during testing did not satisfy VADL requirement 4.4.1.

2.2 Project Plan Changes

Table 4: Changes to the project plan.

Change	Justification
Requirements verification plans and statuses have been updated (Section 7.2).	As the project has matured and key testing and flight demonstrations have taken place, the vast majority of requirements are now verified.
The budget has increased slightly, from \$14,836 to \$14,946 (Section 7.3.2). Launch operations and procedures have been updated for the full-scale vehicle and now include additional safety warnings (Section 6.6).	Two unexpected costs – a new payload parachute and a ground station antenna – led to a slight uptick in the conservatively calculated budget. Spending remains below CDR projections, however. The previous checklists, while serving as a guideline for the full-scale versions, were for the subscale vehicle. Additional safety warnings are necessary to fully encompass all potential hazards encountered on launch day.

3 Vehicle Criteria

3.1 Design and Construction of Launch Vehicle

The mission of the launch vehicle is to launch to an apogee within 5% of 4200 ft, safely retain the payload until jettison at 300 ft, and successfully recover all payload and vehicle sections such that they are reusable without repairs. Section 3 outlines the aspects that ensure the fulfillment of this mission. The full-scale fabricated vehicle is shown in Figure 1 and the corresponding CAD with callouts is shown in Figure 2. The OpenRocket schematic is also provided in Figure 3, indicating the CG and CP locations and the SSM. Key vehicle parameters are summarized in Table 5 and massing breakdowns for both non-jettisoned and jettisoned cases are shown in Figures 4 and 5, respectively.



Figure 1: Fabricated full-scale launch vehicle.

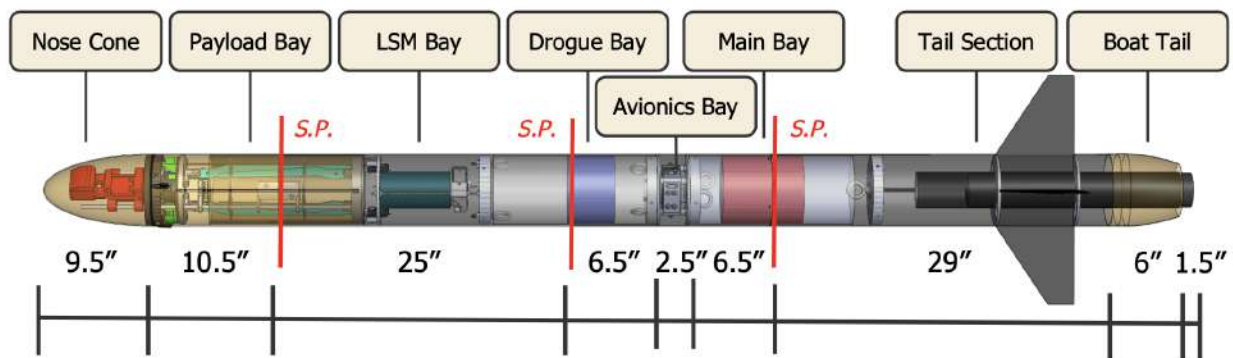


Figure 2: CAD of as-built vehicle with callouts.

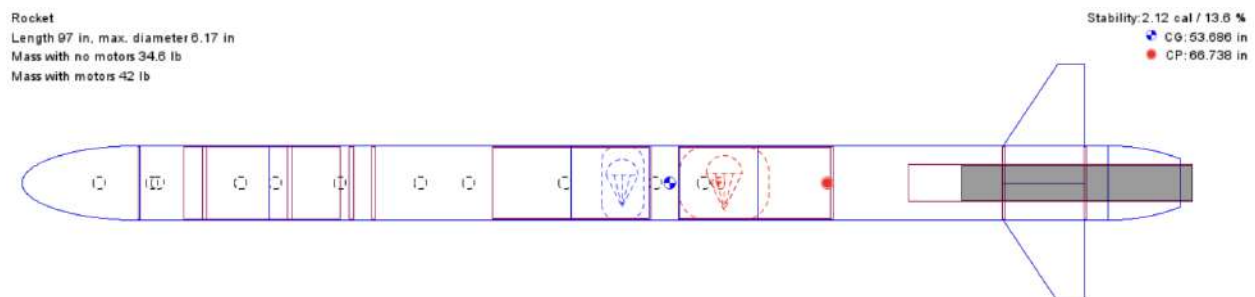


Figure 3: OpenRocket schematic showing CG, CP, and SSM.

Table 5: Key parameters of the as-built vehicle.

Parameter	Value
Target Apogee	4200 ft
Motor	Cesaroni L1720
Length	97 in
Diameter	6 in
Dry Mass (unballasted)	35.8 lbm
Dry Mass (ballasted)	38.1 lbm
Wet Mass (ballasted)	42.0 lbm
Center of Gravity (from fore of rocket)	53.7 in
Center of Pressure (from fore of rocket)	66.7 in
Stability Margin (on pad)	2.12 cal
Airframe Material	Fiberglass

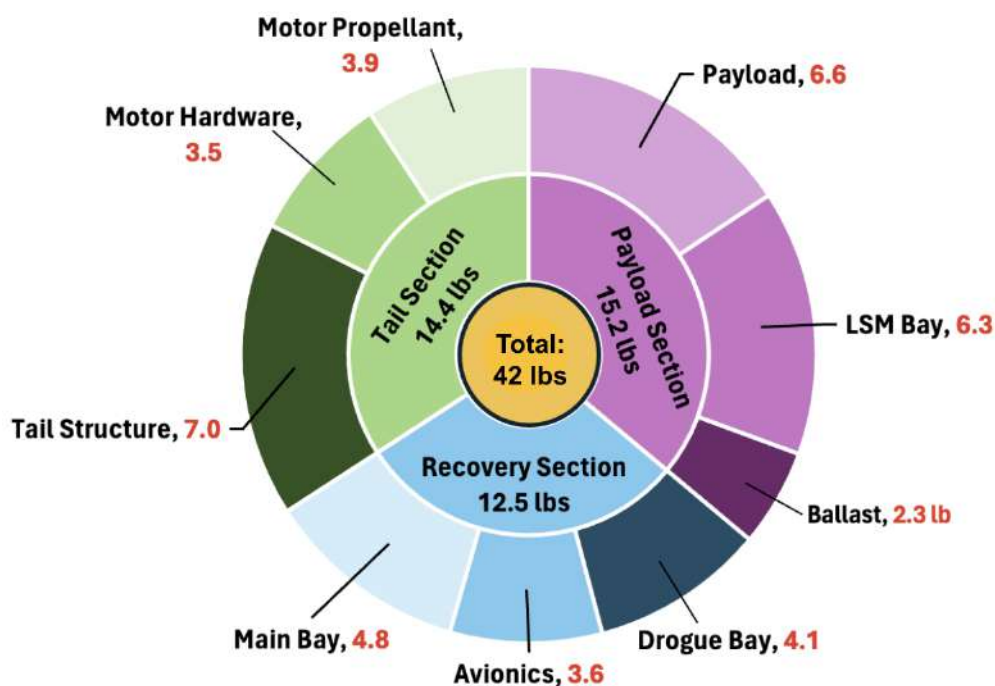


Figure 4: Massing breakdown for the scenario in which the payload is not jettisoned.

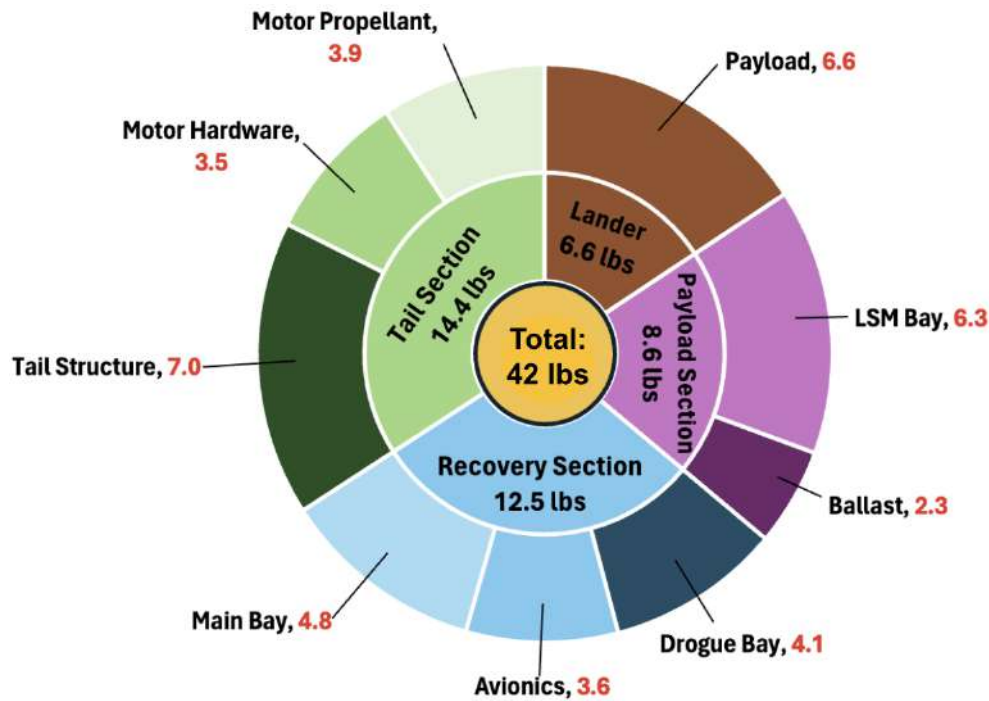


Figure 5: Massing breakdown for the scenario in which the payload is jettisoned.

3.1.1 Vehicle Design Changes Since CDR

Several key changes have been made to the full-scale vehicle since the CDR milestone. These changes were made to accommodate mass changes and other factors as mentioned below. Additionally, all changes were made with consideration to crucial design parameters such as center of gravity, static stability margin, and structural integrity of the vehicle.

The parachute for the lander was switched from an Iris Ultra to Ultralight, thus resulting in a significantly reduced packing volume requirement. For this reason, the parachute tube housed in the Lander Separation Mechanism bay was shortened from 11.7 to 6 in. The team also determined that the top bulk plate of the LSM bay between the parachute tube and electronics housing was unnecessary and could be simplified with a tube lid sandwiched between the parachute tube and the electronics housing. These changes resulted in a shorter length of the LSM mechanism, resulting in the 5.5-inch reduction from the LSM bay airframe. The main and drogue bays were also reduced by approximately 0.5 inches which reduced the overall length of the rocket from 104 inches to approximately 97 inches. This was done to optimize the volume of the two bays, while still remaining compliant with NASA Requirement 2.4.1.

Another key change was recorded in the vehicle recovery system. The fore and aft drogue shock cords were switched from 40/20 ft to 30/15 ft. The lengths were originally selected with the intent of minimizing the intensity of swinging of the payload prior to jettison; however, numerous drop tests have proven successful jettison even in cases of significant swing. These lengths were therefore reduced to lower the acceleration forces experienced at full shock cord extension. Additionally, the aft main shock cord was lengthened from 10 to 15 ft. This shock cord connects to the tail section, which is approximately 3 ft long (including the boat tail). As such, the greater length was selected to more appropriately accommodate the supported section length. All shock cords

were proven to perform as intended during the Vehicle Demonstration flights, covered in Section 5.

As a result of design optimization and the manufacturing process itself, the dry mass of the vehicle came out at 2.3 lbm lower than the originally estimated mass. To ensure the center of gravity stays relatively similar to the initial marking, a 2.3 lbm ballast is added to the rocket in the LSM bay. To creatively adjust and uniformly spread this mass, the team added washers to both of the threaded rods supporting the LSM as seen in Figure 6. The design and construction of the vehicle was verified through two successful vehicle demonstration flights.

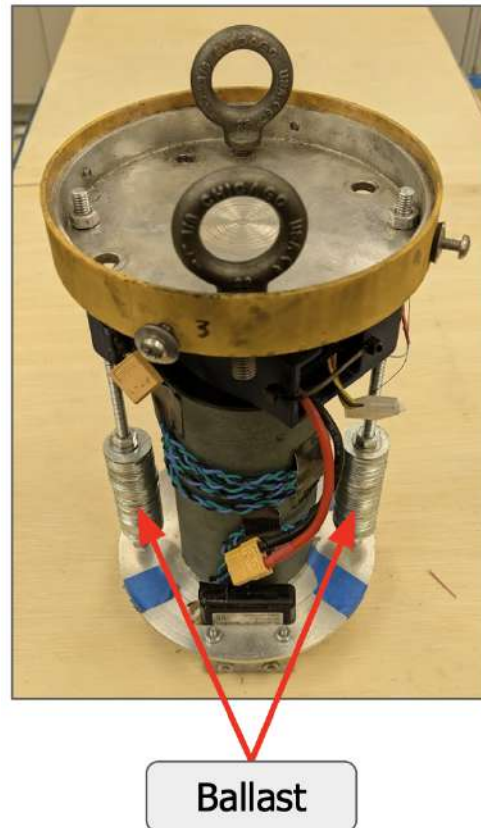


Figure 6: Ballast shown in the assembled LSM.

The lessons learned through the VDFs allowed the team to better understand the recovery system and parachute deployment events. In the first VDF flight, main parachute deployment triggered at 600 ft as intended, but the parachute took roughly 200 ft to unfurl. To ensure that the vehicle reaches a steady terminal velocity prior to payload jettison at 300 ft, the main parachute deployment height has been increased to 750 ft. Simulation results support this shift and indicate that all NASA and VADL imposed requirements will still be met.

3.1.2 Separation Points

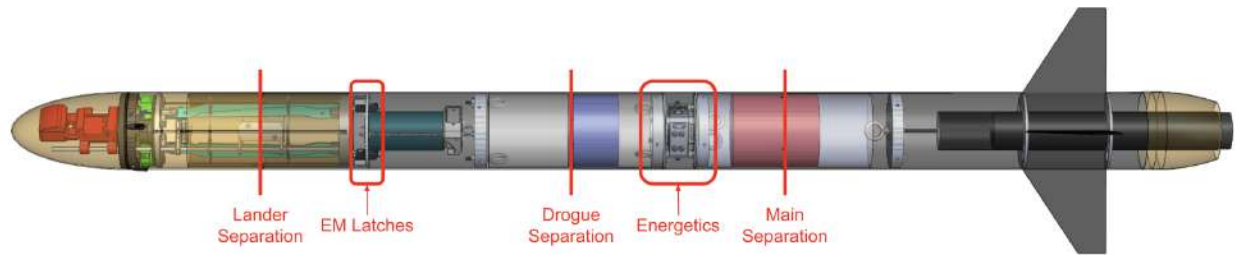


Figure 7: Location of the energetics and main, drogue, and lander separation points.

The vehicle separation points remain unchanged from CDR. The recovery system entails a dual deployment which involves drogue deployment at apogee (backup at apogee + 1s) and main deployment at 750 ft (backup at 700 ft). These deployment events will be implemented through energetics (blast charges) actuated via the altimeters located in the avionics housing. The lander will jettison at 300 ft pending RSO approval and will be achieved through the EM latches in the LSM Bay. The EM latches are a thoroughly tested, safe, and eco-friendly method for separating the lander from the vehicle. The latches have been verified through drop tests and launches conducted throughout the project cycle.

3.1.3 Airframe

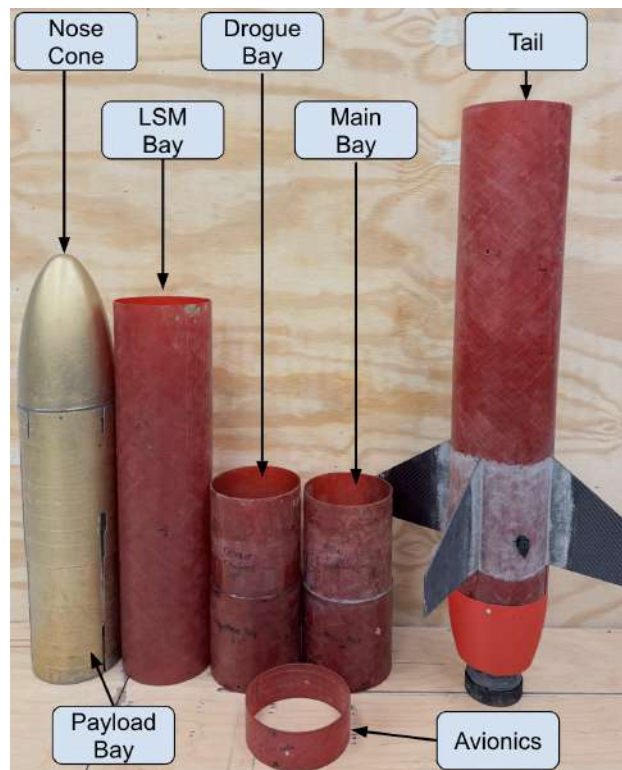


Figure 8: Disassembled vehicle showing all components of the airframe.

The payload objective dictated the airframe selection for the project this year. Since a critical functionality of the payload is to transmit data through RF, the airframe selection came down to compatibility with RF systems and structural strength. The team determined a fiberglass airframe to be the most appropriate material because it fulfills both RF compatibility and strength requirements. As shown in Figure 8, all sections of the airframe were manufactured with fiberglass.

Due to the distance of the fins from the payload, especially in the case that the payload is jettisoned, the team deemed it unnecessary to construct the fins from fiberglass. This decision was made to conserve mass while maintaining high strength and performance. Since DragonPlate has been a trusted material for fin construction in the past, the team determined it to be the optimal material selection for the rocket fins.

The boat tail, a non-load bearing component, did not need to be constructed out of fiberglass. It is located near the motor tube, however, which reaches high temperatures. For this reason, the boat tail is 3D printed out of ABS, which is an impact and heat resistant material—most appropriate for the boat tail. The boat tail is attached to the coupler using four #4-40 shear pins.

The nose cone was also constructed with fiberglass, replicating the manufacturing process used for the subscale nose cone. The team determined that the most logical place to house the STEMnauts was in the nose cone, similar to most space missions. To maximize the space in the nose cone for the SCM and electronics, the team designed an elliptical nose cone which was unavailable commercially. The team took on the challenge of manufacturing the nose cone in house by laying sheets of fiberglass over a 3D printed mold; a novel approach for VADL. As mentioned above, fiberglass was chosen for the nose cone to retain structural integrity while removing the risk of RF interference that could be experienced if carbon fiber was used.

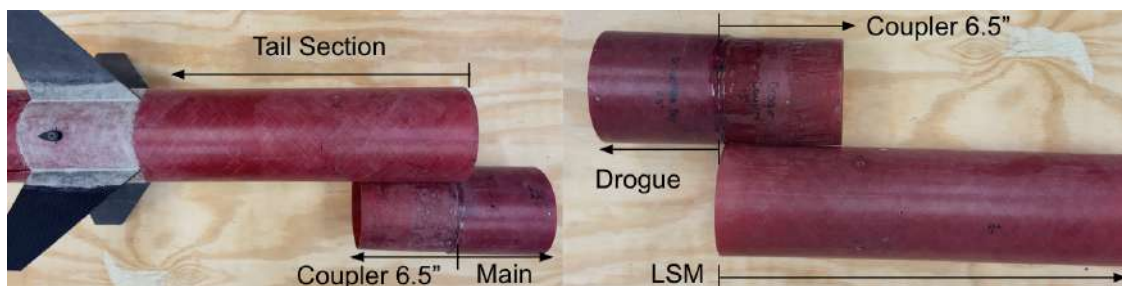


Figure 9: Coupler and airframe sections shown side by side to verify the appropriate overhang. Left shows the main bay and tail coupler. Right shows the drogue bay and LSM bay coupler.

The main and drogue bays were manufactured and the appropriate sized couplers were epoxied to these airframe sections. As seen in Figure 9, the couplers were designed to fulfill NASA Requirement 2.4.1, which states that the coupler at separation points must be at least two airframe diameters in length and should at least be one body diameter of surface contact with each airframe section. This was verified through the design of the rocket, in which each coupler was designated to be 13 in long with 6.5 in of overlap with each airframe section, and careful manufacturing procedures were practiced to ensure all vehicle requirements were met. The coupler and airframe sections are joined together and set in place using #2-56 shear pins which have been verified for deployment through testing.

3.1.4 Tail Section

The tail section of the vehicle also remains unchanged from CDR. The material for the airframe is fiberglass, the motor tube is constructed out of blue tube, the fins are constructed out of Drag-onPlate, and the boat tail is 3D printed using ABS. The geometry of the fins—clipped delta—and the boat tail remains the same as that of the proposed design in CDR.

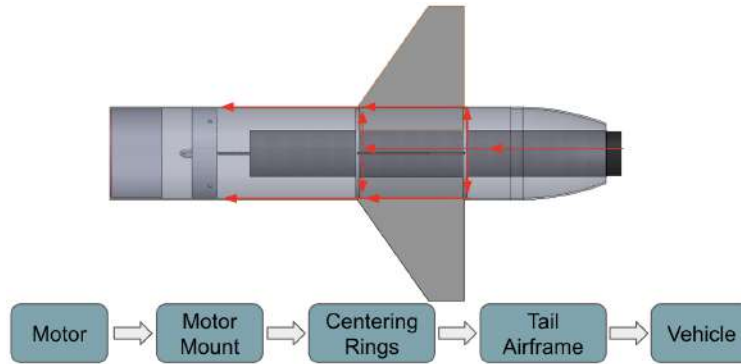


Figure 10: Figure shows the motor thrust load path in the tail section.

The critical load path from motor thrust remains unchanged from the critical load path determined in CDR, see Figure 10. Therefore, the motor retention methodology remains the same as the one proven during the successful subscale flight. This retention design and construction was again verified through the successful VDFs where no damage to any tail components was observed and the vehicle was reflown within two hours.

Since the boat tail does not fall under the critical load path and its main purpose is to reduce pressure drag to optimize aerodynamic performance, it does not need to be constructed out of a high-strength material like fiberglass or carbon fiber. For this reason, the team 3D printed the boat tail out of ABS for the VDFs, which showed promising results as no damage was observed.

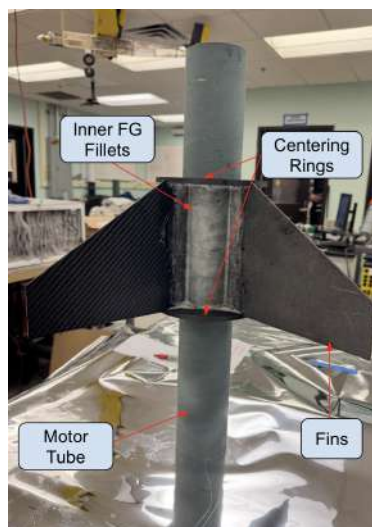


Figure 11: The motor tube, centering rings, and fins assembled and epoxied along with the inner fiberglass fillets between fins for strengthening.

The tail section was manufactured in two parts: the inner tail and the tail casing. The inner tail portion includes the centering rings which sandwich the four fins, fixed around the motor tube. The junction between the Blue Tube motor retention tube and the centering rings was strengthened using epoxy and the junctions of the fins, centering rings, and motor tube were strengthened using fiberglass sheets and epoxy (as seen in Figure 11). The tail construction was inspired by the methodology and lessons learned from the subscale vehicle construction. The manufacturing precision was verified through the performance of the vehicle during the VDFs.

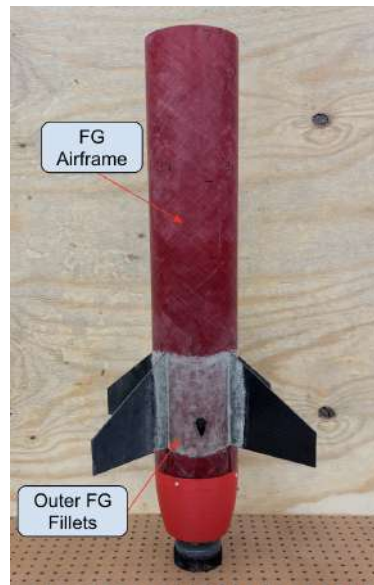


Figure 12: Tail section fully assembled with the airframe and outer fillets. Also shown is the lower rail buttons fixed onto the airframe.

Last, as seen in the complete assembly of the tail section, slots were cut into the main frame to slide the fins and the inner tail portion in. Once they were in, the outer fiberglass fillets were laid using fiberglass sheets and epoxy. The boat tail was then fixed onto the tail using #4-40 shear pins. The motor retainer was also fastened to the motor tube using JB Weld. Lastly, the tail section rail button was attached by carefully measuring and aligning it with the airframe from the tail end, with the aluminum clamp epoxied to the inside of the airframe.

3.1.5 Bulk Plates

Proper bulk plate design and fabrication is essential for ensuring the structural integrity of the vehicle. There are two types of bulk plates used in the final constructed vehicle: flat aluminum disks and recovery plates. Both types were machined using 6061 aluminum, as all are load-bearing. The locations of all bulk plates in the vehicle are shown in Figure 13.

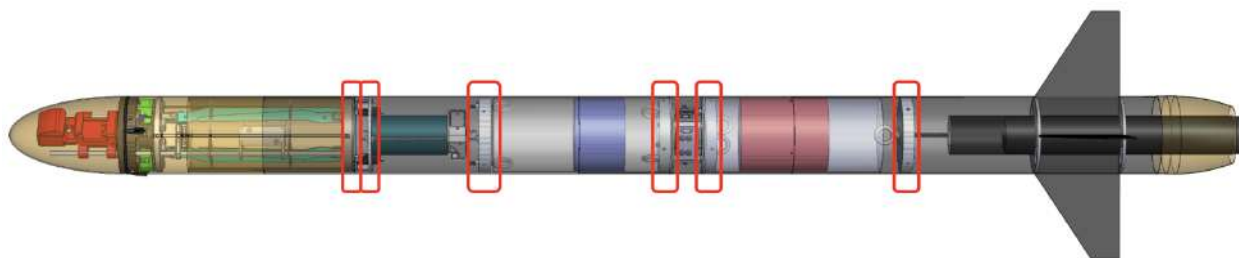


Figure 13: Locations of bulk plates in the final vehicle design.

As seen in Figure 13, there are five aluminum plates in the full-scale vehicle. The two foremost plates are located at the aft of the payload and fore of the lander separation mechanism and form the retention interface for the lander to the rest of the rocket during flight. Both plates were machined using 1/4 in aluminum plate stock. The remaining four bulk plates represent the anchoring points for the drogue and main parachutes. All recovery bulk plates are retained to the airframe via four 10-32 radial bolts and washers. The plates located in the LSM and tail sections were machined using 6 in diameter round bar stock, while the two avionics plates were cut using 6.5 in stock to accommodate the lips matching the outer diameter of the rocket. The LSM and payload bulk plates will be covered in further detail in Sections 4.2.1 and 4.2.6, respectively.

The avionics bulk plates are responsible for attaching the avionics bay to the drogue and main parachutes. For ease of manufacturing and integration, the fore and aft avionics bulk plates are identical. The schematic of the as-machined avionics bulk plates is shown in Figure 14. As mentioned, 6.5 in aluminum 6061 round bar stock was used for these plates and was cut using a CNC machine. The radial bolt holes were then drilled and tapped by hand. Figure 15 shows one of the avionics bulk plates during the machining process.

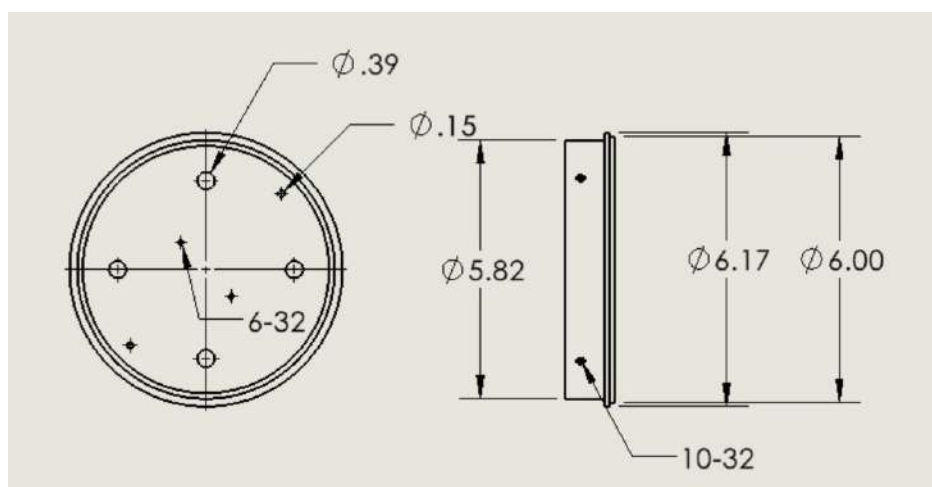


Figure 14: Schematic of the as-built avionics bulk plate.

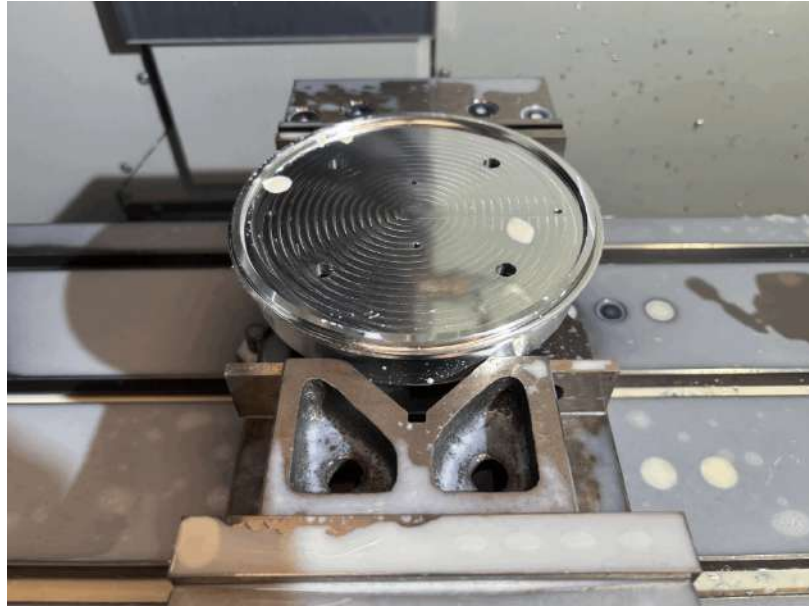


Figure 15: Avionics bulk plate during machining.

The tail bulk plate is responsible for attaching the tail section to the main parachute. The final fabricated bulk plate schematic is shown in Figure 16. As mentioned, the bulk plate was machined out of 6 in aluminum 6061 round bar stock using a CNC machine. Like for the avionics bulk plates, the radial bolt holes were drilled and tapped by hand. The bulk plate is retained within the tail section is shown in Figure 17.

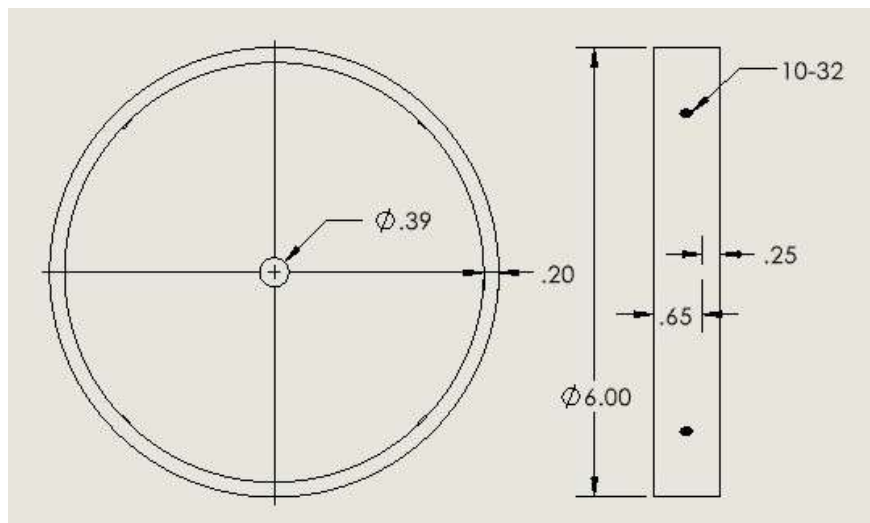


Figure 16: Dimensioned drawing of the as-built tail bulk plate.

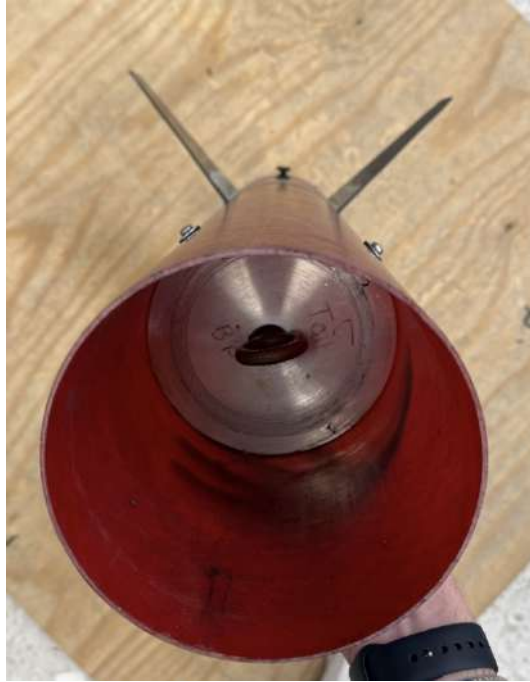


Figure 17: Tail bulk plate assembled with the airframe and anchoring hardware.

3.2 Recovery Subsystem

The recovery subsystem is paramount for ensuring the safety and reusability of the launch vehicle. Figure 18 provides an overview of the recovery sequence. The recovery mission begins at apogee with the deployment of the 18” Fruity Chutes Classic Elliptical drogue parachute. At 750 ft, the 96” Fruity Chutes Iris Ultra main parachute is deployed, further slowing the vehicle down. At 300 ft, pending RSO approval, the payload is jettisoned and descends under its own 60” Fruity Chutes Iris Ultralight. Recovery parameters have been carefully selected to ensure reliability and compliance with NASA and VADL requirements and are summarized in Table 6.

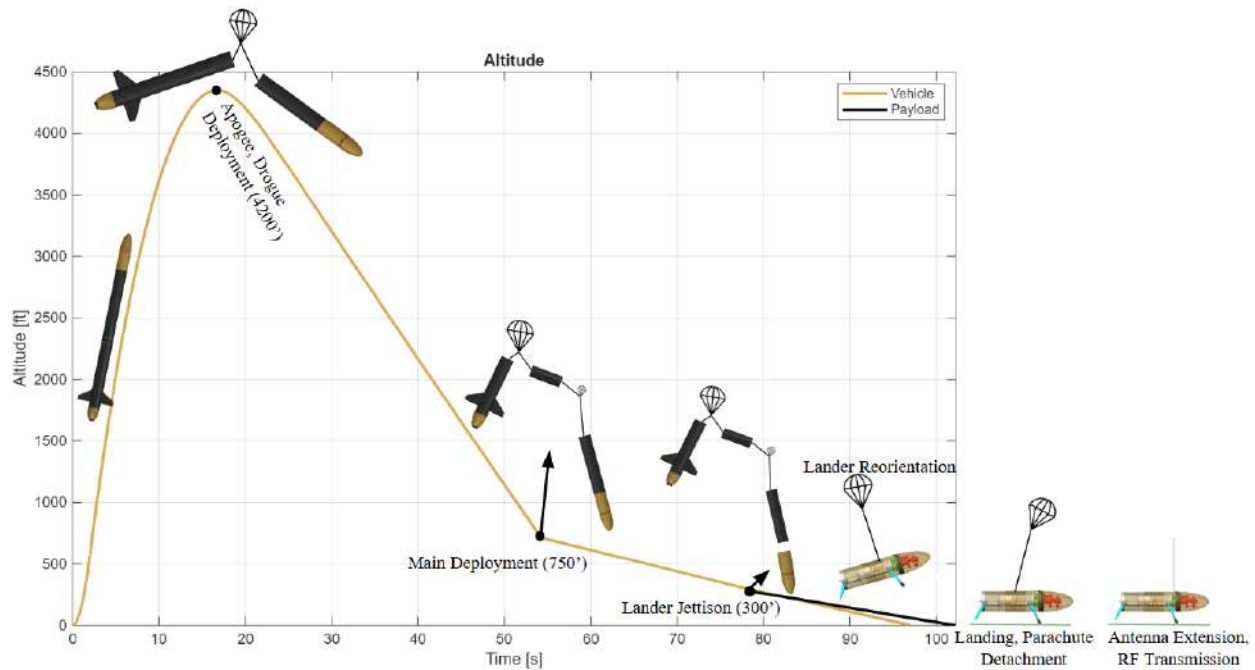


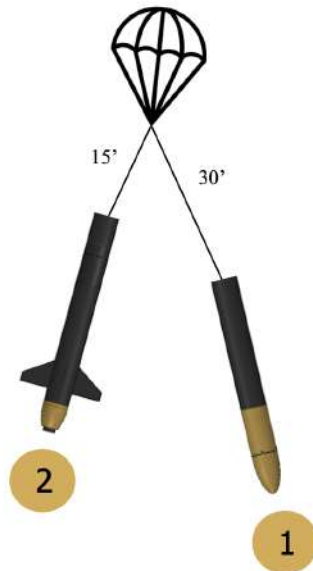
Figure 18: Schematic showing the recovery CONOPs with appropriate altitudes.

Table 6: Recovery system parameters.

Parameter	Drogue Parachute	Main Parachute	Lander Parachute
Diameter	18"	96"	60"
Brand	Fruity Chutes	Fruity Chutes	Fruity Chutes
Type	Elliptical	Iris Ultra	Iris Ultralight
Deployment Altitude	Apogee	750 ft	300 ft
Backup Deployment	Apogee + 1s	700 ft	N/A
Material	Nylon	Nylon	Nylon
Drag Coefficient	1.5	2.2	2.2
Terminal Velocity	101 fps	17.2/15.7 fps	11.7 fps
Shock Cord Lengths	30/15 ft	20/15	20 ft
Shock Cord Thickness	1/2"	1/2"	1/8"

The descent configurations under drogue and main parachutes are depicted in Figure 19. Since CDR, several changes have been made to the shock cord lengths. The fore/aft drogue shock cords have been shortened from 40/20 ft to 30/15 ft. The longer lengths were originally chosen with the aim of minimizing swinging of the payload section prior to jettison. However, numerous drop tests with induced swinging have shown that the payload reliably deploys and descends smoothly even with higher than realistic swing angles. As such, the lengths were shortened to decrease the loading experienced at full shock cord extension. Further, the tail to main shock cord was increased from 10 to 15 ft due to concerns that the 10 ft cord was not sufficiently long for the 3 ft section it is supporting. The altered cord lengths were all flown and fully verified during the Vehicle Demonstration flights.

Drogue Descent



Main Descent

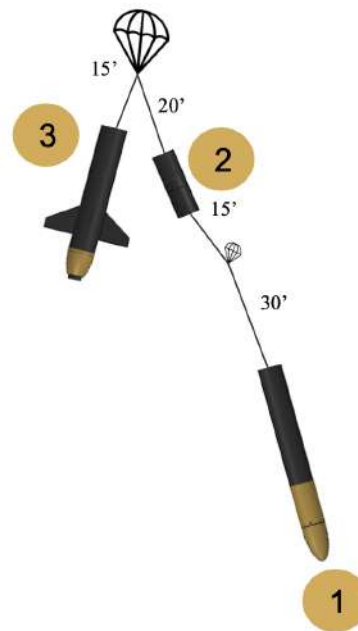


Figure 19: Descent configurations under drogue and main parachutes.

3.2.1 Attachment Hardware

Figure 20 provides a visual overview of the attachment hardware for the drogue and main recovery sections. 1200 lb quick links are used at all connection points. Fireballs are placed at each airframe edge to prevent zippering, fixed in place via zip ties. Shock cords are rated for 6000 lbf, and their lengths were selected such that no sections collide during descent. Additionally, the placement of the main parachute aft of the drogue parachute was done to ensure that the nose cone is the lowest section during descent to facilitate payload jettison. 3/8"-16 eye bolts are used as the anchoring hardware at each recovery bulk plate, selected to ensure sufficient safety factors to withstand the force experienced during main parachute deployment. Last, each parachute contains a swivel link to decouple from the vehicle sections and is wrapped in a Nomex fire blanket to prevent damage from the blast events. The structural integrity of the entire hardware network was verified through two Vehicle Demonstration flights, providing confidence in the system going forward to the Payload Demonstration flight and competition.



Figure 20: Recovery hardware for the vehicle recovery sections.

3.2.2 Parachutes

Proper parachute selection is essential for a safe recovery of the vehicle and compliance with NASA requirements 3.3, 3.11, and 3.12 regarding landing kinetic energy, drift, and descent time, respectively. The parachute selections and descent times are summarized in Table 7 below. The drogue and main parachutes remain unchanged from CDR and were confirmed to perform as intended during the Vehicle Demonstration flights. The lander parachute, meanwhile, has been switched from an Iris Ultra to Iris Ultralight. This switch was motivated primarily by a desire to save mass while maintaining performance and desired descent characteristics. Further, the significantly smaller packing volume of the Ultralight allowed the parachute restraint tube and, subsequently, the LSM bay to be shortened, further reducing mass. The performance of this parachute was verified during the March 1 drop tests, covered in further detail in Section 4.1.3.

Table 7: Parachute selections and descent velocities.

Parachute	Type	Descent Velocity (fps)
Drogue Parachute	18" Fruity Chutes Classic Elliptical	101
Main Parachute (Pre-Jettison)	96" Fruity Chutes Iris Ultra	17.2
Main Parachute (Post-Jettison)	96" Fruity Chutes Iris Ultra	15.7
Lander Parachute	60" Fruity Chutes Iris Ultralight	11.7

3.2.3 Avionics

The avionics system remains largely unchanged from the configuration presented in CDR, with only minor modifications made to improve assembly efficiency during launch day preparations. These modifications primarily focus on optimizing accessibility to critical components and simplifying wiring configurations to streamline the pre-flight setup process.

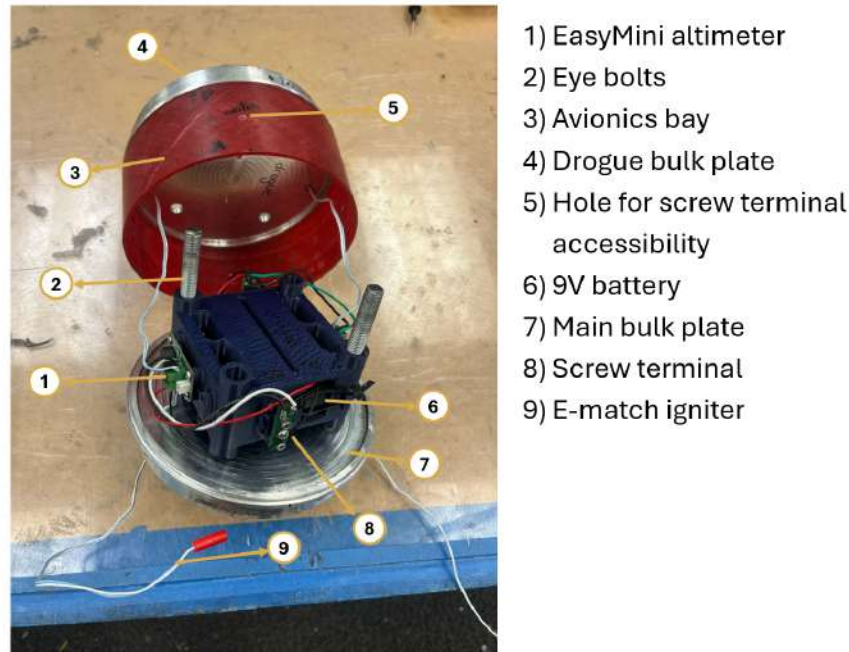


Figure 21: Avionics and Avionics Bay assembly and components.

The system retains the EasyMini altimeters as the primary and backup deployment controllers. These altimeters have demonstrated consistent accuracy and reliability in previous test launches, as further discussed in Section 5.2. The dual altimeter configuration ensures redundancy, with each altimeter powered by a dedicated and isolated 9V battery to limit single point failures.



Figure 22: Assembled avionics bay.

The equation used to determine blast charge masses is shown below. This calculation method was proven to be accurate through successful deployment tests and full-scale flights.

$$m = \frac{SF \cdot P_{\text{burst}} \cdot A_{\text{bulkplate}} \cdot L_{\text{bay}} \cdot g_{\text{lb}}}{T_{\text{burn}} \cdot R_{\text{BP}}} \quad (1)$$

Specific values used for the parameters and calculations are included in Table 8.

Table 8: Key rocket separation parameters.

Parameter	Variable Name	Value
Burst Pressure for Shear Pins	P_{burst}	4.693 psi (Drogue) / 4.775 psi (Main)
Area of Bulkplate	$A_{\text{bulkplate}}$	26.421 in ² (Drogue) / 25.967 in ² (Main)
Parachute Bay Length	L_{bay}	13.5 in (Drogue) / 15 in (Main)
Safety Factor	SF	4
Temperature of Burning Gunpowder	T_{burn}	3307 R
Gunpowder Gas Constant	R_{BP}	265.92 lbf·in/lbm·R
Conversion Factor (grams per pound)	g_{lb}	453.592 g/lb

For drogue parachute:

$$m_{\text{drogue}} = \frac{(4)(4.693)(26.421)(13.5)(453.592)}{(3307)(265.92)} \quad (2)$$

$$m_{\text{drogue}} = 3.45 \text{ g} \quad (3)$$

For main parachute:

$$m_{\text{main}} = \frac{(4)(4.775)(25.967)(15)(453.592)}{(3307)(265.92)} \quad (4)$$

$$m_{\text{main}} = 3.84 \text{ g} \quad (5)$$

Both backup charges have an additional 1.2 factor of safety multiplied by the main charges to ensure separation. The summary of all blast charge parameters can be found in Table 9 below.

Table 9: Blast charge parameters.

Parameter	Drogue Parachute	Main Parachute
Primary Blast Charge (g)	3.5	3.8
Primary Deployment	Apogee	750 ft
Backup Blast Charge (g)	4.2	4.6
Backup Deployment	Apogee + 1s	700 ft

Before the vehicle demonstration flights, the blast charge masses shown above were tested in the Recovery Deployment Test (see Section 7.1.2).

3.2.4 Electrical Elements

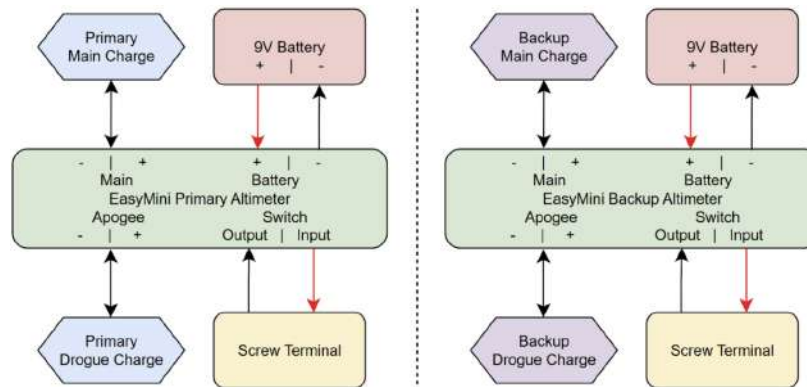


Figure 23: Avionics electronics diagram.

Electrical components used in the recovery subsystem remain mostly unchanged since CDR. The avionics electronics will continue to use the EasyMini Altimeters and 9V batteries as shown in Figure 23.

The LSM electronics, further detailed in Section 4.2.1, continue to use a Teensy 4.1 microcontroller to handle the logic of lander separation when RSO approval is granted and when the altitude drops below 300 feet. The VN100 Rugged IMU is used to detect altitude, the TBS Diversity Nano Rx is used to receive the RSO signal, a 3-cell LIPO battery supplies the 11.1V used to power the latch, a Pololu 5V Step-Down Voltage Regulator supplies power to the Teensy and other devices, and two SouthCo EM Latches connect the payload to the rocket.

The software implements a **finite-state machine** with three main states: **Pre-Launch, Post-Launch, and Post-Apogee.**

- **Pre-Launch State:** This state is entered when the **Teensy** is powered on, and the setup code runs. During this phase, the system:
 - Defines the **ground-level altitude.**
 - Initializes the **pins, serial ports, and SD card.**
- **Post-Launch State:** The system transitions from **Pre-Launch** to **Post-Launch** when the **IMU detects an acceleration of at least 3 G's for 0.5 seconds.**
- **Post-Apogee State:** This state is triggered when the **IMU detects no new maximum altitude for 1 second.** In this state:
 - The **Teensy checks the Primary RSO approval pin**, which goes high when approval is granted.
 - If **RSO approval is granted and the altitude is below 300 ft**, the software executes a conditional statement that **unlatches the latches**, jettisoning the payload.

3.2.5 GPS Transmitters

To comply with NASA requirement 3.13, there are two BigRedBee GPS/APRS Transmitters in the rocket. The one in the main body is attached to the LSM electronics bay; the other is on top of the payload battery bay. These transmitters are powered by their own 3.7 volt, 1 cell LIPO, which will power the device for over 8 hours, well above the NASA requirement 2.6 minimum of 3 hours. To avoid any RF interference with other teams, the GPS transmission frequency is being adjusted from the default value of 433.92 MHz to 426.82 MHz.

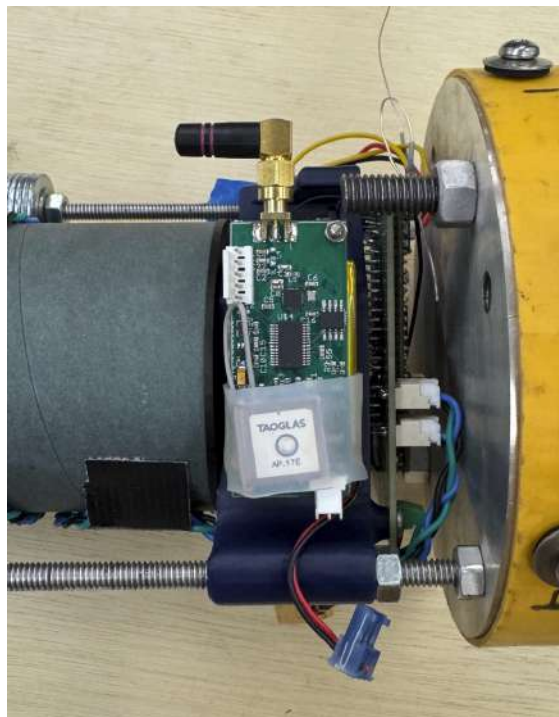


Figure 24: GPS transmission.

During the full-scale launch, these devices were configured to store GPS data on their onboard flash at its fastest sample rate, 1 Hz, and to transmit the data every 5 seconds. These transmissions were offset using a time-slotting function that can be programmed on the devices so that there is a 3 second offset from the payload GPS transmission and the rocket GPS transmission. As seen in Figure 25 below, taken from vehicle demonstration flight 1, the devices were also configured to transmit each package with a call sign and identifying number. The call sign belongs to one of the team members and is KF8DFX. The identifying number for the rocket is “-1” and the identifying number for the payload is “-2”. Coordinates are transmitted in degrees and decimal minutes, where the first two latitude digits are the degrees and the first three longitude digits are degrees while the rest of the digits represent the decimal minutes. The “A” value represents altitude in feet.

```
AFSK1200: fm KF8DFX-2 to APBL10-0 UI^ pid=F0
!3642.64N/08659.75W*/A=000607*
AFSK1200: fm KF8DFX-1 to APBL10-0 UI^ pid=F0
!3642.64N/08659.75W-/A=000607*
AFSK1200: fm KF8DFX-2 to APBL10-0 UI^ pid=F0
!3642.64N/08659.75W*/A=000610*
AFSK1200: fm KF8DFX-1 to APBL10-0 UI^ pid=F0
!3642.64N/08659.75W-/A=000607*
AFSK1200: fm KF8DFX-2 to APBL10-0 UI^ pid=F0
!3642.63N/08659.75W*/A=000610*
AFSK1200: fm KF8DFX-1 to APBL10-0 UI^ pid=F0
!3642.63N/08659.75W-/A=000607*
```

Figure 25: GPS Transmission Screenshot

During vehicle demonstration flight one, transmissions were being received infrequently after launch. This has been identified as a receiver issue. The ground station for GPS APRS reception on launch day consisted of a laptop, an RTL-SDR USB Dongle, and a 6 inch rubber duck antenna. This 6 inch rubber duck antenna was inconsistent in receiving packets once the rocket traveled far from the ground station. This antenna will be replaced with a yagi antenna, which will require further testing of signal strength. This was not an issue in full-scale flight two, as the rocket landed closer to the ground station and transmission was successful.

3.3 Mission Performance Predictions

VADL uses flight simulations to inform design decisions in alignment with safety requirements imposed by NASA and VADL. FRR simulations are used to reaffirm the **as-built rocket** will function as expected and to analyze the effects of the launch day environmental conditions. Validation of VADL simulations is achieved by cross-referencing results from RockSim, OpenRocket, and VADL’s in-house Declarative Rocket Simulation Software (DRSS; [5]). Extensive flight data analysis (Section 5.3) obtained from launch testing has provided a definitive benchmark that verifies VADL’s theoretical frameworks. Unless noted otherwise, all simulations performed in this section will utilize the Competition Day conditions and as-built vehicle measurements detailed in Table 10.

Table 10: Competition day simulation parameters with as-built vehicle measurements.

Launch Site	Huntsville, AL
Wet Mass (lbm)	42
Wet CG (in)	53.7
CP (in)	66.7
SSM on Pad	2.12
Ground Wind Speed (mph)	11
Launch Rail Angle (deg)	5
Launch Site Elevation (ft)	600
Launch Site Temp. (F)	90
Ballast (lbm)	2.3
Effective Launch Rail Travel (in)	132

3.3.1 Flight Profile Simulations and Relevant Requirements

Flight trajectory simulations guide design decisions concerning safety, material and hardware selection, and overall vehicle design. A successful mission must satisfy requirements imposed by NASA and VADL, as detailed in Section 7.2 and summarized in Table 11.

Table 11: Relevant NASA and VADL Requirements.

Requirement ID	Requirement
NASA 2.1	The vehicle shall deliver the payload to an apogee altitude between 4,000 and 6,000 feet above ground level (AGL). Teams flying below 3500 feet or above 6500 feet on their competition launch will receive zero altitude points toward their overall project score and will not be eligible for the Altitude Award.
NASA 2.14	The launch vehicle shall have a minimum static stability margin of 2.0 at the point of rail exit.
NASA 2.15	The launch vehicle shall have a minimum thrust to weight ratio of 5.0 :1.0.
NASA 2.17	The launch vehicle shall accelerate to a minimum velocity of 52 fps at the rail exit.
VADL 2.1.3	The full scale and subscale launch vehicles shall have an average Thrust-to-Weight ratio of 5.5 minimum.
VADL 2.3.5	Flight simulations in the case of denied jettison shall be conducted.
VADL 3.2	In the event of failed jettison, the descent velocity of the vehicle shall not exceed 20 fps upon landing.
VADL 3.11	The terminal descent velocity under drogue parachute of the rocket shall not exceed 115 fps.

Continued on the next page

Requirement ID	Requirement
NASA 3.3	Each independent section of the launch vehicle shall have a maximum kinetic energy of 75 ft-lbf at landing. Teams whose heaviest section of their launch vehicles, as verified by vehicle demonstration flight data, stays under 65 ft-lbf will be awarded bonus points.
NASA 3.11	The recovery area shall be limited to a 2500 ft radius from the launch pads.
NASA 3.12	Descent time of the launch vehicle shall be limited to 90 seconds (apogee to touch down). Teams whose launch vehicle descent, as verified by vehicle demonstration flight data, stays under 80 seconds will be awarded bonus points.

Four Competition Day scenarios are considered to evaluate safety compliance and assess overall mission performance. Unless stated otherwise, the predicted values provided below are based on the VADL-defined baseline conditions: an **11 mph headwind and a 5° launch rail angle**.

- **Scenario A: Official Launch Conditions.** This scenario describes expected launch-day conditions at the official event location in Huntsville, AL. The team's designated **Official Competition Launch Target Altitude** is 4200 ft.
- **Scenario B: Denied Jettison Scenario.** This scenario assesses the effects of the RSO denying payload jettison, which necessarily leads to increased landing masses and kinetic energies. Simulations validate compliance with NASA/VADL safety standards despite these increased risks.
- **Scenario C: No-Wind Scenario.** This scenario evaluates performance under conditions of zero wind, verifying adherence to maximum descent duration and relevant recovery safety parameters.
- **Scenario D: High Wind Conditions.** This scenario addresses a higher wind environment (20 mph) to examine if the rocket maintains compliance with specified recovery radius requirements and related safety constraints.

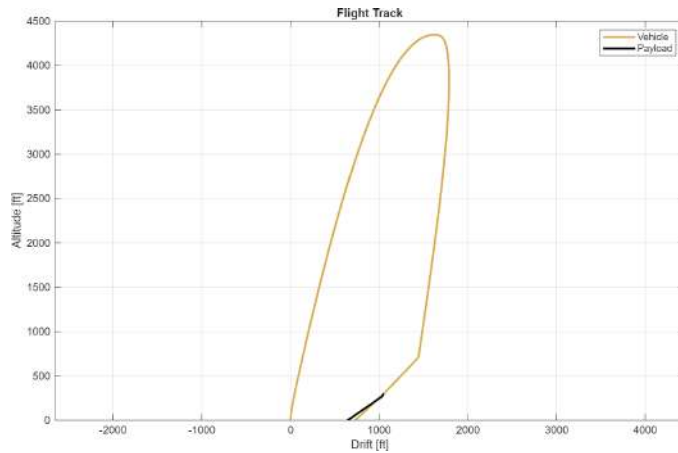
Table 12 outlines the performance outcome metrics of these scenarios; VADL verifies all scenarios are within the vehicle safety and recovery requirements set by NASA and VADL.

Note: Landing velocities differ from pre-flight VDF predictions (Section 5.3) due to expected higher ambient temperatures and correspondingly lower air densities on Competition Day (Table 10). MPP simulations use a main parachute deployment altitude of 750 ft instead of the VDF's 600 ft to match the intended VADL Competition Day parameters.

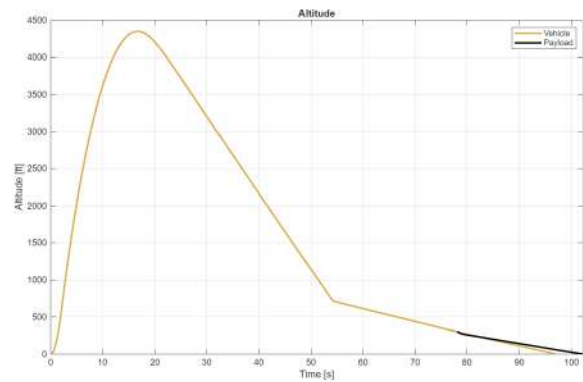
Table 12: Flight performance summary for the as-built vehicle across the evaluated scenarios. Values complying with NASA/VADL requirements are highlighted in green.

Predicted Values and Relevant Requirements.	Scenario A (Baseline)	Scenario B	Scenario C	Scenario D
Static Stability Margin at LRE (NASA 2.14; cal)	2.14			
Static Stability Margin on Pad (VADL 2.1.2; cal)	2.12			
Instantaneous Thrust-to-Weight Ratio at LRE (NASA 2.15; VADL 2.1.3)	8.5 to 1			
Rail Exit Velocity (NASA 2.17; fps)	DRSS: 76.0 RockSim: 75.0 OpenRocket: 75.1			
Apogee (AGL; NASA 2.1; ft)	DRSS: 4349 RockSim: 4509 OpenRocket: 4053 Average: 4300		DRSS: 4534 RockSim: 4478 OpenRocket: 4152 Average: 4390	DRSS: 4100 RockSim: 4494 OpenRocket: 3936 Average: 4180
Vehicle Landing Velocity / Descent Velocity Under Main Parachute (VADL 3.2; fps)	15.7	17.2	15.7	15.7
Descent Velocity Under Drogue Parachute (VADL 3.11; fps)	101		101	101
Payload Landing Velocity / Descent Velocity Under Payload Parachute (VADL 4.1.2; fps)	11.7	N/A	11.7	11.7
Vehicle Drift (descent time × wind speed; NASA 3.11; ft)	1293	1266	0	2287
Payload Drift (descent time × wind speed; NASA 3.11; ft)	1379	N/A	0	2444
Vehicle Descent Time after Apogee (NASA 3.12; s)	80.2	78.5	82.0	78.0
Payload Descent Time after Apogee (NASA 3.12; s)	85.4	N/A	87.3	83.3
Sectional Landing Kinetic Energies (NASA 3.3; lb-ft)	Fore: 32.6 Mid: 47.7 Aft: 39.8 Payload: 13.9	Fore: 69.9 Mid: 57.7 Aft: 48.2 Payload: N/A	Fore: 32.6 Mid: 47.7 Aft: 39.8 Payload: 13.9	Fore: 32.6 Mid: 47.7 Aft: 39.8 Payload: 13.9
Sectional Kinetic Energies under Drogue Parachute (lb-ft)	Fore: 2404 Aft: 3643		Fore: 2404 Aft: 3643	Fore: 2405 Aft: 3642

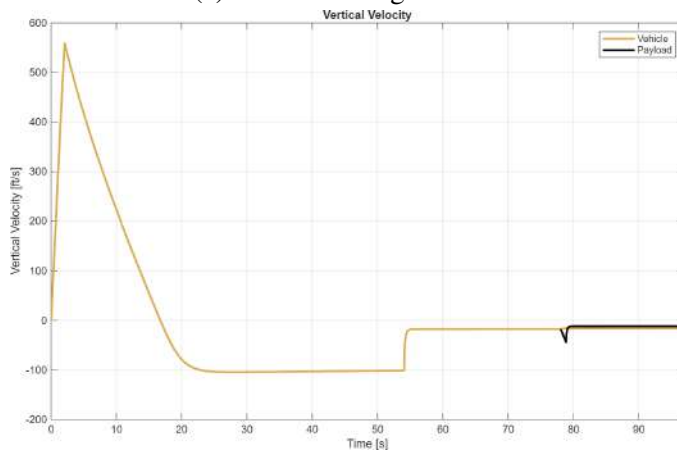
The flight profile for the baseline **Scenario A** scenario are presented below; due to VADL's customized vehicle descent methodologies, the figures below were produced by DRSS.



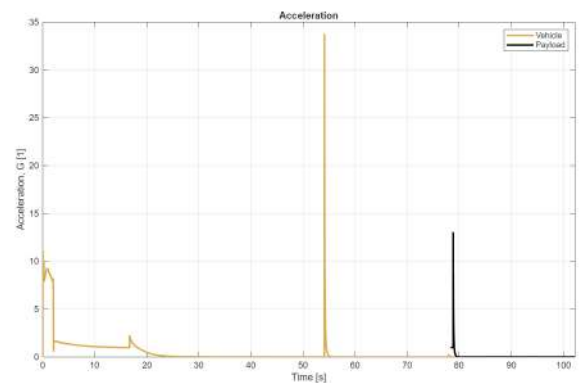
(a) Simulated Flight Track.



(b) Simulated Altitude (apogee = 4349 ft).



(c) Simulated Velocity (max = 560 fps).



(d) Simulated Acceleration.

Figure 26: Scenario A (Baseline) Flight profiles.

Comparison with RockSim/OpenRocket

RockSim and OpenRocket results are included for flight ascent metrics (e.g., velocity at LRE and predicted apogee) in the above flight profiles. As detailed extensively in VADL's CDR Milestone Report, this decision was primarily driven by VADL's highly customized descent modeling of transient parachute unfurling dynamics, and effects of vehicle angular swing on frontal and side drag profiles. VADL acknowledges that additional source-level modifications to the OpenRocket/RockSim software would be required to bring those third-party descent model into more precise alignment with the empirically validated refinements implemented in DRSS. The full RockSim and OpenRocket trajectory projections were conducted and are not graphically nor exhaustively presented here.

For vehicle ascent metrics, results from DRSS closely align with those from the industry standard software RockSim for Competition Day simulations (Table 12) and more so for the VDF pre-flight predictions (Sections 5.3.1 and 5.3.2).

Table 13: Sample comparison of predicted performance metrics using DRSS vs. OpenRocket vs. RockSim. A non-jettison, 0 mph wind condition was selected.

	Apogee (ft)	% Diff from DRSS	Max Vertical Velocity (fps)	% Diff from DRSS	Ascent Time (s)	% Diff from DRSS	Final Descent Rate (fps)	% Diff from DRSS
CFD-DRSS	4534	N/A	571	N/A	17.1	N/A	17.2	N/A
OpenRocket	4152	-8.4%	559	-2.1%	16.4	-4.1%	17.4	1.2%
RockSim	4478	-1.2%	563	-1.4%	17.1	-0.1%	17.4	1.2%

Concurrent with the VDF analysis and further discussed in later sections, the discrepancies in apogee predictions between DRSS and OpenRocket are a result of divergent estimations of vehicle drag coefficient. The DRSS model utilizes drag estimations supplied from Computational Fluid Dynamics (CFD) software, with detailed methodology further discussed in Sections 3.3.6 and 3.3.7.

Further modeling differences between DRSS and OpenRocket — such as numerical integration methods, wind modeling, aerodynamic theory, and other simulation assumptions—are addressed in Sections 3.3.2 through 3.3.7.

Section 3.3.6 verifies key elements of vehicle design via simulated and computed values obtained from multiple software tools (Solidworks, DRSS, and OpenRocket), which converge to verify against relevant NASA and VADL requirements to ensure mission safety.

Recovery Radius

The benefit of using DRSS for estimating drift over a simple multiplication of descent time and wind velocity is to assess the bounding of flight trajectory within recovery radius **at all times of flight**, in addition to final landing drift. The below figure shows the DRSS-generated flight tracks for wind conditions from 0 to 20 mph in both head and tail wind. **The purpose of drift simulation is not to produce a conclusive prediction on Competition Day drift but to provide another layer of verification to ensure recovery radius compliance.**

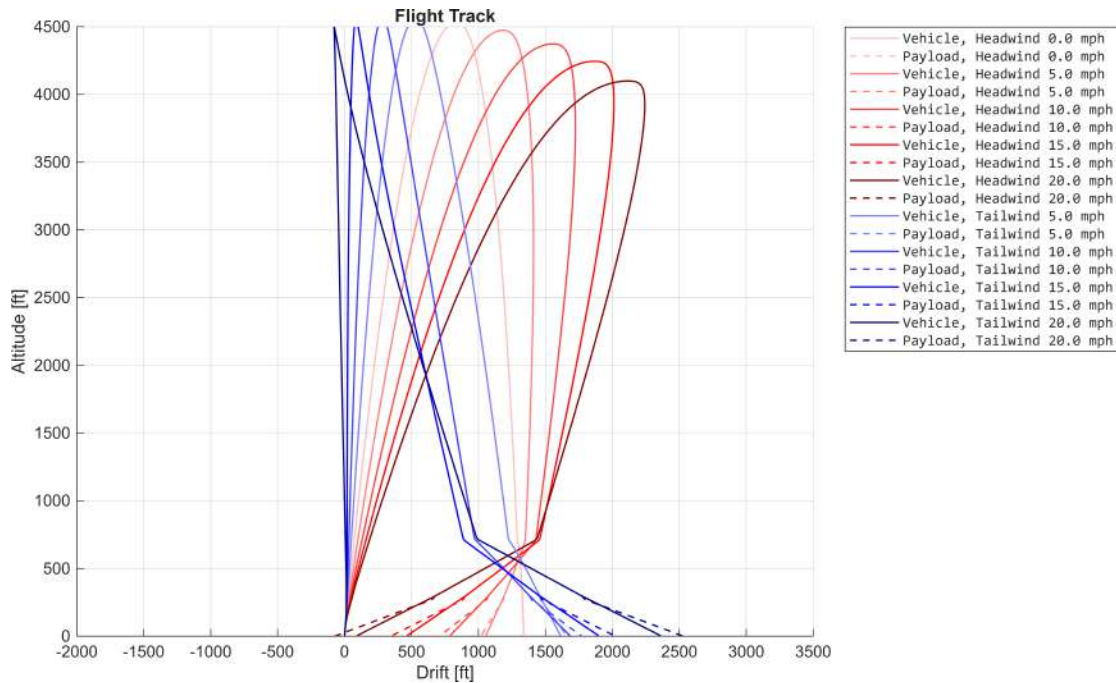


Figure 27: Flight Tracks of Select Wind Conditions for 5-Degree Launch Angle of Full-Scale Vehicle.

Table 14: Drifts at landing using the DRSS model.

Wind Speed (mph)	Payload Drift from Pad (ft)	Payload Drift from Apogee (ft)	Rocket Drift from Pad (ft)	Rocket Drift from Apogee (ft)	Rocket Descent Time (ceiled; s)
0	1343	507	1338	502	83
Headwind, 5	1018	168	1055	130	82
Headwind, 10	709	851	789	771	81
Headwind, 15	346	1531	468	1409	80
Headwind, 20	73	2197	94	2031	78
Tailwind, 5	1661	1129	1614	1082	83
Tailwind, 10	1771	1487	1683	1400	83
Tailwind, 15	2023	1937	1897	1812	82
Tailwind, 20	2529	2609	2360	2440	82

Figure 27 and Table 14 show that, under all headwind conditions, the trajectories of both the vehicle and the payload remain strictly within the 2500 ft recovery radius limit **throughout the entire flight duration and upon landing**. Additionally, the descent times of both the payload and the vehicle comply with NASA Requirement 3.12, pertaining to the maximum allowable descent time.

While the previously discussed model ensures trajectory bounds for all time during flight, it assumes a finite, non-zero time constant for the rocket to adapt to wind speed after reaching apogee. To calculate the maximum theoretical drift **at landing** and confirm compliance with NASA Requirement 3.11 regarding maximum recovery radius, Figure 28 and Table 15 present drift values obtained using a simplified approach. This method calculates drift by multiplying total descent time by wind speed, effectively assuming the rocket is positioned directly above the launch rail at

apogee and instantly aligns its descent velocity with wind speed (i.e., final drift downrange minus apogee downrange). These results are summarized below.

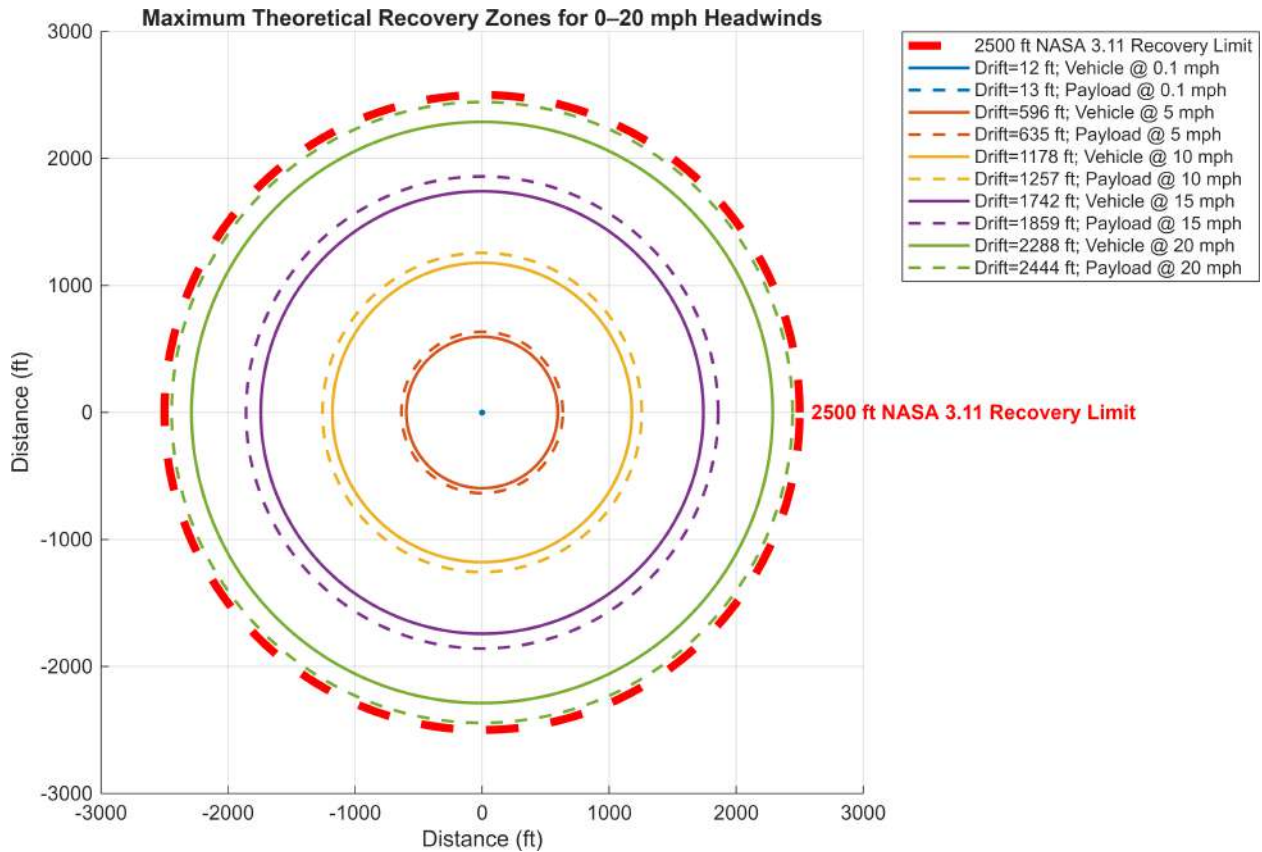


Figure 28: Maximum Theoretical Drift Conditions for 5-Degree Launch Angle of Full-Scale Vehicle.

Table 15: Maximum theoretical drifts ((Descent Time) $\times$ (Wind Speed)).

Wind Speed (mph)	Vehicle Descent Time (s)	Vehicle Recovery Radius (ft)	Payload Descent Time (s)	Payload Recovery Radius (ft)
0	82.1	0	87.4	0
5	81.3	596	86.6	635
10	80.3	1179	85.7	1257
15	79.2	1742	84.5	1859
20	78.0	2288	83.3	2444

Using both DRSS and the maximum theoretical drift methods, the above results verify compliance against NASA 3.11 and NASA 3.12 to ensure safe vehicle and payload recovery.

3.3.2 Numerical Integration Methodology

VADL’s in-house DRSS utilizes the **variable-step, variable-order, Adams-Bashforth-Moulton PECE solver of orders 1 to 13** to solve the differential equations describing kinematics and ki-

netics of the vehicle and payload. The new solver was implemented during VADL’s PDR Milestone. The maximum allowed integration step was 50 ms, the relative tolerance was limited to 0.0000001%, and the minimum integration step determined and utilized by DRSS to achieve this tolerance was ~**200 femtoseconds**. Optimization of DRSS allows computation of this low tolerance to be performed under 800 ms of runtime per flight simulation. In OpenRocket, the time step of 10 ms was used under the default 6-th order Runge-Kutta solver. In RockSim, 1000 samples per second were computed using its default Euler’s Method solver. VADL is confident that simulation resolution does not realistically impose any numerical artifacts that would have otherwise negatively impacted the accuracy and precision of the team’s simulations and analyses.

3.3.3 Parachute and Descent Dynamics

Parachute Unfurling Model

As extensively detailed in VADL’s CDR Milestone Report, empirical payload drop tests confirm that shock cords exceeding 20 ft unravel slower than parachutes under ~6 ft diameter. Unlike the PDR’s exponential unfurling model, the CDR and FRR methodology assumes the shock cord tightens when parachute unfolds, which essentially yields a conservative maximum instantaneous loading for structural analysis, airframe design, and mission predictions. This approach was validated through instrumented drop tests showing strong agreement between predicted and measured shock loads in the CDR Report, and is applied uniformly in this report for structural and safety assessments. DRSS simulations implement this by setting parachute opening time to negative infinity. Maximum theoretical deployment shock loads are summarized below.

Table 16: Simulated parachute deployment loading with as-built massing.

Parachute	Maximum Shock Loading G	Maximum Total Shock Cord Tension
Drogue	8.1	309 lbf
Main	31.4	1197 lbf
Payload	16.3	108 lbf

The above simulated loading were confirmed in the CDR Milestone Report to yield a Factor of Safety of >2.5 for the swivel link and >3.0 for other mission-critical recovery hardware such as the quick link, bulk plates, eye bolts and shock cords. As noted in Section 5, there were no structural damage in the fullscale vehicle after two consecutive Vehicle Demonstration Flights.

Drag Adjustments due to Swinging

As extensively detailed in VADL’s CDR Milestone Report, the subscale flight demonstrated that the drag induced by the rocket and parachute sides must account for the nondeterministic swinging motion of the rocket body during descent. To reflect this effect on drift and descent drag characteristics, the DRSS model for the full-scale vehicle has been revised using a refined methodology. This updated approach considers partial exposure of the parachute canopy to lateral wind forces as a function of the rocket body’s angular swinging.

It should be emphasized that the MPP drift calculation ultimately employs the more conservative approach, to verify compliance with NASA Requirement 3.11 regarding recovery radius (see previous subsection on Recovery Radius). Conversely, the DRSS model adjustments accounting for drag due to vehicle swinging primarily serve to enhance estimation of the flight track profile (Figure 26a) and provide additional validation by extending the interpretation of Requirement 3.11 to include the entire flight trajectory (i.e., for all times during flight), not just the final landing position (Figure 27).

Additionally, modifications have been incorporated into the vehicle's frontal drag modeling to account for the vertical exposure of the vehicle's side profile resulting from significant angular swinging.

As described in VADL's CDR Milestone Report, theoretical scaling analyses involving blast charge chemical potential energy, shock cord design, and rocket inertial characteristics indicate an approximately 40% reduction in angular swinging when scaling from subscale to fullscale vehicles. Dual deployment design and longer descent time on drogue descent additionally dissipates energy relative to the subscale vehicle. This reduction has been appropriately factored into the updated modeling approach described above.

3.3.4 Launch Rail Dynamics

Theoretical modeling and analyses conducted by Flanigan et al. [8] demonstrate that launch rail friction may lead to a reduction of up to **4% in the delivered impulse of a rocket motor**, particularly when the vehicle is exposed to crosswind conditions, which increase the normal forces between the rocket and the launch rail. Additionally, **NFPA regulations permit a total impulse variation of $\pm 20\%$** for manufactured rocket motors under standard room conditions, thereby complicating precise apogee predictions.

Historical IMU flight data collected by VADL for the L1720 motor and similar motors consistently indicate impulse delivery losses, causing deviations of up to 5% from manufacturer-specified values. To account for these observed variations, a conservative friction factor of 3.5% in motor thrust was applied across all VADL flight simulations within this report. This adjustment has been empirically validated through the close agreement between theoretical predictions and recorded apogee during VADL's subscale flight testing (See CDR Milestone Report) and the VDF (Section 5.3). The adjustment was implemented by uniformly scaling down the nominal L1720 thrust curve by 3.5% within both DRSS, RockSim and OpenRocket simulations.

The adjusted thrust curves for the L1720 motor, obtained using the described methodology, are presented below.

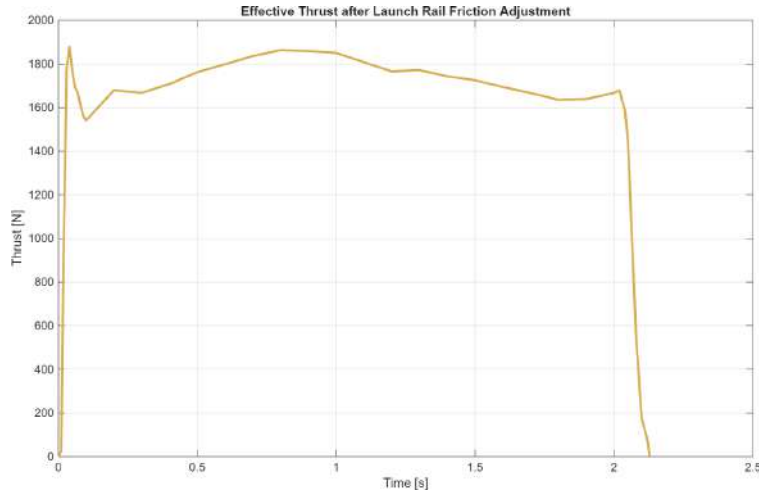


Figure 29: L1720 Adjusted Thrust Curve.

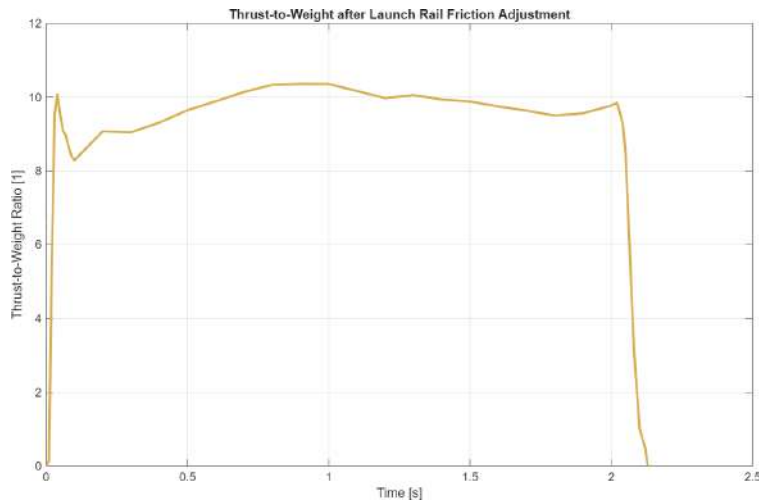


Figure 30: L1720 Adjusted Thrust-to-Weight Ratio.

3.3.5 Environmental Modeling

Planet Boundary Layer

DRSS models wind speed as a function of the rocket's altitude above surface level (ASL) by accounting for viscous friction within a planetary atmosphere boundary layer. The wind speed as a function of altitude is computed using Equation 6:

$$w = w_r \left(\frac{z}{z_r} \right)^{\frac{1}{\xi}} \quad (6)$$

w is the local wind speed at z height above ground, w_r is the reference wind speed at a reference altitude z_r , and ξ is a dimensionless parameter representing atmospheric stability. For these simulations, the reference altitude z_r corresponds to the launch site altitude of 600 feet ASL, and w_r is the specified wind speed at this altitude. ξ is set to 7 for neutral stability conditions.

For example, a base wind speed w_r of 10 mph at 800 feet and a target altitude z of 4800 feet yield the computed wind speed as follows:

$$w = 10 \left(\frac{4000}{800} \right)^{\frac{1}{7}} \approx 12.9 \text{ mph}$$

Air Density Variation

Variation in air density as a function of altitude ASL is modeled using Equation 7:

$$\rho(z) = \frac{\rho_0}{R(T_0 - \beta z)} \left(1 - \frac{\beta z}{T_0} \right)^{\frac{g_0}{R\beta}} \quad (7)$$

T_0 is the reference temperature at the launch site, ρ_0 is the air density at the reference altitude, R is the specific gas constant for air, g_0 is the acceleration due to gravity, and z is the altitude. A linear temperature lapse rate, β , is assumed due to the flight occurring within the troposphere. For the simulations, $\beta = 6.50 \times 10^{-3} \text{ K/m}$, $R = 287 \text{ J/(kg}\cdot\text{K)}$, $T_0 = 288.15 \text{ K}$, and $\rho_0 = 1.225 \text{ kg/m}^3$.

Motor Exhaust Friction

Before MECO, the inertia of the propellant exhaust against the motor housing induces rotational damping, generating a moment opposing the rocket's tilting motion [4]. This behavior is characterized by Equation 8, where $\dot{m}$ represents the propellant mass flow rate, $\dot{\theta}$ is the angular velocity, L denotes the total rocket length, and L_{CG} is the distance from the nose cone tip to the rocket's center of gravity (CG). Thrust data utilized in the simulation are extracted from reference [3] in CSV format, with linear interpolation employed to calculate thrust values at each simulation time step.

$$M = \dot{m}\dot{\theta}(L - L_{CG})^2 \quad (8)$$

For example, assume the following parameters for a powered ascent scenario:

- $\dot{m} = 2 \text{ kg/s}$ (propellant mass flow rate)
- $\dot{\theta} = 0.1 \text{ rad/s}$ (angular velocity)
- $L = 2.5 \text{ m}$ (rocket length)
- $L_{CG} = 1.5 \text{ m}$ (distance to CG)

Substituting these values into Equation 8:

$$M = 2 \cdot 0.1 \cdot (2.5 - 1.5)^2 = 0.2 \text{ Nm}$$

The resulting moment would be 0.2 Nm under this scenario, which would provide a thrust damping effect during powered ascent.

Air Compressibility

Compressibility effects on aerodynamic coefficients become non-negligible when the Mach number (M) exceeds approximately 0.3. These effects are modeled using Equation 9, which accounts for the adjustment of the baseline coefficient C due to air compressibility:

$$C' = \frac{C}{\sqrt{1 - M^2}} \quad (9)$$

Here, C' represents the adjusted coefficient, C is the baseline coefficient, and M is the Mach number.

For instance, assume the following parameters for a scenario where compressibility effects are considered:

- $C = 0.5$ (baseline aerodynamic coefficient)
- $M = 0.4$ (Mach number)

Substituting these values into Equation 9 yields:

$$C' = \frac{0.5}{\sqrt{1 - (0.4)^2}} = \frac{0.5}{\sqrt{0.84}} \approx \frac{0.5}{0.9165} \approx 0.595$$

The adjusted coefficient C' is approximately 0.595, which has non-negligible difference against the nominal $C = 0.5$ value due to air compressibility effects at $M = 0.4$.

Gravity Modeling

The team's declared apogee is approximately 4200 ft. Gravity decreases non-negligibly with increasing altitude as the distance from the Earth's center grows. This is characterized using Equation 10. g_0 is the standard gravitational acceleration at sea level, r_E is the Earth's radius, and z is the altitude above sea level.

$$g = g_0 \left(\frac{r_E}{r_E + z} \right)^2 \quad (10)$$

Assume the following parameters for a scenario considering altitude effects on gravity:

- $g_0 = 9.80665 \text{ m/s}^2$ (gravitational acceleration at sea level)
- $r_E = 6371 \text{ km}$ (Earth's radius)
- $z = 3 \text{ km}$ (altitude)

Substituting these values into Equation 10:

$$g = 9.80665 \left(\frac{6371}{6371 + 3} \right)^2 \approx 9.794 \text{ m/s}^2$$

, which yields an approximately -0.13% decrease in gravitational pull. Within DRSS, the local gravity of the Alabama area of $9.7964m \cdot s^{-2}$ was applied (Department of Physics, Montana State University [1]). In OpenRocket, gravity modeling was additionally refined by specifying the approximate coordinates of the launch site.

3.3.6 Vehicle Verification

Mass Ballast Effects & Competition Day Variability

The primary purpose of including ballast in the vehicle is to provide a stable reference mass for simulations and ensure flight consistency as final manufactured hardware mass may fluctuate. A ballast mass of 2.3 lbm was flown during VDF to accommodate possible mass creep between the FRR Milestone and Competition Day. Concretely, if the final, fully assembled rocket weighs less than the VDF mass, then the difference in mass is supplemented by ballast. This ensures that the flight profile closely matches the model used for stability and apogee predictions. VADL does not plan to exceed the VDF mass, and a small margin — up to 0.5 to 1.0 lbm beyond 2.3 lbm — may not be expected on Competition Day due to final vehicle adjustments.

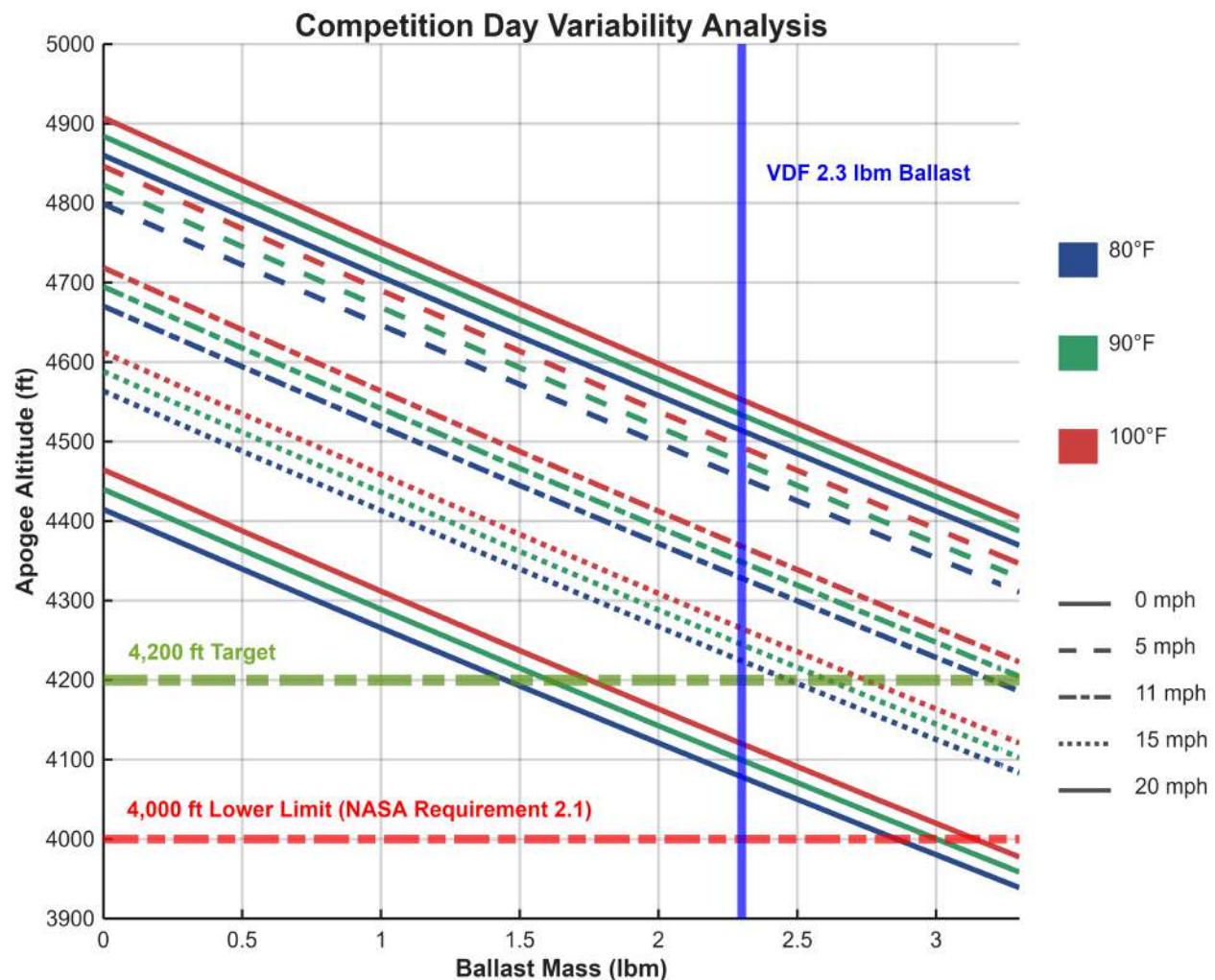


Figure 31: Competition Day environmental variability and its effect on predicted apogee.

The vehicle's apogee demonstrates predictable variation with ballast mass (Figure 31). Cesaroni Technology Incorporated guarantees a $\pm 5\%$ impulse response with the selected L1720 motor. VADL does not expect nor plan to exceed the VDF massing. In the worse-case, 20-mph wind scenario, the 2.3 lbm VDF massing provides 100 ft margin above the 4,000 ft lower limit (NASA Requirement 2.1) at 90°F. Competition day pre-flight simulations may be performed in-the-field to apply ballast and launch angle adjustments as deemed necessary. Figure 31 suggests a reasonable degree of robustness of the VDF massing and fullscale vehicle design such that sensitivity to Competition Day environmental conditions do not strongly suggest major deviations from the 4,200 ft declared target apogee nor possibility of flying below the 4,000 ft limit as set by NASA Requirement 2.1 under reasonable wind conditions.

Probabilistic Apogee Model

The analysis of interest here assumes competition-day ballast mass fixed at the VDF configuration:

$$B = 2.3\text{lbm} \quad (\text{VDF Massing}) \quad (11)$$

The environmental uncertainty model becomes:

$$\begin{aligned} W &\sim \mathcal{N}(\mu_W = \text{mean}, \sigma_W = \text{STD.}) \\ T &\sim \mathcal{N}(\mu_T = \text{mean}, \sigma_T = \text{STD.}) \\ B &\equiv 2.3, \text{lbm} \quad (\text{deterministic parameter}) \end{aligned} \quad (12)$$

The apogee system response surface $A(W, T; B)$ is constrained to the VDF mass configuration, reducing the system to a bivariate stochastic model as a function of: wind speed W and ambient temperature T . The environmental parameters are assumed to follow Gaussian distributions with typical values substituted below:

$$W \sim \mathcal{N}(\mu_W = 11, \text{mph}, \sigma_W = 3, \text{mph}), \quad T \sim \mathcal{N}(\mu_T = 90^\circ\text{F}, \sigma_T = 5^\circ\text{F}) \quad (13)$$

where the standard deviations represent majority coverage of expected competition-day conditions.

The system response surface $A(W, T)$ is constructed from $N = 1925$ discrete DRSS simulations spanning the parameter space. Statistical moments are computed via numerical quadrature:

$$\mathbb{E}[A] = \sum_{i=1}^N \sum_{j=1}^M A(w_i, t_j) \cdot f_W(w_i) \cdot f_T(t_j) \cdot \Delta w \Delta t \quad (14)$$

$$\sigma_A^2 = \sum_{i,j} [A(w_i, t_j) - \mathbb{E}[A]]^2 \cdot f_W(w_i) \cdot f_T(t_j) \cdot \Delta w \Delta t \quad (15)$$

where $\Delta W = 2 \text{ mph}$, $\Delta T = 5^\circ\text{F}$ are grid spacings, and f_W, f_T are the probability density functions (PDFs) of wind and temperature, respectively.

Threshold exceedance probabilities employ indicator functions:

$$P(A \geq A_0) = \sum_{i,j} \mathbb{I}_{A(w_i, t_j) \geq A_0} \cdot f_W(w_i) \cdot f_T(t_j) \cdot \Delta w \Delta t \quad (16)$$

A tri-linear interpolation scheme ensures continuous evaluation between grid points while maintaining conservation of probability mass. Numerical stability is confirmed through Richardson extrapolation with relative error <0.5%.

$$\mathbb{E}[A] = 4334 \text{ ft} \quad (3\% \text{ above target})$$

$$\sigma_A = 69.7 \text{ ft}$$

$$P(A \geq 4000, \text{ft}) > 0.99 \quad (3\sigma \text{ margin to requirement})$$

$$P(A \geq 4200, \text{ft}) \approx 0.97 \quad (\text{Declared target apogee})$$

$$P(A \geq 4400, \text{ft}) \approx 0.22 \quad (\text{Conservative performance cap})$$

$$P(A \geq 4600, \text{ft}) < 0.01 \quad (\text{Physically infeasible regime})$$

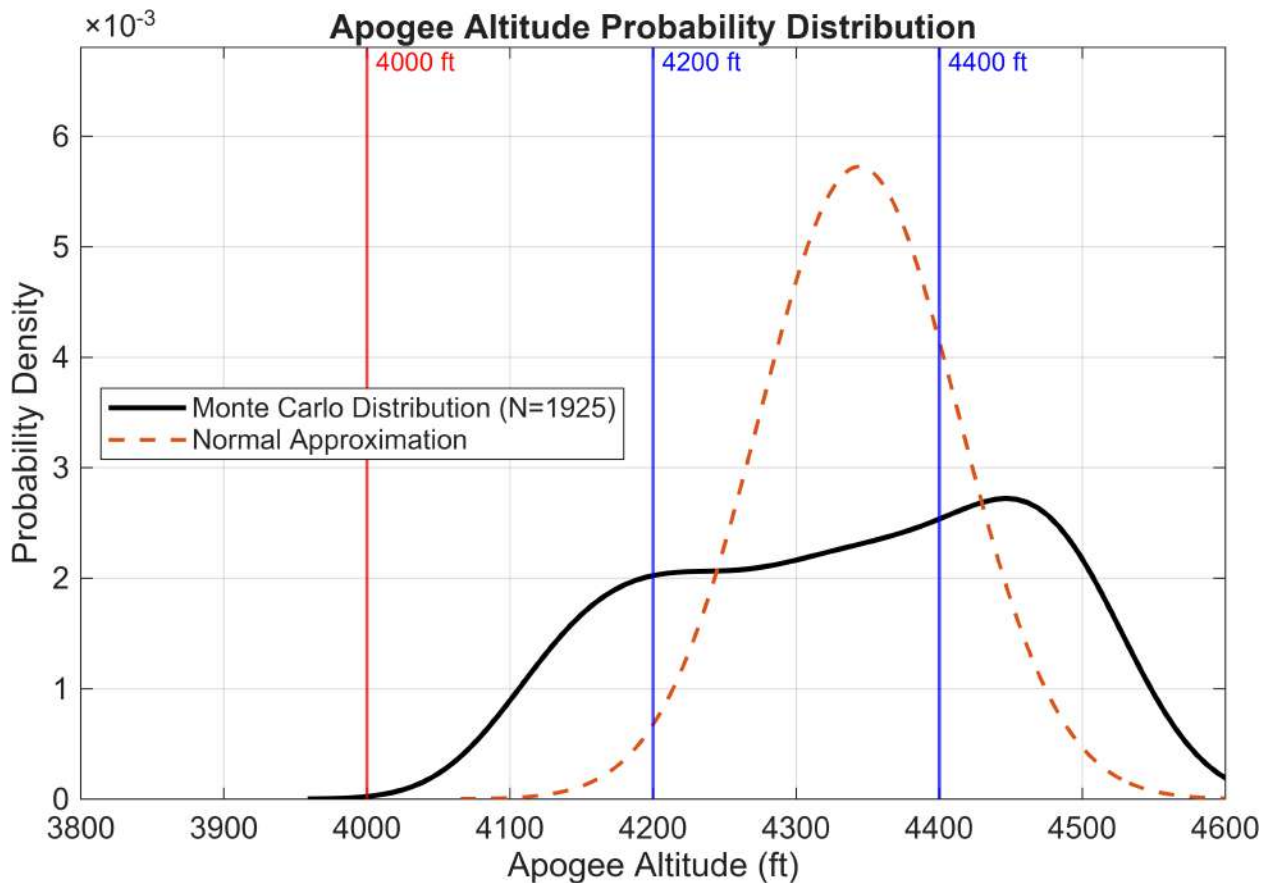


Figure 32: Probability density function of predicted apogee altitudes showing Monte Carlo distribution (black; $N = 1925$) versus normal approximation (orange). Vertical lines indicate critical altitude thresholds.

The Richardson extrapolation scheme for numerical integration error estimation follows: (1) let $I(h)$ be the integral estimate at grid spacing h , (2) assume error scales as $\epsilon(h) = Kh^p$, (3) with two grid resolutions (h_1, h_2):

$$I_{\text{exact}} = I(h_2) + \frac{I(h_2) - I(h_1)}{r^p - 1} + \mathcal{O}(h_2^{p+1})$$

where $r = h_1/h_2 = 2$ for the grid refinement. For second-order accurate quadrature ($p = 2$):

$$I_{\text{rich}} = \frac{4I_{\text{fine}} - I_{\text{coarse}}}{3}, \quad \epsilon_{\text{est}} = \frac{|I_{\text{fine}} - I_{\text{rich}}|}{3}$$

Richardson extrapolation confirms numerical convergence:

Grid Resolution	25×25	13×13	Extrapolated	Error	
$\mathbb{E}[A]$ (ft)	4344.1	4344.3	4344.0	0.0	
$P(A \geq 4000)$	1.000	1.000	1.000	0.000	(17)
$P(A \geq 4200)$	0.968	0.965	0.970	0.000	
$P(A \geq 4400)$	0.215	0.220	0.213	0.001	

The analysis demonstrates:

1. Near 100% probability of exceeding NASA's 4,000 ft lower limit (Requirement 2.1).
2. High likelihood of achieving an apogee equal to or greater than the 4200 ft target.
3. Altitudes exceeding 4600 ft require unphysical conditions (e.g., $W < 0$ mph, $T > 120^\circ\text{F}$).

Stability Margin Verification

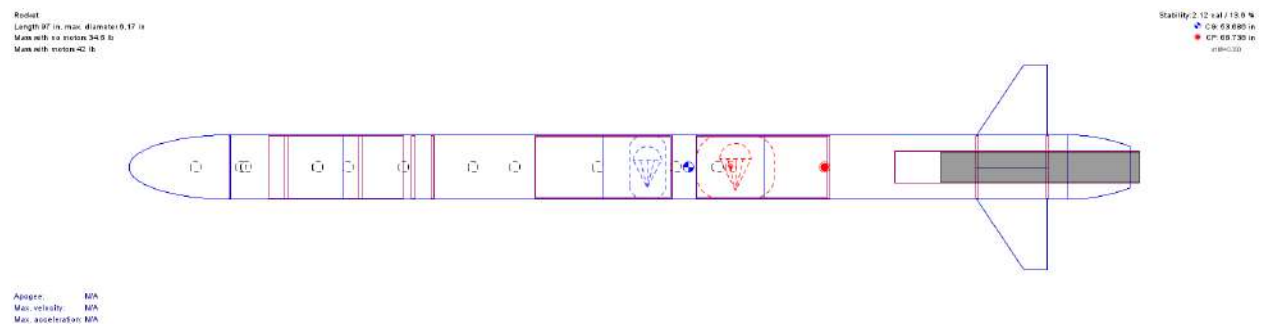


Figure 33: Full-scale rocket design from OpenRocket showing the center of gravity (CG) and the center of pressure (CP).

To ensure vehicle compliance against NASA 2.14 on minimum static stability margin (SSM), the vehicle Center of Pressure (CP) is independently computed using Rocksim, OpenRocket and DRSS from vehicle geometry, which all converge to verify compliance against SSM as shown below.

The vehicle Center of Pressure (CP) is independently computed using OpenRocket and DRSS from vehicle geometry, which all converge to verify compliance against SSM as shown below.

Table 17: Comparison of Rocket CP Measurement Methods.

Software	Rocket CP from tip of Nose Cone (in)	SSM on pad / Minimum SSM (cal.)
RockSim	66.5	2.29
OpenRocket	66.7	2.12
DRSS	66.8	2.12

As noted in Section 5, the measured CG for the as-built rocket taken together with the CP estimations yielded a 2.12 cal SSM on the launch pad. As noted in the above sections, the static mass ballast serves two primary functions: maintaining mass properties consistent with the VDF configuration and ensuring adequate static stability margins throughout the operational envelope (Figure 31). The SSM sensitivity analysis quantifies these relationships through a parametric sweep of mass loading conditions presented in Figure 34.

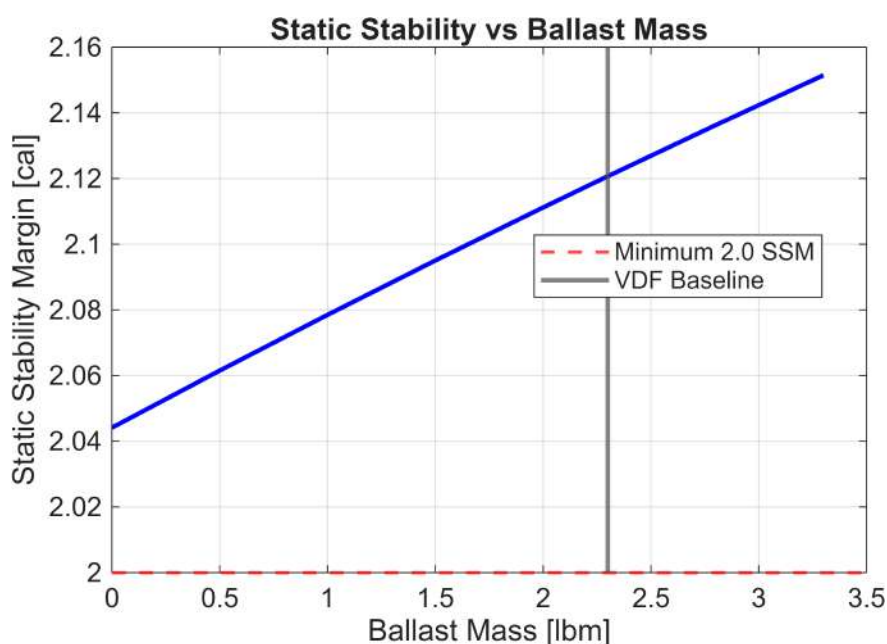


Figure 34: Variability of SSM (on pad) as a function of mass ballast.

While VADL does not intend to fly below the VDF Baseline massing (2.3 lbm ballast), Figure 34 confirms the vehicle SSM on pad remains above the 2.0 limit for the ballast range of 0 to 3.3 lbm. The SSM necessarily increases as a function of ballast mass due to the anterior placement of the ballast in the payload/LSM bay relative to vehicle CG. Since VADL expects the Competition Day vehicle to equal or exceed VDF massing, the >0.12 cal margin provides robustness against unmodeled aerodynamic uncertainties.

Vehicle Drag Calculations

Drag coefficient for the full-scale vehicles was calculated using the CFD software ANSYS Fluent (Canonsburg, Pennsylvania, U.S.). The resulting coefficient of drag of 0.3138 was referenced in the DRSS model using Box’s method [2], which is grounded in the United States Air Force’s DATCOM aerodynamic theory. DRSS additionally corrects for angle-of-attack variations and other environmental factors, with the drag coefficients further decoupled into normal and axial components. The below table details the differences in the vehicle C_D parameters used by OpenRocket, CFD-DRSS, and RockSim.

Table 18: Predicted Coefficients of Drag of the Fullscale Vehicle.

	Predicted Coefficient of Drag
OpenRocket	0.42
ANSYS Fluent	0.31
RockSim	0.31

A similar divergence in predicted drag characteristics was observed when conducting fullscale pre-flight simulations. As extensively analyzed in Section 5.3, the actual fullscale apogee matches within 1% to 3% of the RockSim and DRSS models while OpenRocket falls short with >9% relative error (Tables 34, 35, 40 and 41). As required by the NASA USLI Handbook, estimation of the drag coefficient of the full-scale vehicle was enabled by the VDF flight data, which shows high resemblance in experimental versus predicted drag coefficients (Figures 99 and 109).

3.3.7 Vehicle Ascent Methodology

In DRSS, the rocket’s trajectory simulation incorporates side wind force F_N , air resistance F_A , thrust T , gravitational force mg , angle of attack α , and free-stream velocity v . Lift and drag forces are calculated via Equation 18, with i representing N (normal) or A (axial), ρ as fluid density, C_i the aerodynamic coefficient, and A the rocket’s nose cone base cross-sectional area:

$$F_i = \frac{1}{2} \rho v^2 C_i A \quad (18)$$

The normal force coefficient C_N is calculated using Box’s method with reference values from CFD, considering angle of attack, Reynolds number, and rocket geometry. The drag coefficient C_D also utilizes CFD reference values. C_A is calculated using Equation 19:

$$C_A = \frac{C_D \cos(\alpha) - 0.5 C_N \sin(2\alpha)}{1 - \sin^2(\alpha)} \quad (19)$$

In Equation 19, α , representing the rocket’s angle of attack, is derived using Equation 20. Here, θ denotes the rocket’s angle relative to the vertical, and ϕ represents the angle of the free-stream velocity relative to the vertical:

$$\alpha = \phi - \theta = \tan^{-1} \left(\frac{v_x + w}{v_z} \right) \quad (20)$$

Utilizing Equations 18–20, a sample calculation for determining the normal and axial forces is provided below. A baseline case with a 10 mph wind and a predicted rail exit velocity of 60 fps is considered for a perpendicular launch rail. The angle of attack is first calculated:

$$\alpha = \phi - \theta = \tan^{-1} \left(\frac{v_x + w}{v_z} \right) = \tan^{-1} \left(\frac{15 \text{ fps}}{60 \text{ fps}} \right) = 14.0^\circ$$

Assume Box’s Method yields a C_N value of 2.1 under these conditions. Assume CFD analysis provides an average C_D value of 0.32. Substituting these into Equation 19 results in a C_A value of 0.30. Finally, Equation 18 is used to compute the axial and normal forces. At rail exit, the normal force F_N , representing the force inducing wind-cocking, is determined as follows:

$$F_N = \frac{1}{2} \rho v^2 C_N A = (0.5)(1.2, \text{ kg/m}^3)(18.3, \text{ m/s})^2(2.1)(0.0220, \text{ m}^2) = 9.28, \text{ N} = 2.09, \text{ lb}$$

The above calculation illustrates the magnitude of forces present at the exit of the rail, which must be accounted for in trajectory modeling to ensure accuracy in predicting vehicle behavior.

CFD Methodology

Full-scale vehicle structure was modeled in SolidWorks. A detailed mesh, including 10 inflation layers around the rocket body, is then generated in the CFD software ANSYS Fluent (Canonsburg, Pennsylvania, U.S.). Flow simulations use Reynolds-Averaged Navier-Stokes (RANS) equations with the $k-\omega$ turbulence model. Appropriate boundary conditions, derived from field observations, were applied to accurately define the simulation domains (e.g., walls, outlet pressure, and inlet velocity).

Due to variable atmospheric conditions encountered during ascent, altitude-dependent boundary conditions are averaged to constant values. This method simplifies simulations and reduces computational effort while maintaining a reasonable degree of accuracy. The simulation temperature is set to 288.16 K. Dynamic viscosity and air density are fixed at 1.822×10^{-5} Pa·s and 1.196 kg/m^3 , respectively. Parameters like Y^+ for heat transfer remain at default values, as they minimally impact drag or pressure accuracy. Additional constants, including the specific heat ratio ($\gamma = 1.4$) and gauge pressure ($P = 0 \text{ Pa}$), are based on validated experimental data.

Table 19: Boundary conditions utilized on CFD simulation. Values such as viscosity and density are determined from air temperature.

Boundary Condition	Numerical Value
Area [m ²]	0.02115
Density [kg/m ³]	1.196
Enthalpy [J/kg]	0
Length [m]	2.46
Gauge Pressure [Pa]	0
Temperature [K]	295
Viscosity [kg/ms]	1.82E-05
Ratio of Specific Heats	1.4
Y^+ for Heat Tran. Coef	300

The $k - \omega$ turbulence model was selected due to its proven accuracy in predicting flow behavior in near-wall regions and its effectiveness in handling flows with adverse pressure gradients. Furthermore, the $k - \omega$ -SST (Shear Stress Transport) variation was utilized for its accuracy in simulating complex geometries, as it can effectively account for the significant curvature gradients present in rocket designs.

To ensure simulation accuracy, an optimized mesh configuration was developed. The number of mesh elements was increased incrementally until a balance between simulation convergence and accuracy was achieved. Additionally, inflation layers were incorporated to more accurately represent fluid behavior in the boundary layer near the rocket's surface. The thickness of the inflation layers was determined based on a prior literature review and calculations performed using online mesh-optimization tools.

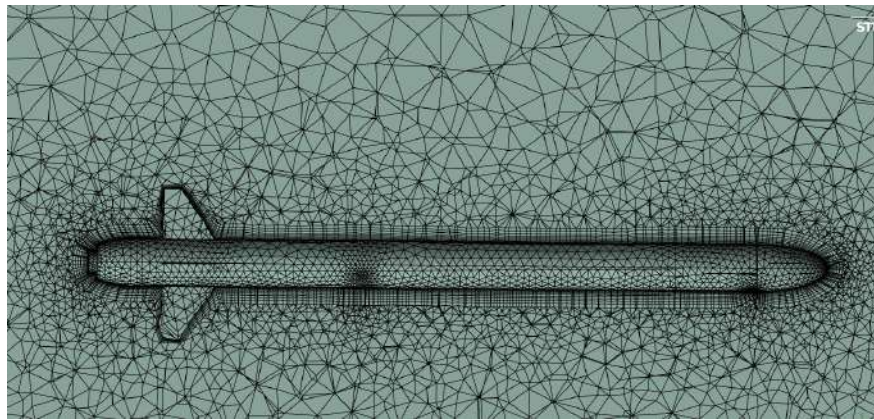


Figure 35: Side view of full-scale mesh.

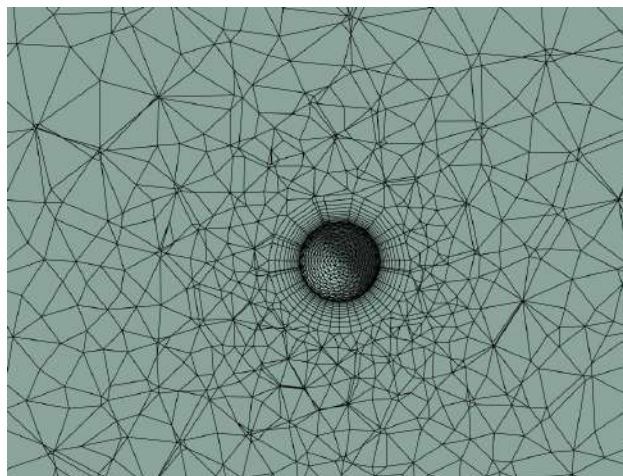


Figure 36: Front view of full-scale mesh; cross section measured 246.38 cm from tip of nose cone.

As seen below, the drag coefficient of the fullscale vehicle generally decreases with increasing velocity and converges at 0.295 as velocity approaches infinity. The mean and median c_d was determined as .3149 and .314, respectively.

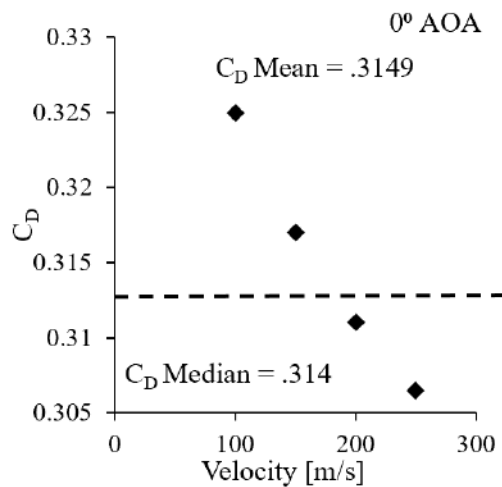


Figure 37: Full Scale coefficient variation as a function of vertical velocity for 0 degree angle of attack.

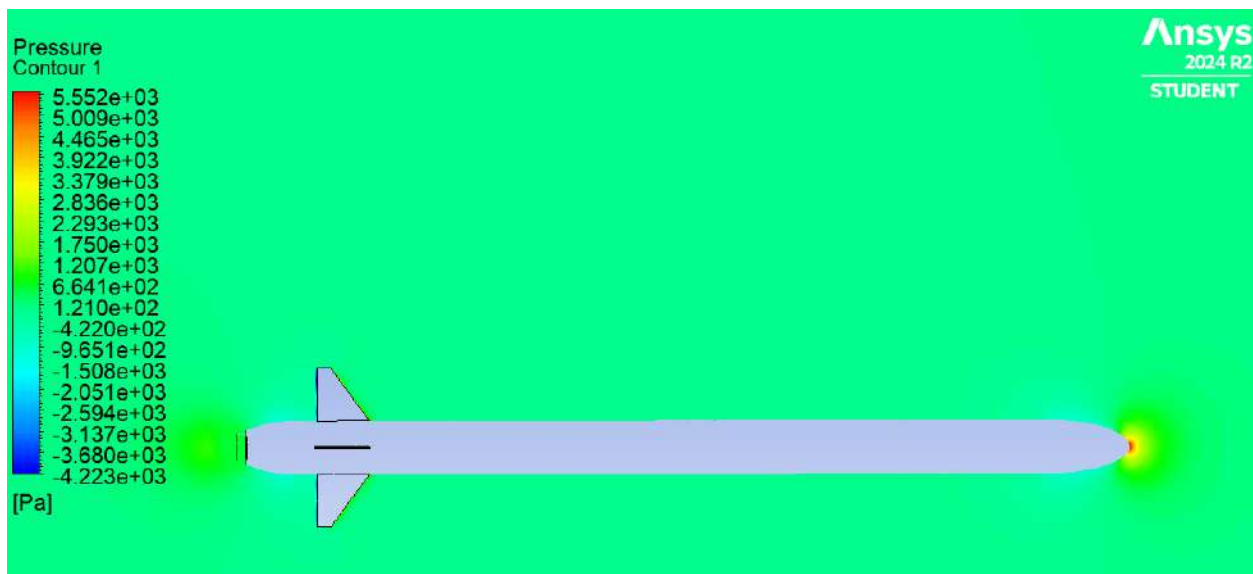


Figure 38: Static pressure contour plot for full-scale traveling at 100 m/s and 0 deg angle of attack.

3.4 Final Assembly and Flight Reliability

The final design, fabrication, and testing of the vehicle subsystems described in the preceding sections has resulted in a robust, structurally secure launch vehicle. Two successful full-scale flights have further proven the reliability of the vehicle, shown in its fully assembled configuration in Figure 39. The vehicle assembly process is covered in detail in the procedures outlined in Section 6.6.



Figure 39: Fully assembled vehicle prior to launch.

1. Powered Ascent
2. Stable Flight
3. Drogue Parachute Deployment
4. Main Parachute Deployment
5. Payload Retention
6. Safe Recovery
7. Avionics Data Acquisition

All of these criteria were verified in the Vehicle Demonstration flights, covered in further detail in Section 5. Criteria 1-3 and 5-7 were fully verified in both flights; the main parachute deployed in both flights, but an issue with shroud line entanglement prevented the main parachute from fully opening in the first flight. This issue was resolved for the second flight, which fully verified all seven criteria. Additionally, a robust vehicle testing protocol (covered in Section 7.1.2) has verified all vehicle requirements in terms of structural integrity, recovery performance, and more. The team is confident in the flight reliability of the launch vehicle moving forward into the Payload Demonstration flight and competition.

4 Payload Criteria

4.1 Payload Design and Testing

4.1.1 Payload Design Changes Since CDR

The team went through multiple iterations of the lander design to ensure compliance with all NASA and VADL imposed requirements. Among these changes include adjustments to the PDM and LDM systems, as well as mass optimization by removing two of the aluminum rods from the internal skeleton. The impetus behind these design changes are outlined in further subsections within Section 4.1.3. These changes ultimately led to a mass reduction of 1.5 lbm, taking the original CDR iteration mass of 8.1 lbm to 6.6 lbm. No changes were made to any component within the payload without thorough testing. The final version of the payload in this FRR cycle is the version that was flown on the February 22nd VDF launch.

4.1.2 Payload Design Overview

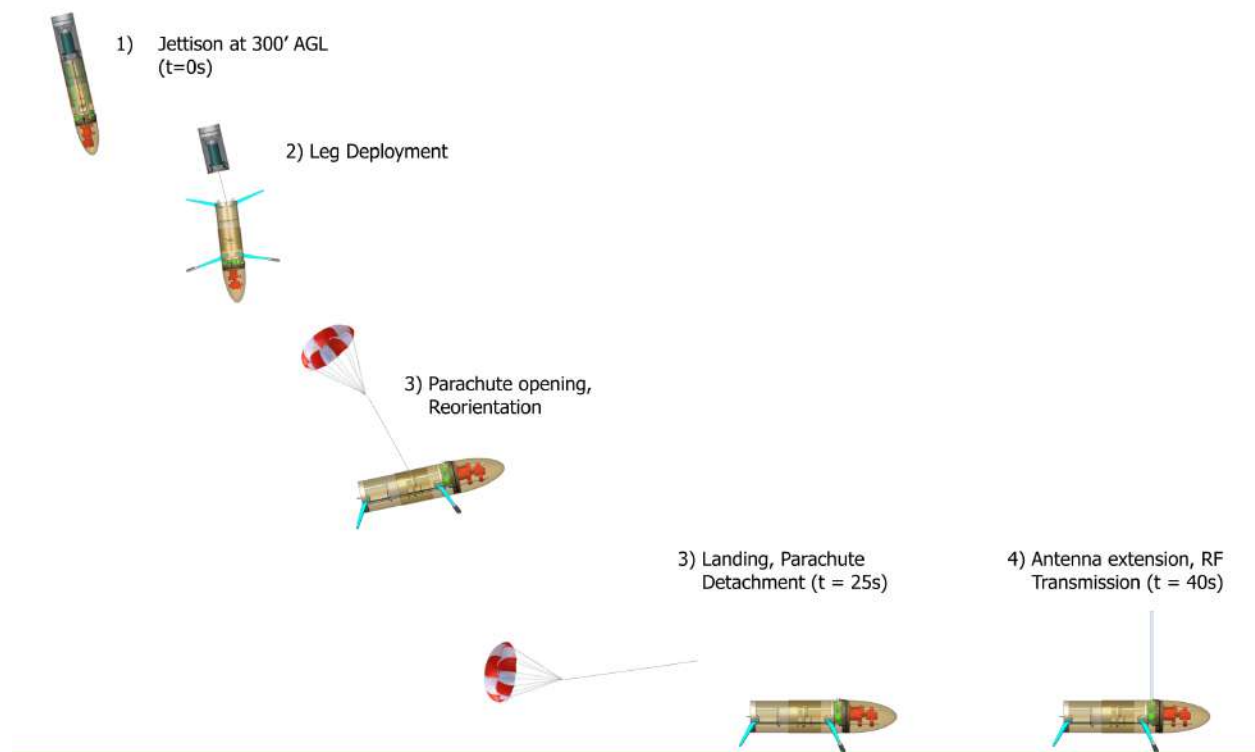


Figure 40: Payload mission concept of operations.

The payload consists of a parachute-arrested lander that separates from the fore-section of the launch vehicle during main parachute descent at an altitude of 300 ft AGL. The lander separation mechanism (LSM) retains the payload through 2 electromechanical rotary latches throughout the rigors of launch and vehicle recovery before detecting the proper altitude and clearance from the RSO in accordance with NASA requirement 4.3.3. The lander parachute has a diameter of 5 ft and

a drag coefficient of 2.2. It is connected to the lander via a 20 ft length of 1/8 in tubular kevlar shock cord.

Upon separation the leg deployment mechanism (LDM) actuates with torsion springs pushing 4 aluminum legs out through slots within the fiberglass airframe. These legs have 2 purposes: to soften the landing impact to protect the STEMnauts and payload electronics, and serve as the ground plane for the quarter-wave ground plane antenna system. Immediately after separation the lander also reorients from a vertical orientation (nose cone towards ground) to a horizontal orientation (nose cone parallel to ground) along with parachute opening.

The payload descends safely underneath a 5 ft diameter parachute and upon landing detaches the parachute shock cord within 1 second. In the next 10 seconds the flight telemetry is compiled and sent to the light APRS board. After an additional 5 seconds the antenna extension mechanism (AEM) is actuated allowing the antennas to unfurl from slots within the airframe and begin RF data transmission.

The pieces of data chosen to transmit are landing site temperature, apogee, battery percentage left, orientation of STEMnauts, landing time in local time, maximum lander vertical velocity, landing velocity and G-forces sustained, and STEMnaut crew survivability using the Dynamic Response Index (DRI).

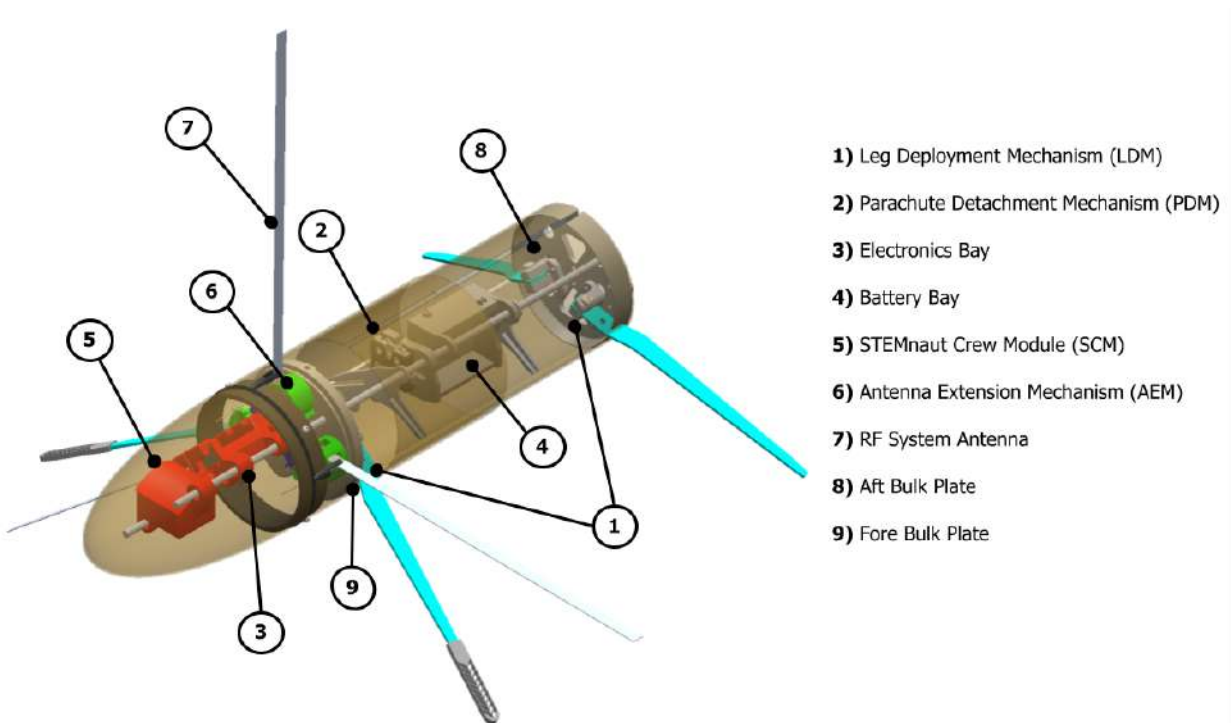


Figure 41: Payload overview with key components highlighted.

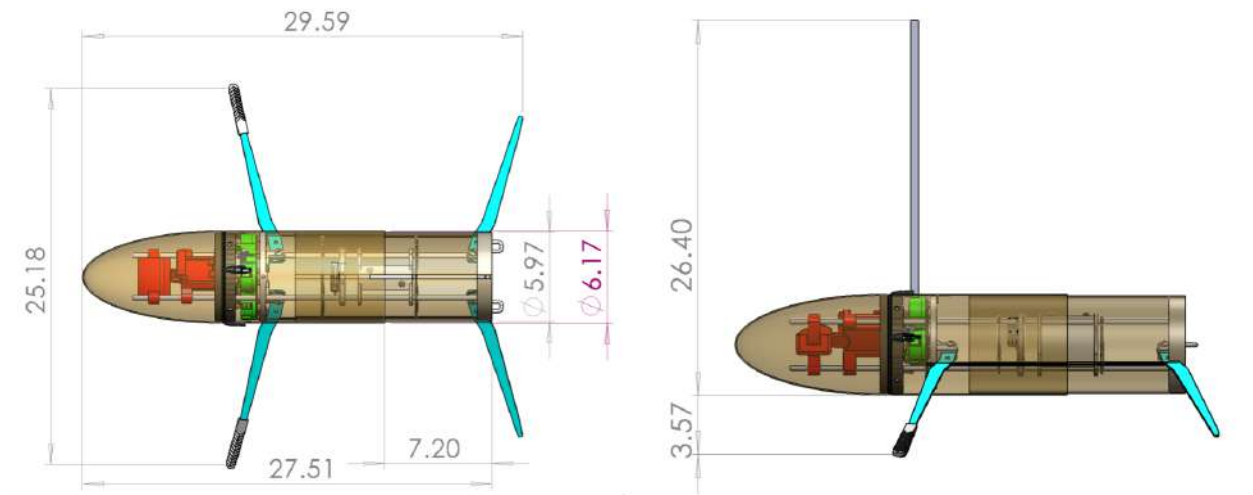


Figure 42: Key as-built payload dimensions in inches.

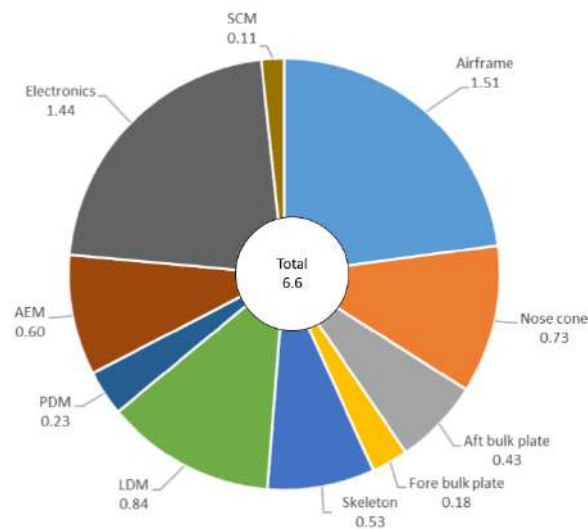


Figure 43: Current mass breakdown of payload.

4.1.3 Payload Testing Overview

VADL has striven to optimize and verify the payload design through the use of materials testing and functionality testing. In the FRR cycle the team has conducted 3 rounds of full-scale drop testing from 300 ft AGL, on top of the VDF to verify aspects of payload performance from functionality to structural durability.

Hexacopter Drop Tests

Round 1 - January 25, 2025

The first round of drop tests was conducted on January 25th, 2025. Average wind was measured at around 10 mph blowing from the South. The payload mass was measured in the field to be 8.1 lbm. This test drove the decision to move to a single point attachment PDM instead of the dual attachment, as well as changes to the LDM.



Figure 44: Round 1 of drop tests

Table 20: Summary of results from Round 1

Payload Function	Drop 1	Drop 2	Drop 3	Drop 4
Lander Separation	Successful	Successful	Partial	Successful
Leg Deployment	Partial	Partial	N/A	N/A
Landing Detection	Failed	Successful	Successful	Successful
Data Collection	Successful	Successful	Successful	Successful
Parachute Detachment	Failed	Partial	Partial	Successful
Antenna Extension	Failed	Successful	Successful	Successful
RF Transmission	Failed	Successful	Successful	Successful
Upright Landing	Failed	Failed	Successful	Successful

Table 21: Round 1 Data

Trial ID	Avg Descent Rate (fps)	Landing KE (lb-ft)	Drift (ft)	Landing Spinal G	DRI	Injury Risk	Injury Risk
1	14.6	26.8	290	14.9	9.24	Low	Low
2	14.1	25.0	110	8.6	10.19	Low	Low
3	14.2	25.4	270	9.9	9.90	Low	Low
4	18.7	44.0	320	11.2	11.17	Low	Low

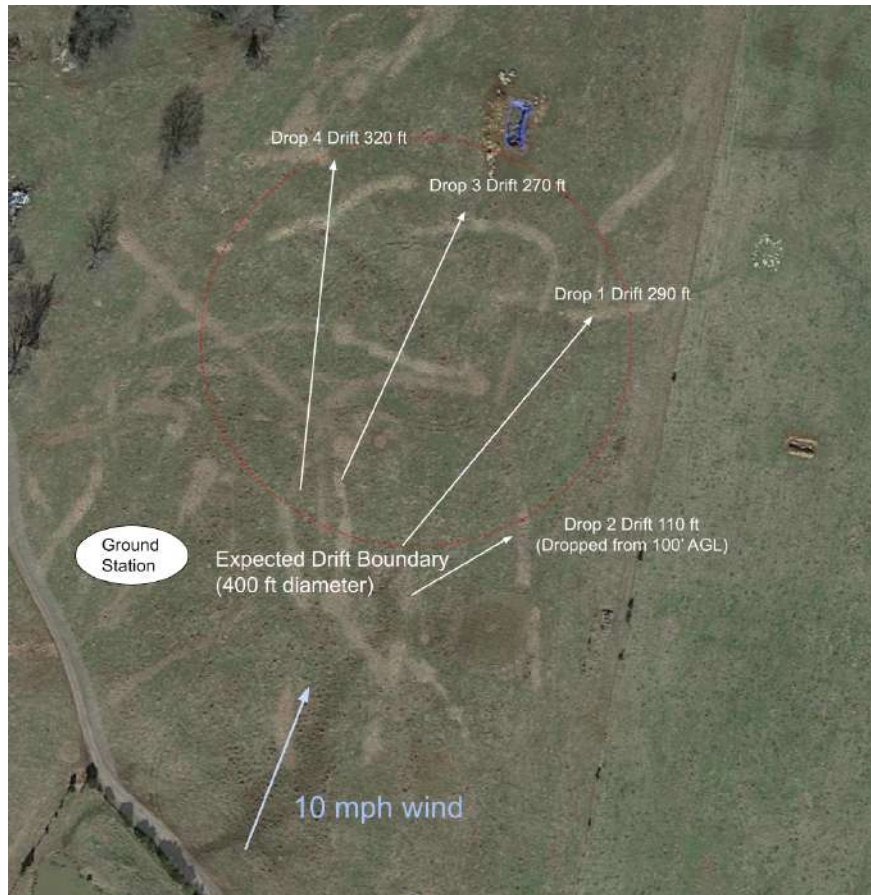


Figure 45: Drift overview for Round 1. Drop 2 had a lower drop to verify electronics.

On the first drop, the electronics did not detect landing due to running test code that consists of three distinctive data logging methods. The intention was to test out launch detection and post-landing autonomous data truncation. However, simultaneously writing to three files hinders the real-timeliness of the system, and as a result the computer missed landing detection. The test code was not run on the remaining drops and the electronics performed as expected. The relatively poor parachute performance with a much lower CD than expected drove the team to consider both acquiring a new 5 ft parachute and a 6 ft parachute. In the next round of drop tests, the 6 ft parachute was compared to the 5 ft one used in round 1.

LDM Adjustments

The LDM iteration used in round 1 was consistent with the final CDR version. The legs were 0.5 in wide and extended 1.7 in from the belly of the lander. This was accomplished by extending the leg lengths from 10 to 12 in. For this test, 6063 aluminum was used instead of the previous 2024 material due to issues with stock availability. It was observed that certain landing orientations caused the legs to plastically deform, such as when the payload had significant angular momentum leading to a single leg striking the ground prior to any other part. The team decided to prevent this by switching to the 7075 alloy and redesigning the legs to increase the width of the legs.



Figure 46: Leg redesign with previous iteration (bottom) for comparison.

Another issue observed in the field was inconsistent LDM deployment. The tolerance of the leg guides was determined to be too large, and as a result the legs could cross over each other causing them to bind. Because of this, on the first 2 drops only 2 of the legs deployed causing the lander to roll over after landing. For this reason, the last 2 drops were flown with the legs pre-deployed. To fix these issues, the leg guides were redesigned with a smaller tolerance from 0.25 in to 0.20 in.



Figure 47: LDM guide redesign with previous iteration (left) for comparison.

PDM Adjustments

In the dual attachment system, the shock cord loop goes through 2 hooks before attaching to the PDM latch. This design has issues with the shock cord failing to release completely from the latch hook. Previously during the CDR design cycle a non-load bearing plastic piece was attached to the top of the latch to guide the shock cord off of the latch hook. However, due to the significant force of the shock cord pulling on the sides of the hook, the plastic guide failed to properly dislodge the shock cord. For this reason, a single point attachment system was tested in the field on the final drop of the day to verify that it would not adversely affect the landing stability during the reorientation from vertical to horizontal. This test was successful, so the single point attachment was chosen as the parachute detachment method moving forward from this test.

Round 2 - February 1, 2025

The second round of drop tests was conducted on February 1st, 2025. During this drop test, the new leg material and 6 ft parachute were tested. All electronics performed nominally. The payload mass remained at 8.1 lbm.



Figure 48: Round 2 of drop tests.

Table 22: Summary of results from Round 2.

Payload Function	Drop 1	Drop 2	Drop 3	Drop 4
Lander Separation	Successful	Successful	Successful	Successful
Leg Deployment	Successful	Successful	N/A	N/A
Landing Detection	Successful	Successful	Successful	Successful
Data Collection	Successful	Successful	Successful	Successful
Parachute Detachment	Partial	Successful	Successful	Successful
Antenna Extension	Successful	Successful	Successful	Successful
RF Transmission	Successful	Successful	Successful	Successful
Upright Landing	Failed	Failed	N/A	N/A

Table 23: Round 2 Data

Trial ID	Parachute Diameter (ft)	Avg Descent Rate (fps)	Landing KE (lb-ft)	Drift (ft)	Landing Spinal G	DRI	Injury Risk
1	6	12.7	20.3	340	15.8	13.36	Low
2	5	10.9	15.0	260	15.5	11.57	Low
3	5	14.6	26.8	280	15.5	16.91	Low
4	6	9.4	11.1	330	15.6	16.30	Low

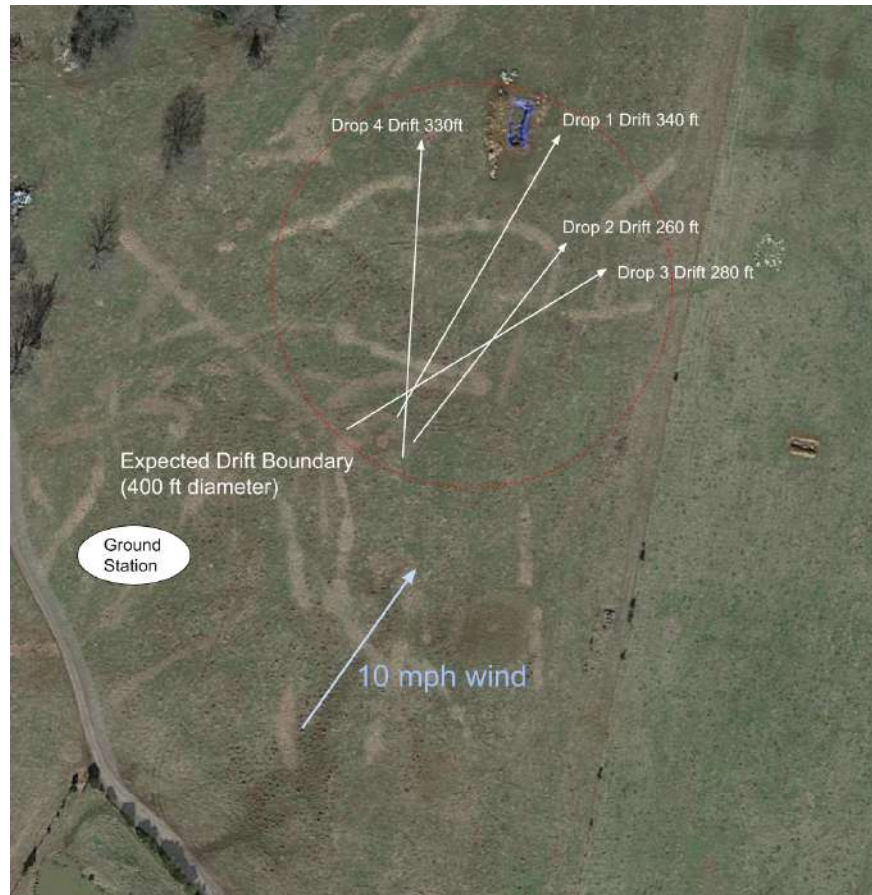


Figure 49: Drift overview for Round 2.

In the last 2 drops, the LDM was restrained to simulate an in-rocket landing configuration in preparation for VDF in which detachment would not occur. For this reason upright landing is not applicable to those drops. It was observed that the slot for the parachute shock cord needed to be slightly widened to allow for the loop to pass cleanly through upon detachment. After this fix was done, the remaining drop tests had no issues with parachute detachment. This round of tests drove the decision to add a delay to the AEMs to prevent payload rolling after impact from interfering with antenna extension. During all drops at least 1 antenna was still able to extend, however the team decided to add a delay to ensure all 3 antennas extend successfully. This delay was set to 15 seconds at which point both AEM extension and RF transmission actuate together.

LDM Adjustments

There were high winds observed during this test with steady wind up to 10 mph, and this led to high horizontal velocity upon landing. On some of the drops this led to the lander impacting the ground backwards, hitting the aft bulkplate before the legs impacted the ground. This led to bouncing and the lander flipped over. The team later decided to fix this by moving the PDM slightly aft of the CG to ensure a slight nose down position during descent. It was chosen to have a nose down configuration as the nose cone provides a round surface that doesn't catch on the ground as easily as the aft bulkplate.

Due to the high wind the lander had significant horizontal velocity upon impact, which led to

the excessive lander bouncing and at times flipping over. The team decided that to improve the landing stability additional damping was needed by changing the material back to 6063 aluminum, increasing the distance that the legs protrude below the belly, as well as adding plastic shoes that deform and/or break upon impact to dissipate energy and prevent the payload from bouncing. The change back to a softer material prevents excessive bouncing, and also improves damping by allowing the legs to plastically deform. The breakable shoes are fastened to the front legs with a tight friction fit aided by a thin layer of cyanoacrylate glue to prevent the parachute opening from dislodging the shoes. These shoes are considered wear items that are replaced after each flight.

Parachute Selection

The 6 ft parachute did not provide any benefits in terms of landing stability or descent rate with the high winds observed. The lander was blown around quite strongly during descent, leading to inconsistent descent velocities. For this reason, the team decided to move forward with a new 5 ft Iris Ultralight parachute. This parachute also provides a much smaller packing volume and mass, which benefit vehicle flight.

Vehicle Demonstration Flight - February 22, 2025

To ensure a nominal rocket launch, modifications were made to the internal payload structure in order to optimize the mass. This included redesigning bulk plates and reducing the number of aluminum rods in the skeleton from 4 to 2. These changes reduced the payload mass from 8.1 to 6.6 lbs. During VDF, the payload was not detached from the rocket in order to simulate a case where the lander jettison is denied by the RSO. For this reason the landing orientation could not be guaranteed and it was prudent that all 3 antennas be deployed to ensure that at least 1 would be pointed upwards.

Table 24: Summary of payload performance in VDF

Mission Aspect	Requirement	Result
Detect payload landing	VADL 4.7.2	Successful
Collect flight telemetry and landing site data	VADL 4.7.3 - 4.7.9	Successful
Actuate parachute detachment latch 1s after landing	VADL 4.4.1	Successful
Extend antennas	VADL 4.5.1 - 4.5.5	Successful
Transmit RF data	VADL 4.6.1	Successful (470 ft transmission range)
Cut-off RF Transmission after 5 minutes	VADL 4.6.3	Successful

The payload performed nominally during the VDL launch, and met all NASA mission requirements for the Student Launch problem. Details on the VDF performance are outlined in Section 5.4.

Round 3 - March 1, 2025

The final round of drop testing for this cycle was conducted on March 1st, 2025. The aim of this

test was to determine the landing profile for the payload using both the adjusted leg angle and the plastic shoes. This test would also confirm the use of the Fruity Chutes Iris Ultralight 5 ft diameter parachute that had not been flown previously. The lander mass remained at 6.6 lbm.



Figure 50: Round 3 of drop tests

Table 25: Round 3 data

Trial ID	Avg Descent Rate (fps)	Landing KE (lb-ft)	Drift (ft)
1	10.3	10.9	370
2	10.4	11.1	330
3	13.6	19.0	280
4	6	9.4	11.1

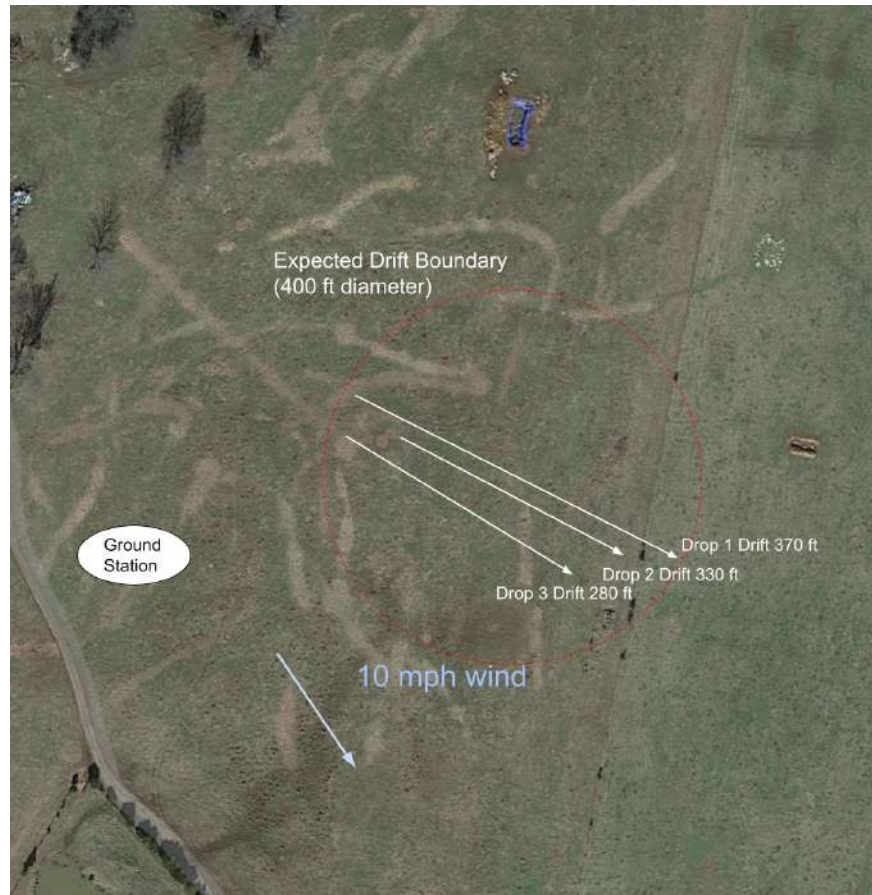


Figure 51: Drift overview for Round 3.

The team adjusted the lander balance such that it has a 15 degree nose down attitude upon landing. Minimal bouncing was observed on all drops, verifying that the LDM changes met VADL imposed requirements for landing orientation (VADL requirement 4.1.7). This gives the team confidence in the lander's readiness for payload demonstration flight (PDF) to occur in late March.

RF Range Testing

The team has conducted multiple rounds of RF range testing up to 1 mile in distance to confirm VADL requirement 4.6.2. The two long range tests were conducted with extended antennas and collapsed antennas. RF test 1 was conducted with the receiver atop a hill with the transmitter in a park located to the North. During this test, the team members present verified a line of sight between the transmitter and receiver. The transmitter was moved around and behind obstacles such as small hills in order to see whether or not the receiver would stop receiving telemetry. Only by moving the transmitter behind a building did the receiver fail to receive any data.

RF test 2 was conducted at a local airfield with all antennas collapsed within the AEMs to simulate a situation where the AEMs do not deploy. During this test there was no visible line of sight between the transmitter and the receiver as a treeline blocked the receiver position. During this test it was observed that by placing the payload behind a small berm the receiver was unable to receive

telemetry. This validates the need for the AEM. RF Range testing is described in detail in section 4.3.2.

4.1.4 Payload Flight Reliability

VADL has proven that the payload design is capable of reliably performing its mission under even the harshest of conditions. All mechanical structures withstood the rigors of vehicle launch, recovery and landing during VDF. Additionally, some of these structures had previously been used and are continued to be used in full-scale drop tests from 300 ft AGL. For example, the same AEM assemblies have been utilized throughout drop tests in the PDR and CDR cycles, and the subscale launch. All electronics have survived the same tests unscathed and are continuing to be used.

Following any test all payload components are carefully scrutinized for any structural or functional damage. Strict adherence to payload checklists ensures that no mechanism or component is overlooked.



VADL Vehicle Demonstration Flight
2/22/2025

Payload Checklist

2

☐ Connect to main vehicle body

WARNING: Failure to complete previous steps will result in payload section separation and destruction

☐ **Final Software Check**

☐ Check I2C connectivity

☐ Double-check threshold values

☐ Arm software in the background

☐ Check process running

☐ Check I2C connectivity

☐ Check RF not transmitting

☐ **Re-Launch Procedure**

☐ Remove recovery components

☐ Remove nose cone

☐ **Recover flight data**

☐ Stop the software process

☐ Download flight data

☐ Plot data using Jupyter Notebook script

☐ Give permission to power off safely

☐ Document special events during the test

☐ Disconnect battery

☐ Respool antennas

☐ Verify all fasteners and structure

☐ Change the battery (if needed, unlikely) and verify charge using a multimeter/onboard ADC

☐ **Power on electronics when ready for re-launch**

☐ Adjust the ground-level parameter

☐ Verify threshold parameters (multiple people)

☐ Verify IMU connectivity (check quaternion changes)

☐ Verify servo and latch actuation

☐ Clear APRS decoder output

☐ Attach skeleton to fore bulk plate

☐ Pull antennas into slots

☐ Connect recovery components

☐ Reattach nose cone

☐ Connect to main vehicle body

4 of 5



VADL Vehicle Demonstration Flight
2/22/2025

Payload Checklist

2

WARNING: Failure to complete previous steps will result in payload failure or complete destruction

☐ **Final Software Check**

☐ Check I2C connectivity

☐ Double-check threshold values

☐ Arm software in the background

☐ Check process running

☐ Check I2C connectivity

☐ Check RF not transmitting

Name	Signature
Rais Nurhidajat	
Zachary Stutsman	
Patrick Chen	

5 of 5

Figure 52: Excerpt from post-launch payload checklist.

Testing done to individual components, mechanisms, and materials is outlined in detail in Section 7.1.2.

4.2 Mechanical Subsystems

4.2.1 Lander Separation Mechanism (LSM)

The lander separation mechanism enables the release of the payload at the desired altitude during rocket descent. The LSM is located in the payload bay section of the rocket, and is the only system located in this area besides the payload itself. The aft cap of the LSM bay is a bulk plate facing the drogue bay which has two eye bolts to which one of the drogue shock cords is attached. Two 1/4-20 aluminum rods are attached to opposing sides of this bulk plate and run the length of the LSM. Secured to these two rods and directly beneath the bulk plate is the electronics housing for the LSM. This is a 3D printed housing that contains a battery slot, mounting holes on the sides for a VectorNav VN100 and a BigRedBee GPS transmitter, and mounting holes on top to retain the electronics needed for data processing and latch actuation. Directly beneath the electronics housing there is a 6" long piece of 3" diameter blue tube which acts as the parachute bay for the payload parachute. This blue tube section has a thin cap that is clamped beneath the electronics bay to ensure that the parachute is fully contained within the tube. Further, the LSM is assembled such that there is always a clamping force being applied to the top of the blue tube from the electronics bay, thus ensuring that the tube is held firmly in place. In addition to this downward clamping, the blue tube piece is glued about its base to prevent shifting during flight. The forward section of the LSM is a 0.2" thick aluminum plate that serves as the base for the parachute bay and has two slots through which the EM latches can access the aft end of the payload. These latches are mounted to two aluminum blocks which are fastened to the aluminum plate. Also included in the LSM bay is the ballast being flown on the rocket, which was 2.3 lb for the VDF. The fully assembled LSM can be seen below with components called out.

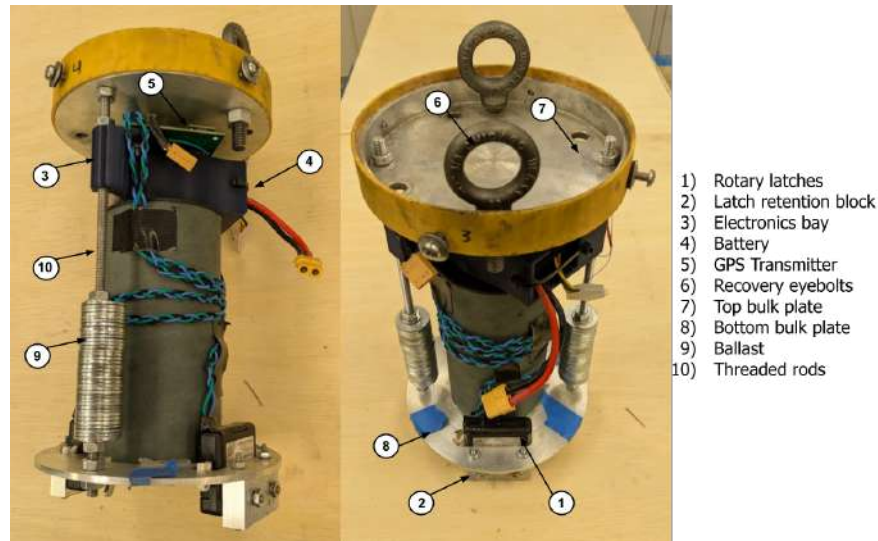


Figure 53: LSM Assembly

The yellow ring that can be seen around the top bulk plate is a spacer ring which allows it to better fit within the rocket section. This piece is 3D printed and is completely non-load bearing. This design choice was driven primarily by the materials which were already in stock in the lab. The team had some aluminum round bar which was sized properly for a coupler section of the rocket but which was a little small for the main body section. Rather than purchase a new piece of round bar, the team elected to use what was in stock and add the spacer. This design was successfully

proven through the two vehicle demonstration flights conducted by the team, and no damage was observed anywhere on or around this bulk plate.

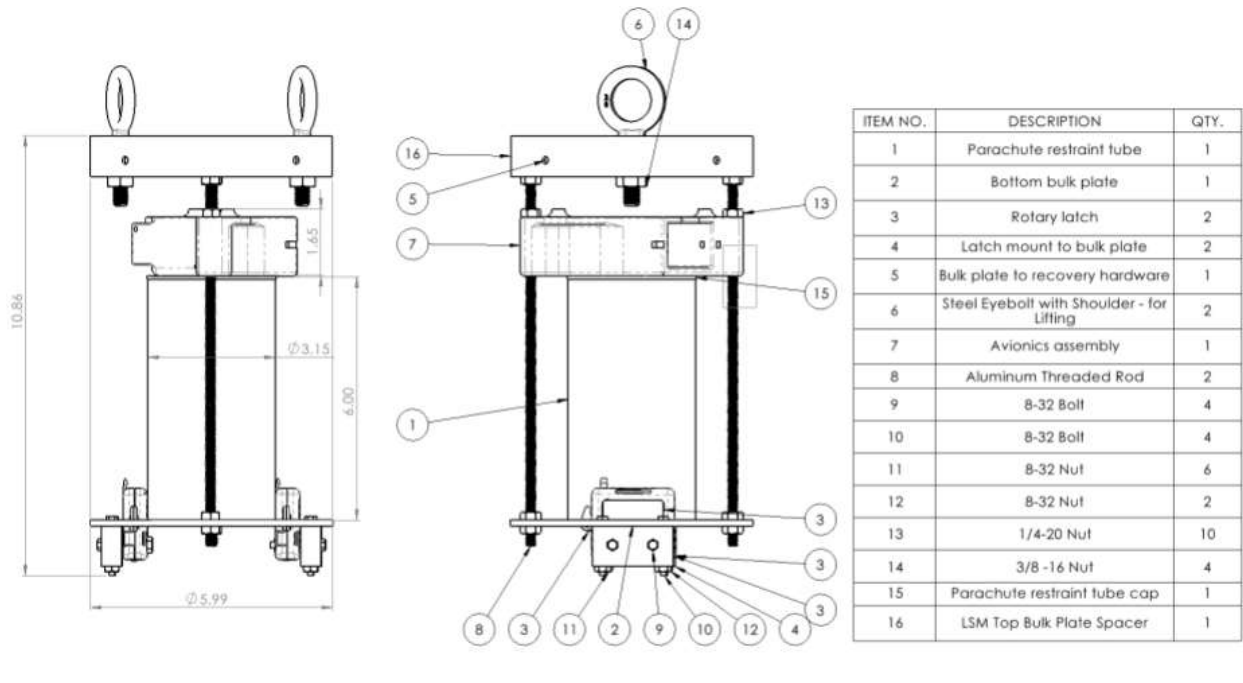


Figure 54: LSM assembly with as-built dimensions in inches.

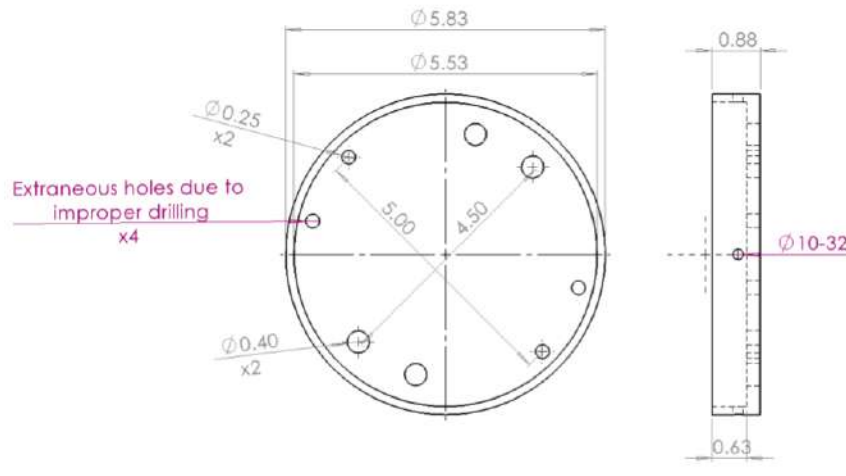


Figure 55: Top LSM bulk plate drawing with as-built dimensions in inches.

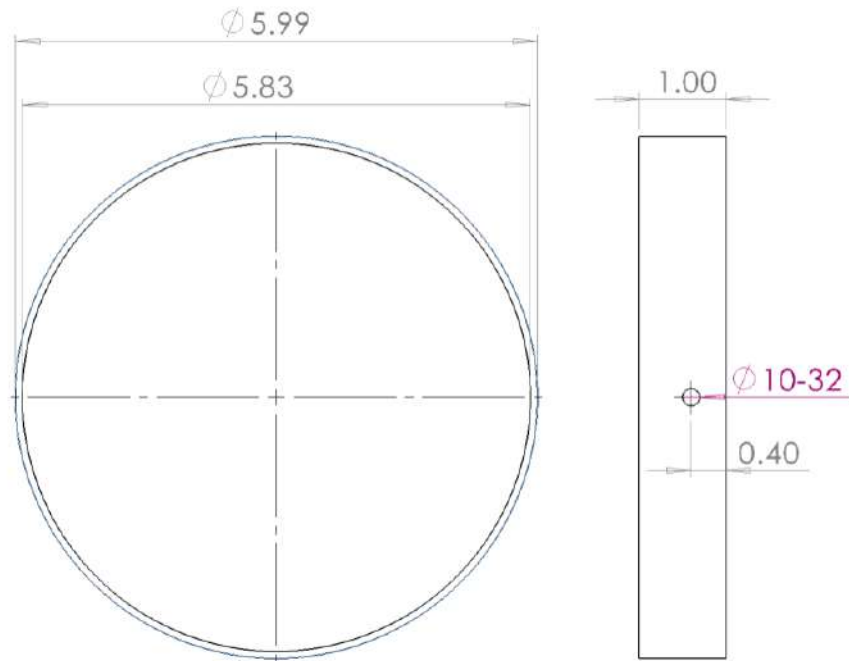


Figure 56: Top LSM bulk plate spacer drawing with as-built dimensions in inches.

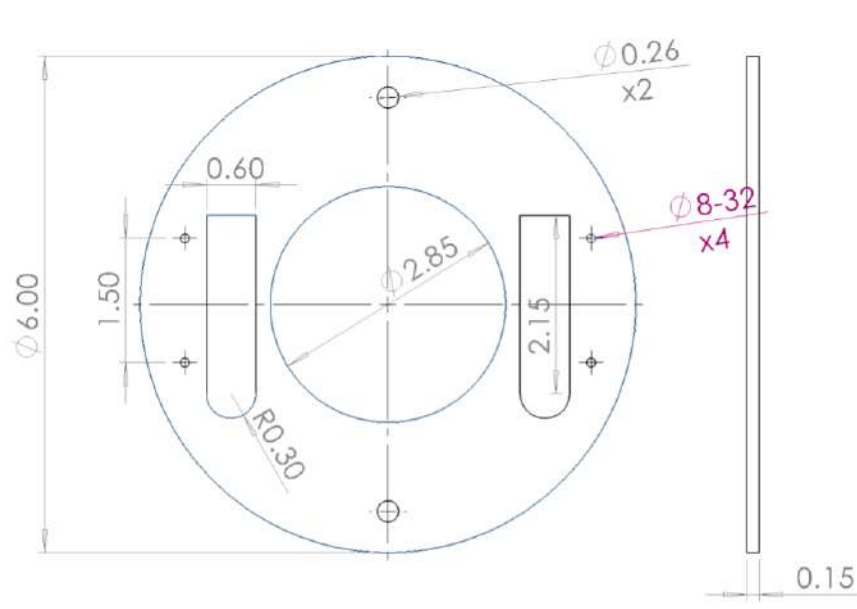


Figure 57: Bottom LSM bulk plate drawing with as-built dimensions in inches.

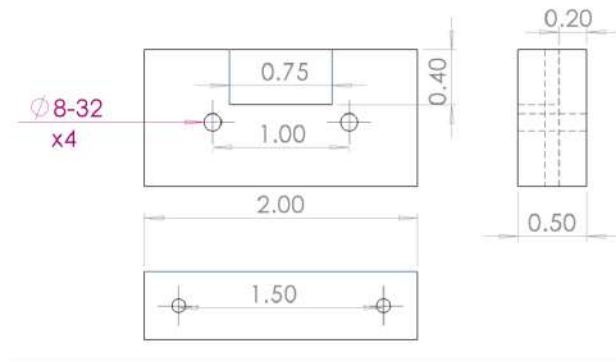


Figure 58: Rotary latch retention block.

The bulk plates and retention blocks are all made of 6061 aluminum. All mechanical components have been verified with 2 full-scale rocket flights and the team is confident in continued success of those components at the Student Launch competition flight.

The LSM receives the release approval by way of a TBS Crossfire Nano Rx receiver. This is transmitted from the ground station once the RSO approves release of the payload. We have two methods of sending RSO approval on the transmitter end. The Primary RSO approval signal is controlled by a switch and may only be received by the onboard Teensy in the Post Apogee State of the state machine coded into the Teensy. In the event that the Teensy fails to reach this Post Apogee State, through some failure to detect launch or apogee, there is also a Backup RSO approval signal that is controlled by both a trigger and a switch being flipped simultaneously. This Backup RSO signal is received by the Teensy regardless of its current state. Once it receives this approval, the LSM is programmed to wait until the rocket descends below an altitude of 300 ft AGL, at which point it will open the latches and release the payload. The altitude is monitored by a VectorNav VN100 IMU which is onboard the LSM. A diagram of the electrical system and signal path for the LSM is shown below in Figure 59.

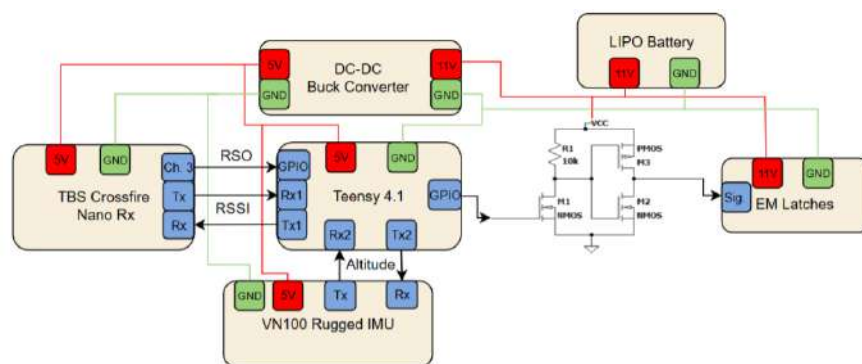


Figure 59: LSM approval system with amplification circuit.

The electrical design of the LSM now uses a custom made PCB with through holes to attach the peripheral devices and JST connectors to send the jettison signal to the EM latches. The amplification of the signal from the Teensy to the latch has also been changed to use a transistors

which shift the output voltage from the 3.3 volts Teensy GPIO pin to 8 volts, well within the 4.5 to 16 volt specifications of the control signal for the latch.

The interface between the LSM and the payload can be seen in Figure 60 below. The payload has two hooks on its top bulk plate which slot into the EM latches on the LSM. This attachment method has been rigorously proven through both drop tests and launches.

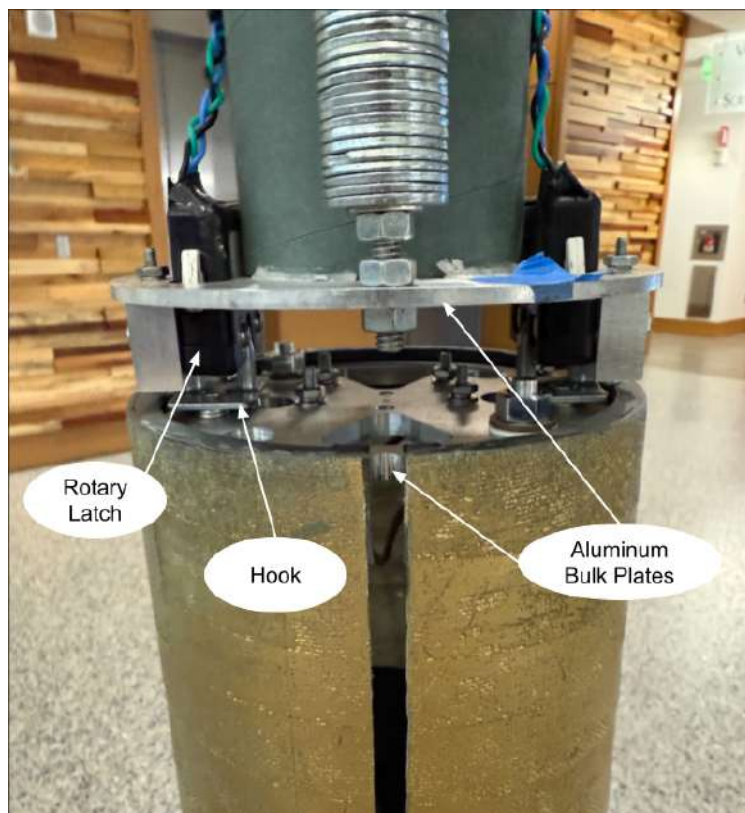


Figure 60: Payload to launch vehicle retention mechanism.

4.2.2 Leg Deployment Mechanism (LDM)

The leg deployment mechanism consists of four legs and brackets which are fastened to bulk plates on opposing ends of the lander. All of these legs are angled downward from the lander and are deployed via torsional springs. This deployment occurs once the lander is jettisoned from the rocket; during flight the legs are held in place by the overlap of the coupler portion of the payload with the body of the rocket.

The LDM has undergone a few changes since the CDR milestone. All of these changes have been driven by testing results, with the intent of creating the most reliable lander possible. All of these changes and the test results that drove them are detailed in Section 4.1.3.

The current landing legs are made out of 6063 aluminum and have a wider body than the design used at CDR. The legs on the nose cone side of the lander have a breakable foot which is friction fit onto the leg. With the foot, these legs are a total of 14” long. The legs on the opposing side of the lander are 12” long and do not have a foot. The nose cone side legs are longer because the lander

is weighted such that the nose cone is angled downwards during descent. This allows the nose cone side legs to hit the ground first and helps ensure that the lander lands upright. Having these legs impact first also allows the breakable shoe to absorb some of the landing force. The force that is not absorbed by the breaking of the shoe is either absorbed by the landing legs through plastic deformation or by the rubber inserts on the bulk plates. The legs are intentionally designed to plastically deform, and are easily replaceable in the field. Thus, damage to the legs does not affect expedient re-flight capability. The leg brackets are of the same design used in CDR, and are 3D printed out of carbon fiber infused onyx. The leg components can be seen below in Figure 61.

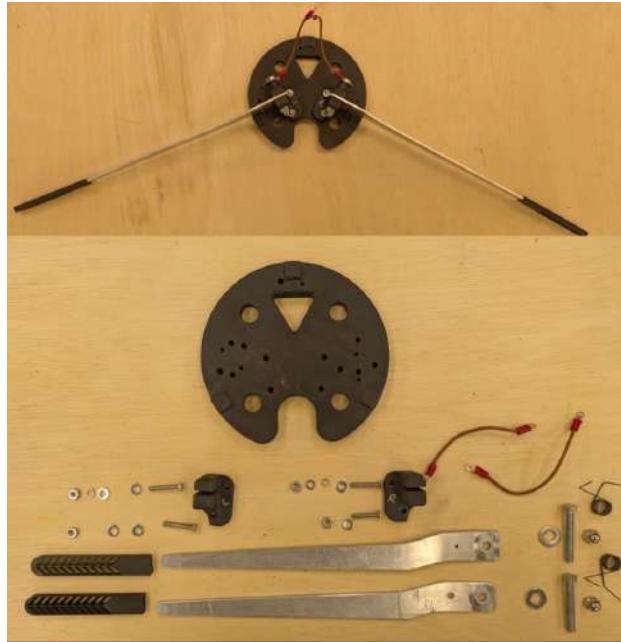


Figure 61: Fore LDM Assembly and parts

Important parts of the LDM are the leg guides seen in Figure 62 below. These are thin 3D printed pieces through which the legs run when they are stowed. They help to ensure that the legs do not get crossed over each other while inside the payload, therefore preventing them from binding. They also stretch nearly to the outside edge of the lander, ensuring that the legs successfully exit the lander and do not get caught on the inside edge of the airframe.



Figure 62: LDM guide

The breakable shoe for the leg can be seen in Figure 63 below. These shoes fit over the tip of the aluminum leg and are designed to crumple when they impact the ground to absorb some of the landing energy and prevent the lander from tipping over. The increase in damping also helps prevent bouncing thereby increasing the survivability of the STEMnauts. These shoes are printed out of Onyx filament and the dimensions can be seen below.

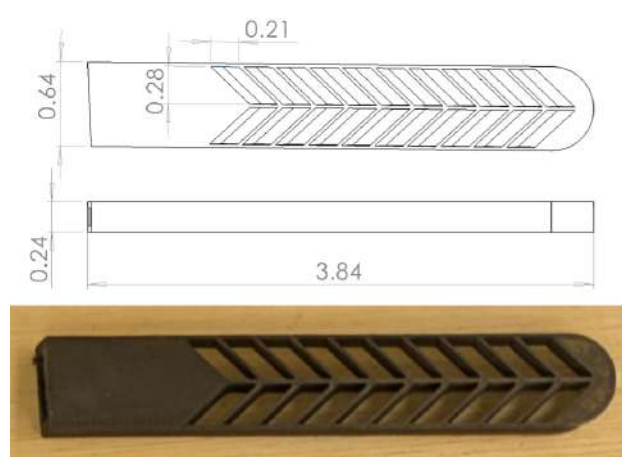


Figure 63: Breakable shoe

In order to quantify the effectiveness of the breakable shoe, several drop tests were conducted. In these tests, the payload was held at a steady orientation 5 ft above the ground. The payload was then dropped repeatedly from this point and the onboard IMU measured the landing acceleration.



Figure 64: LDM Drop test showing with-shoe configuration

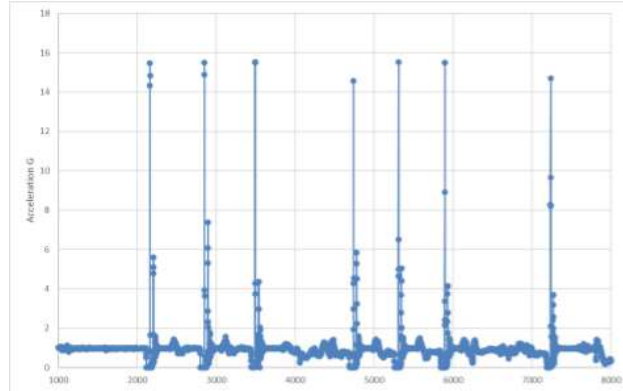


Figure 65: No shoe drop data

The set of control tests was done with a 14” long solid aluminum leg which matches the length of the leg with the foot. This allows the two scenarios to be accurately compared. Figure 65 shows the raw data of landing G force for the no shoe drops and Figure 66 shows the raw data for the drops with the landing shoe. A breakdown of the landing G force experienced in each test as well as the average landing G force can be found in Table 26 below.

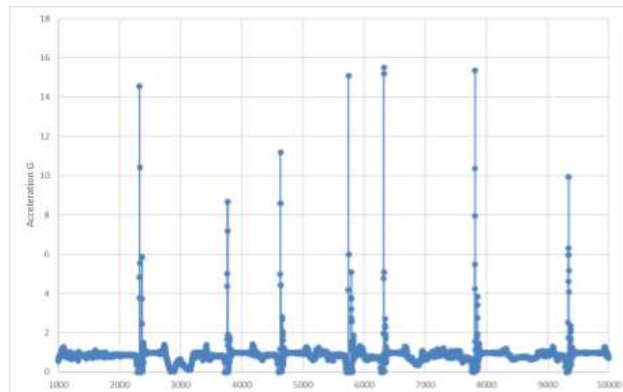


Figure 66: With shoe drop data

Table 26: Leg Damping Comparison

Drop #	No Shoe Acceleration (G)	With Shoe Acceleration (G)
1	15.47	14.55
2	15.50	8.68
3	15.50	11.20
4	15.56	15.10
5	15.51	15.51
6	15.49	15.34
7	14.71	9.93
Avg	15.39	12.90
Std Dev	0.30	2.88

After conducting these tests, it was confirmed that the shoe provided additional damping during landing. A cause for the large variance in the shoe damping is likely the non-quiescent nature of the tests where different parts of the lander would impact the ground first. In actual full-scale drop tests this was observed as the swinging of the parachute causes the lander to impact the ground differently every time. It was observed that the shoes yielded the lowest impact accelerations when the lander struck the ground with the shoes pointed towards the ground at a steep angle. This is expected as the shoes are primarily designed to reduce the chance that the lander flips over after striking the legs. An unpaired t-test yields a p-value of 0.03, which is less than 0.05. This suggests that the reduction in the average impact acceleration is indeed statistically significant. The average acceleration experienced during landing is 2.5 G's lower with the shoes than without, which led the team to keep them in the design.

The other primary functionality of the landing legs is creating a ground plane for the antenna system. This is achieved through an electrical connection from each leg to a common antenna plate. The antenna is located in the forward portion of the payload, near the nose cone, and grounding wires run from this point to each of the four legs. These wires are attached to the arm of the torsional spring, which allows them to move with the legs as they swing out, providing a reliable connection throughout operation. This design is optimal because it removes the risk of pulling out the grounding wires upon leg deployment, or of leg deployment being restricted by attachment to the grounding wires. Another benefit of the aluminum legs is that they serve as the ground plane without the addition of wires along the length of the leg. This greatly simplifies the manufacturing process, making the system more modular and quicker to assemble.

4.2.3 Parachute Detachment Mechanism (PDM)

The PDM's main purpose is to detach the lander parachute on impact to prevent the parachute from dragging the payload around and causing it to tip over. The PDM consists of a single electromechanical latch bolted to a carbon fiber infused Onyx 3D printed bulk plate. The bulk plate attaches to both aluminum rods in the skeleton and is restrained by 4 shaft collars, 1 on either side of the bulk plate.

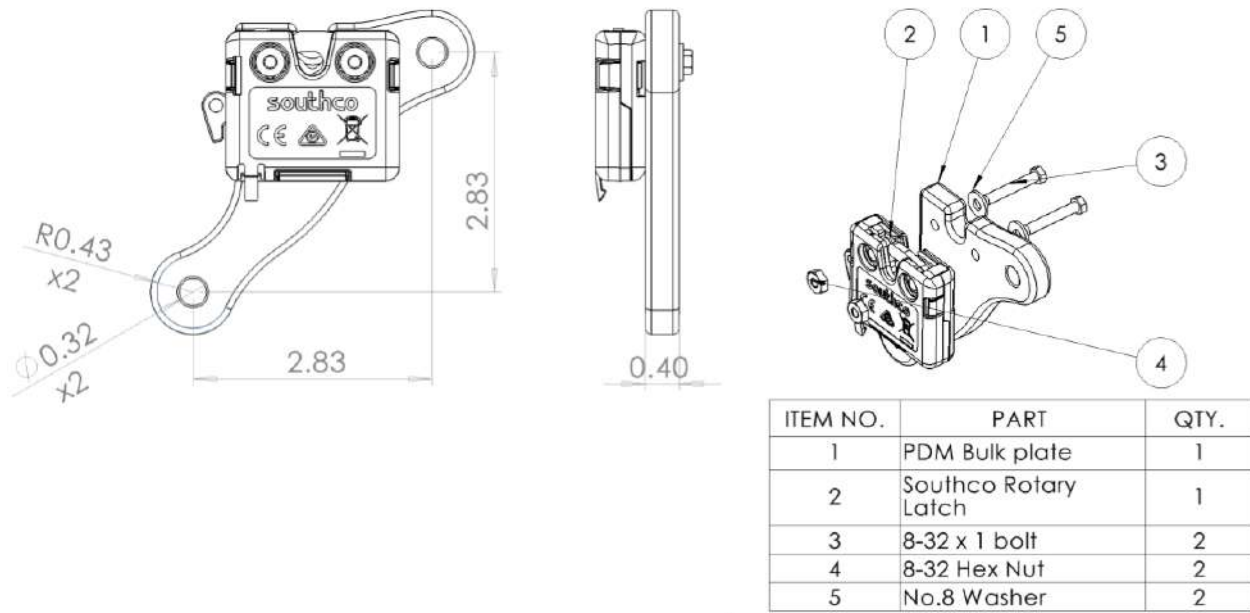


Figure 67: PDM assembly drawing

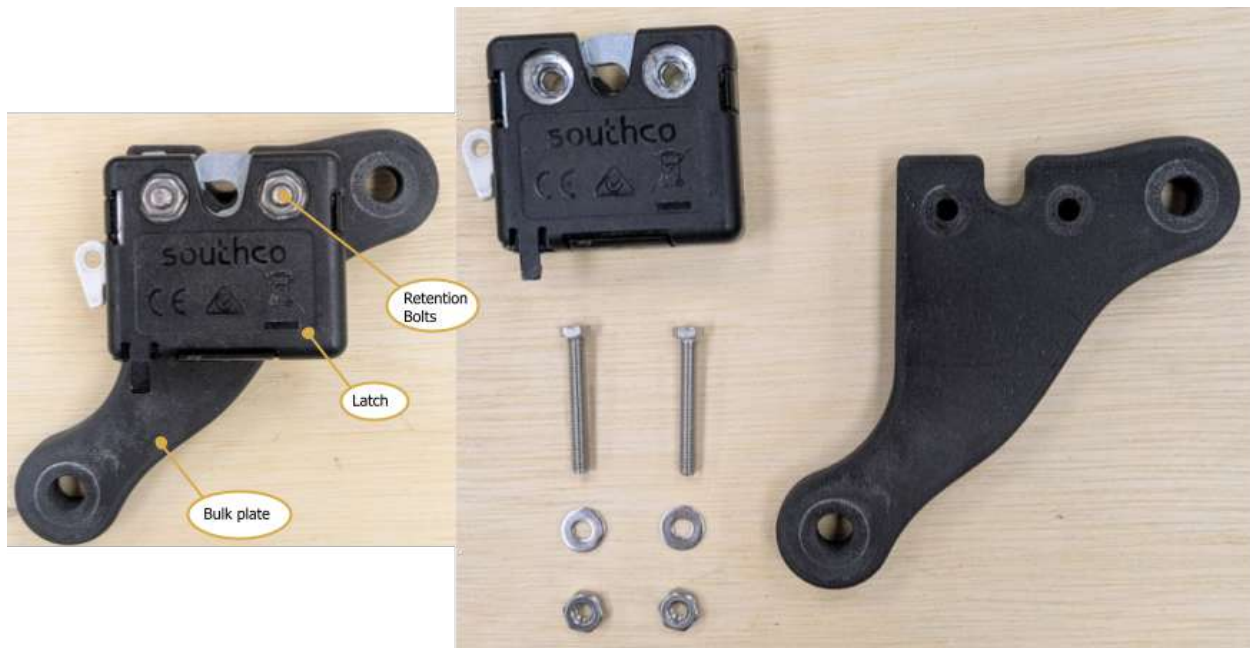


Figure 68: PDM assembly and components

The PDM has been proven on full-scale drop tests where it satisfies VADL requirement 4.4.1 by releasing the parachute within 1 second of landing. The PDM plate and latch have proven to be structurally sound and have sustained no damage across any of the drop tests conducted. According to the manufacturer specifications, the latch can withstand a force of 1061 lbf before opening [6]. The team has tested the latch using a drop test outlined in Section 7.1.2.

4.2.4 Antenna Extension Mechanism (AEM)

The AEM's main purpose is to raise the antenna above the landing surface in order to maintain a line of sight with the ground station receiver. In nominal situations where the lander is allowed to jettison from the launch vehicle only 1 antenna is needed to maintain this line of sight. However, in the event that separation is not allowed there are 2 more antennas to ensure at least 1 is unfurled upwards regardless of the landing orientation.

Each AEM contains its own antenna/tape spring that is wrapped around a rotating cylinder. During flight this cylinder is locked by a lever arm that is released by a 9g servo. Upon landing, this servo unlocks the cylinder and allows the antenna to unfurl to a height of 24 in above the ground. The entire mechanism is contained within a case, and two parts of the AEM prevent the antenna from binding itself within the case. The cylinder rests on a rotary damper providing a damping torque of 2.88 in-oz. This prevents the tape from unfurling too fast and binding on itself before exiting the AEM case. Additionally, a strip of Teflon tape is attached to the side of the airframe opening to reduce the friction between the antenna and the airframe. Combined, these 2 factors ensure the antennas unfurl smoothly and quickly after actuation. The AEMs are structurally and functionally the same as the final CDR iteration, with the only change being the electrical connectors for the 9g servos. Instead of a friction fit, they now have clips on their connectors to prevent disconnection during flight.

The AEMs were inspired by passively deploying composite booms (DCBs) soon to be utilized in satellites. DCBs allow for large structures to be deployed from small satellites, thus lowering payload size and mass. Keeping with the spirit of the NASA Student Launch Competition, the AEM and RF system were designed with DCBs in mind.

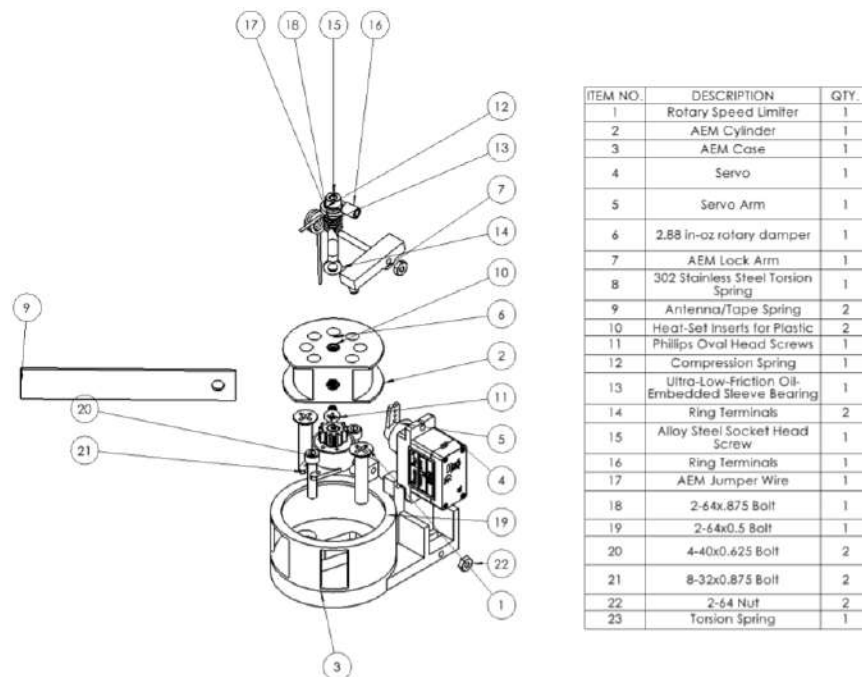


Figure 69: Individual AEM exploded drawing.

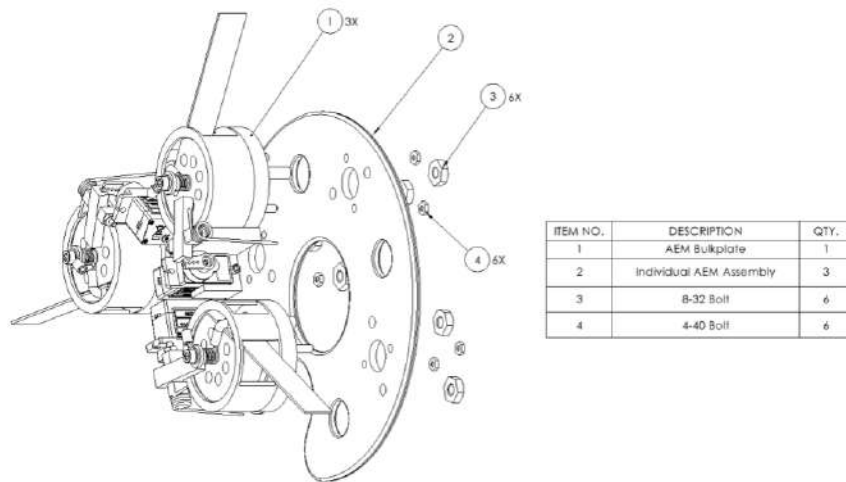


Figure 70: High level AEM assembly exploded drawing.

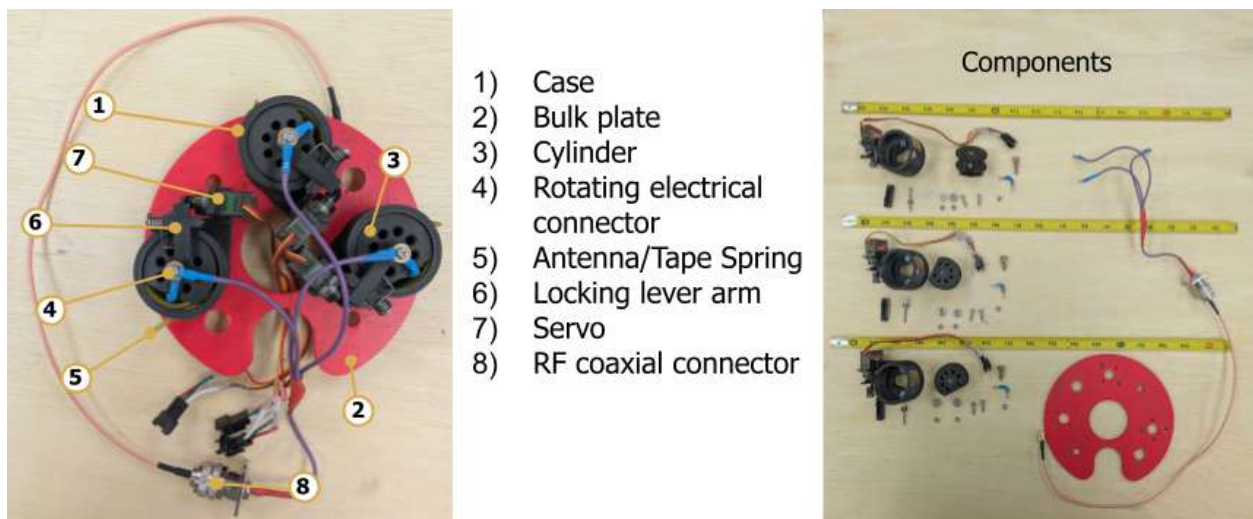


Figure 71: AEM assembly and components

In the above figure, the new snap connectors for the AEM servos can be seen above the RF coaxial connector. These connectors are significantly more robust and secure for in flight use.

4.2.5 STEMnaut Crew Module (SCM)

The purpose of the SCM is to meet the NASA requirement of retaining 4 STEMnauts. To meet that requirement 4 Lego minifigures act as the STEMnauts and are retained in the nose cone of the lander, mimicking actual crewed launch vehicles.

The SCM consists of 2 3D printed pieces: the capsule containing the STEMnauts, and the roof. The STEMnauts are fixed to the bottom of the SCM using circular studs that slot into the holes in the minifigures. The entire assembly is attached to the skeleton with rubber inserts around the aluminum rods to dampen the rigors of landing on the STEMnauts. The STEMnauts are oriented such that they are sitting upright upon landing, and facing upwards during launch.

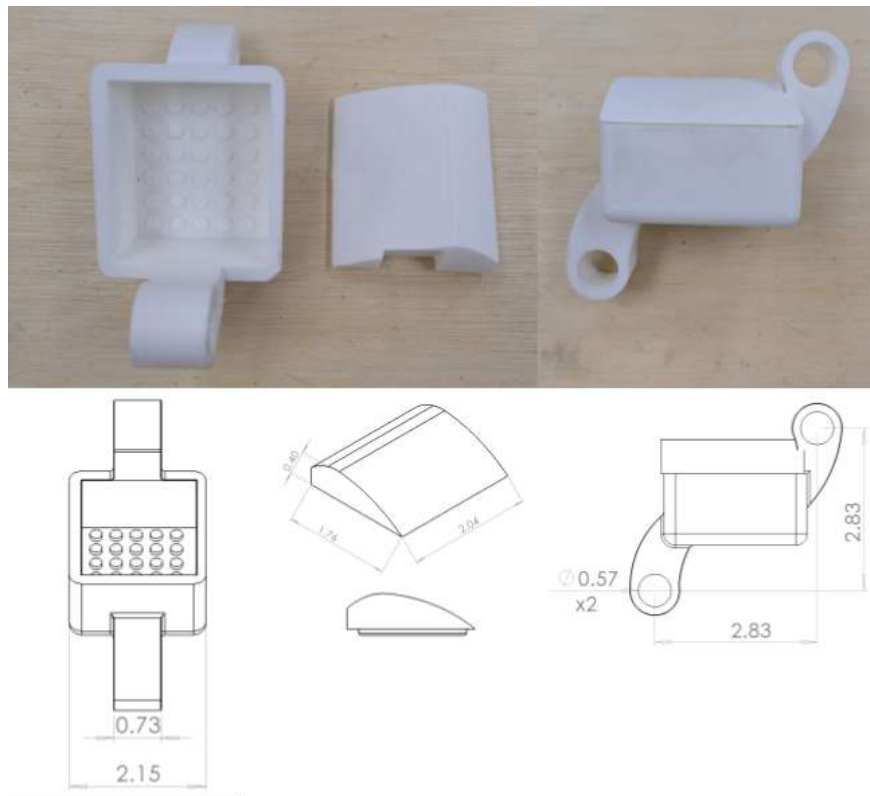


Figure 74: SCM assembly with as-built dimensions in inches

4.2.6 Integrated Lander Structure

Modifications have been made to the internal structure since the CDR iteration of the payload. To optimize mass for VDF the team decided to reduce the number of aluminum rods in the skeleton from 4 rods to 2 rods. This change has driven a redesign of internal bulk plates such as the PDM bulk plate and the battery bay. In the CDR iteration, the electronics bay were located to the rear of the payload, while now they are moved to the nose cone right behind the SCM. This change was made to shift the CG of the lander forward and facilitate a nose-down orientation during landing. The team has verified that this change aids in landing stability as it allows the curved surface of the nose cone to slide along the ground rather than catch during landings with high horizontal velocities.



Figure 75: Rendering of STEMCRaFT structure.

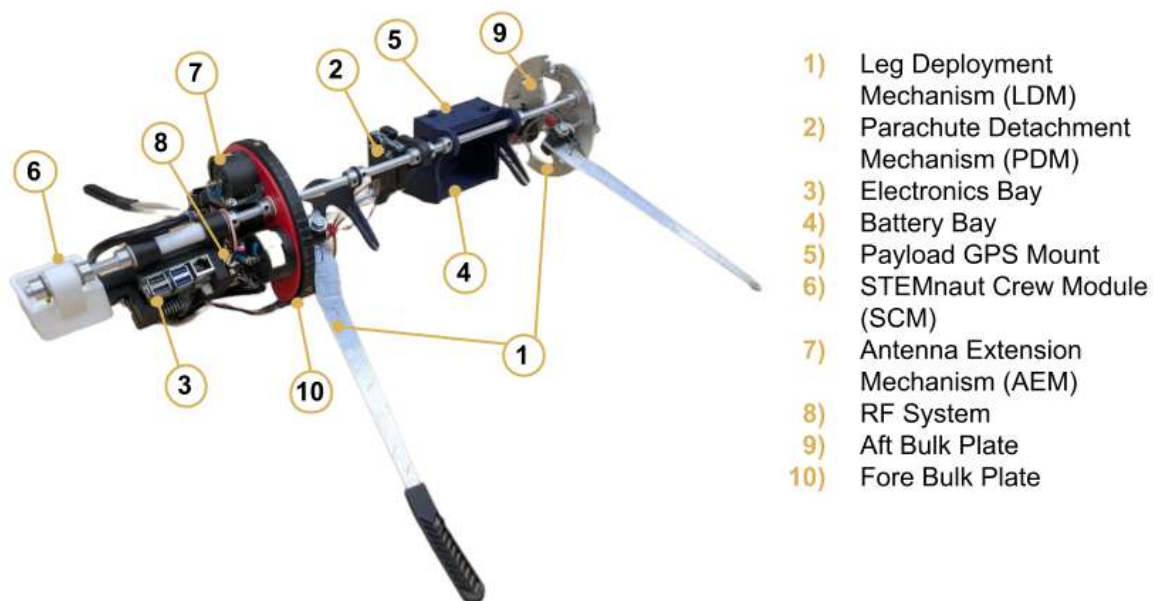


Figure 76: STEMCRaFT skeleton without airframe.

Aside from the reduction in rods, the overall structure remains the same. The lander consists of a fiberglass airframe and nose cone connected to an internal skeleton made of aluminum rods. The skeleton is attached to 2 bulk plates fore and aft which in turn attach to the airframe. The fore bulk plate attaches via 3 8-32 radial bolts while the aft bulk plate slots into a 3D printed retention ring epoxied to the rear of the airframe. This retention ring is made up of Onyx filament inlaid with a continuous carbon fiber strand. The combination of these two attachment points ensure that the airframe is securely retained by the skeleton and LSM.

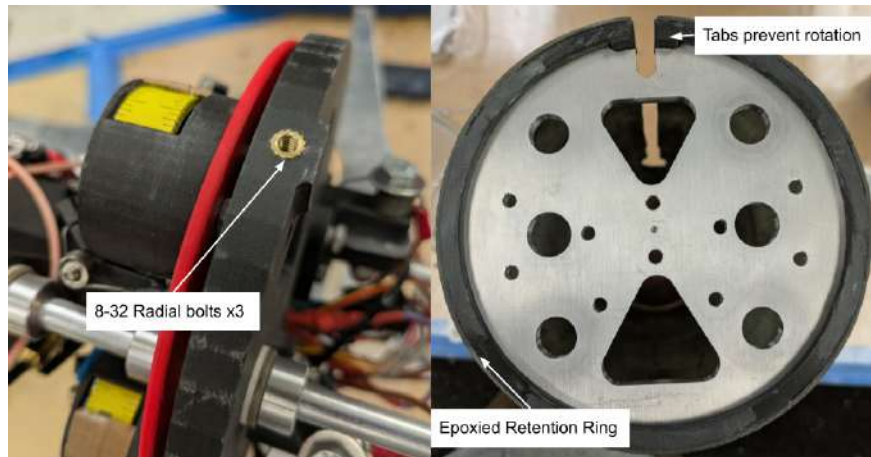


Figure 77: Bulk plate to airframe retention methods. Fore is shown left and is 0.4 in thick, aft is shown right and is 0.25 in thick.

Due to the reduction in internal rods, the team decided to move away from using shaft collars to retain the aft bulk plate as this part withstands the highest load during parachute opening. Instead of a pair of shaft collars on either side of the bulk plate on each rod, the rod is now threaded with 5/16-28 threads to attach 2 nuts on either side to keep the bulk plate retained.

Rubber tubing was added to the aluminum rods in the skeleton in order to further protect electronics and other components. This tubing is made of 50A medium-soft rubber, and has a wall thickness of 1/8 in. The tubing was added to the fore and aft bulk plates, the SCM, and the electronics bay.

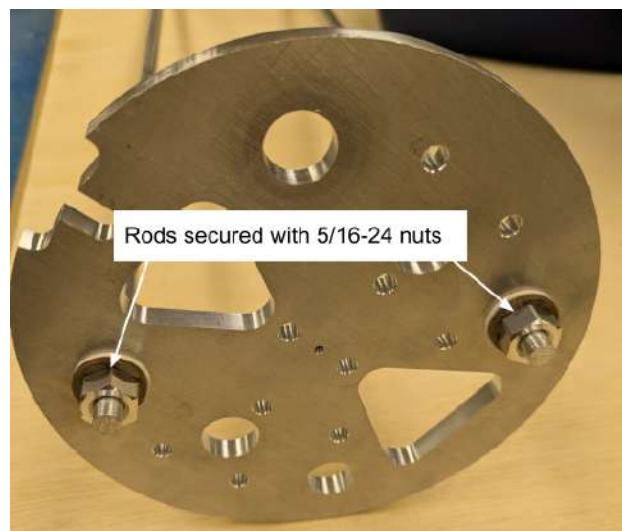


Figure 78: Aft bulk plate rod retention. This bulk plate contains the hooks which retain the payload during flight.

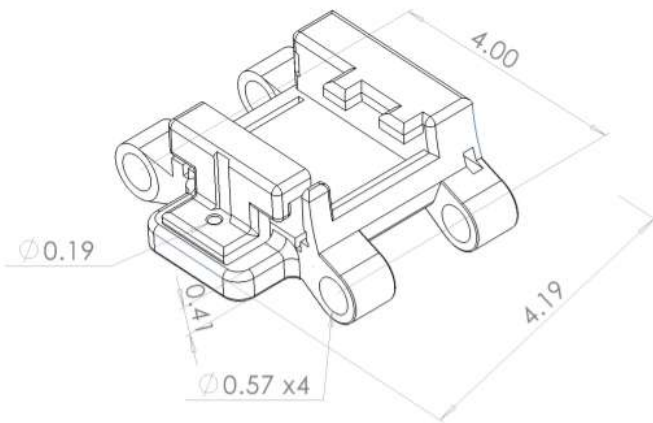


Figure 79: Electronics bay. Dimensions in inches.

The electronics bay sits within the lander nose cone right behind the SCM. The Raspberry Pi 5 flight computer is securely retained using a clip that is fixed to the bay using a 10-32 bolt. The entire assembly is made of 3D printed Onyx filament and has been validated in numerous drop tests and launches.

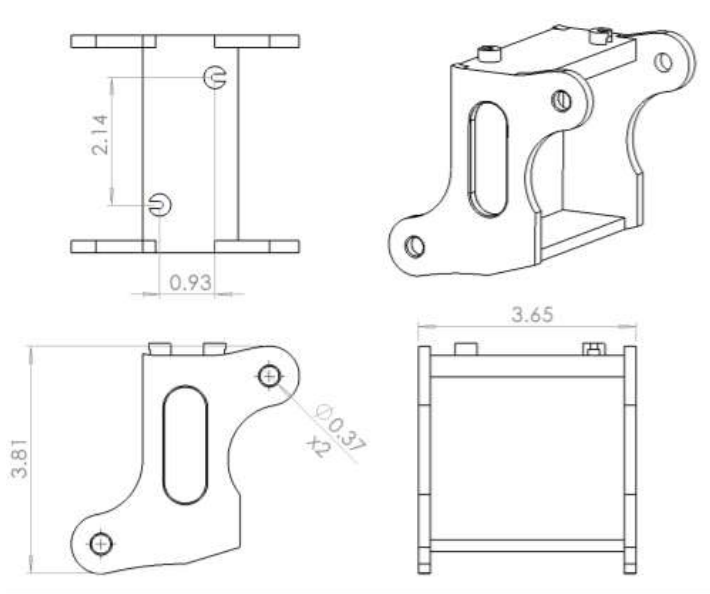


Figure 80: Battery bay. Dimensions in inches.

The battery bay sits in the rear of the lander closest to the aft bulk plate. The battery bay also retains the payload GPS board on its top. This ensures that the GPS transmitter is as far as possible from the main RF system and prevents interference issues. This rearwards placement also allows it to act as a counterweight to the electronics bay and balance out the lander.

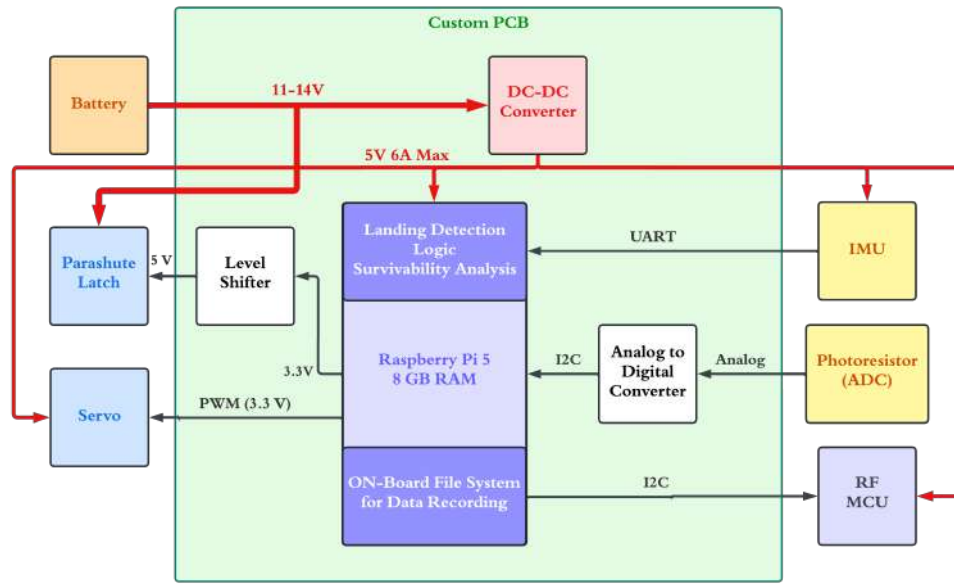


Figure 81: Electronics System Overview

4.3 Electrical Subsystems

The onboard electrical design has remained unchanged since the CDR, and an updated figure provides more detail. A custom PCB now integrates the DC-DC buck converter, level shifter, ADC, and wire connectors. The Raspberry Pi handles multi-threading for data collection, real-time analysis, sensor communication, and RF coordination.

Actuators include a parachute latch to reduce drag after landing and a servo for antenna extension. The RF microcontroller manages transmissions under the Pi's control. Reliability is ensured through component derating, safety margins, obeying NASA's Workmanship Standards, and repeated tests.

Past issues—power disconnects, RF dropouts, wire management, EMF interference, and sensor anomalies—were thoroughly considered. This section presents detailed design changes and improvements that addressed all potential failure points and culminated in a successful full-scale launch operation.

4.3.1 RF Telemetry Format

This year, the VADL Team has decided to transmit all eight types of data to the NASA receiver.

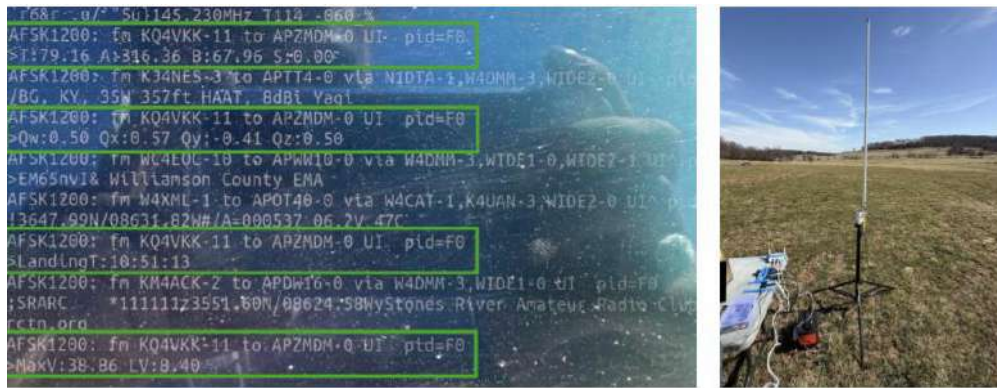


Figure 82: Field decoding in progress. At the time this picture was taken, survivability data was still in development. The payload transmission is displayed in the green boxes (left), and the ground station setup, consisting of a Diamond X50 antenna and an RTL-SDR decoder, is shown on the right.

The RF data is transmitted in four structured packages, as shown in Figure 82. Each package contains specific parameters. Table 27 provides a breakdown of the symbols and their corresponding meanings:

Symbol	Meaning
T	Ambient Temperature (°F)
A	Apogee Altitude (ft)
B	Battery State of Charge (%)
S	Survivability Index (Dynamic Response Index)
Qw, Qx, Qy, Qz	Orientation (Quaternions)
LandingT	Landing Time (HH:MM:SS UTC)
LG	Landing Acceleration (m/s^2)
MaxV	Maximum Velocity (ft/s)
LV	Landing Impact Velocity (ft/s)

Table 27: Symbols and Meaning of Parameters in RF Packages

An example of an RF data package is shown below:

```
>T:79.16 A:316 B:100 S:1.2
>Qw:0.1 Qx:0.1 Qy:0.1 Qz:0.1
>LandingT:10:10:10
>MaxV:10.00 LV:8.40 LG:100
```

This structured format ensures efficient transmission and decoding of critical flight data.

The landing acceleration will be added to the system soon.

4.3.2 RF Transmission Tests

Reliable RF transmission and decoding are vital for project success. Following VADL Requirement 4.6.2, the system was tested to ensure reliable transmission over a minimum distance of 1

mile (Applying a safety factor of 2). To this end, six field transmission tests were conducted, both qualitatively and quantitatively. Successful is the most important metric for these tests, as it ensures that the RF system is capable of actually transmitting the telemetry. In summary, the payload is capable of reliably transmitting all telemetry packages over 1 mile under the worst possible conditions. The general test procedure is shown below:

Setup

1. The RF board is programmed to transmit telemetry packages at 10-second intervals.
2. The ground station is positioned in a safe location, away from roads and interference.
3. The ground station antenna is set up with proper alignment and safety considerations.

Transmission & Data Collection

1. The payload team starts at the ground station, making the initial data reception.
2. The payload, actively transmitting, is carried away from the station.
3. At set intervals, the team stops, allowing the ground station to collect 10 received packets before proceeding further.
4. The payload team records GPS coordinates at each stop for later distance calculation.

Ground Station Reception & Signal Strength Measurement

1. The ground station operator, using an RTL-SDR connected to a computer, runs RTL-SDR's APRS decoder on the 2M band via Linux WSL.
2. Once packet reception is verified, the operator closes the decoder (USB access is limited to one process at a time).
3. SDRUno is launched to measure and record RSSI (signal strength) in dBm.

Post-Test Data Processing

1. Both teams reconvene in the lab.
2. GPS data is processed to calculate precise transmission distances.
3. Packet reception rates and RSSI values are analyzed to assess transmission reliability.

Test	Range (ft)	Strength (dBm)	Decoding	Radio Sound
0	5	-2	Success	Clear
1	180	-32	Success	Clear
2	338	-40	Success	Clear
3	880	-59	Success	Clear
4	1232	-62	Success	Clear
5	1434	-68	Success	Clear
6	1565	-71	Success	Clear
7	1781	-71	Success	Clear
8	1892	N/A	Success	Clear
9	2011	N/A	Success	Clear
10	2263	N/A	Success	Clear
11	2505	N/A	Success	Clear
12	2669	N/A	Success	Clear
13	2925	N/A	Success	Clear
14	3146	N/A	Success	Clear
15	3316	N/A	Success	Clear
16	3568	N/A	Success	Clear
17	3839	N/A	Success	Clear
18	4063	N/A	Success	Clear
19	4300	N/A	Success	Clear
20	4566	N/A	Success	Clear
21	4815	N/A	Success	Clear
22	5068	N/A	Success	Clear

Table 28: Transmission and Decoding Result from the Centennial Park Test

The results of the Centennial Park test are summarized in Table 28. Beyond approximately 1781 ft, equipment limitations caused the APRS signal to be embedded in a noise region around -70 dBm. Despite this, both decoding performance and radio clarity remained robust. As a result, the test shifted from a quantitative to a qualitative evaluation focusing on successful decoding and clear radio reception.

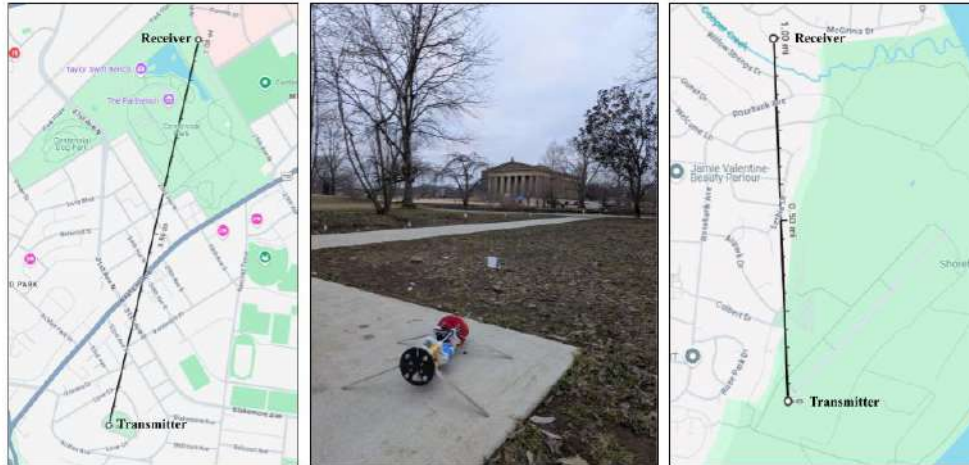


Figure 83: Left: Map of the Centennial Park Test (Fully Extended Condition). Middle: Photo from the Centennial Park Test. Right: Map of the Cornelia Fort Air-park Test (Collapsed Configuration).

In addition to the Centennial Park Test, another important test was conducted at Cornelia Fort Air-park to evaluate RF performance in a constrained configuration. This test simulated scenarios where the antenna fails to fully extend.

Despite operating without a clear line of sight due to dense tree cover, the system consistently successfully decoded data in the in-rocket configuration, demonstrating the transmission system's robustness under restricted deployment conditions.

4.3.3 PCB Design and Verification

Following the initial concept presented in the CDR report, the team has successfully brought the design to reality. Two iterations of the PCB have been developed. The first lacked battery measurement circuits, while the current version integrates them and significantly improves reliability and EMF shielding.

The custom PCB is a four-layer design with a large continuous ground plane. This ensures robust signal integrity and minimizes EMI. It is designed to fit on the Raspberry Pi 5 like a HAT and integrates the following key subsystems: Onboard buck converter, Level shifters, ADC circuits for the photoresistor, State LED indicators, and Battery measurement circuits

By consolidating all systems onto a single PCB, the payload achieves greater reliability than the previous subscale design. This design change eliminates excessive external modules and reduces failure points caused by complex wiring. For instance, overuse of prefabricated modules in earlier versions led to loose connections and interference issues.

All wiring is directed through Molex connectors, offering secure and vibration-resistant connections. This further enhances robustness by preventing loose wire terminations under dynamic conditions. Hot glue will be applied to enhance reliability further.

The PCB design follows industry standards to ensure manufacturability and reliability. Key rules include a minimum trace width of 6 mil (0.152 mm) to prevent signal degradation, minimum trace spacing of 6 mil (0.152 mm) to avoid crosstalk, via size of 0.3 mm drill with a 0.6 mm

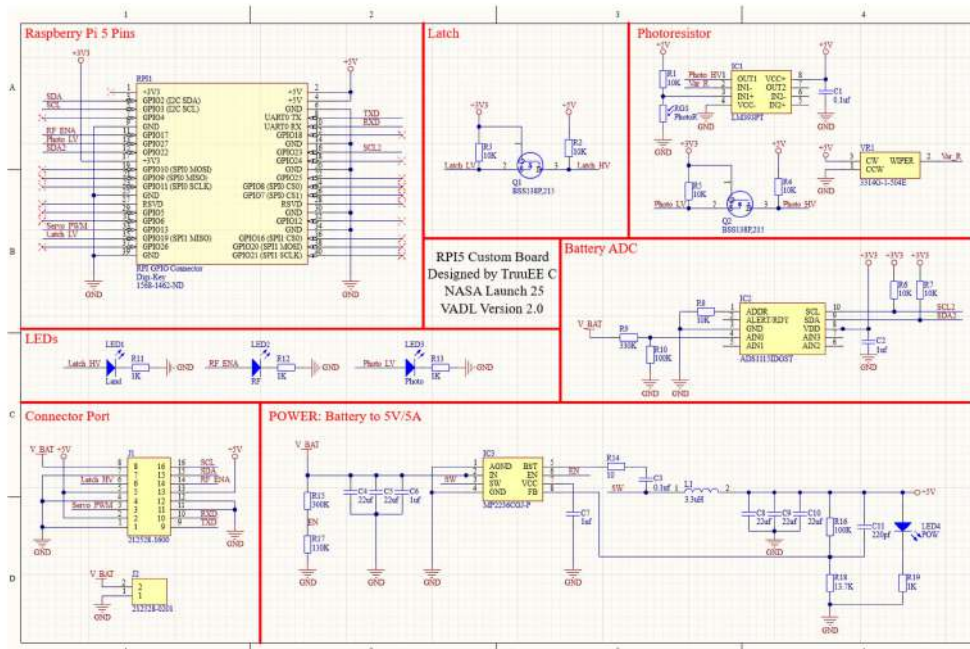


Figure 84: Schematics of the PCB.

pad for robust electrical connections, and solder mask clearance of 3 mil (0.075 mm) to prevent unintended solder bridging. These design constraints ensure compliance with aerospace-grade reliability standards.

Soldering of SMT components strictly follows **NASA-STD-8739.2**:

“The solder joint surface is smooth, nonporous, undisturbed, with a finish varying from satin to bright. The fillet completely wets all elements to the periphery of the connection and is concave.”

Extensive tests were performed on the power circuits to verify reliable voltage output:

- The MP2236 buck converter maintains stable voltage regulation.
- When the MP2236 output was at **0V**, the ripple measured via oscilloscope was **26mV**.
- Under normal operation, the chip temperature stabilizes at **57.7°C (135.86°F)** while supplying power.

This design ensures **high reliability**, reduced EMI, and seamless integration, significantly improving on previous iterations.

The PCB has successfully passed all **drop tests**, **hard landing tests**, and **full-scale launch evaluations**, demonstrating its superior reliability compared to previous designs.

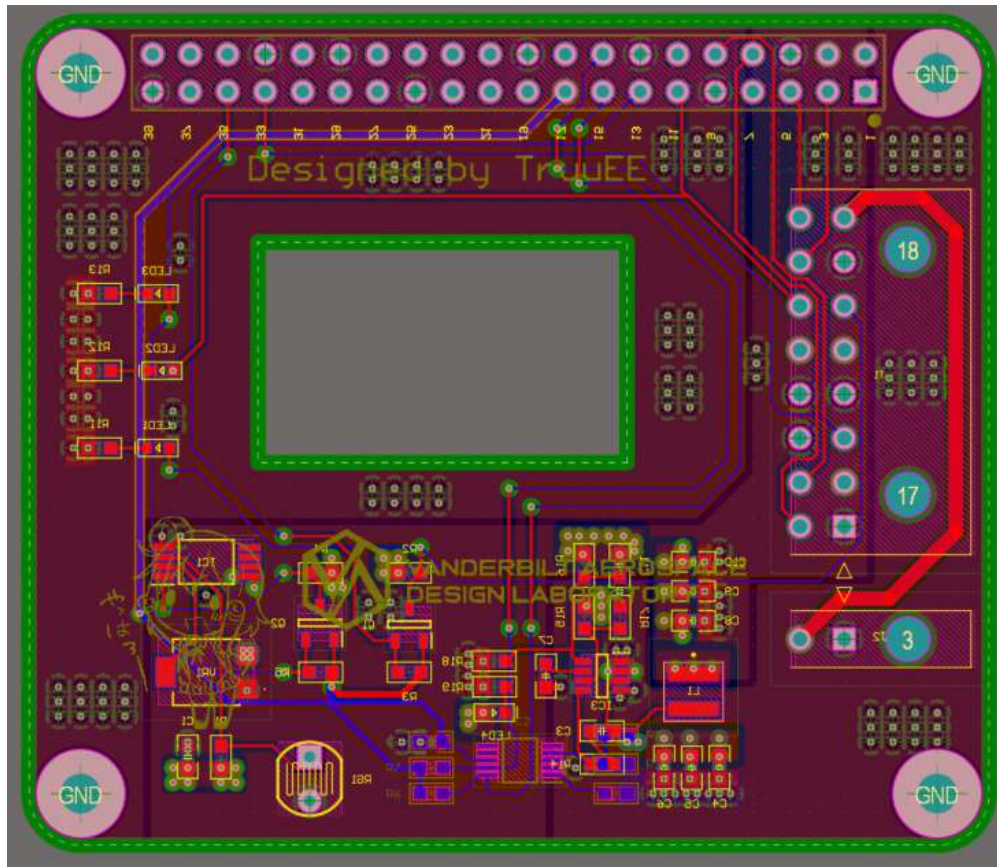


Figure 85: Layout of the PCB.

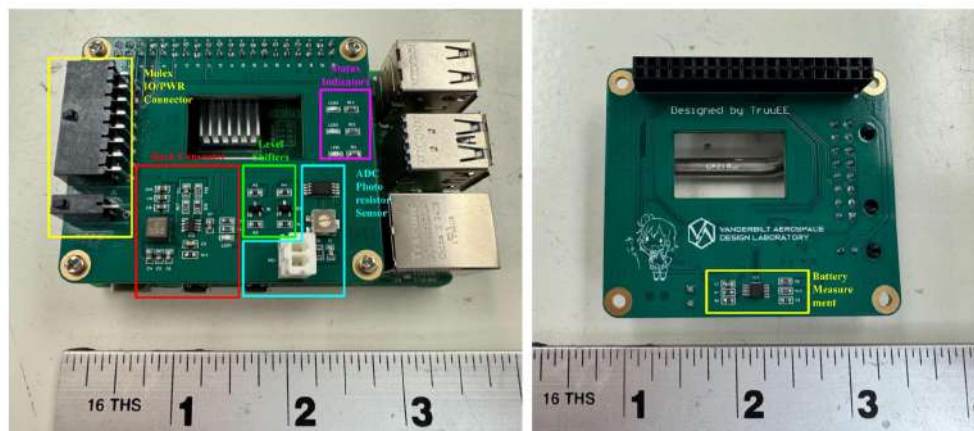


Figure 86: Assembled PCB with major subsystems labeled.

4.3.4 Reliable Landing Detection

The redundancy-embedded landing detection logic presented in the CDR design cycle report remains unchanged. To further validate the system's ability to detect landing events, tests were conducted both in the lab and in the field. While the drone test has been detailed in previous reports, this section focuses on lab-based validation methods.

Two controlled tests were conducted to validate landing detection in a repeatable and efficient manner. Unlike the **drone test**, which requires additional personnel and has operational constraints, these methods provide a more practical way to verify the system's accuracy.

Elevator Test

The payload is placed in an elevator at ground level. The program starts, calibrating the initial altitude. The elevator ascends to the 8th floor and then descends. Around the 2nd or 1st floor, the payload is shaken to simulate impact. After landing detection, post-landing procedures run, data is decoded, and communication integrity is verified.

Pressure Cooker Test

The payload electronics, without the full frame, are placed inside a pressure cooker. The existing system pulls air out, reducing pressure to simulate altitude gain. Once the target pressure is reached, valves are closed, and pressure is gradually restored, simulating landing. The payload is then removed and shaken to mimic landing acceleration.

Key Advantages of These Tests:

- Provides a **controlled, repeatable environment** for validation.
- Simulates altitude changes and impact conditions efficiently.
- Avoids logistical constraints of drone testing while maintaining reliability.

4.3.5 Shielding and Interference

During the elevator test, I2C communication consistently failed when the RF system started transmitting. The issue was traced to two main factors: the design change that moved the antenna closer to the electronics bay and the elevator boosting the RF signal, which generated significant EMF interference that disrupted I2C communication. To prevent this issue in the field, especially if other teams are transmitting at high power, multiple mitigation strategies were implemented. The new PCB version incorporates ground poly pours on every layer, eliminating dead poly islands. This industry-standard practice ensures lower impedance grounding, improving overall circuit stability, while also enhancing EMF shielding to reduce RF interference with sensitive signals.

It was also determined that grounded shielded wires provide superior EMF protection compared to standard wires. Shielded wires block external electromagnetic noise from coupling into signal lines and reduce radiated emissions. During the full-scale launch, shielded wires were manually fabricated using copper tape and heat tape. In the current iteration, commercially shielded wires are used instead. The conductive shielding is properly grounded, eliminating floating metal that could act as an unwanted EMF source. The entire manufacturing process follows NASA-STD-8739.4, which ensures compliance with aerospace-grade assembly standards.

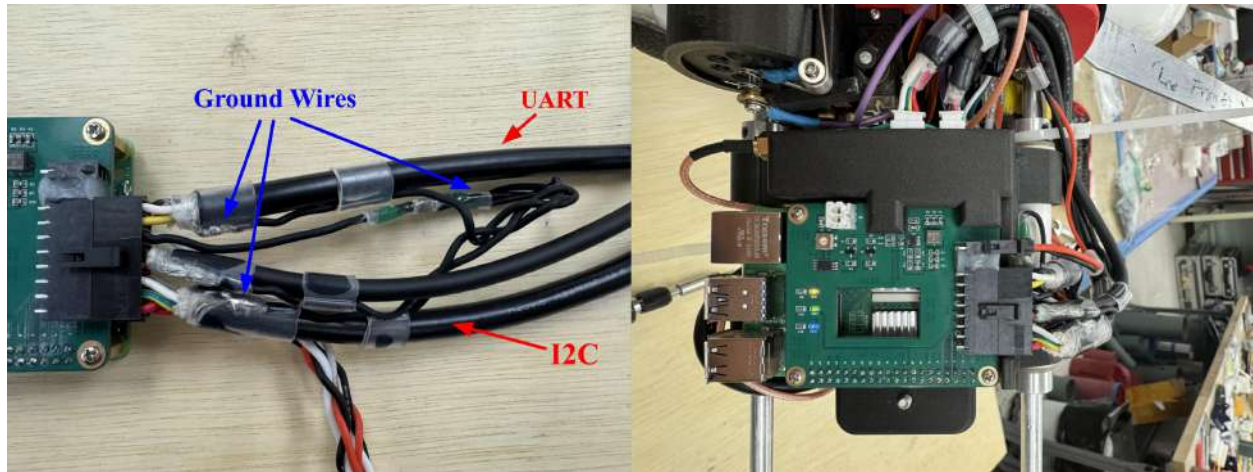


Figure 87: Shielded wires for improved EMF protection.

To further safeguard I2C communication, a 15-second transmission delay was introduced in the software. This allows I2C to first transfer data to the RF board. This will ensure that even if RF transmission disrupts I2C sync, the data is already stored on the RF board. This fail-proof approach guarantees data integrity, even in high RF activity environments.

These combined hardware and software measures effectively eliminate RF-induced I2C failures and ensure robust and field-ready performance.

4.3.6 Survivability Estimation

The **Dynamic Response Index (DRI)** is used to assess spinal compression and predict the likelihood of significant injury during landing. This concept was introduced in the CDR report. DRI has been used by NASA in numerous critical missions like the Apollo and Artemis programs for ejection seat design. The computation follows a second-order differential equation that models the human spine as a **damped dynamic system under shock acceleration**:

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \ddot{z}(t) \quad (21)$$

where:

- $x(t)$ = relative deflection of the spine
- $\dot{x}(t)$ = relative velocity
- $\ddot{z}(t)$ = shock acceleration input from IMU
- $\omega_n = 52.90$ rad/s = spinal system natural frequency
- $\zeta = 0.224$ = damping ratio

The maximum value of $x(t)$ is then used to compute **DRI**:

$$DRI = \frac{\omega_n^2}{g_{surface}} X_{\max} \quad (22)$$

A **DRI of 17.7** corresponds to a **10% probability** of significant injury, classified as **low risk**. A probability between **10% and 30%** is considered **moderate risk**, while values exceeding **30%** are categorized as **high risk**. A **DRI of 21** equates to a **30% chance** of significant injury.

State-Space Representation The solution to this second-order differential equation can be obtained using the **state-space representation**:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \ddot{z}(t) \quad (23)$$

The corresponding **state-space matrices** are:

$$A = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (24)$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \end{bmatrix} \quad (25)$$

In the Python code, the function **control.ss(A, B, C, D)** constructs a **state-space system** using the derived matrices. Then, the control solver simulates the system's response to the input spinal acceleration using the **state-space equations**, with **zero initial conditions**. In these cases and in the actual payload, the spinal acceleration was used to calculate DRI on its own.

MATLAB DRI	Real-Time DRI	% Error
21.5827	21.5909	0.04%
28.7033	29.0278	1.13%
16.7244	15.9697	4.51%
9.2527	9.5962	3.71%
24.4998	24.12	1.55%
16.171	16.34	1.05%

Table 29: Real-time DRI Estimation vs. MATLAB Simulated DRI from Pressure Cooker In-Lab Test

To validate the robustness of the proposed method, tests were conducted in a pressure cooker to simulate launch and landing conditions, as described in Section 4.3.5. The real-time DRI values computed onboard were compared against MATLAB post-processed DRI values. The results show a percent error of less than **5%**, confirming the legitimacy of transmitting real-time DRI during flight.

Further validation will be performed through **drone drop tests** and **full-scale launches** to reinforce the reliability of the survivability estimation system under actual flight conditions.

4.3.7 Battery Measurements

Due to the non-linear relationship between battery percentage and voltage, **machine learning** was used to estimate battery life instead of assuming a linear model.

The first step is obtaining an accurate voltage reading from the battery. This was achieved using a simple yet reliable **voltage divider circuit** along with an ADC IC ADS1115. The voltage divider consists of $R_1 = 330K\Omega$ and $R_2 = 100K\Omega$, with a maximum input voltage of 12.60V. The output voltage is calculated as:

$$V_{\text{out}} = 12.60V \times \frac{100K\Omega}{330K\Omega + 100K\Omega} = 2.93V \quad (26)$$

This value is within the ADC's safe operating range of $< 3.3V$, ensuring it will not damage the electronics.

The second step is mapping voltage to battery percentage. Using the **manufacturer's voltage-to-battery percentage table**, a **polynomial regression** was performed to model the non-linear relationship. The resulting equation is:

$$y = 740.79x - 28.87x^2 - 4653.88 \quad (27)$$

where y represents battery percentage and x is the measured voltage.

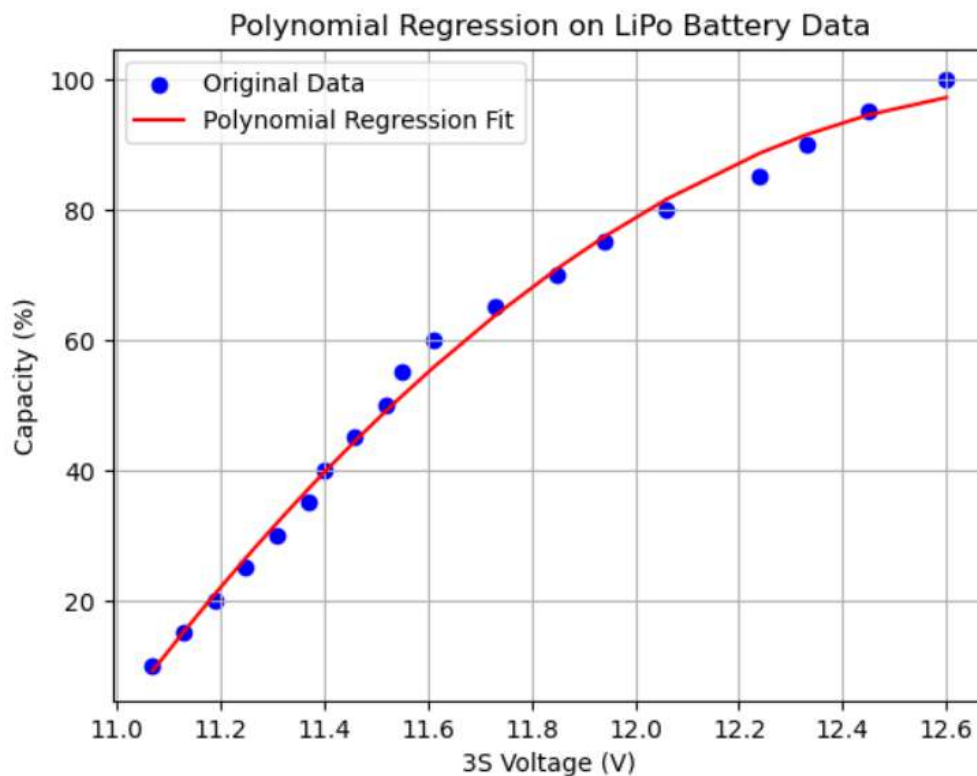


Figure 88: Polynomial regression results on LiPo battery data, based on manufacturer-provided values.

Finally, the trivial software implementation is straightforward, allowing real-time battery percentage estimation. This system provides visibility into payload power levels and helps the team evaluate constraints and optimize battery usage.

4.4 Payload Demonstration Flight

There are two major milestones remaining with respect to payload testing. The first is a final hexacopter drop test in which the fully functional payload will be demonstrated. All functional parts have been demonstrated at different points throughout testing, but a test has not yet been conducted where full functionality is achieved on 100% of drops on a test day. Therefore, on March 22nd the team will be dropping the full, finalized payload design with integrated electronics from the hexacopter drone. This will enable the team to prove that upright landing, parachute detachment, antenna extension, and data transmission can all be achieved jointly. This will give the team confidence moving into the payload demonstration flight.

The following weekend, on March 29th, the team will conduct its payload demonstration flight (PDF). The team plans on flying the rocket twice on this day in order to gain as much data and experience as possible. For PDF, the rocket will not be flown on the L1720 competition motor. Rather, it will be flown on an Aerotech K1100T, which the team has in its inventory. This decision was made in order to conserve the L1720 inventory for competition and to enable the team to fly PDF at a lower powered launch site.

5 Vehicle Demonstration Flights

On Saturday Feb 22, 2025, VADL flew two full-scale test flights in Olmstead, KY to satisfy the Vehicle Demonstration flight requirements. The team flew its as-built full-scale competition vehicle using the Cesaroni L1720 motor for both flights. Launch conditions for both flights and the launch day timeline are summarized in Table 30 and 31, respectively. Conditions for both launches were appropriate for rocket flight. The team was able to prepare the vehicle for flight in under two hours for both flights and refly the same day without repairs to the launch vehicle, thus satisfying NASA Requirements 2.3 and 2.5. The full-scale, functioning competition payload was flown for both flights, fully powered to complete the mission.

Table 30: Conditions for both flights in Olmstead, KY.

Condition	Flight 1	Flight 2
Temperature (F)	42	45
Average Wind Speed (mph)	3	1
Launch Rail Angle (deg into the wind)	5	1
Cloud Ceiling (ft above ground level)	6000	9000
Elevation (ft above mean sea level)	600	600
Effective Launch Rail Travel (in)	132	132

Table 31: Launch day timeline highlighting key events.

Time	9:30 AM	10:00 AM	11:45 AM	12:30 PM
Event	Arrived at launch field	Launch waiver opened / vehicle preparation started	Vehicle preparation completed	Flight 1
Time	1:00 PM	1:15 PM	2:45 PM	3:15 PM
Event	Vehicle sections retrieved	Vehicle re-preparation started	Vehicle preparation completed	Flight 2

The full-scale vehicle on the launch rail is shown in Figure 89, and key vehicle parameters are summarized in Table 32. The team elected to fly with 2.3 lbm of ballast to more closely resemble the predicted CDR mass and achieve desired flight performance metrics for apogee, descent time, etc.



Figure 89: Full-scale vehicle on the launch rail for flight 2.

Table 32: Key vehicle parameters for the Vehicle Demonstration flights.

Parameter	Value
Length	97 in
Motor	Cesaroni L1720
Dry Mass (unballasted)	35.8 lbm
Dry Mass (ballasted)	38.1 lbm
Wet Mass (ballasted)	42.0 lbm
CG (with motor)	53.7 in
CG (without motor)	46.5 in
CP	66.7 in
SSM (on pad)	2.12

5.1 Flight Objectives

For the Vehicle Demonstration flights, VADL aimed to accomplish several key objectives as outlined in Table 33. Primary vehicle objectives, as informed by NASA requirements for a successful VDF, are listed first and highlighted in gold. Beyond these success criteria, VADL also imposed the requirements of demonstrating the full payload mission from an in-rocket configuration, or in other words, a case where the lander does not jettison and comes down with the launch vehicle. An in-rocket configuration means that the LSM and LDM are not actuated. The PDM does actuate, however does not separate the parachute as the parachute is still retained inside the launch vehicle. The AEM, and electronic systems work nominally to transmit data over RF to the ground station.

Table 33: VDF success criteria and relevant requirements, with vehicle objectives shown in dark gold.

Success Criteria	Relevant Requirements
Achieve L.R.E. velocity >52 fps	NASA 5.17
Reach apogee within 5% of simulated	NASA 2.1; VADL 2.1.1
Demonstrate nominal recovery	NASA 2.19.1.1, 3.1; VADL 3.2, 3.11
Retain payload through flight	NASA 4.1; VADL 4.2.4
Achieve descent time <90 s	NASA 3.12
Drift <2500 ft	NASA 3.11
Achieve landing K.E. <75 lb-ft for all sections	NASA 3.3
Record altimeter and GPS data	NASA 2.19.1.8.1, 2.19.1.8.3; VADL 3.8
Prepare vehicle in under two hours	NASA 2.5
Incur no structural damage	NASA 2.3; VADL 2.1.4
Detect payload landing	VADL 4.7.2
Collect flight telemetry and landing site data	NASA 4.1; VADL 4.7.1-9
Actuate parachute detachment latch	VADL 4.4.1
Extend antennas	VADL 4.5.1-5
Transmit RF data	NASA 4.2.1, 4.2.4, 4.2.6; VADL 4.6.1, 4.6.2

5.2 Flight Results

5.2.1 Flight 1 Results

Key events throughout flight 1 are illustrated in Figure 91 below, and the results of the mission success criteria are depicted in Figure 92. The first flight saw a clean launch and recovery of the vehicle. However, upon reviewing the altimeter data following the flight, the descent velocity under the main parachute was higher than predicted. Closer inspection of the flight video revealed that one of the shroud lines got wrapped around the parachute, thus preventing all 12 gores from fully opening. This issue was resolved with more careful packing for the second flight which, as covered in Section 5.2.2, was fully successful.

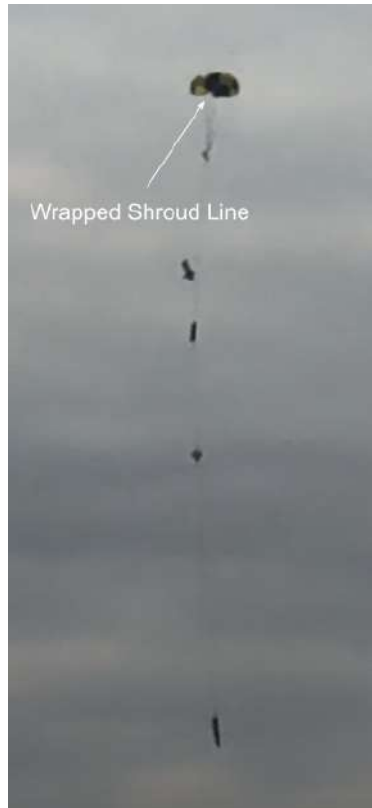


Figure 90: Parachute opening on Flight 1 descent. One of the shroud lines was wrapped over a parachute gore upon deployment preventing full opening.

Additionally, the team-imposed secondary objective of completing the payload mission via an in-rocket configuration was not fully successful due to an electrical disconnect for the payload flight computer during flight. This issue was resolved for the second flight, which saw a fully successful payload mission. The payload performance during the Vehicle Demonstration flights is covered in further detail in Section 5.4.

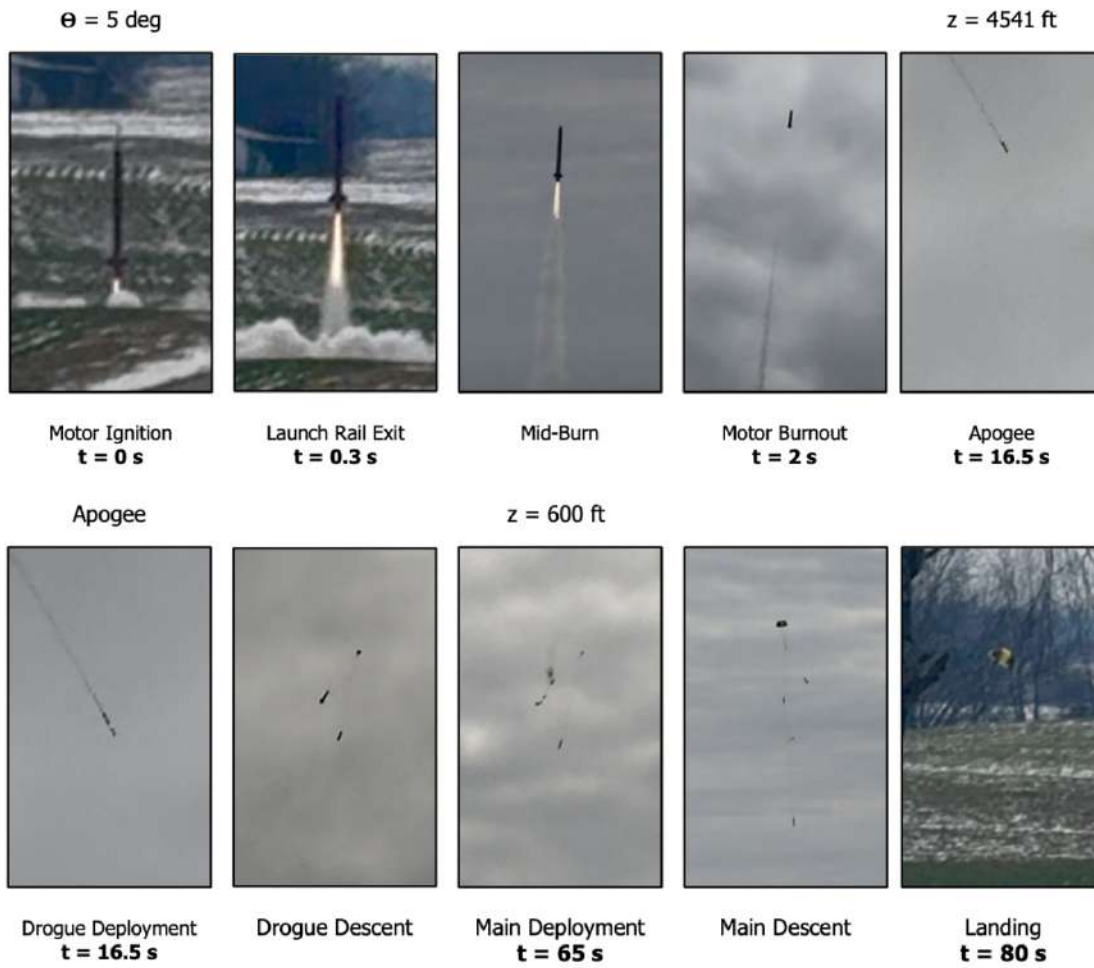


Figure 91: Key events throughout flight 1.

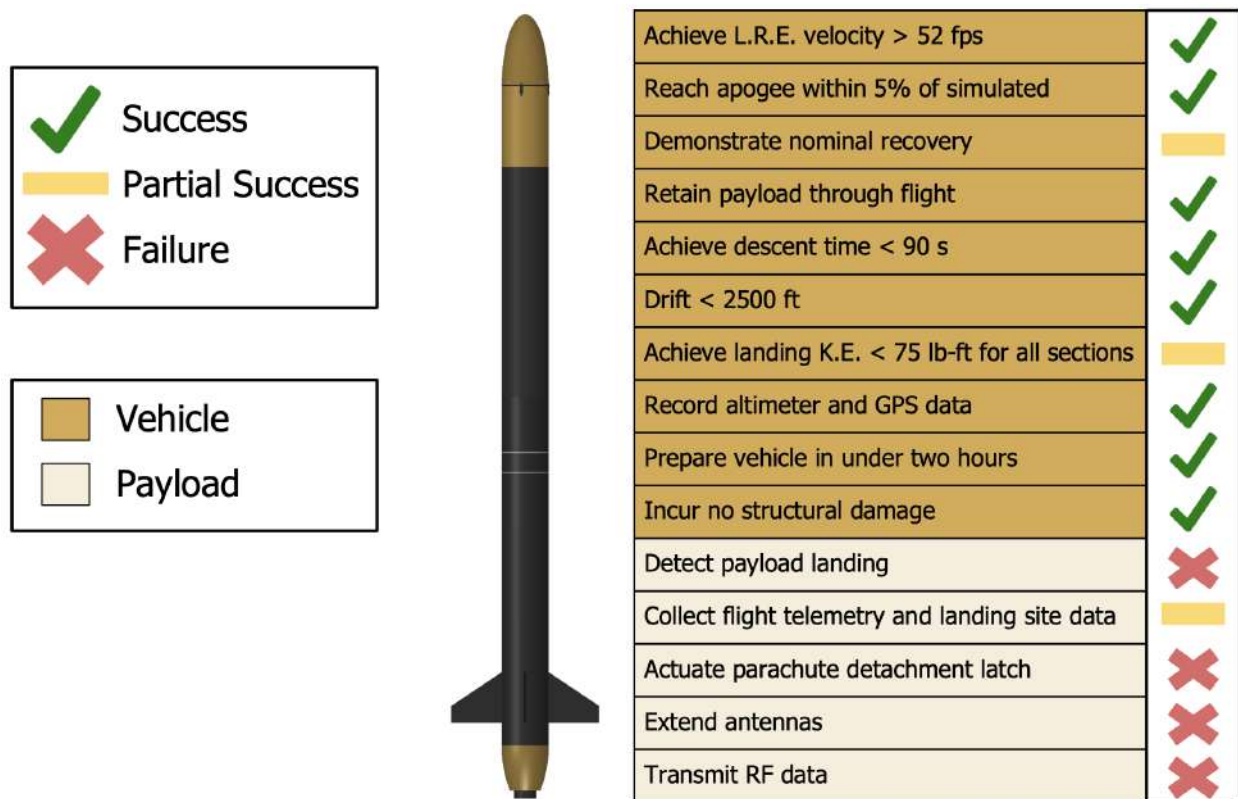


Figure 92: Outcomes of mission success criteria for flight 1.

The landed configurations of the three vehicle sections are shown in Figure 93. As seen, no structural damage was incurred despite landing in an icy area. All three sections were certified for reflight, reprepared for flight in under two hours, and reflown the same day without any repairs.



Figure 93: Landed configurations of tail section (left), recovery section (middle) and payload section following flight 1.

5.2.2 Flight 2 Results

The second flight key events and outcomes are summarized in Figures 94 and 95 below. As noted, the minor issues that occurred during the first flight were quickly mitigated, allowing the team to reprepare the rocket within two hours and relaunch the same day. All primary and secondary objectives were met in this second flight, as depicted in Figure 95. The team is confident in having verified all requirements for the launch vehicle and the NASA payload mission, and it looks forward to verifying the team-imposed requirements of payload jettison and landing during the Payload Demonstration flights at the end of March.

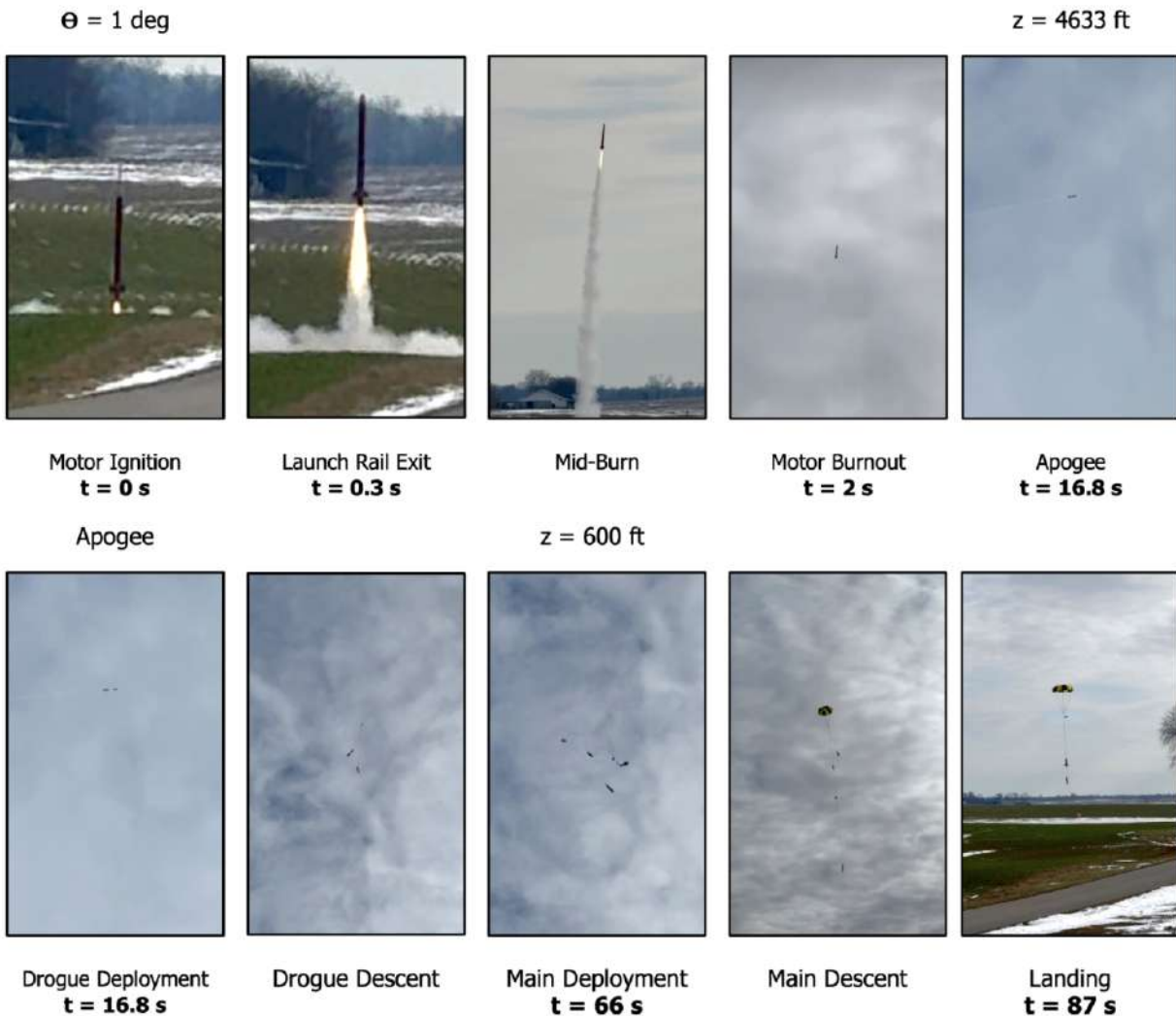


Figure 94: Key events from flight 2.

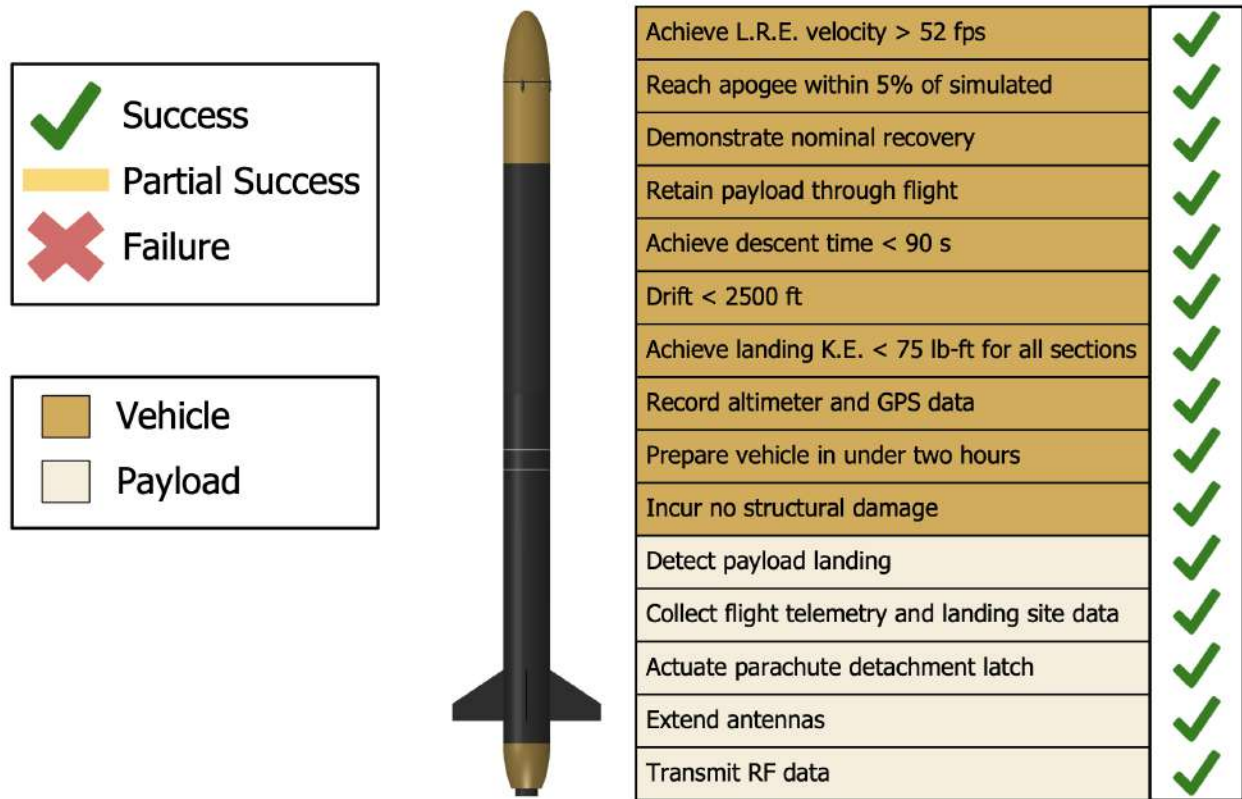


Figure 95: Results from flight 2, demonstrating full mission success for both the vehicle and payload.

The landed configurations of the three vehicle sections after flight 2 are shown in Figure 96. Again, no structural damage was incurred to any of the sections. Additionally, the payload is shown with antennas extended as part of its fully successful payload mission completion.



Figure 96: Landed configurations of all three sections following flight 2. From left to right: tail section, recovery section, and payload section featuring deployed antennas.

5.3 Flight Data and Analysis

5.3.1 Flight 1 Analysis

In Flight 1, the L1720 motor and the fullscale vehicle delivered successful and up-to-spec mission performance during rocket ascent. Analyses of motor thrust delivery, launch rail exit velocity, achieved apogee, and vehicle drag characteristics detailed in this section show their compliance with VDF success criteria and relevant requirements outlined in Table 33. Close alignments between theoretical and experimental results validated that the rocket was manufactured well up-to-spec, both in terms of structural rigidity and aerodynamic performance. During descent, vehicle recovery was achieved without damage to any vehicle structure. Deviations in key descent outcome metrics from pre-flight predictions are precised using IMU and altimeter data, which are also discussed in this section. Unless otherwise noted, pre-flight models refer to simulations conducted before each flight using environmental and vehicle measurements obtained on launch day (Tables 30 and 32) and in line with the theoretical methodology outlined in Section 3.3.

Altimeter Data Analysis

Table 34: Launch 1 Pre-flight model predictions versus actual achieved apogee.

<i>* with 3.5% launch rail friction (pre-flight model)</i>				
Flight 1	Apogee (ft)	Predicted Apogee (ft) & Relative Error		
		Rocksim	CFD-DRSS	OpenRocket
		4404	4403	4025
		-3.1%	-3.1%	-12.8%

Figures 97 and 98 present raw AGL altimeter data during Flight 1. Table 34 show the rocket achieved an apogee of 4541 ft. The pre-flight prediction of 4400 ft was the target altitude for this launch. Pre-flight simulations using the industry standard software RockSim and VADL's in-house DRSS software [5] with a 3.5% launch rail friction factor underpredicted apogee by only ~3%. Launch rail friction is a highly non-deterministic phenomenon. Current literature has shown that launch rail friction may account for up to 4% loss in effective motor delivered impulse [8]; the 3.5% loss factor was selected in VADL's pre-flight model to account for friction at the rail. While VDF launch day winds were expected to not exceed 5 mph, for consistency with the theoretical modeling of the Competition Day simulations (Section 3.3) and to ensure compliance with NASA's minimum apogee requirement of 4000 ft, the 3.5% factor was conservatively retained in VDF pre-flight models. While vehicle modeling, environmental parameters and motor specification were consistent across the three softwares, OpenRocket's deviation in apogee prediction was due to differences in drag estimation performed by the software relative to RockSim and DRSS.

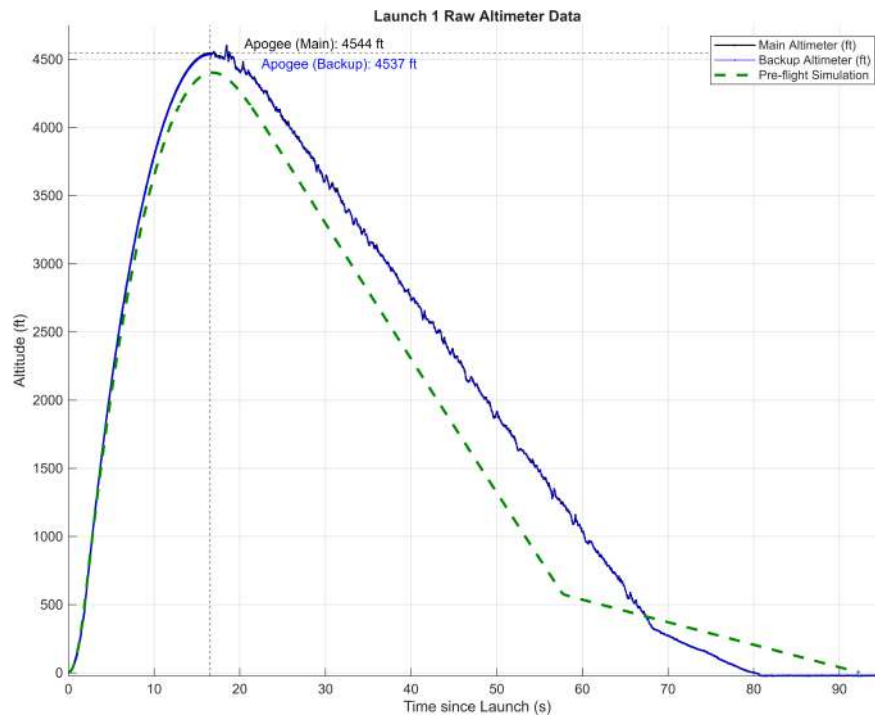


Figure 97: Raw Avionics Altimeter Data vs. Simulation.

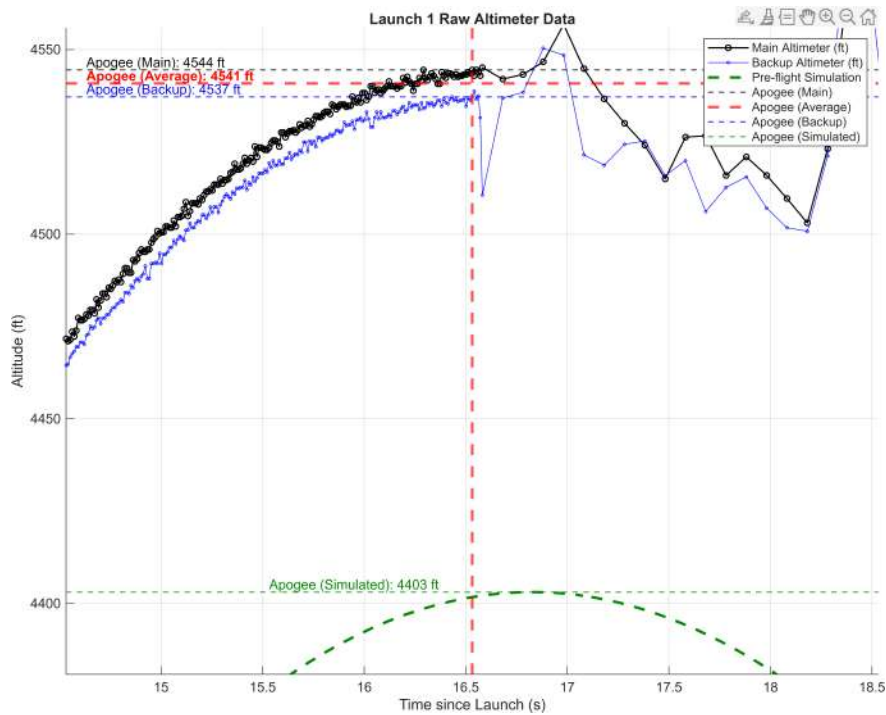


Figure 98: Raw Avionics Altimeter Data vs. Simulation (zoomed in).

Adjusting the friction factor to 1.5% significantly improved model accuracy: Rocksim and DRSS predictions aligned within $\pm 1\%$ of measured apogee (Table 35), confirming the notion that side wind is proportional to launch rail friction. While rail friction is not the sole variable behind vehicle ascent metrics, as detailed in subsequent analyses, VADL is confident that other ascent aspects such as drag characteristics and motor impulse delivery are unlikely causes behind the marginal deviations in apogee prediction — that is, launch rail friction presents a most significant reason behind the $\sim 3\%$ underprediction in apogee. While the experimental vehicle drag was remarkably close to the CDF results, the simulations in Table 35 utilize the empirically obtained Cd estimates from Figure 99.

Table 35: Launch 1 Adjusted model predictions with 1.5% launch rail friction and experimental Cd versus actual achieved apogee.

** with 1.5% friction factor (adjusted model)*

	Apogee (ft)	Predicted Apogee (ft) & Relative Error		
		Rocksim	CFD-DRSS	OpenRocket
Flight 1	4541	4559	4535	4153
		0.4%	-0.1%	-9.3%

To characterize vehicle drag during ascent, the LSM flight computer IMU data and avionics flight computer altimeter data were analyzed to reconstruct estimated drag coefficients. Equation 28 below was applied during coast ascent (no thrust forces):

$$C_D = \frac{2m(g_0 \cos \theta + a_{\text{axial}})}{\rho A v_{\text{axial}}^2} \sqrt{1 - \left(\frac{v_{\text{axial}}}{c_{\text{sound}}} \right)^2} \quad (28)$$

Axial velocity was calculated as:

$$v_{\text{axial}} = \frac{v_{\text{vert}}}{\cos \theta} + v_{\text{wind}}(z) \cdot \sin \theta \quad (29)$$

Due to minimum wind cocking effects in low wind conditions, the velocity component reconstructed in the rocket's axial direction accounted for side wind velocity as a function of altitude (Equation 6) and angle of the launch rail.

Atmospheric density ρ followed the hypsometric equation using the 1976 US Standard Atmosphere $T_0 = 288.15$ K, $P_0 = 101325$ Pa and lapse rate $\beta = 6.5 \times 10^{-3}$ K/m.

The compressibility term $\sqrt{1 - \left(\frac{v_{\text{axial}}}{a_{\text{sound}}} \right)^2}$, provided Prandtl-Glauert corrections for Mach numbers below 0.8.

As shown in Figure 99, the experimental drag coefficient closely aligns with pre-flight CFD simulations. This is further supported by a close alignment between altimeter and predicted vertical velocity during vehicle ascent (Figures 100a and 100b). For this reason, vehicle drag is assessed to not be a likely cause behind the initial discrepancy in predicted apogee. The alignment confirms that VADL's fullscale vehicle construction was up-to-spec relative to its designed geometry.

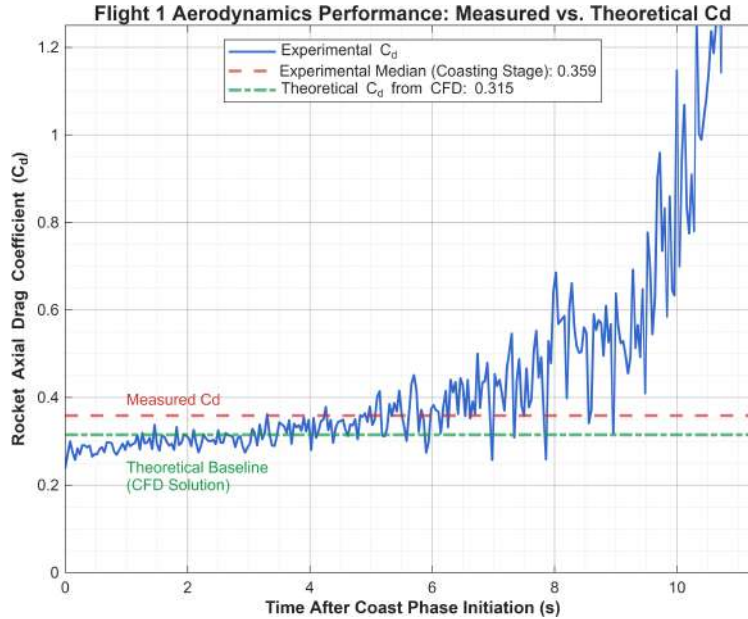
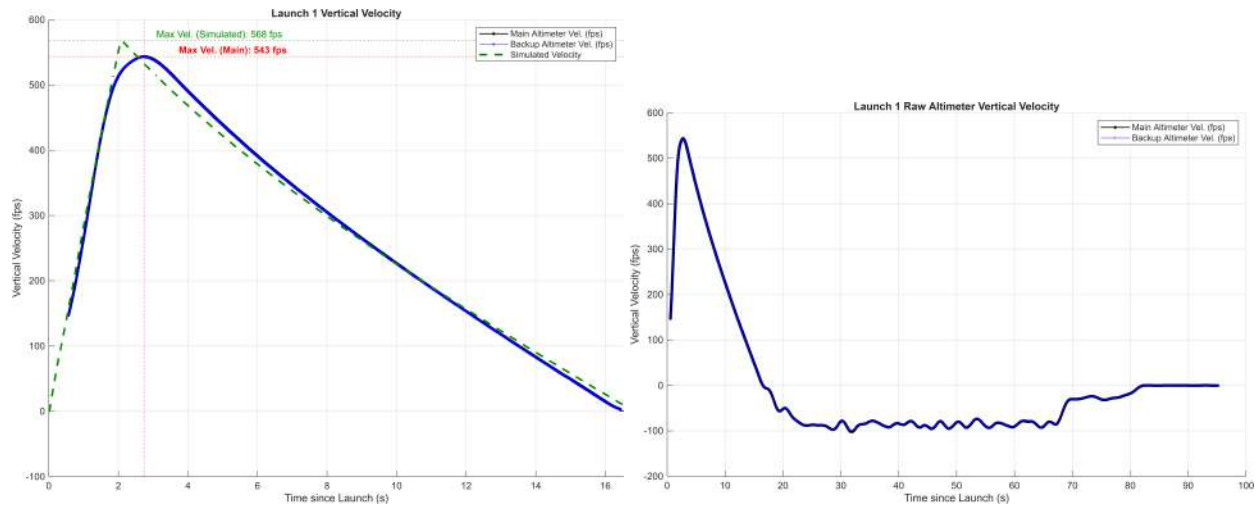


Figure 99: Estimated vehicle drag computed from IMU and altimeter data during coasting phase (burn out to apogee).



(a) Ascent only.

(b) Ascent and descent.

Figure 100: Raw Altimeter Velocity Data. Data points unavailable during the first second of flight was due to excessive data cropping performed by the EasyMini altimeters, which utilize an aggressive altitude change threshold for launch detection to prevent mistriggers.

Motor Thrust Analysis

The LSM flight computer IMU provides data for computation of motor thrust delivery. It is important to note that IMU-measured kinetics data necessarily deviates from the nominal L1720 motor thrust curve; this is due to two reasons: (1) as mentioned in previous sections and informed by current literature [8], launch rail frictional forces may impose up to 4% of effective motor thrust, and (2) non-negligible vehicle drag before MECO further reduce IMU-measured acceleration. VADL therefore addresses these two factors by comparing the actual IMU measurements against simulated IMU data while factoring in rail friction and vehicle aerodynamic drag. In Figure 101, launch rail friction is apparent in the reduced peak in measured force data (red curve) in the first 0.2 seconds after motor ignition, and the tapering trend observed in the last 50% of motor burn time directly correspond to increasing vehicle drag, as simulated by DRSS in the purple dotted curve. Motor burn time is graphically verified to remain consistent with expectations (Figure 101).

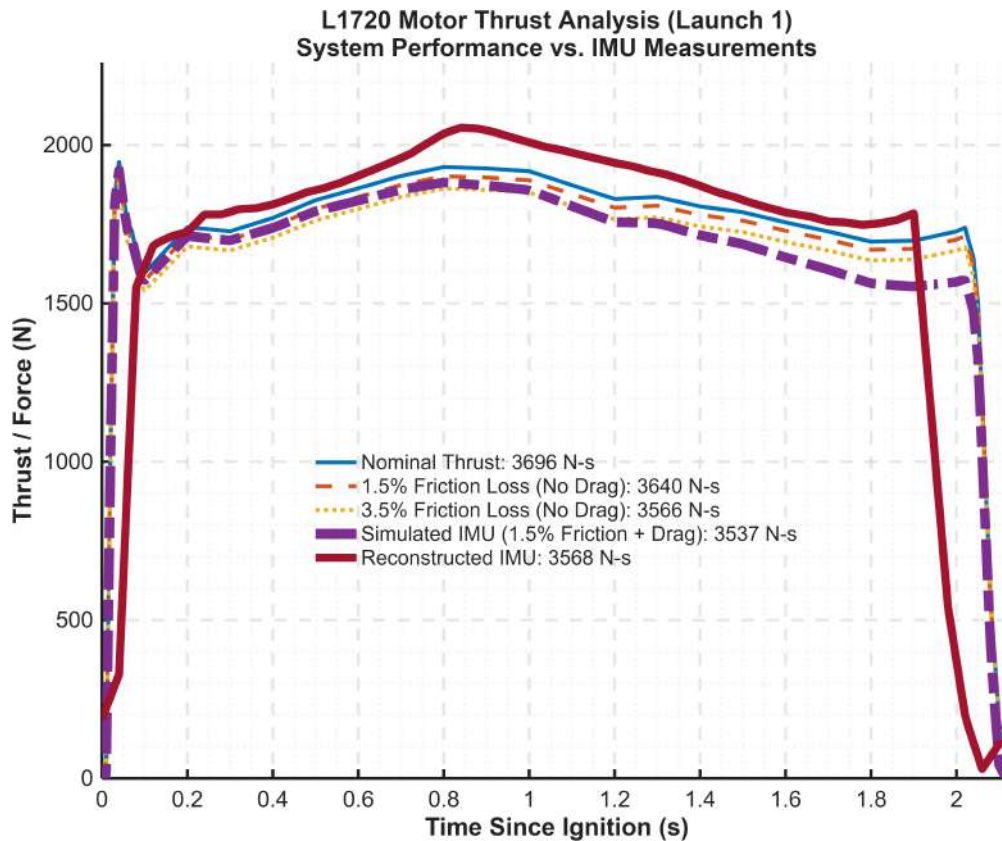


Figure 101: Flight 1 Effective Motor Impulse Delivery.

The L1720 motor manufacturer, Cesaroni Technology, guarantees a maximum variation in impulse of 5%. IMU-measured motor impulse closely matched simulations using the 1.5% friction model (3537 N-s predicted vs. 3568 N-s measured) with a remarkable 1% relative error (Table 36), highlighting the importance of accounting for vehicle drag and launch rail friction in analyzing IMU measurements. Without vehicle drag and launch rail friction corrections, the nominal L1720 thrust curve was within a 4% margin relative to the measured impulse. The results validated the L1720 motor’s ability to deliver impulse up-to-spec, and is consistent with the apogee results discussed earlier in the section; the verification of motor impulse delivery additionally validates the expected Thrust-to-Weight ratio of approximately 9:1 as computed in Section 3.3.

Table 36: Flight 1 Actual IMU impulse during motor burn relative to predictions.

	Measured IMU Impulse (N-s)	Simulated IMU (1.5% launch rail friction factor w/ vehicle drag)	Nominal L1720 Thrust Curve
Flight 1	3568	3537	3696
		-1%	4%

Due to the short timescale of launch rail clearance, the response time, sampling rate and noise uncertainty of altimeter data are insufficient to accurately determine the Launch Rail Exit (LRE) velocity. Forces due to motor thrust are expected to be relatively smooth, without abrupt jerks, and to satisfy the Nyquist Criteria relative to IMU sampling time; for short time spans with known

initial conditions, IMU data integration may therefore be appropriate to estimate velocity. To characterize the ability of using IMU acceleration data in computing LRE velocity, $N = 340$ samples of stationary acceleration were taken from the LSM flight computer IMU (VN100, VectorNav Technologies, Dallas, TX, USA) post landing to compute noise and uncertainty propagation. The measured samples resulted in a standard deviation of $\sigma = 0.0127 \text{ m} \cdot \text{s}^{-2}$, which remains in line with the manufacturer's specified uncertainty of $0.14 \text{ mg}/\sqrt{\text{Hz}} \approx 0.02 \text{ m} \cdot \text{s}^{-2}$. Propagating this per integration step for the duration of rail exit yields:

$$\text{Var}(v(n)) = \sum_{i=1}^n (\Delta t)^2 \sigma_a^2 = n(\Delta t)^2 \sigma_a^2 \quad (30)$$

$$\sigma_v = \sqrt{\text{Var}(v(n))} = \sigma_a \Delta t \sqrt{n} \quad (31)$$

$$\approx (0.2 \text{ m} \cdot \text{s}^{-2})(0.36 \text{ s})\sqrt{9} \approx 0.22 \text{ m} \cdot \text{s}^{-1} \quad (32)$$

The result of Equation 32 suggests the propagation of noise uncertainty in IMU integrated LRE velocity is on the order of magnitude of $0.3 \text{ m} \cdot \text{s}^{-1}$ at the point of Launch Rail Exit and, therefore, does not realistically present resolution concerns for informing the actual vehicle LRE velocity, which is presented in Figure 102.

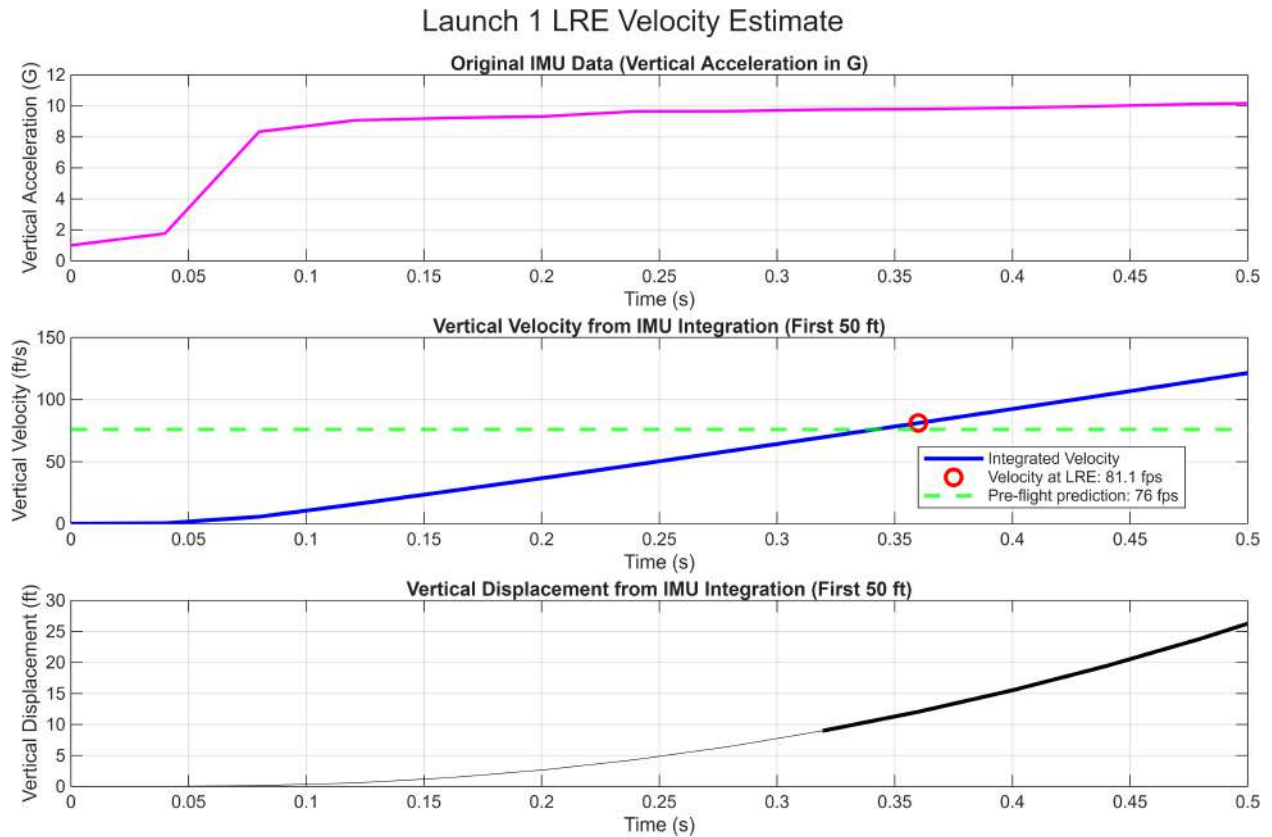
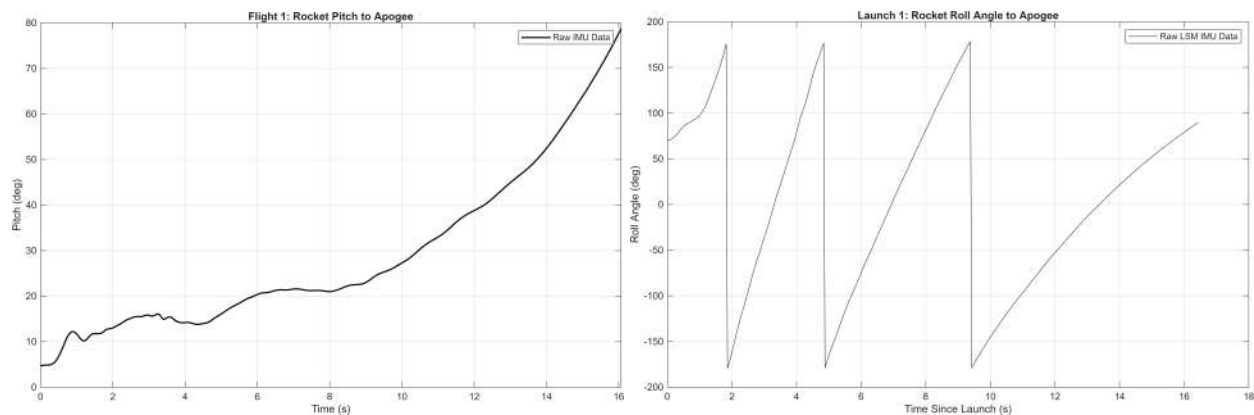


Figure 102: Flight 1 IMU-integrated LRE Velocity Estimate.

IMU Data

IMU gyroscopic data computed rocket pitch and roll shown in Figures 103a and 103b, respectively. The relatively low number of oscillations in rocket pitch suggest minimal wind negotiation in the

first few seconds of flight, consistent with the low wind conditions on launch day. Both roll and pitch data do not suggest issues in vehicle aerodynamic stability, structural rigidity, asymmetrical thrust, nor fin fluttering concerns. Coupled with a nominal delivery of motor impulse and an apogee that was well within pre-flight predictions, IMU gyroscopic data supports the notion that Flight 1 was successful in all aspects of vehicle ascent metrics.



(a) IMU Pitch Data to Apogee.

(b) IMU Roll Data to Apogee.

Figure 103: Flight 1 IMU Gyroscopic Data.

The raw total IMU acceleration data from the LSM flight computer is presented in Figure 104.

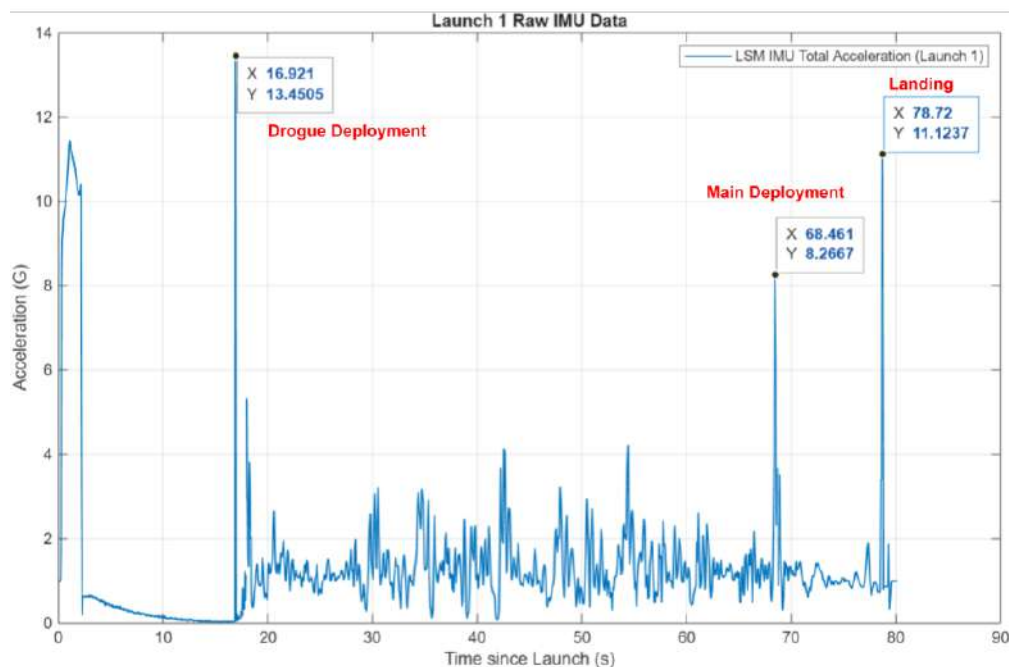


Figure 104: Flight 1 LSM Flight Computer IMU Raw Acceleration Data (Euclidean Norm).

Droge and main parachute deployments are a highly nondeterministic phenomenon that cannot be accurately predicted using deterministic modeling. Nevertheless, the IMU data shows deployment loading within the theoretical expectations established by the load analyses conducted during VADL's CDR milestone, as detailed in the team's CDR Report. In previous years, the acceleration

experienced during parachute deployment was calculated simply by taking the velocities before and after deployment, divided by the time required for the parachute to open; this calculates the average acceleration during opening rather than peak loading. However, it has been discovered through empirical testing that this is not representative of the true load experienced. Corroborated by extensive drop test data, it is understood that the parachute opening time is significantly less than full shock cord extension; as such, the actual jerk comes from the shock cord itself becoming taut. During the CDR milestone, VADL therefore applied a conservative approach to determine a maximum instantaneous loading imposed by drogue and main deployments; the Flight 1 actual main deployment load of 8.3 G remains below the 36 G prediction, which was a conservative over-estimation for structural analysis purposes, and the data therefore validates the structural safety of VADL’s fullscale vehicle design.

Notably, the LSM flight computer is positioned distally in the vehicle’s fore section relative to the nose cone. In contrast, the payload flight computer’s IMU better captures landing loads due to its proximal placement in the same vehicle section during descent. As mentioned in earlier sections, due to a payload electronics disconnect during vehicle ascent, payload IMU data was unavailable. Nonetheless, the drogue and main deployment forces were accurately and adequately represented using data solely from the LSM flight computer, reinforcing the value of VADL’s redundant design for mission-critical electronics. Landing load measured for Flight 1 using LSM IMU was 11.2 G (Figure 104).

Descent Rates

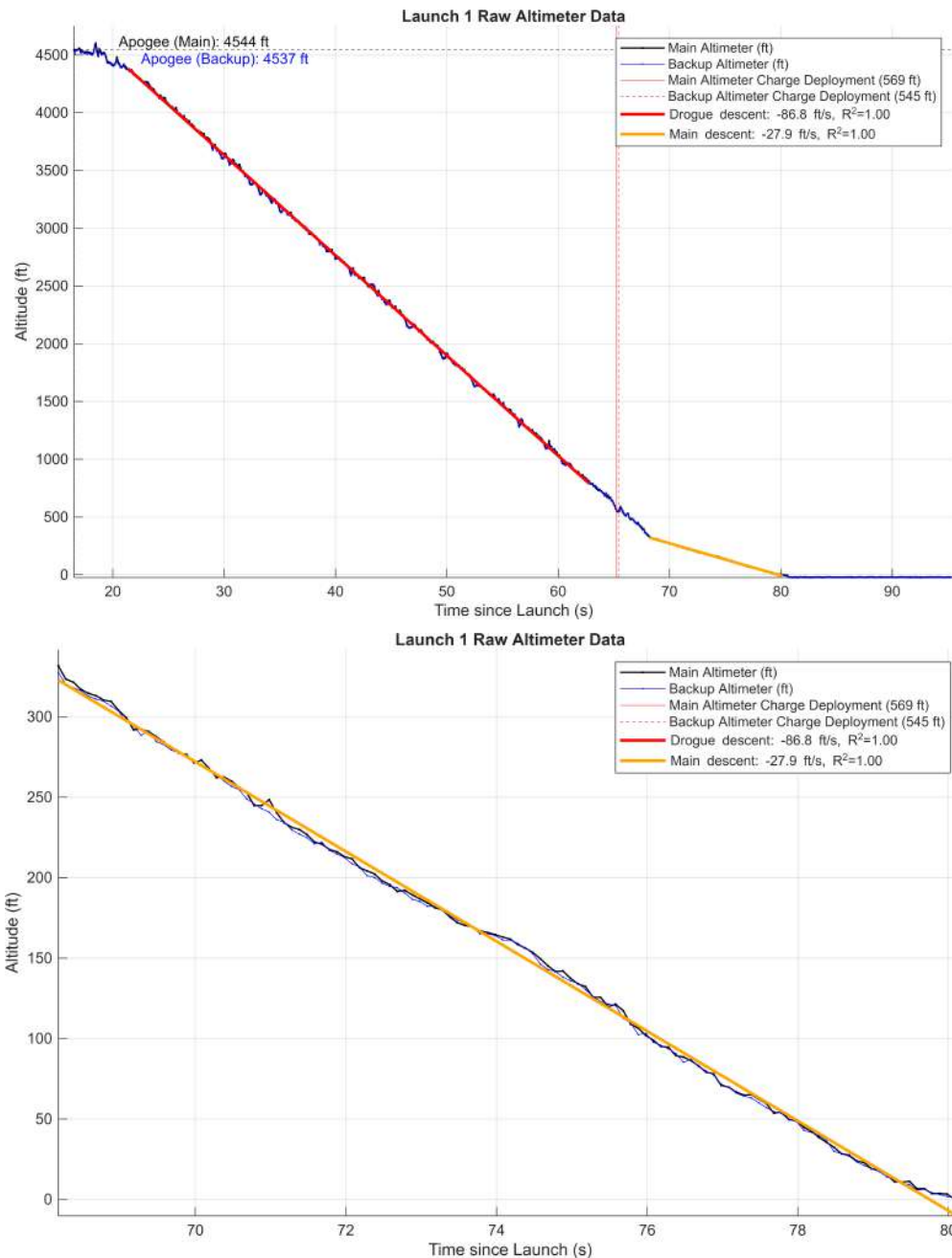
Table 37: Flight 1 Descent Rates.

	Drogue Descent Rate (fps)	Predicted Rate (fps) & Relative Error	Main Descent Rate (fps)	Predicted Rate (fps) & Relative Error	Max Sectional Landing KE (lb-ft)	Descent Time (s)	Predicted Time (s) & Relative Error
Flight 1	86.8	96.2 11%	27.9	16.5 -69%	99.6	64.3	75.2 17%

Table 38: Flight 1 Landing KEs.

	Landing KE (lb-ft)		
	Fore	Mid	Aft
Flight 1	99.6	67.9	47.3

During vehicle descent, vehicle recovery was achieved without damage to any vehicle structure. Altimeter data reviewed immediately upon vehicle retrieval suggest the main parachute descent rate was higher than expected (Table 37). As detailed in Section 5.2.1, flight video revealed that one of the shroud lines was tangled/wrapped around the parachute, thus preventing all 12 gores from fully opening. The elevated descent rates under this partially inflated main parachute resulted in a 93.4 lb-ft maximum sectional landing KE (Table 38). This issue was resolved with more careful packing for the second flight which, as covered in Section 5.2.2, was fully successful.



(b) Main descent only.
Figure 105: Flight 1 altimeter-based descent rates.

The descent rates in Table 37 were computed using linear regression of altimeter data collected by the avionics flight computer. The data, shown in Figure 105, confirm that descent rates stabilized to time-invariant values after the parachute fully unfurled. However, Figure 105 also reveals a 200 ft vertical displacement between main parachute charge deployment and steady-state descent. This extended stabilization period — likely caused by shroud line entanglement and other unpredictable dynamic factors — exceeded initial expectations.

While the main parachute achieved stable descent rates above the 300 ft payload jettisoning altitude, the observed lag in stabilization suggests a risk of interference between parachute deployment

and payload jettison. To mitigate this, VADL will increase the main parachute deployment altitude from 600 ft to 750 ft for all future full-scale flights. This adjustment moves to ensure sufficient vertical margin to accommodate stabilization delays and to maintain an optimal payload mission.

As detailed extensively in VADL’s CDR Milestone Report, VADL’s subscale flight revealed that the rocket and parachute side drag must account for the nondeterministic swinging of the rocket body during descent. To address potential underestimations of drift and descent drag characteristics, the full- scale vehicle’s DRSS model partially exposes the top of the parachute canopy to side wind as a function of the expected angular swinging of the rocket. Similarly, the vehicle’s frontal drag model includes adjustments to its side profile exposure in the vertical axis to reflect observed oscillatory motion of the rocket sections. These model updates address limitations in a deterministic descent framework used in third-party softwares such as OpenRocket and Rocksim, which assumes idealized, straight-body dynamics akin to ascent phases.

Flight 1 data indicated a drogue descent rate ~11% higher than pre-flight predictions (Table 37). While the DRSS model initially allocated 25% of drag to side-profile exposure to approximate swing effects, flight video analysis revealed sustained multi-axis tumbling, with vehicle sections rotating over themselves during drogue descent. This motion resulted in greater side-profile exposure to vertical airflow than originally modeled. Table 39 shows updated descent rates using an empirically determined 55:45 (side:front) drag ratio from the pre-flight 25:75 parameter to better align with observed flight dynamics.

Table 39: Flight 1 Descent rates with post-hoc corrections applied.

** with 55/45 side/frontal vehicle drag instead of 25/75*
*** with 0.74 fitted Cd instead of 2.2 (w/ nominal diameter)*

	Drogue Descent Rate (fps)	Predicted Rate (fps) & Relative Error *	Main Descent Rate (fps)	Predicted Rate (fps) & Relative Error
Flight 1	86.8	87.0	27.9	27.9 **
		0%		0.0%

It is important to note that the Launch Day simulations outlined in Section 3.3 retain the 25:75 (side:front) drag ratio to conservatively predict the drogue descent rate to assess the structural integrity and safety aspects of vehicle design during main parachute deployment. During drogue deployment, the rotational energy induced by the release of chemical deployment charges inherently results in highly chaotic vehicle swinging during drogue descent. The difficulty to account for this nondeterministic phenomenon in a deterministic simulation model does not warrant the MPP (Section 3.3) to confidently utilize this empirically obtained 55:45 ratio.

GPS Data

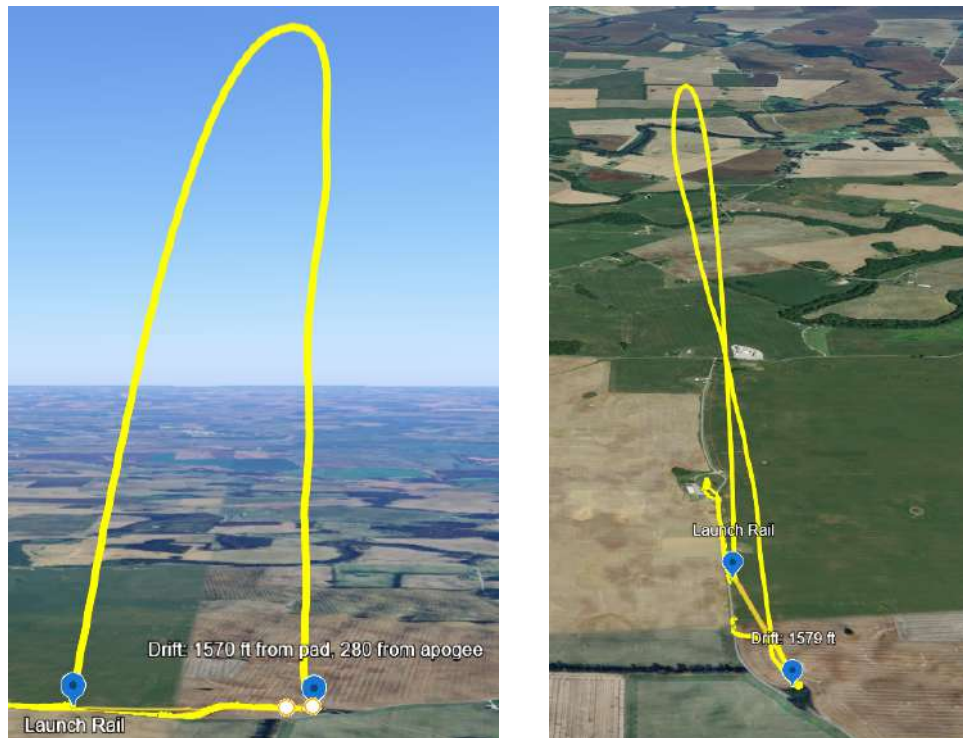


Figure 106: Flight 1 vehicle path as interpreted by GPS.

The GPS transmitter was validated during VADL's vehicle demonstration flight. The battery was plugged into the transmitter at 11:12 AM on launch day. The launch vehicle was installed on a launch rail approximately 1500 ft from the RF ground station. A stable RF connection was maintained up until launch, after which APRS packets were received less frequently by the ground receiver. This is further discussed in Section 3.2.5. After recovery, the transmitter was verified to still be transmitting, and the battery was left connected until the rocket was disassembled between flights, at 1:14 PM. The path of the launch vehicle tracked by GPS is plotted below in Figure 106. The final vehicle drift was 1560 ft from launch pad and 280 ft from apogee, which all remain below the 2500 ft recovery radius per NASA Requirement 3.11. The expected maximum theoretical drift informed by the pre-flight model is as below:

$$\text{Max. Theoretical Drift} = (\text{Descent Time}) \times (\text{Wind Speed}) \quad (33)$$

$$= 75.2 \text{ s} \times 3 \text{ mph} \quad (34)$$

$$\boxed{\approx 330 \text{ ft}} > 280 \text{ ft} = \text{Actual Drift} \quad (35)$$

5.3.2 Flight 2 Analysis

In Flight 2, a successful and nominal full-scale flight was conducted, meeting all ascent and descent requirements as outlined in Table 33. Similar to Flight 1, the L1720 motor delivered performance consistent with design specifications during rocket ascent. Close alignments between theoretical and experimental results validated that the rocket was manufactured well up-to-spec, both in terms

of structural rigidity and aerodynamic performance. During descent, the rocket recovery system successfully landed all vehicle sections without damage and met all bonus criteria, achieving a maximum sectional landing kinetic energy below 65 lb-ft and a total descent time of under 80 seconds. No major deviations in key outcome metrics from pre-flight predictions were observed in IMU and altimeter data. Unless otherwise stated, data processing methodologies are consistent with those described in Section 5.3.1. Pre-flight modeling references simulations conducted using environmental and vehicle measurements obtained on launch day (Tables 30 and 32), following the theoretical framework detailed in Section 3.3.

Altimeter Data Analysis

Table 40: Launch 2 Pre-flight model predictions versus actual achieved apogee.

* with 3.5% launch rail friction (pre-flight model)

	Apogee (ft)	Predicted Apogee (ft) & Relative Error		
		Rocksim	CFD-DRSS	OpenRocket
Flight 2	4633	4442	4475	4086
		-4.3%	-3.5%	-13.4%

Figures 107 and 108 present raw AGL altimeter data during Flight 2. While the wind conditions remained similar between flights (Flight 1: 3 mph, Flight 2: 1 mph; Table 30), the launch rail angle was modified from 5° to 1° for the purpose of obtaining experimental data under varied launch conditions. Table 40 show the rocket achieved an apogee of 4633 ft. The pre-flight prediction of 4450 ft was the target altitude for this launch. Pre-flight simulations using the industry standard software RockSim and VADL’s in-house DRSS software [5] with a 3.5% launch rail friction factor underpredicted apogee by ~4%.

As mentioned in Section 5.3.1, launch rail friction is a highly non-deterministic phenomenon that cannot be accurately modeled in a deterministic framework. Analyses conducted by Flanigan et al. [8] show that launch rail friction may impose a reduction up to 4% in the delivered impulse of a rocket motor. The 3.5% loss factor was selected in VADL’s pre-flight model to account for friction at the rail. While VDF launch day winds were expected to not exceed 5 mph, for consistency with the theoretical modeling of the Competition Day simulations (Section 3.3) and to ensure compliance with NASA’s minimum apogee requirement of 4000 ft, the 3.5% factor was conservatively retained in VDF pre-flight models. OpenRocket’s deviation in apogee prediction was due to differences in drag estimation performed by the software relative to RockSim and DRSS.

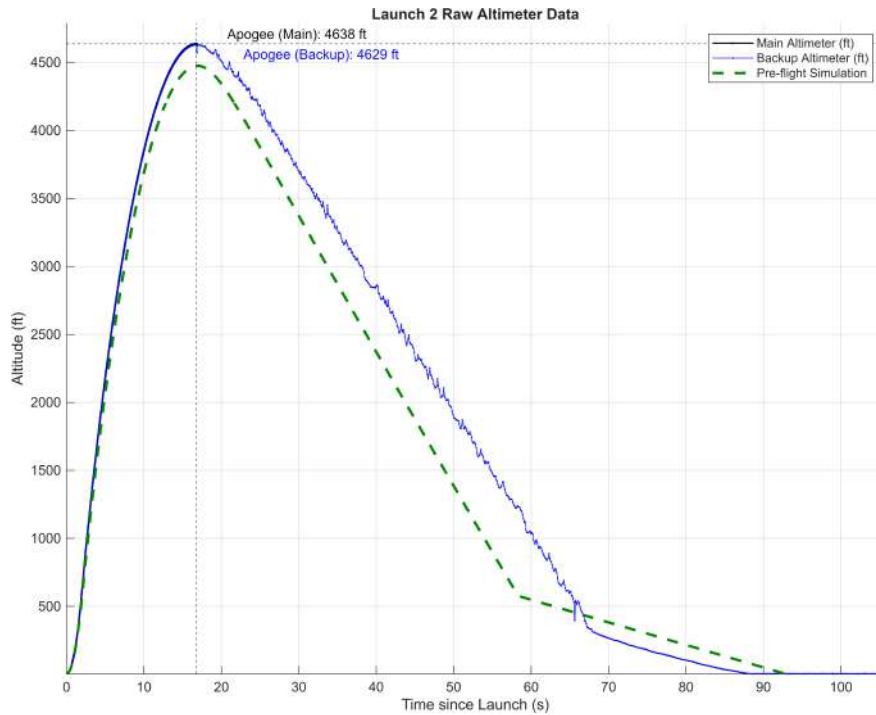


Figure 107: Flight 2 Raw Avionics Altimeter Data vs. Simulation.

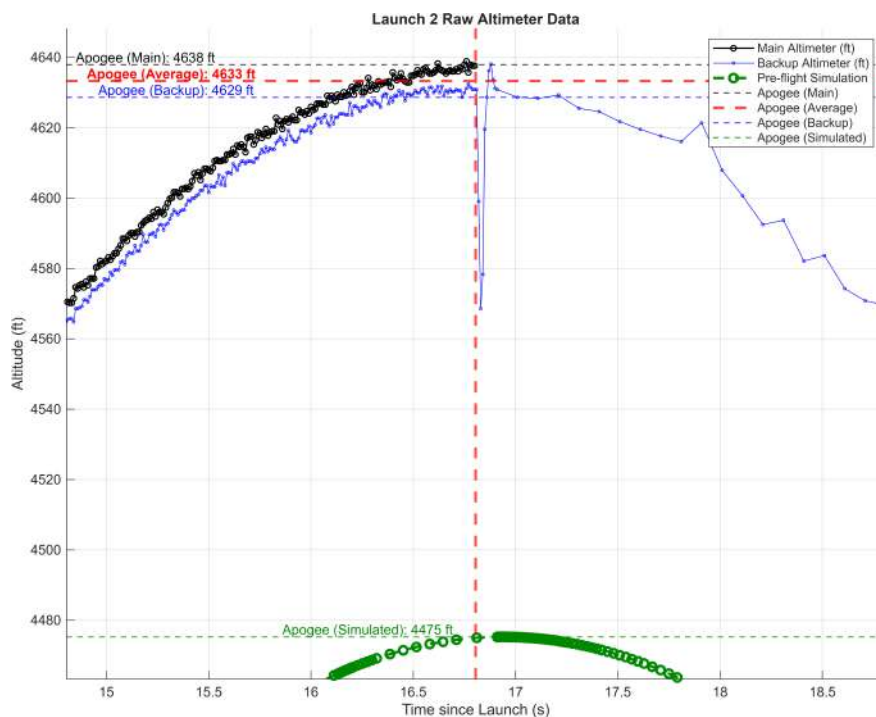


Figure 108: Flight 2 Raw Avionics Altimeter Data vs. Simulation (zoomed in).

A post-hoc, 1.5% friction factor model was shown with improved model accuracy: Rocksim and DRSS predictions aligned within 0.8% of measured apogee (Table 41), confirming the notion that side wind is proportional to launch rail friction. As mentioned in Section 5.3.1 other ascent aspects

such as vehicle drag and motor impulse delivery are unlikely causes behind the marginal deviations in apogee prediction, and launch rail friction presents a most significant reason behind the marginal deviations between actual apogee versus the pre-flight predictions. The simulations in Table 41 utilize the empirically obtained Cd estimates from Figure 109.

Table 41: Launch 2 Adjusted model predictions with 1.5% launch rail friction and experimental Cd versus actual achieved apogee.

* with 1.5% friction factor (adjusted model)

	Apogee (ft)	Predicted Apogee (ft) & Relative Error		
		Rocksim	CFD-DRSS	OpenRocket
Flight 2	4633	4595	4615	4215
		-0.8%	-0.4%	-9.9%

Flight computer data from the LSM IMU and the avionics altimeter were utilized to reconstruct estimated drag coefficients. Using methodology extensively detailed in Section 5.3.1, Equation 28 was applied during coast ascent (no thrust forces). Figure 109 suggests the experimental drag coefficient aligns closely with pre-flight CFD simulations. This is further supported by a close alignment between altimeter and predicted vertical velocity during vehicle ascent (Figures 110a and 110b). For this reason, vehicle drag is assessed to not be a likely cause behind the initial discrepancy in predicted apogee. The alignment confirms that VADL’s fullscale vehicle construction was up-to-spec relative to its designed geometry. The alignment between CFD and experimental C_d in Flight 2 (Figure 109) was observed to be marginallyl closer than that in Flight 1 (Figure 99); this is possibly due to the near-straight launch angle of the Flight 2, which necessarily reduces the effects of gravity-turn and other pitch-related dynamics on vehicle drag estimation.

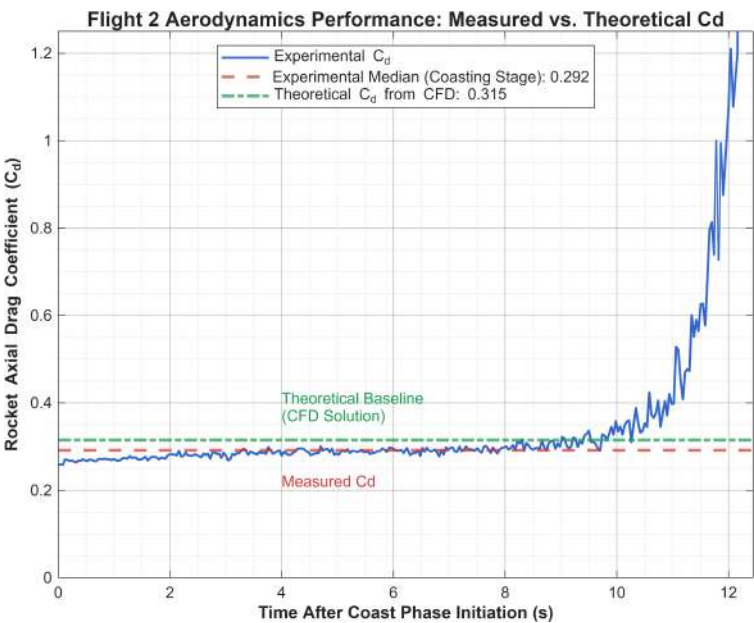
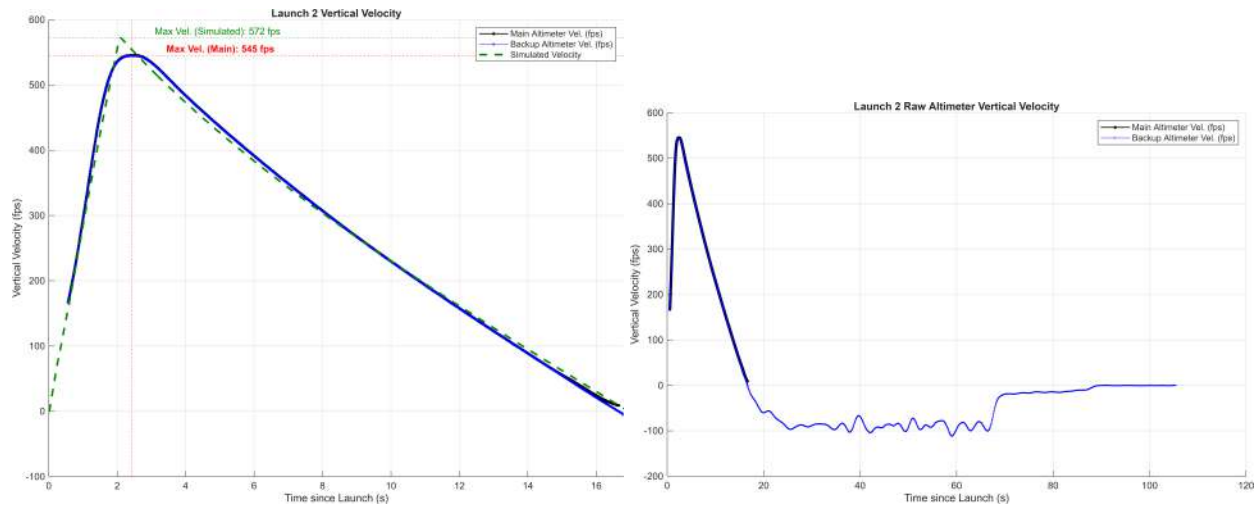


Figure 109: Flight 2 estimated vehicle drag computed from IMU and altiemter data during coasting phase (burn out to apogee).



(a) Ascent only.

(b) Ascent and descent.

Figure 110: Flight 2 Raw Altimeter Velocity Data. Data points unavailable during the first second of flight was due to excessive data cropping performed by the EasyMini altimeters, which utilize an aggressive altitude change threshold for launch detection to prevent mistriggers. A power disconnect during drogue deployment prevented the main altimeter from collecting descent data.

Motor Thrust Analysis

The LSM flight computer IMU data from Flight 2 further corroborates the methodology established in Section 5.3.1 for computing motor thrust delivery. As with Launch 1, IMU-measured kinetics data deviates from the nominal L1720 motor thrust curve due to two primary factors: (1) launch rail friction, which imposes up to 4% of effective motor thrust loss, and (2) vehicle aerodynamic drag prior to MECO, both of which reduce IMU-measured acceleration. Following the approach in Flight 1, VADL compares actual IMU measurements against simulated data while incorporating rail friction and drag corrections. Figure 111 demonstrates analogous trends to Flight 1, with rail friction evident in the suppressed thrust peak during the initial 0.2 seconds post-ignition (red curve), and the gradual thrust reduction in the latter half of motor burn time aligning with DRSS-simulated drag effects (purple dotted curve). Motor burn duration remains consistent with nominal expectations, as graphically confirmed in Figure 111.

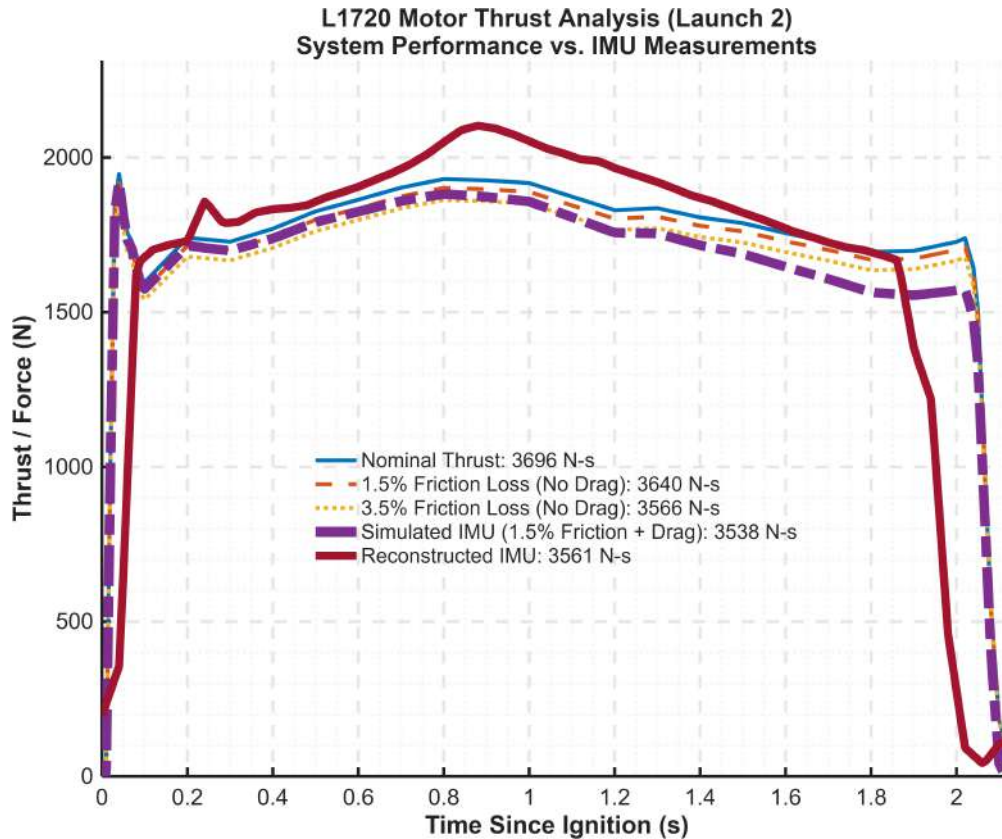


Figure 111: Flight 2 Effective Motor Impulse Delivery.

Consistent with Cesaroni Technology’s guaranteed 5% impulse variation, IMU-measured impulse for Launch 2 showed strong agreement with simulations using the 1.5% friction model (3538 N-s predicted vs. 3561 N-s measured), yielding a relative error of 1% (Table 42). This further highlights the necessity of accounting for rail friction and aerodynamic drag, as uncorrected data exhibited a 4% deviation from the nominal thrust curve. Taken together with Flight 1’s Motor Thrust Analysis (Table 36), these results validated the L1720 motor’s reliability in delivering specified impulse. The validated impulse delivery also supports the 9:1 Thrust-to-Weight ratio predicted in Section 3.3, demonstrating consistency across both flights.

Table 42: Flight 2 Actual IMU impulse during motor burn relative to predictions.

	Measured IMU Impulse (N-s)	Simulated IMU (1.5% launch rail friction factor w/ vehicle drag)	Nominal L1720 Thrust Curve
Flight 2	3561	3538	3696
		-1%	4%

Building on the methodology outlined in Section 5.3.1, IMU acceleration data integration was similarly employed to estimate Launch Rail Exit (LRE) velocity for Flight 2. As with Flight 1, altimeter resolution limitations persist for LRE velocity determination due to rapid rail exit timescales. To quantify velocity uncertainty, $N = 340$ post-landing stationary IMU samples were analyzed, yielding a noise standard deviation of $\sigma = 0.0131$ m/s consistent with the VectorNav

VN100's specified uncertainty of $0.14 \text{ mg}/\sqrt{\text{Hz}}$. As detailed in Section 5.3.1, propagating this uncertainty over the rail exit duration via Equation 32 results in $\approx 0.22 \text{ m} \cdot \text{s}^{-1}$.

This uncertainty magnitude remains low for practical LRE velocity resolution, which is presented in Flight 2's IMU-integrated velocity profile (82.7 fps; Figure 112). The results exhibit similar magnitudes with Flight 1 (81.1 fps; Figure 102).

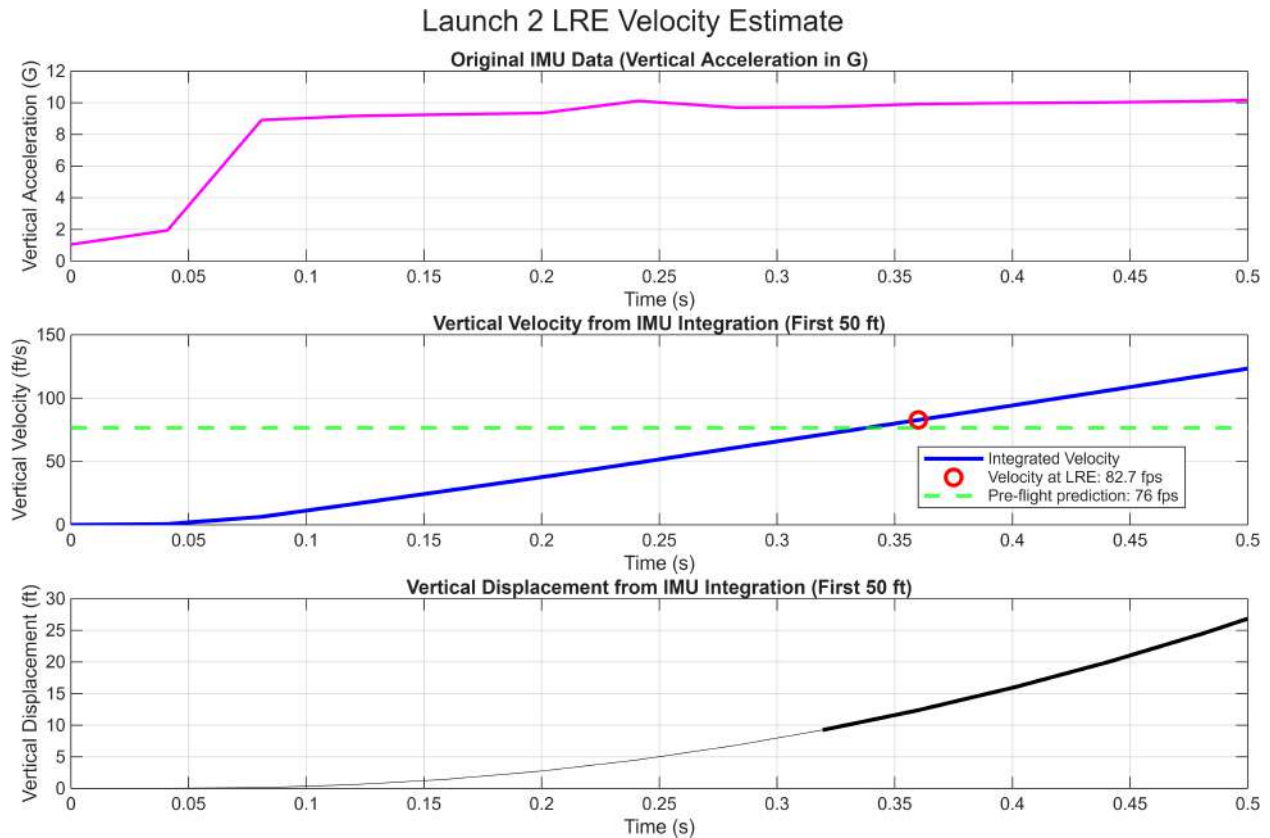
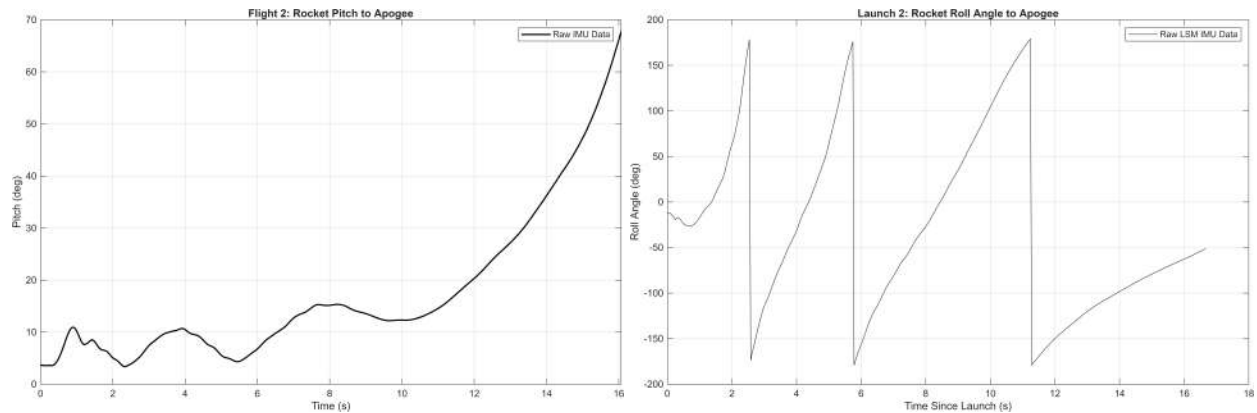


Figure 112: Flight 2 IMU-integrated LRE Velocity Estimate.

IMU Data

IMU gyroscopic data for Flight 2, shown in Figures 113a and 113b, demonstrates comparable performance to the Flight 1 results analyzed in Section 5.3.1. The trajectories suggest no indications of structural compliance, thrust vector irregularities, or fin-induced vibrations. Combined with nominal motor performance and apogee achievement matching preflight simulations, the IMU data validates Flight 2's nominal ascent profile.



(a) IMU Pitch Data to Apogee. (b) IMU Roll Data to Apogee.
Figure 113: Flight 2 IMU Gyroscopic Data.

The complete IMU acceleration profile for Flight 2 appears in Figure 114. Parachute deployment dynamics followed similar stochastic patterns to those described in Section 5.3.1, with measured loads remaining well below the more conservative CDR predictions. Main deployment peaked at 18.7 G versus the 32 G design limit / pre-flight simulation.

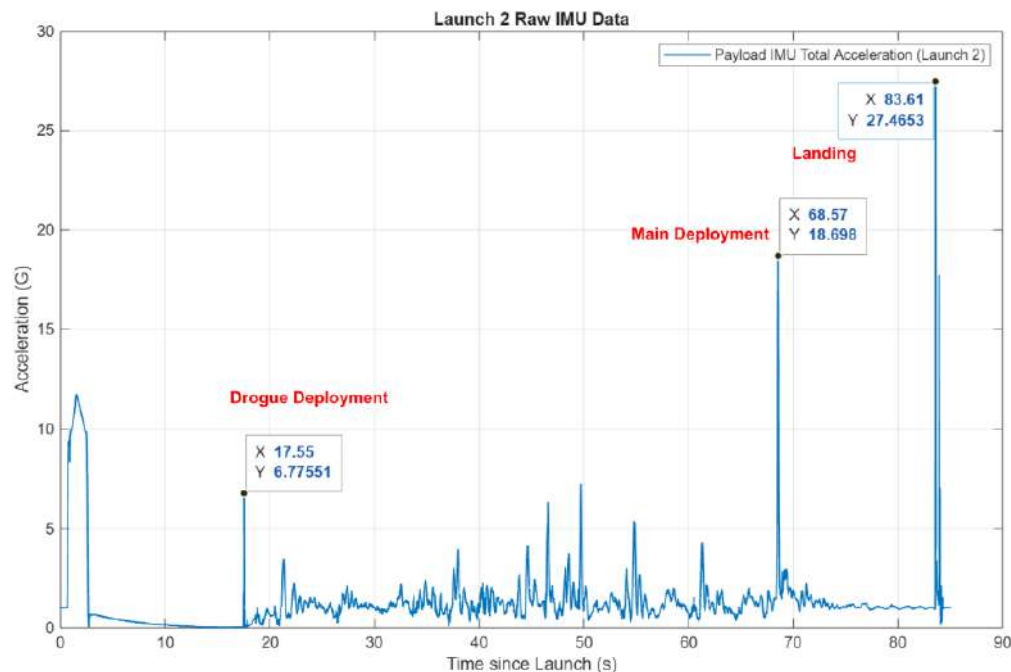


Figure 114: Flight 2 Payload Flight Computer IMU Raw Acceleration Data (Euclidean Norm).

Flight 2 did not present any electrical disconnect issues seen in Flight 1, and the payload IMU data was fully available. The payload flight computer's positioning in the vehicle fore section provided more adequate resolution for landing impact measurements relative to the LSM flight computer. Landing loads of 27.5 G (Figure 114) were recorded, which are relatively higher than the LSM IMU data from Flight 1 (Figure 104). Outside of nondeterministic impact dynamics, this difference is most likely due to an order-of-magnitude greater data collection rate on the payload IMU (>100 Hz versus 12 Hz). Notably, this is the first time in VADL history that such high-

frequency IMU data collection has been performed, and a higher observed landing acceleration therefore does not indicate an impact more abrupt than any historical landings. As detailed in the following section, the landing velocities and sectional kinetic energies of Flight 2 were remarkably low (<55 lb-ft for all sections), no data support the notion that a harder-than-average impact was experienced by the vehicle. Using methodology detailed in Section 4.3.6, the DRI index using the payload flight computer IMU data was computed to be 13.5. Vertical spinal compression of this DRI magnitude is assessed to be of “Low Injury Risk”, which is the lowest category defined by current literature [7].

Descent Rates

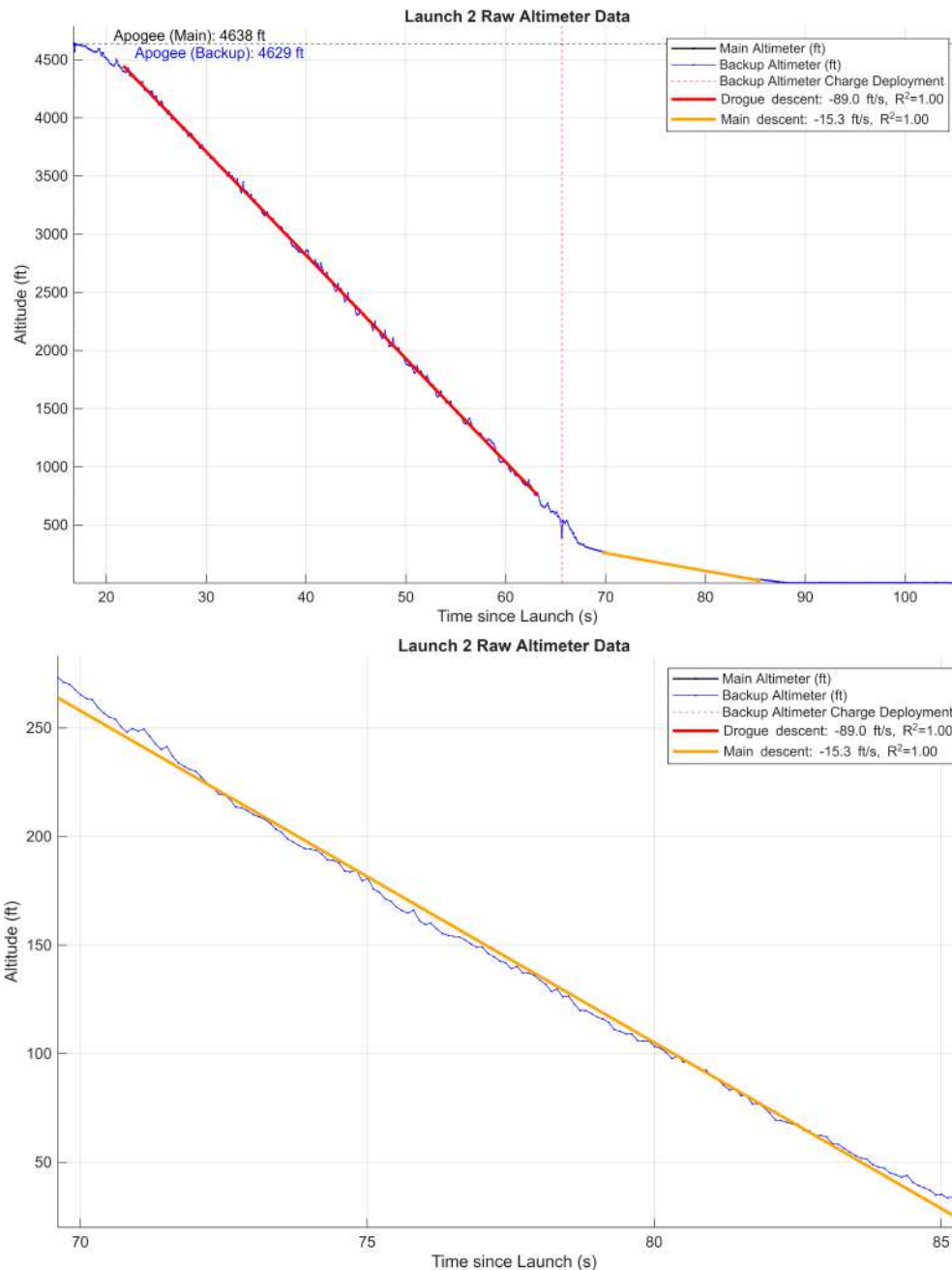
Table 43: Flight 2 Descent Rates.

	Drogue Descent Rate (fps)	Predicted Rate (fps) & Relative Error	Main Descent Rate (fps)	Predicted Rate (fps) & Relative Error	Max Sectional Landing KE (lb-ft)	Descent Time (s)	Predicted Time (s) & Relative Error
Flight 2	89.0	96.3 8.1%	15.3	16.5 7.8%	54.6	71.5	75.7 6%

Table 44: Flight 2 Landing KEs.

	Landing KE (lb-ft)		
	Fore	Mid	Aft
Flight 2	54.6	37.2	25.9

Flight 2 conclusively demonstrated the vehicle recovery system’s ability to land the vehicle sections without damage and meet all bonus criteria, achieving a remarkable maximum sectional landing KE below 55 lb-ft and a total descent time of under 75 seconds. No major deviations in key outcome metrics from pre-flight predictions were observed. As Table 43 substantiates, the implemented parachute packing improvements eliminated Flight 1’s shroud line entanglement issue in the main parachute, yielding a final descent rate of 15.3 fps with only a 7.8% deviation from pre-flight predictions. This nominal performance reduced maximum sectional landing kinetic energy to 51.2 lb-ft (Table 44), 43% below Flight 1’s values and well within the 65 lb-ft bonus limit per NASA Requirement 3.3.



(b) Main descent only.
Figure 115: Flight 2 altimeter-based descent rates.

The improved descent profile is evident in Figure 115, which shows reduced stabilization distance (75 ft vs. Flight 1's 200 ft) following main deployment. While the planned altitude margin increase to 750 ft remained unimplemented for Flight 2, the successful canopy deployment rendered this precaution unnecessary. However, VADL will retain the increased deployment altitude for future missions as a risk mitigation measure against potential stabilization delays.

Flight 2 data validated the 55:45 side/frontal drag ratio empirically derived from Flight 1's drogue descent analysis (Table 39). As shown in Table 45, implementing this updated ratio reduced drogue rate prediction error from 8.1% to -2.3%.

Table 45: Flight 2 Descent rates with post-hoc corrections applied.

** with 55/45 side/frontal vehicle drag instead of 25/75 (fitted from Flight 1)*

*** with 2.55 fitted Cd instead of 2.2 (w/ nominal diameter)*

	Drogue Descent Rate (fps)	Predicted Rate (fps) & Relative Error *	Main Descent Rate (fps)	Predicted Rate (fps) & Relative Error
Flight 2	89.0	87.0	15.3	15.3 **
		-2.3%		0.0%

As mentioned in Section 5.3.1, the MPP (Section 3.3) maintains the conservative 25:75 drag assumption for drogue analysis. This preserves safety margins despite the validated 55:45 ratio's accuracy, as chaotic tumbling effects remain challenging to fully quantify deterministically. Possibly due to local gusting effects, a 2.55 main Cd values was derived from Flight 2 data (Table 45), which remains close within the 2.2 nominal specification.

GPS Data

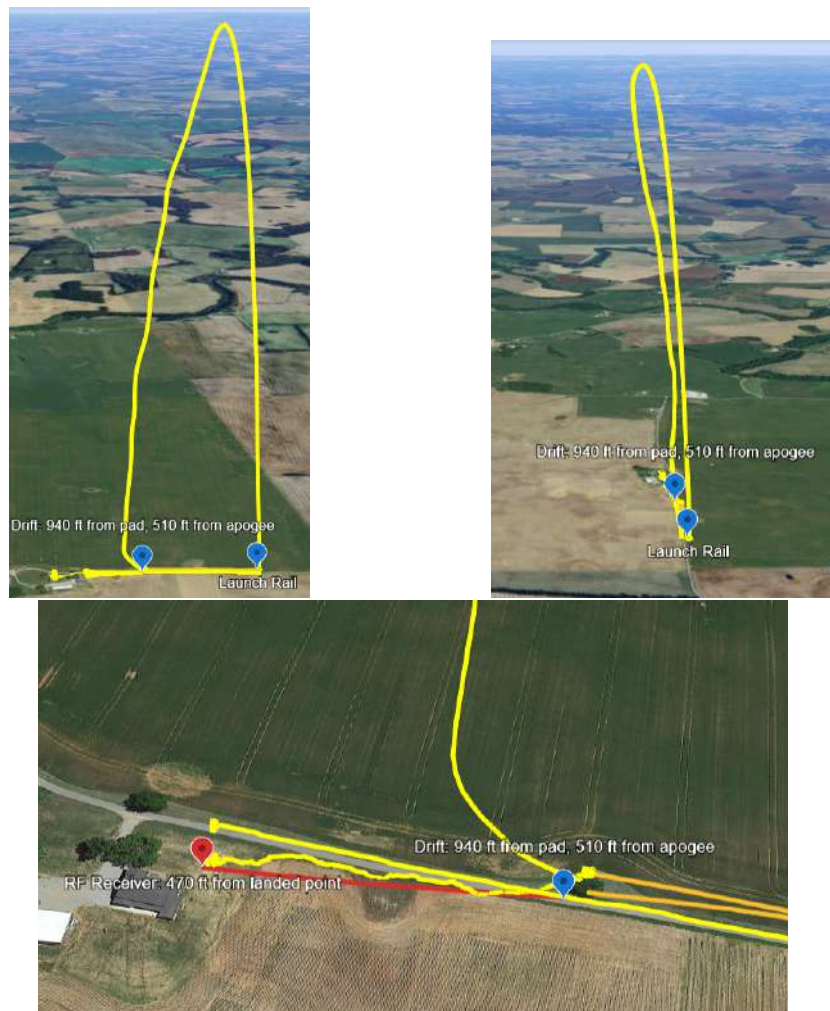


Figure 116: Flight 2 vehicle trajectory reconstruction using triangulated GPS data.

GPS telemetry performance showed marked improvement from Flight 1, with continuous RF link maintenance for the entire flight. After retrieving the KML files and clearing the flash from both GPSs, a process that took 20 minutes, the GPS were powered back on at 4:10 pm. Both the rocket and payload RF transmission was confirmed through successful post-landing signal decoding at 470 ft from the recovery site over flat terrain. The GPSs were unplugged at 4:50 pm, following the retrieval of the rocket, proving the transmitter was able to transmit over a combined 2 hours and 40 minutes over the course of two flights. This is only a fraction of the 10 hours of transmission the GPSs are capable of. As shown in Figure 116, the Flight 2 final vehicle position was 940 ft downrange from the launch pad and 510 ft from apogee — both metrics remain well within NASA Requirement 3.11’s 2500 ft recovery radius.

Theoretical drift calculations using conservative wind estimates show:

$$\text{Max. Theoretical Drift} = (\text{Predicted Descent Time}) \times (\text{Max. Wind Speed}) \quad (36)$$

$$= 71.5 \text{ s} \times 1 \text{ mph} \quad (37)$$

$$\boxed{\approx 105 \text{ ft}} < 510 \text{ ft} = \text{Actual Drift} \quad (38)$$

This discrepancy arises from the inherent sensitivity of drift calculations to low baseline wind speeds. While pre-launch anemometry recorded steady 1 mph winds, small time-variant deviations during descent — such as brief 5 mph gusts — translate to large relative errors in practice. A 4 mph absolute increase over the measured wind speed translates to a 400% relative error in the input parameter, directly amplifying drift predictions by 5×:

$$\text{Adjusted Drift} = 71.5 \text{ s} \times (1 + 4) \text{ mph} \approx 524 \text{ ft} \quad (39)$$

The 510 ft observed drift aligns with this adjusted scenario, highlighting two critical limitations of static wind measurements: (1) low-magnitude wind speeds ($\ll 2$ mph) disproportionately magnify relative errors from minor absolute fluctuations, and (2) transient gusts during descent phases are not captured by pre-launch ground-level anemometry. Taken together, these effects provide a robust explanation behind the divergence between predicted and actual drift while reaffirming compliance with NASA safety margins.

5.3.3 Subscale Comparison

Key flight results from the subscale and Vehicle Demonstration Flights are compared below in Table 46. The on-pad SSM’s of the two flights were very similar. As detailed in the team’s CDR Milestone Report, the issues with the altimeters from the subscale flights led to the switch to the EasyMini for the full-scale flights. The EasyMini altimeters were successfully flown during VDF with nominal recovery. The subscale fabrication process also informed fullscale construction. Many of the same methods were used in both manufacturing processes.

Table 46: Comparison of results from second flights for subscale and VDF.

Parameter	Subscale 2	VDF 2
Length	61 in	97 in
Diameter	4 in	6 in
Mass (On Pad)	13.1 lbm	42.0 lbm
SSM (On Pad)	2.21 cal	2.12 cal
T/W	5.8	8.5
LRE Velocity	77 fps	83 fps
Apogee	1076 ft	4633 ft
Terminal Descent Velocity	19.0 fps	15.3 fps
Drag Coefficient	0.40	0.29
Vehicle Performance	Fully Successful	Fully Successful
Payload Performance	Not Successful	Fully Successful

5.4 Payload Performance

As discussed in Section 5 the team flew the full-scale payload on the rocket in order to complete the outlined NASA Student Launch mission in a non-jettison case otherwise known as the in-rocket configuration. VADL is committed to fulfilling mission requirements in any possible circumstance, and decided to test an in-rocket configuration to ensure the rocket and payload would perform nominally without jettison. The team has previously flown components that are used in the full-scale payload on the subscale launch vehicle, and has also dropped the payload from a 300 ft AGL drop under a hexacopter. That testing allowed the team to certify the payload as ready to fly on the full-scale vehicle, and it successfully completed the Student Launch mission required.



Figure 117: Payload during Vehicle Demonstration Flight 2

Table 47: Summary of payload performance in VDF

Mission Aspect	Requirement	Result
Detect payload landing	VADL 4.7.2	Successful
Collect flight telemetry and landing site data	VADL 4.7.3 - 4.7.9	Successful
Actuate parachute detachment latch 1s after landing	VADL 4.4.1	Successful
Extend antennas	VADL 4.5.1 - 4.5.5	Successful
Transmit RF data	VADL 4.6.1	Successful (470 ft transmission range)
Cut-off RF Transmission after 5 minutes	VADL 4.6.3	Successful

The payload performed nominally during this launch and fulfilled all NASA Student Launch mission requirements. The payload successfully withstood launch, recovery, and landing, collected telemetry, and transmitted the telemetry back towards the team ground station over the 2M band until cutting off transmission 5 minutes after landing. There was an electrical disconnect experienced during flight 1, which prevented full completion of the payload mission during this flight. This issue was resolved for the second flight by replacing the cable, resulting in the fully successful payload performance during flight 2.

All VADL imposed requirements related to this launch were also met. All 3 antennas in the RF system were fully extended, which was critical to maximizing RF range as the orientation of the payload could not be controlled with the in-rocket configuration. Since payload jettison is purely a VADL imposed requirement, by completing this launch VADL has demonstrated its ability to successfully complete the Student Launch mission in its entirety.

Mission requirements unrelated to the payload performance such as landing kinetic energy were also met. The payload is the heaviest section on the launch vehicle, and had a landing kinetic energy below 75 lb-ft which satisfies NASA requirement 3.3.

Table 48: Payload VDF results

Avg Descent Rate (fps)	Landing KE (lb-ft)	Drift (ft)	Landing G	DRI	Injury Risk
15.3	51.2*	292	16.1	13.5	Low

*This is the landing KE of the entire fore section, as it was not jettisoned **A DRI value below 17.7 corresponds to an injury risk of “low”. Further detail on the DRI (Dynamic Response Index) method for calculating survivability is outlined in Section 4.3.6.

To validate the accuracy of the telemetry transmitted to the ground station, the lander altimeter data was analyzed. During the VDF launch the lander transmitted a landing velocity value of 17.9 ft/s. A linear regression analysis of the altimeter data yielded a value of 16.6 ft/s. This is a percent error of 7.3%.

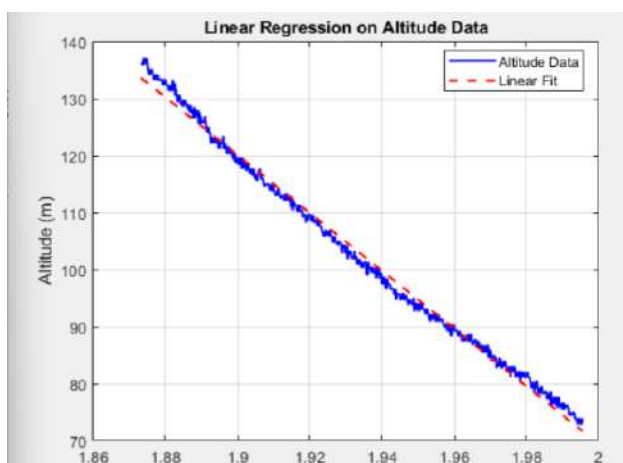


Figure 118: Linear regression analysis on altitude data to verify landing velocity transmitted to ground station.



Figure 119: Payload being prepared during VDF launch.

All mechanical components and mechanisms survived both launches unscathed, thus verifying major structural changes made to the payload since the CDR design cycle. In particular, the change from 4 aluminum rods to 2 in the payload skeleton did not affect the payload performance or its structural integrity throughout launch, recovery, and landing. Likewise, the shaft collars managed to hold position and did not noticeably shift. The AEMs are the same as those utilized on the subscale flight, which makes this the 4th rocket launch that they have survived without any failure to function or structure. The only 2 mechanical mechanisms not tested were the PDM and LDM. The team shall deploy both during a rocket flight during the Payload Demonstration Flight (PDF).

Given the successful payload performance in the VDF, the team is confident that the payload is ready for PDF to verify every single aspect of payload functionality.

5.5 Lessons Learned

The Vehicle Demonstration flights, while successful, taught the team several important lessons. Summarized in Table 49 below, these learnings and their associated action items will be carried forward as the team prepares for Payload Demonstration flights and the competition.

Table 49: Lessons learned and associated mitigations/action items from the Vehicle Demonstration flights.

Lesson Learned	Mitigation/Action Item
Parachute folding and packing must be performed with the utmost attention and care.	The parachute folding and packing process will be rehearsed numerous times prior to flight, and opening tests will be performed to ensure the parachute fully unfurls as intended.
All electrical connections must be certified to withstand the rigors of flight.	To prevent wires from backing out from the connector, the Terminal Position Assurance component is currently being investigated for implementation. Having this added component will prevent wire back-out.

6 Safety and Procedures

6.1 Safety Overview

The Vanderbilt Aerospace Design Lab adheres to a strict safety protocol for all activities conducted during the USLI competition. Safety is a top priority for the team and is central to all manufacturing, testing, and launch operations. The day-to-day safety of the team is overseen by the Student Safety Officer, whose responsibilities will be discussed at greater length in Section 6.1.2. Lab safety is guided by a collection of documents, including the VUSE Shop Safety Guidelines, the Material Safety Data Sheets (MSDS), the Equipment Operation Approval Sheet, and the VADL Cardinal Safety Rules. All of these guides are stored in the safety binder - which is easily accessible in the lab - and the Cardinal Rules are additionally posted around the lab for easy reference. Launches are governed by the above listed documents, as well as the NAR and FAA guidelines. Extensive mitigation strategies for potentially dangerous activities have been prepared by the team, all of which can be found in Section 6.2 through Section 6.5. These strategies and their organization is heavily guided by the MIL-STD-882, which is the standard system safety document for the Department of Defense. These sections are also influenced by the NASA System Safety Handbook, which helped to supplement the guidelines of the MIL-STD-882.

6.1.1 VADL Safety Structure

VADL follows a straightforward approach to safety in the lab, which is guided by the documents mentioned in Section 6.1 and by the VADL safety personnel covered in Section 6.1.2. This structure is detailed in the flowchart below.

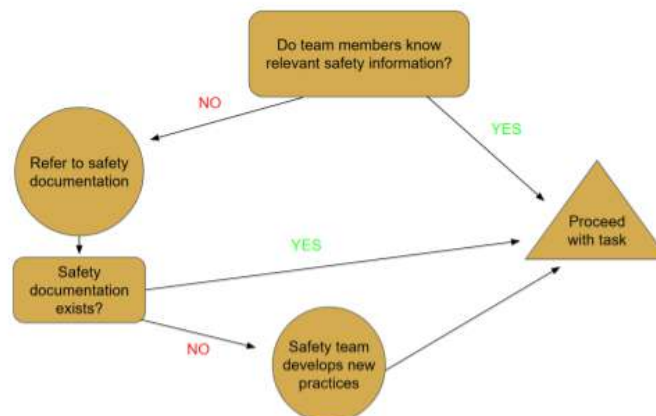


Figure 120: Safety flowchart.

The primary purpose of this process is to take a preemptive approach to lab safety, with an emphasis on stopping accidents before they happen. By referring to safety protocols for each step in the manufacturing, testing, and launch process, the team can prevent most accidents from occurring in the first place. In the event that an accident does occur, the safety guidelines will be carefully reviewed by the safety officer, and adjustments to lab practices will be made to avoid such accidents in the future. These amended practices are then added to the safety binder.

6.1.2 Safety Personnel

While all team members are responsible for maintaining a safe environment, three individuals hold the ultimate responsibility for directing and ensuring this safety. These are the Student Safety Officer, Assistant Safety Officer, and Safety Mentor. The Assistant Safety Officer is Mark Thelen. He is responsible for overseeing all of the mechanical engineering lab spaces at Vanderbilt, which includes the Aerospace Design Lab. Mark works closely with the team throughout the project cycle and is regularly consulted about lab equipment and practices. Chris Dondanville is the Safety Mentor for the team. He is responsible for preparing the rocket motor in the field prior to each launch, and is also responsible for approving the construction of the rocket. Chris makes himself available to the team during the design process and serves as an invaluable resource for information pertaining to recovery systems, motor selection, and general rocket construction. The Student Safety Officer (SSO) this year is Zachary Stutsman. His primary role is to maintain a safe lab, testing, and launch environment throughout each phase of the project. This is accomplished through a variety of means, which are discussed below.

Each Monday, the SSO gives a weekly safety briefing to the team detailing potential hazards that could be encountered during the manufacturing and testing activities planned for that week. This serves as an important preventative measure, as it provides immediately relevant safety information to each team member and thus greatly reduces the likelihood of accidents occurring. In addition to these weekly briefings, the SSO gives an additional briefing prior to each launch or drop test, as these scenarios involve numerous potential safety hazards. These briefings are an important component of the mitigation strategies which will be discussed in more detail in Section 6.2 through Section 6.5. The SSO is responsible for keeping the Material Safety Data Sheets, NAR, FAA, and Vanderbilt regulations up to date in the Safety Binder, and for ensuring that all team members are aware of its location and contents. Another responsibility of the SSO is to regularly check the first aid kits, PPE, and fire extinguishers in the lab to ensure that they are in their proper place and are well-stocked. As an extension of these tasks, the SSO should be well versed in all safety related practices, and should be able to address any safety related questions posed by the team members or mentors. This can involve either directly addressing the question or knowing which resources to utilize to find an answer.

6.2 Safety Matrix Design

Safety matrices are used as a means to succinctly yet methodically list, categorize, and address hazards and failure points that could be encountered at any point throughout the design, building, and testing cycle. The matrices are split into three main sections: personnel hazards, failure modes, and environmental hazards.

The structure of these matrices draws heavily on the standards set by MIL-STD-882, especially with regard to severity classification, probability levels, and risk assignment. The descriptions of each severity and probability level have been adjusted to fit within the context of the USLI competition, but the structure of this system remains largely consistent with the MIL document. This is done in an effort to acknowledge and implement industry best-practices, thus ensuring a robust and proven safety framework for the project.

6.2.1 Severity Classifications

An important part of the categorization of hazards is assigning severities to them. This allows for hazards and failure points to be systematically sorted based on the criteria that they meet. Table 50 below lists the severity levels associated with different criteria.

Table 50: Severity classifications.

Severity Classifications		
Severity	Classification	Result of Mishap
Catastrophic	1	Could result in loss of life or severe injury, extreme environmental impact, or complete loss of rocket or payload.
Critical	2	Could result in injury to personnel, significant impact to environment, or substantial damage to the payload or rocket.
Marginal	3	Could result in minor injury to personnel or reversible moderate damage to the rocket, payload, or environment.
Negligible	4	No injury risk to personnel, and chance of minor reversible damage to rocket, payload, or environment.

This classification system encompasses all possible hazards or failure modes that could be encountered in the project, allowing for a thorough evaluation of each potential risk.

6.2.2 Probability Levels

Probability levels are the second core component of meaningful risk assessment. This allows for events to be sorted by the likelihood that they will occur. These range from frequent to improbable, with a separate eliminated category that addresses a risk which has been removed through mitigation measures. This classification is described in Table 51 below.

Table 51: Probability levels.

Probability Levels		
Probability	Classification	Description
Frequent	A	Likely to occur often
Probable	B	Will likely occur several times
Occasional	C	Will likely occur at some point
Remote	D	Unlikely but could possibly occur
Improbable	E	Unlikely to occur
Eliminated	F	Incapable of occurrence

With respect to probability, the goal of mitigation strategies is to lower the chance of occurrence as much as possible. By minimizing the likelihood of an event, it is possible to offset the impact of more severe hazards, even if it is not possible to lower the severity of these hazards.

6.2.3 Risk Assessment Matrix

The risk assessment matrix assigns a risk level to each hazard or failure mode based on its probability and its severity. These levels are color coded for ease of reading and are listed in Table 52 below.

Table 52: Color key.

Color Key	
	High
	Substantial
	Moderate
	Low
	Eliminated

As can be seen, there are five risk levels, ranging from high to eliminated. A high risk event is one that has both high severity and probability, while a low risk event is one that has both low severity and probability of occurrence. There is also a section for eliminated risks, which accounts for hazards that are removed through mitigation strategies. Ideally, all hazards would be completely eliminated, but this is not a realistic expectation. Therefore, the aim of the mitigation strategies conducted by the team is to reduce the risk level as much as possible by lowering the severity and/or probability of an event occurring. The risk assessment matrix is shown in Table 53 below.

Table 53: Risk assessment matrix.

Risk Assessment Matrix				
	Catastrophic (1)	Critical (2)	Marginal (3)	Negligible (4)
Frequent (A)	1A	2A	3A	4A
Probable (B)	1B	2B	3B	4B
Occasional (C)	1C	2C	3C	4C
Remote (D)	1D	2D	3D	4D
Improbable (E)	1E	2E	3E	4E
Eliminated (F)	Eliminated			

As mentioned above, the goal of mitigation measures is to shift hazards and failure modes as close to the bottom right corner of this table as possible. This risk level classification will be used throughout the safety matrices in Sections 6.3 through 6.5 to denote the risk level of each hazard or failure mode.

Each hazard will be assigned two risk levels. One is the risk level prior to mitigation and the second is the risk level after mitigation. The risk level before mitigation represents the risk associated with that hazard, assuming that all of its potential causes represent the actual conditions in the lab or field. Many of these will be higher level risks than desired. The post-mitigation risk assessment, however, represents the risk after all mitigation measures have been applied. This represents the actual risk associated with these activities as they are conducted by the team, and these should all be within the team-defined acceptable risk range. An acceptable risk is defined as a hazard having a post-mitigation risk level of moderate, low, or eliminated. Hazards will be continually evaluated based on experience gained and/or safety incidents. Once re-evaluated, the hazard mitigations will be revised as necessary to ensure the post-mitigation risk level remains at moderate or lower.

6.2.4 Hazard and Failure Mode Key

The safety matrices cover personnel hazards, environmental hazards, and failure modes. Each of these high level areas comprise specific risks which can occur across many phases of the project. For that reason, each safety matrix is broken into subsections pertaining to the project phase in which they can occur. These subsections are described in Table 54 below.

Table 54: Safety matrix subsections.

Safety Matrix Subsections		
Personnel and Environmental Hazards		
Designation	Meaning	Description
M	Manufacturing	Safety hazards that could arise during the manufacturing process.
L (S)	Launches	Safety hazards associated with the launch and recovery of the rocket. All hazards apply to the full-scale launch vehicle, and the subset that also applied to the subscale launch vehicle are denoted by an (S).
DT	Drop Testing	Safety hazards that could arise during drop testing. This designation specifically refers to testing conducted on the hexacopter drone owned by the team.
LT	Lab Testing	Safety hazards associated with testing beyond the hexacopter drop tests.
Failure Modes		
P	Payload	Failure modes associated specifically with the payload portion of the mission.
V	Vehicle	Failure modes associated specifically with the vehicle portion of the mission.
T	Team (team functioning)	Failure modes associated with team or organizational hazards.

The personnel and environmental hazard matrices will have the same subsections, as they both are concerned with hazards that can arise during the project. The failure modes table will have its own set of subsections, as it is concerned with project failure modes instead of hazards. These subsections make the safety matrices more readable and provide a framework to more thoroughly define risks.

Hazards or failure modes that apply to one or more of these subsections will have all relevant subsections listed in the ‘Designation’ column of the matrix. For ease of reading, all hazards which have multiple designations will be listed at the bottom of the matrix, following the items which have only a single designation.

6.3 Personnel Hazard Analysis

The matrix below details the risks to any personnel associated with the project throughout the duration of its cycle. Personnel in this sense primarily entails VADL team members but also includes team mentors, spectators, and occupants of the areas surrounding both laboratory and field sites. Each hazard is assigned a pre- and post-mitigation risk level, as detailed in Section 6.2.3. Higher risk levels, in this table, indicate the potential for more severe injuries to personnel, with death being a possibility in the highest risk cases.

Section 6.2.4 can be referred to for information about the designation of each hazard. This helps to organize the matrix based on the phase of the projection which a hazard could occur. All hazards with more than one designation can be found at the bottom of the table.

Several of the verification columns in this matrix mention that only approved team members are allowed to use certain equipment. This refers to the Equipment Operation Approval Sheet, which was discussed in depth in PDR. In short, this sheet lists which team members are approved to use each piece of power equipment in the lab. In order to be approved, individuals must be trained by a team member who is already approved on the equipment and then be checked off by the SSO.

It is the ultimate responsibility of the SSO to regularly check that mitigation strategies are being followed. The verification measures listed below and in the subsequent matrices have been established by the SSO to help ensure that each mitigation measure is being met every day. This forces the entire team to take personal responsibility for safety, which greatly increases the overall effectiveness of the preventative measures. The weekly safety briefings also serve as an essential part of mitigation for every hazard discussed in the matrices. Each week brings a unique combination of risks, and the SSO uses the briefings to review relevant mitigation strategies with the team.

Personnel Hazard Analysis							
Designation	Hazard Description	Causes of Hazard	Hazard Effect	Risk Level	Mitigation Strategy	Verification of Mitigation	Post-Mitigation Risk
M	Tripping and falls	- Untidy lab spaces - Power cords strung across walkways	- Cuts or bruises - Potential head injury	3B	- Power cords will be kept clear of walkways - Lab will be regularly cleaned and walkways will be kept clear - Team members will be alert while moving about lab spaces	- Rais is assigned to check the lab each day for cords or cluttered walkways - Weekly safety briefings will include reminders about this and other pertinent details	3E
M	Power drill related injuries	- Improper fixturing of work piece - PPE not worn - Hands or other body parts too close to work area	- Cuts or puncture wounds - Eye injury or loss of vision from flying debris	2B	- Eye protection and work gloves will be worn when operating drills - Respirator will be worn as well when drilling into fiberglass - Work pieces will be firmly restrained while drilling - Other team members will stand clear of the work area	- Eye protection, gloves, and respirators are readily available in the lab at all times and their location is known by every team member - Clamps are also available and their location is known by all team members - Weekly safety briefings have/will include this information when drilling is expected during that week - Only approved team members will use drills	4D
M	Chop saw related injuries	- Improper fixturing of work piece - PPE not worn - Hands or other body parts too close to work area	- Loss of fingers or limbs - Severe cuts - Eye damage or loss of vision from debris - Inhalation of damaging particulates - Hearing damage	1C	- Gloves, safety glasses, ear protection, and respirator will be worn throughout saw operation - Hands will be kept well clear of blade whenever saw is plugged in - Work piece will be firmly clamped in place	- Gloves, eye protection, ear protection, and respirators are available in the lab and their location is known by all team members - Clamps are easily accessible - Only approved members will use the chop saw - Two or more members will be present during saw usage - Weekly safety briefings have/will include mitigation information when saw usage is expected	3D
M	Dremel related injuries	- Improper fixturing of work piece - PPE not worn - Hands or other body parts too close to work area	- Cuts or abrasions - Eye damage or loss of vision from debris - Inhalation of damaging particulates	2C	- Eye protection, gloves, and respirator will be worn while using dremel - Hands will be kept clear of work area and piece will be firmly clamped - Other team members will be kept clear of work area	- Eye protection, gloves, and respirators are available in the lab - Only approved team members will use the dremel - Weekly safety briefings have/will include this mitigation information when dremel usage is expected	4D
M	Table sander related injuries	- Loose clothing or hair near sanding belt - Inattention while sanding - PPE not worn	- Severe abrasions - Inhalation of harmful particulates - Damage to hearing	2C	- Eye, ear, and hand protection will be worn while operating sander - Operator will be fully focused on sanding - Hands will be kept clear of sanding belt	- Gloves, ear, and eye protection are easily accessible in the lab - Only approved team members will use the table sander - Weekly safety briefings have/will include this mitigation information when table sander usage is expected	4D
M	Electrical shock	- Frayed or damaged power cords - Cutting of power cord during tool operation - Improper wiring of electronics	- Electric burns - Muscle spasms - Nerve damage - Irregular heartbeat - Cardiac arrest	1C	- Power cords will be regularly checked to ensure they are insulated properly - Power cords will be kept out of harms way during tool operation - Electronics layouts will be checked by at least two team members and only trained members will wire electrical components - Power sources will be kept to the minimum viable strength	- Rais is responsible for checking power cords for damage each week - Patrick is responsible for overseeing electronics checks - Cord management is included as part of power tool training	3E
M	Inhalation of damaging fumes or particulates	- Cutting, drilling, or sanding of fiberglass without a respirator - Use of epoxy, polyurethane, or other chemical without proper ventilation or mask - Soldering with Leaded Solder	- Brain or lung damage	2B	- No chemicals are to be used without adequate ventilation - All members near fiberglass manufacturing will wear respirators and other appropriate PPE	- MSDS are easily accessible in the lab and are consulted prior to use of chemicals - Respirators and other PPE are easily accessible - Fiberglass manufacturing best practices have been discussed at length with full team	3D
M	Mill or lathe related injuries	- Inattention or improper PPE while working on mill or lathe - Loose clothing; dangling jewelry or hair - Hands or other body part too close to work piece - Improper fixturing of tools or work piece	- Severe injury or death if pulled into machine - Eye or body wounds from shrapnel or debris	1C	- Loose clothing will not be worn in the shop - Dangling jewelry will be removed and hair will be put up - Eye protection will be worn at all times in the shop - Only team members who have completed the machining course or have approval from the shop supervisor can use the mill or lathe - Any and all additional requirements of the shop supervisor will be followed	- Special approval by shop supervisor is required to use machining equipment - Shop supervisor will directly supervise any team members working in the machine shop and will be available for any questions	3D
M	Injury during blast charge preparation	- Preparation of charges near heat or electrical source - Ematch wires not wrapped together - Preparer is not grounded prior to assembling charges	- Ignition of charge leading to burns on hands, body, or face of preparer - Potential damage to eyes	2C	- Charges will be prepared far from any source of heat or electricity - Ematch wires will be wrapped together prior to placement in charge to ensure electricity cannot flow through match - Preparer will ground themselves prior to charge preparation	- SSO will personally prepare blast charges to ensure safety measures are met - Blast charge preparation procedure will be carefully followed to ensure all mitigation strategies are adhered to	F
M	Injuries associated with electrical system preparation	- Soldering iron left on or held too close to skin - Heat gun left on or held too close to skin - Molten solder not allowed to cool before handling - Poor labeling of system leading to unknown live wires	- Potential burns, especially on hands, from soldering iron, heat gun, or molten solder - Electrical shock from live wires	3B	- Soldering irons and heat guns will be kept off and unplugged when not in use - Electronics will not be handled until solder has had sufficient time to cool - Electrical systems will be constructed in accordance with documented diagrams, and will only be handled by those familiar with the layout	- Only approved team members will use soldering equipment - All electrical engineers must complete solder training prior to participation in manufacturing activities - Patrick is responsible for approving electrical diagrams prior to circuit construction	3E
M	Fire in lab space	- Exposed electrical wiring - Overheating of power tools - Hot box left on - Heat gun or soldering iron left on - Exposure of energetic material to heat	- Minor to major burns on team members - Damage to or destruction of lab and possibly of Olin Hall	1C	- Hot box, heat gun, and soldering irons will always be turned off and unplugged before leaving - Extension cords and other wires will be regularly checked to ensure proper insulation - Energetic material will be kept in MAGloc box whenever not in use; heat sources will be kept away from these boxes and away from energetics during usage - Power tools will be allowed to cool after use prior to storage - Fire extinguishers are easily accessible and regularly inspected	- Fire extinguishers will be inspected by the SSO each month - Team members are reminded of extinguisher locations each week in the safety briefing - Rais is responsible for checking extension cords for damage each week - MAGloc boxes are locked at all times - Cord management is included in all power tool training	3E
M	Explosions in lab	- Pressure cooker left on or allowed to exceed safe operating pressure - Black powder or fuel grains exposed to heat - Black powder or fuel grains stored outside of proper containers	- Shrapnel strikes team members or damages other lab equipment - Shockwave damages hearing or vision - Fire is started by explosion of energetics	1C	- Pressure cooker will be kept under operating limit by at least a safety factor of 1.5 - Energetics will be stored in MAGloc boxes when not in use and will be kept clear of heat sources when being used - MAGloc boxes will be kept locked at all times	- Vehicle lead will oversee operation of pressure cooker - MAGloc boxes will be locked at all times - Removal of energetic material from MAGloc boxes will be overseen by either the SSO or the team president, both of whom are well versed in the risks associated with energetic material handling	F

L (S)	Burns during launch or launch preparation	- Blast charge or motor exposed to heat source during preparation - Multimeter used to check electrical connections on prepared altimeter/blast charge system - Improper wiring or coding of avionics leading to premature ignition of blast charge - Motor ignition charge exposed to electricity or heat during installation - Motor ignition charge installed before rocket placed on launch rail	- Minor to severe burns on face or body of team members - Loss of fingers - Chance of death	1C	- Blast charges will be prepared in a clear area away from heat or electrical sources - Only essential personnel will be near blast charge/motor preparation and arming - Avionics will be thoroughly tested and calibrated prior to wiring of blast charges - Multimeters will not be used to check electronic connections of energetics at any point - Motor ignition charge will not be installed until rocket is on rail and properly oriented and will only be installed by the rocketry mentor, Chris Dondanville	- SSO will oversee avionics preparation and motor assembly - Avionics will be assembled by the same team member each time, who has been specially trained on safe avionics assembly - Chris Dondanville will be the only person who handles the assembled motor - SSO will ensure area is clear prior to the placement of the motor ignition charge by Chris Dondanville - The avionics checklist, launch procedure, and safety checklist will be carefully followed on launch day	2E
L (S)	Tripping and falls	- Ground tarp not staked down - Stakes left sticking up above ground level - Inattention of team members while moving about rocket preparation area - Toolboxes or other items left in walkways - Rocket preparation area near large rocks or potholes - Wet conditions leading to slick surfaces	- Scrapes, bruises, cuts - Potential broken bones or head injury	3B	- Rocket preparation area will be selected away from ground hazards such as rocks or pot holes - Tables, chairs, and toolboxes will be kept in designated areas and won't be left in walkways - Team members will be attentive while moving about rocket preparation area - Stakes will be fully hammered into ground and tarp will be fully staked down - Launch day procedures will be thoroughly rehearsed, reducing chance of mishaps	- Launch rehearsal will be attended by all team members - Notes will be taken on any hazards noticed during rehearsal - Each team member will be reminded of their personal responsibility for safety during the launch day safety briefing	4C
L (S)	Back injury	- Using improper form while lifting heavy objects - Lifting more than physically capable of - Risk objects include: launch rail, tool chests, tent, tables, and MAGloc boxes	- Acute or sustained pain in back - Muscular or nervous system injury	2C	- Heavy objects will be moved by two or more team members, as necessary - Team members will not try to lift more than they are capable of - Heavy objects will be lifted using legs and while maintaining a straight back	- Dangers of heavy lifting were discussed by full team at beginning of year - Weekly safety briefings have/will include a reminder of this information prior to launch week, as launch week generally contains most of the heavy lifting done during the project	3D
L (S)	Team members or spectators struck by rocket	- Launch rail angled incorrectly with respect to wind - Team members located too close to launch rail during launch - Strong wind gusts or gusts of changing direction - Failed parachute deployment - Improper rocket assembly leading to falling debris or rocket sections - Improper sizing of rocket fins	- Minor to severe injury to team members or spectators - Possible death	1C	- Launch rail angle will be checked and approved by the SSO and rocketry mentor prior to launch; launch rail will be properly secured - Team members will be located at least 300 feet away from the launch rail, as designated by the NAR codes. - Launch will not proceed in winds higher than 20 mph, or in unpredictably gusty scenarios - Rocket fins will be sized to provide an appropriate stability margin of approximately 2.2 - Rocket will be carefully assembled following a strict procedure - Blast charges will be carefully prepared and sized according to data from both calculations and deployment tests - All team members and spectators will keep eyes on the rocket from launch until landing	- Launch must be approved by the SSO, team president, and Rocketry Mentor - Wind speed will be measured in field with an anemometer to ensure compliance - All relevant checklists and procedures leading up to launch will be carefully followed - Rocket construction must be approved by Chris Dondanville prior to launch	3E
L (S)	CATO	- Improper spacing or preparation of motor grains - Use of cracked or otherwise damaged grains - Introduction of foreign material into motor chamber - Improper fastening of nozzle - Improper fastening of end cap - Improper insertion of ignition charge	- Team members struck by flying debris	3C	- Motor will be prepared exactly as shown in the instruction booklet - Motor assembly will be done by the rocketry mentor and overseen by the SSO - Grains will be carefully checked for cracks or other damage prior to assembly - Nozzle and end cap will be securely fastened - Motor casing will be checked for damage prior to each launch - Metal blast deflector will be placed beneath motor exhaust prior to each launch - Team members will maintain at least the NAR minimum distance from the launch rail and will maintain visual contact with the rocket during launch operations	- SSO will review motor assembly instructions prior to launch day - SSO will provide rocketry mentor with motor components and will remain with him until motor is fully assembled; this ensures that oversight is provided at each step - Grain inspection will be conducted by the SSO and the rocketry mentor	4D
DT	Loss of control of hexacopter drone	- High winds render drone control unreliable - Incompetence of drone pilot - Drone runs out of battery during flight	- Significant injury from impact of drone with team member - Potential death	1C	- Hexacopter drone will ONLY be flown by trained and designated drone pilot - Drone will not be flown if winds are too high - Drone batteries will be checked prior to each flight to ensure sufficient charge - Team members will stand clear of drop point	- SSO will consult drone pilot prior to each flight to ensure wind speeds are manageable and batteries are charged - Drone pilot has flown the hexacopter drone extensively and is very capable of controlling the drone	2E
DT	Injury from impact with drone blades	- Team member standing too close to drone during takeoff or landing - Inattention from team members or drone operator	- Minor to severe cuts or lacerations	2C	- Team members will stand well clear of drone during takeoff and landing - All parties will remain aware of their surroundings at all times - Drone operator will issue verbal notice prior to turning drone on	- SSO and drone pilot must visually scan the area surrounding the drone and issue a verbal warning prior to startup of drone	F
LT	Load frame related injuries	- Debris thrown when airframe pieces fail - Team members standing too close to load frame - Team members handling test piece or other portions of load frame during use	- Eye injury or vision loss from debris - Crushing of fingers, hands, or arms	1C	- Plexiglass shield will be placed in front of load frame prior to use - All team members will wear eye protection and stand clear of load frame - Piece will be securely fastened to frame - Frame will not be touched at any point during operation	- Load frame operation procedures will be carefully followed - Load frame will only be operated when Mark Thelen is present to oversee operation - Eye protection is readily available in the load frame facility	4E
LT	Leg loading tests	- Team members standing too close to test area - Weights fastened to legs insecurely - Leg breaks during testing	- Bruising - Chance of minor cuts - Chance of broken fingers or toes	3C	- Team members will keep feet clear while weight is being applied to legs - Weight will be securely fastened to legs and will be steadily and carefully applied - Foam will be placed beneath leg to soften any falling weight	- SSO approved testing setup and plan prior to tests beginning	4D
LT	Debris or falling components during small scale drop tests Debris or falling components during parking garage drop testing	- Team or passerby standing too near to drop point - Improper sizing of shock cord - Improper attachment of shock cord to payload or to fixed point - Poor construction of payload leading to breakage upon shock cord extension - Improper length of shock cord used leading to higher than expected forces	- Head or eye injury to team members or passerby - Bruising - Minor cuts	2B	- Team members will stand at least 30 feet from point beneath falling payload (SMALL SCALE DROPS) - Team members will stand at least 75 feet from point beneath falling payload (GARAGE DROPS) - Sufficient number of team members will be present to keep lookout and ensure area is clear of passerby - Drop tests will be conducted outside of peak foot traffic hours - Calculations will be carefully conducted and payload will be constructed to proper specifications - Team members will maintain visual contact with payload throughout drops	- SSO must be present to oversee garage drops - At least two members are required to be present for small scale drops; more are required if the area is moderately populated by passerby - At least two team members are required at the drop point and an additional minimum of two members must be on the ground for garage testing. This ensures that eyes are on the payload at all times and no steps are overlooked - Prior to any drop testing, a procedure will be written out, walked through, and then carefully followed for testing	3E

M, L (S), DT, LT	Exhaustion/fatigue	- Long work hours - Insufficient sleep - High temperatures at launch or tests - Lack of food/water	- Heat exhaustion - Inattention or forgetfulness leading to a variety of unsafe conditions - Dehydration	2C	- Milestones will be reasonably set and work will steadily be done to reach them - No one team member will be responsible for work exceeding their capacity - Food and water will be brought to every launch or field test - Team members will be intentional about their sleep schedule	- Sufficient food and water was brought to subscale launch and the same will be done for full scale - The importance of sleep will be regularly mentioned by the SSO during weekly safety briefings	3D
L (S), DT	Hypothermia or frostbite	- Cold weather during outdoor activities - Lack of proper clothing - Extended exposure to the elements - Wet or windy conditions	- Team members experience hypothermia, frostbite, or both - Discomfort and distraction, potentially leading to safety oversights in other areas	3C	- Launches will not be held if weather is excessively cold, especially if also rainy and/or windy - Team members will dress properly for weather conditions - Time outside will be limited to that necessary for the testing or launch - Hand warmers and emergency blankets will be on hand when cold weather is expected	- Safety briefing have/will cover cold weather hazards if expected for launch or drop test - Hand warmers and emergency blankets are included on packing list for launches and drop tests	4D
L (S), DT	LiPO battery fire	- Battery overheats from temperature or sun exposure - Battery is punctured during flight or landing	- Potential burns incurred by team members - Potential of falling debris if fire occurs in air	2D	- Battery will be checked for bloating or puncture after each flight - Safe clearance will be maintained throughout drone operation - Team members on ground will remain attentive throughout drone operation	- SSO will inspect all batteries used after each launch or drop test - Payload and vehicle leads are responsible for ensuring their respective LiPO batteries are transported in the proper LiPO container - Bloating or otherwise damaged LiPOs will be discarded immediately as specified in the MSDS	3E
L, DT	Payload failure	- Payload unable to withstand parachute deployment forces - Falling debris from broken lander legs or other broken components	- Bruising and potential head injury from falling components	2C	- Team members will maintain a safe clearance from drop point at all times - Payload will be thoroughly tested prior to field drop test - Visual contact with payload will be maintained throughout drop	- SSO and team president will both individually verify that area is cleared prior to launches or drops - Verbal warnings will be issued before both drops and launches by the team president - Payload lead will ensure that proper testing has been conducted on the payload prior to launch and that the construction of the payload is adequate	F
L, DT	Team members struck by payload	- Parachute packed incorrectly leading to rapid descent of payload - Quick link not secured properly - Team located too close to drop point - Excessive or unpredictable wind	- Bodily injury, ranging from minor to significant - Slim chance of death	2C	- Parachute packing will be done carefully and overseen by at least two team members - Quick link will be fully tightened and checked for attachment prior to payload flight - Team will be located sufficiently far from drop point. This location will be determined based on wind speed and drop height - Team members will maintain visual contact with payload throughout drop	- Payload lead will be in charge of parachute packing, with oversight from one of several team members who are familiar with the parachute packing procedure - Dry runs of parachute packing will be conducted before each launch - Payload lead is responsible for checking quick links for damage prior to each flight; checks must be approved by SSO before flight - Wind speed will be measured with an anemometer prior to drop or launch - Team president is responsible for providing a verbal warning prior to drop or launch	3E
M, L (S), DT, LT	Team members experience damage to mental health	- Excessive working without breaks - Insufficient communication of needs among members	- Exacerbation of mental health issues - Loss of motivation - Decrease in team productivity	2B	- Mental health will be prioritized by all team members - Honest and respectful communication will be practiced by all team members - Breaks will be taken as needed by each individual	- The team lead will promote a culture of mental health awareness - The team lead will regularly check in with each team member to ensure they are in a healthy mental state	2D
L (S), DT	Excessively snowy launch/drop area	- Launch or drop conducted soon after snowfall	- Increased risk of frostbite/hypothermia for team members - Increased risk of slips and falls	3B	- Extra caution will be taken while moving about snowy areas - Team members will wear adequate clothing and the team will bring hand warmers to the field	- Team members have successfully conducted launches and drop tests without experiencing any trips or falls - Frostbite and hypothermia have been entirely avoided	3D
L (S), DT	Payload landing on frozen bodies of water	- Launch or drop site too close to water hazards - Excessive wind during launch or drop test	- Potential injury of team member during retrieval	2C	- Vehicle will only be launched in locations approved by the team rocketry mentor and SSO - Wind speeds will be checked with an anemometer prior to each launch to ensure rocket will not drift too close to water bodies - In the event of a water landing, the RSO and team mentor will be consulted prior to any retrieval efforts - All possible methods of retrieval from shore will be exhausted before any team member is permitted to go onto the ice - If ice appears unsafe in any way, no team members will be allowed onto it regardless of retrieval success through other methods	- Landings in frozen bodies of water were fully avoided during winter testing - Launch and drop sites have been and will continue to be aptly chosen with sufficient clearance from water bodies - No injury was sustained by any team member due to water retrieval efforts	3D

6.4 Analysis of Failure Modes

The matrix below details potential failure modes for the vehicle, payload, and team as a whole. The vehicle and payload portions are focused primarily on failures that could occur during the competition rocket flight, whereas the team failure modes cover team or organizational level issues that could prevent the production of a viable final product. For the purposes of this matrix, failure is defined as an event or occurrence that leads to full or partial incompleteness of the mission. The risk level assigned to each failure mode reflects whether that particular failure represents a full or partial incompleteness of the mission, with full incompleteness yielding higher risk classifications.

Analysis of Failure Modes							
Designation	Failure Mode	Causes of Failure Mode	Effect of Failure Mode	Risk Level	Mitigation Strategy	Verification of Mitigation	Post-Mitigation Risk Level
P	Antennas fail to extend	- Improper winding of AEM - Blockage of AEM opening - Loss of electrical connection to servo - Failure to detect landing - Loss of power prior to landing - Servo strength insufficient to release AEM - Misalignment of payload within rocket body - Premature extension before rocket settles	- Higher likelihood of ground interference with transmission - Potential mission failure	2B	- Landing detection software will be thoroughly tested through lab tests, drop tests, and launches - AEMs will have a delay period after landing to allow rocket to settle in no-jettison case - Electrical components will be thoroughly ruggedized and tested - Battery will be sized to provide at least 3 hours of power to full system - Rocket will be carefully constructed - Assembly will be practiced many times prior to launch and a careful assembly procedure will be laid out - Servo will be tested in both lab and field settings - Three AEMs will be included in the payload to render blockage of all improbable	- Antennas have been tested under a variety of operating conditions - Antennas have proven capable of extending from within an airframe - Antennas have been successfully reset in the field multiple times - Transmission has been verified up to 1 mile with AND without extension	3D
P	Parachute does not deploy	- Tape is not removed before packing parachute - Parachute is packed in a manner that does not expose it to aerodynamic forces - Parachute is improperly folded - Shock cord is wrapped around parachute	- Payload is not slowed in its descent - Major damage or complete destruction of payload - Mission failure	1C	- Parachute will be folded carefully along each gore to ensure swift opening - Shock cord will not be wrapped around chute - Folding and packing will be practiced prior to launch and a thorough procedure will be followed each time	- Payload and vehicle leads will fold and pack their respective parachutes - Parachute folding and packing will be overseen by at least one other team member familiar with the process - Parachute folding has been successfully completed many times by multiple team members	1E
P	Legs fail to deploy	- Misalignment of payload within airframe - Insufficient strength of torsional springs to push legs out - Excessive friction with airframe or with other payload components	- Failure to land upright - Partial mission failure	3B	- Legs will be carefully constructed - Assembly will be practiced prior to launch and all components will be tested to ensure proper fit - Leg deployment will be thoroughly tested in both lab and field settings	- Payload lead has conducted deployment testing on the legs - Legs have been proven to work in a variety of conditions, and have been assembled and disassembled many times without losing any performance	3E
P	LSM bulk plate failure	- Excessive loading forces during main parachute deployment - Insufficient thickness of LSM attachment bulk plate - Shaft collars are attached poorly or are of insufficient strength to withstand main parachute deployment	- Premature release of payload from rocket - Partial destruction of LDM - Possible partial or full destruction of electronics - Potential drifting outside of transmission range - Possible mission failure	2C	- LSM bulk plate will be constructed from aluminum - FEA analysis will be performed on LSM bulk plate and hooks to ensure that it can withstand loading - Sufficient safety factors will be used in bulk plate construction - Shaft collars will be tested to ensure they can withstand forces greater than the expected loading\	- Payload lead is responsible for ensuring that proper strength testing has been conducted on the LSM bulk plate - FEA analysis shows a factor of safety greater than 3 for the LSM bulk plate - LSM bulk plate performed nominally during VDF and shock testing	2E
P	Improper Handling of Electronics	- Wet conditions introduce moisture to electronics - Excessive or unpredicted launch or landing forces	- Short circuiting or breakage of electronics - Failure to detach parachute, collect data, extend antennas, and transmit data	1C	- Electronics will be prepared under a tent to protect from rain/weather - Launch will not be held in excessively rainy conditions - Electronics will be extensively ruggedized and tested in field conditions - Battery will be tested to ensure sufficient power is provided	- Electronics preparation has been practiced in the lab in order to streamline the field preparation - Electronics have been ruggedized a sufficient amount to withstand drop testing - Electronics have been proven to withstand the rigors of full-scale flight	1E
P	Software failure	- Improper setting of altitude thresholds - Logic error in code - Insufficient testing of software under operating conditions - Improper integration of sensors into central logic	- Failure to detect landing - Failure to collect data - Failure to extend antennas and/or transmit data	1C	- Software will be thoroughly tested in the lab and in the field while integrated with the electronics system - Full sensor suite will be integrated and tested prior to launch - Software logic will be checked by multiple team members	- Software has been tested in both lab and field settings and has performed nominally - Altitude setting procedure has been practiced in the field multiple times with success - All sensors have been integrated into the software and the majority have been tested - All sensors/software will undergo integrated testing during the PDF	1E
V	CATO	- Improper spacing or preparation of motor grains - Use of cracked or otherwise damaged grains - Introduction of foreign material into motor chamber - Improper fastening of nozzle - Improper fastening of end cap - Improper insertion of ignition charge	- Significant damage to rocket and possibly payload - Possible complete destruction of rocket and possibly payload - Likely removal of ability to relaunch - Likely complete mission failure	1C	- Motor will be prepared exactly as shown in the instruction booklet - Motor assembly will be done by the rocketry mentor and overseen by the SSO - Grains will be carefully checked for cracks or other damage prior to assembly - Nozzle and end cap will be securely fastened - Motor casing will be checked for damage prior to each launch	- SSO will review motor assembly instructions prior to launch day - SSO will provide rocketry mentor with motor components and will remain with him until motor is fully assembled; this ensures that oversight is provided at each step - Grain inspection will be conducted by the SSO and the rocketry mentor during motor assembly	1E

V	Improper stability margin	- Location of CP is not calculated correctly - Location of CG is too close to or far from CP	- Unstable flight - Possible destruction of rocket	1C	- CP will be carefully calculated, checked by multiple team members, and approved by the rocketry mentor prior to flight - CG will be carefully measured - Stability margin of approximately 2.2 will be achieved through sizing of fins and manipulation of CG	- Rocketry Mentor will check and approve rocket CP and CG prior to flight - CFD and flight simulations have been conducted and indicate stable flight of the launch vehicle - As assembled CP and CG were adequate; vehicle performed nominally during VDF	2E
V	Fin flutter	- Fins are poorly constructed - Improper sizing of fins - Poor choice of fin material	- Breakage of fins - Potential crashing of rocket	1C	- Fins will be constructed from carbon fiber and will be firmly attached to airframe - Fin flutter will be calculated to ensure that critical values are not met	- Fins performed nominally during full-scale flight - Apogee was in range of the expected value - No damage was observed on the fins or in the surrounding area of the rocket	2D
V	Airframe failure	- Airframe material too weak - Bulk plate bolts placed too close to edge of airframe - Greater launch or separation forces than expected - Blast charges blow apart airframe	- Loss of rocket and/or payload - Possible mission failure	1D	- Airframe will be constructed from fiberglass - Bolts will be at least 1.5 to 2 bolt diameters in distance from the edge of the airframe to minimize risk of tearout - Load testing will be performed on airframe and bulk plates to ensure that it can withstand all flight forces with a substantial safety factor - Blast charges will be sized based on both calculations and deployment testing	- The team has conducted load frame testing and verified the safety factor for airframe hoop stress, bolt tear out stress, and axial compressive stress are greater than 2.5 - Blast charges have been successfully sized, tested, and implemented for both the subscale and full-scale vehicles	2E
V	Deployment failure	- Blast charges too small to separate sections - Parachute bay not sufficiently sealed from rest of rocket - Excessive friction between overlapping airframe pieces - Altimeter fails to register apogee	- Rocket sections do not separate - Drogue and/or main parachute does not deploy - Probable destruction of rocket and payload - Probable mission failure	1C	- Rocket sections will be sized so that they slide easily into each other - Blast charges will be sized based on both calculations and deployment tests which will verify that sufficient black powder is being used - Openings out of the parachute bay will be blocked with putty prior to each launch - Altimeters will be calibrated prior to launch and will be tested in the subscale and full scale flights	- Blast charges have been successfully sized, tested, and implemented - Putty was successfully used to seal off parachute compartments from the rest of the vehicle - Altimeters were successfully calibrated and deployed the parachutes at the correct altitudes - Rocket construction was adequate for safe flight, recovery, and rapid reusability	1E
V	Sticking on launch rail	- Buttons not in line with each other - Rocket fins in contact with launch rail - Excessive friction between rail buttons and rail - Ice buildup on launch rail or rocket	- Rocket does not reach target apogee - Rocket exits launch rail at angle that is not conducive to safe flight	2C	- Launch rail will be greased with WD-40 prior to each launch - Rail buttons will be carefully placed and will be sized to fit launch rail - Rocket will be dry fitted to rail before launch day to ensure good fit - Launch will be postponed in excessively icy conditions	- WD-40 was successfully used to lubricate the launch rail during full-scale flight - Rail buttons were successfully placed - Rocket behaved as expected on the launch rail	2E
T	Planning failures	- Project timelines not created - Time not set aside for payload testing - Team unaware of goals and milestones - Manufacturing gets behind schedule - Poor communication between team and mentors	- Necessary testing is not completed - Project gets behind schedule - Manufacturing is poor or incomplete - Potential mission failure	2B	- Team will hold meetings with team mentor twice per week - Team will meet nightly for short update meetings - Timelines and goals will be discussed in depth by team and will be adhered to	- Team meets twice weekly with the team mentor - Team has consistently held nightly meetings to update on progress and organize - Timelines have been kept thus far and goals have been clearly communicated - These practices will be continued for the duration of the project	2D
T	Report writing failures	- Poor team participation - Lack of communication on section responsibilities and deadlines - Poor proofreading of sections - Lack of full project knowledge from team members - Not leaving enough time to write	- Poor scores on report sections - Significant negative impact to team standing in competition	2B	- Full team will be involved in report writing process - Team will develop an in depth document where section responsibilities, proofreading responsibilities, and report completion will be tracked - All sections will be thoroughly proofread by multiple team members - Report will be started at least one week in advance of due date - Testing and manufacturing will be well documented, which will help strengthen reports	- The team has successfully produced three reports thus far - Report writing has been streamlined for FRR to ensure efficient and thorough completion - CDR feedback was taken into account to guide adjustments to writing styles and practices	2D
P, V	Loss of power	- Battery does not supply enough power - Battery dies before or during launch - Launch forces cause battery connection to separate - Electrical components disconnect from battery due to launch or landing forces	- No or incomplete collection of data - No or incomplete transmission of data - Failure to disconnect parachute - Failure to trigger blast charges - Failure to complete mission	2B	- Careful power calculations and extensive testing will be done on the chosen batteries - Batteries will be fully charge prior to launch and spares will be brought to the field - All electrical connections will be firmly fastened and thoroughly ruggedized	- Power calculations have been done - Spare batteries have been brought to each field test and launch and will continue to be brought - Electrical connection issues have been identified and fixed - Batteries have proven to be of sufficient power during testing thus far	1E

P, V	Parachute fails during deployment	- Damaged parachute is used - Parachute strings or material is unable to withstand shock loading experienced - Fire blanket is improperly wrapped, leading to burning of chute - Blast charges are too large, leading to excessive deployment forces - Sections of the rocket or payload strike or tangle with parachute during deployment	- Rapid descent of rocket and/or payload - Complete or partial destruction of rocket and/or payload - Mission failure	1C	- Parachutes will be carefully inspected for damage prior to each launch - Deployment loadings will be carefully calculated and parachutes will be chosen to withstand these loads with an added safety factor - Fire blanket will be wrapped fully around blast-deployed chutes - Blast charges will be carefully prepared and sized based on both calculation and deployment testing	- Vehicle and payload leads are responsible for checking their respective parachutes prior to launch or drop - SSO and Rocketry Mentor will oversee packing of vehicle parachutes, including the wrapping of the fire blanket, the attachment of the shock cord, the placement of the fire ball, and the addition of fire retardant	1E
P, V	Shock cord fails during deployment	- Damaged shock cord is used - Loading is higher than predicted - Shock cords do not have enough safety margin	- Complete or partial destruction of rocket and/or payload - Mission failure	1C	- Shock cords will be visually inspected prior to each launch or drop test - Loading will be carefully calculated and verified through testing - Shock cords will be selected with a safety margin of at least four	- Rocketry Mentor must approve shock cords prior to launch - Vehicle and payload leads will check shock cords both in the lab and in the field before launch - Spare shock cord will be brought if available	1E
P, V	Shock cord bulk plate failure	- Excessive loading during parachute deployment - Poor material choice for bulk plate - Insufficient bulk plate thickness	- Partial or complete destruction of payload - Probable mission failure	1C	- FEA analysis will be conducted on plates - Extensive testing of bulk plates and deployment events will be performed - Bulk plates will be sufficiently strong with an added safety factor	- FEA analysis has been conducted on all shock cord bulk plates. Safety factors greater than 2.5 were observed on each, which meets the VADL minimum for recovery components - Completed testing of bulk plates indicates that airframe should fail before bulk plates do	1E
P, V	Landing in unintended areas	- Excessively high launch day winds - Failure to correctly choose launch angle - Improper SSM - Poor choice of launch site	- Vehicle and payload potentially sustain water damage - Vehicle and payload are potentially struck by motor vehicle - Vehicle and payload are potentially stuck in trees	2C	- Launch site will meet NAR guidelines and will be well clear of potential hazards such as water, roadways, and trees - Launch site must be approved by the rocketry mentor prior to launch - Launch angle and rail setup approved by rocketry mentor prior to launch - Launch will not be carried out if winds exceed 20 mph	- Anemometer was used to check wind speeds before launch and will continue to be used - Relevant approvals from Chris Dondanville are and will be received prior to launch - Launches are carried out only at NAR approved sites	3D
P, V	Payload landing on frozen bodies of water	- Launch or drop site too close to water hazards - Excessive wind during launch or drop test	- Potential loss of rocket and/or payload if ice breaks - Cracking of airframe or electrical components due to hard landing surface	2C	- Vehicle will only be launched in locations approved by the team rocketry mentor and SSO - Wind speeds will be checked with an anemometer prior to each launch to ensure rocket will not drift too close to water bodies - In the event of a water landing, the RSO and team mentor will be consulted prior to any retrieval efforts - All possible methods of retrieval from shore will be exhausted before any team member is permitted to go onto the ice - If ice appears unsafe in any way, no team members will be allowed onto it regardless of retrieval success through other methods	- Landings in frozen bodies of water were fully avoided during winter testing - Launch and drop sites have been and will continue to be aptly chosen with sufficient clearance from water bodies - No damage has been sustained by the rocket or payload due to landing in water - No injury was sustained by any team member due to water retrieval efforts	3C
P, V	Excessively wet or muddy launch/drop area	- Launch or drop conducted soon after rain or snow melt - Launch or landing in low-lying areas that do not drain well	- Potential damage to shock cords or electrical components during landing and retrieval of vehicle - Interference with moving components; possible rusting of metallic components if not cleaned well	3C	- Efforts will be made to launch in dry areas - Extra caution will be taken when retrieving or preparing the vehicle in muddy areas - Any components that contact mud will be thoroughly cleaned prior to reflight - Any electrical components which become muddy/wet must be cleaned and tested prior to reflight - Any muddy shock cords must be cleaned and approved by the team rocketry mentor prior to reflight of vehicle	- Drop tests and launches have largely avoided muddy areas - When components have been muddied, cleaning procedures have proved sufficient and reflight was achieved even after muddy landings - Electronics are sufficiently protected from encroachment of mud or other moisture during flight and recovery	4C
P, V	Excessively snowy launch/drop area	- Launch or drop conducted soon after snowfall	- Potential damage to electronics from moisture - Potential for mass creep during reflight if shock cords, other components become waterlogged - Loss of tools in snow around prep area	3C	- Electronics will be centrally located within the rocket airframe - Electronics will not be placed by separation points of the vehicle without proper sealing of bulkplates - Tarps will be used in prep area to prevent small components and tools from being lost in the snow	- Electronics have not been damaged by moisture during launches or drops - Bulk plates have successfully partitioned the open separation points from the rest of the rocket	4C
P, V	RF interference from EMF produced by neighboring rocket motors during launch	- Presence of high-power devices near RF receivers - Improper grounding of electrical subsystems	- Loss of communication with the payload - Corrupted or incomplete data transmission - Potential mission failure if commands are not received	2B	- Shield RF-sensitive components with grounded enclosures - Test for interference before launch - Ensure sufficient physical separation between RF systems and high-power electronics - Use properly shielded cables for critical signal lines	- Altimeter tests were conducted using Baofeng in controlled environment - Data integrity was verified - Pre-flight RF tests were performed at the launch site	3D

6.5 Environmental Hazard Analysis

The following matrix details potential environmental hazards and their associated risks. Higher risk levels on this table indicate more extreme or far-reaching impacts to the environment. This matrix follows the same type of designations as the Personnel Hazard Analysis matrix and contains some overlap in hazard causes. However, the result of hazard, risk levels, and mitigation strategies differ significantly. The goal of mitigation strategies in this table is to reduce the risk to the environment as much as possible. In the event that mitigation strategies for environmental factors would result in unsafe conditions for personnel, personnel safety will always be prioritized. However, no facet of the project will be undertaken if its personnel, failure, or environmental risk level exceeds the VADL-defined upper limit of moderate.

Environmental Hazard Analysis							
Designation	Hazard Description	Causes of Hazard	Hazard Effect	Risk Level	Mitigation Strategy	Verification of Mitigation	Post-Mitigation Risk Level
M	Fire in lab	- Exposed electrical wiring - Overheating of power tools - Hot box left on - Heat gun or soldering iron left on - Exposure of energetic material to heat	- Expulsion of toxic smoke into atmosphere - Spillage of chemicals - Significant physical damage to area surrounding Olin Hall	1C	- Hot box, heat gun, and soldering irons will always be turned off and unplugged before leaving - Extension cords and other wires will be regularly checked to ensure proper insulation - Energetic material will be kept in MAGloc box whenever not in use; heat sources will be kept away from these boxes and away from energetics during usage - Power tools will be allowed to cool after use prior to storage - Fire extinguishers are easily accessible and regularly inspected	- Operating procedures for hot items have been successfully followed so far and will continue to be adhered to - Rais has regularly checked the extension cords and will continue to do so - Energetic material is locked at all times. SSO oversees any handling of energetic material - Power tools continue to be discussed in safety briefings when relevant	2E
M	Explosions in lab	- Pressure cooker left on or allowed to exceed safe operating pressure - Black powder or fuel grains exposed to heat - Black powder or fuel grains stored outside of proper containers	- Damage to building and natural areas immediately surrounding it in extreme cases	3C	- Pressure cooker will be kept under operating limit by at least a safety factor of 1.5 - Energetics will be stored in MAGloc boxes when not in use and will be kept clear of heat sources when being used - MAGloc boxes will be kept locked at all times	- Vehicle lead oversees any operation of pressure cooker - MAGloc boxes are locked at all times. SSO oversees any handling of energetic material	F
L (S)	Fires/explosions during launch or launch preparation	- Blast charge or motor exposed to heat source during preparation - Multimeter used to check electrical connections on prepared altimeter/blast charge system - Improper wiring or coding of avionics leading to premature ignition of blast charge - Motor ignition charge exposed to electricity or heat during installation - Motor ignition charge installed before rocket placed on launch rail	- Scorched ground or destroyed plantlife surrounding burn/blast area - Cratering in the event of a blast	2C	- Blast charges will be prepared in a clear area away from heat or electrical sources - Avionics will be thoroughly tested and calibrated prior to wiring of blast charges - Multimeters will not be used to check electronic connections of energetics at any point - Motor ignition charge will not be installed until rocket is on rail and properly oriented and will only be installed by the rocketry mentor, Chris Dondanville	- SSO will oversee avionics preparation and motor assembly - Avionics will be assembled by the same team member each time, who has been specially trained on safe avionics assembly - Chris Dondanville will be the only person who handles the assembled motor - SSO will ensure area is clear prior to the placement of the motor ignition charge by Chris Dondanville - The avionics checklist, launch procedure, and safety checklist will be carefully followed on launch day	3E
L (S)	Rocket impact with landscape, plants, or animals	- Launch rail angled incorrectly with respect to wind - Strong wind gusts or gusts of changing direction - Failed parachute deployment - Improper rocket assembly leading to Falling debris or rocket sections - Improper sizing of rocket fins	- Damage to trees or other plant life near launch site - Damage to landscape around launch site, primarily through impact craters - Pollution of land immediately surrounding launch site in the event that debris is scattered to the point that team is unable to fully clean it	3C	- Launch rail angle will be checked and approved by the SSO and rocketry mentor prior to launch; launch rail will be properly secured - Launch will not proceed in winds higher than 20 mph, or in unpredictably gusty scenarios - Rocket fins will be sized to provide an appropriate stability margin of approximately 2.2 - Rocket will be carefully assembled following a strict procedure - Blast charges will be carefully prepared and sized according to data from both calculations and deployment tests	- Chris Dondanville approves the launch rail assembly and angle prior to each launch - Wind speed is checked with an anemometer prior to each launch to ensure compliance - Rocket construction and SSM was adequate for subscale launch. Similar construction and design procedures will be followed for full scale - Blast charges have been successfully prepared, tested, and implemented	3E
L (S)	CATO	- Improper spacing or preparation of motor grains - Use of cracked or otherwise damaged grains - Introduction of foreign material into motor chamber - Improper fastening of nozzle - Improper fastening of end cap - Improper insertion of ignition charge	- Scorching of ground surrounding launch rail - Start of fire near launch rail - Release of toxic fumes from burning plastic components - Spreading of debris about launch site - Potential cratering immediately beneath rocket	2C	- Motor will be prepared exactly as shown in the instruction booklet - Motor assembly will be done by the rocketry mentor and overseen by the SSO - Grains will be carefully checked for cracks or other damage prior to assembly - Nozzle and end cap will be securely fastened - Motor casing will be checked for damage prior to each launch - Metal blast deflector will be placed beneath motor exhaust prior to each launch	- SSO will review motor assembly instructions prior to launch day - SSO will provide rocketry mentor with motor components and will remain with him until motor is fully assembled; this ensures that oversight is provided at each step - Grain inspection will be conducted by the SSO and the rocketry mentor	3D
DT	Loss of control of hexacopter drone	- High winds render drone control unreliable - Incompetence of drone pilot - Drone runs out of battery during flight	- Harm or death of wildlife - Damage to plant life or test field - Puncturing of LiPO (discussed in detail below)	2C	- Hexacopter drone will ONLY be flown by trained and designated drone pilot - Drone will not be flown if winds are too high - Drone batteries will be checked prior to each flight to ensure sufficient charge - Area will be visually scanned for wildlife prior to each flight	- SSO will consult drone pilot prior to each flight to ensure wind speeds are manageable and batteries are charged - Drone pilot has flown the hexacopter drone extensively and is very capable of controlling the drone	2E
LT	Debris or falling components during small scale drop tests Debris or falling components during parking garage drop testing	- Animal life too near to drop point (GARAGE DROPS) - Improper sizing of shock cord - Improper attachment of shock cord to payload or to fixed point - Poor construction of payload leading to breakage upon shock cord extension - Improper length of shock cord used leading to higher than expected forces	- Damage to landscape from payload impact - Harm to animals from impact of whole or parts of payload	3D	- Payload will be thoroughly tested prior to field drop test - Drop area will be visually checked for animal presence prior to launch	- SSO must be present to oversee garage drops - At least two members are required to be present for small scale drops - At least two team members are required at the drop point and an additional minimum of two members must be on the ground for garage testing. This ensures that eyes are on the payload at all times and no steps are overlooked - Prior to any drop testing, a procedure will be written out, walked through, and then carefully followed for testing	F

M, L (S), DT, LT	Improper disposal of waste	- No trash bags available in field - Windy conditions blow trash away from prep area - Negligence from team members - Chemicals or hazardous material not disposed of through proper channels	- Potential harm or death to wildlife - Pollution of land or bodies of water near test sites or near garbage dump - Potential contamination of drinking water supply	2B	- Trash bags will be on hand for every launch or field test - Team members will be reminded about trash disposal prior to each field test as a part of the safety briefing - Any team members handling chemicals or hazardous materials will refer to the appropriate MSDS for disposal instructions	- Trash has been properly discarded at every field test thus far and will continue to be taken care of - MSDS have been successfully followed so far. They will continue to be consulted when chemicals are being handled - Safety briefings have covered relevant trash and chemical handling and will continue to do so	F
L (S), DT	LiPO battery fire	- Battery overheats from temperature or sun exposure - Battery is punctured during flight or landing - Drone battery is punctured during landing or in loss of control event	- Fire starts prior to take off or after landing - Plants and animals damaged or destroyed as a result - Spillage of harmful chemicals or fumes	2C	- Battery will be checked for bloating or puncture before and after each flight - Sun exposure will be limited as much as possible - Launch/drop will only be conducted when area is free from animal life - Launch/drop site will be chosen in area as free from plant life as possible	- SSO will inspect all batteries used after each launch or drop test - Payload and vehicle leads are responsible for ensuring their respective LiPO batteries are transported in the proper LiPO container - Bloating or otherwise damaged LiPOs will be discarded immediately as specified in the MSDS	2E
L, DT	Payload failure	- Payload unable to withstand parachute deployment forces	- Pollution of launch site from falling debris - Potential impact with animals	3C	- Payload will be thoroughly tested prior to field drop test - Launch or drop area will be visually checked for animal presence prior to launch	- Payload lead is responsible for ensuring that payload is well constructed - Payload lead is responsible for ensuring that necessary strength testing has been completed - SSO and team president are responsible for scanning the drop or launch area for wildlife prior to each launch or drop	4E
L, DT	Payload impact with landscape, plants, or animals	- Parachute packed incorrectly leading to rapid descent of payload - Quick link not secured properly - Drop area not checked for wildlife before drop - Excessive or unpredictable wind	- Injury or death of animals - Damage to trees or other plants - Damage to field in which drop is conducted	3C	- Parachute packing will be done carefully and overseen by at least two team members - Quick link will be fully tightened and checked for attachment prior to payload flight - Area will be visually checked for wildlife prior to each drop or launch	- Payload lead will be in charge of parachute packing, with oversight from one of several team members who are familiar with the parachute packing procedure - Dry runs of parachute packing will be conducted before each launch - Payload lead is responsible for checking quick links for damage prior to each flight; checks must be approved by SSO before flight - Wind speed will be measured with an anemometer prior to drop or launch - SSO and team president are individually responsible for scanning the area prior to launch or drop	3E

6.6 Launch Operations Procedures

It is imperative to have detailed documentation of procedures and safety considerations on launch day. As such, the Student Safety Officer has compiled checklists covering the following categories: avionics, payload, recovery, motor, general vehicle, launch rail, launch procedure, igniter installation, general safety, post-flight procedures, launch day responsibilities, and a troubleshooting guide. Each checklist details the required materials, personnel, and assembly procedure for all launch day tasks. Further, the responsible personnel must sign off on each sheet pending the procedure's completion. There is also a master checklist that lists a range of miscellaneous items which are necessary to complete rocket assembly and launch. This checklist also contains boxes to record some of the high level launch day data such as pressure, temperature, wind speed, and assembled vehicle parameters.

Note that the preparation of the electrical systems is detailed in the payload and avionics checklists, as these are the only sections that contain electrical components. The launch pad setup procedure is detailed in the launch procedure and launch rail checklists. The checklists are shown below in their entirety.



VADL Competition Flight 5/3/2025

Master Checklist

0

General Information

- Date: 5/3/2025
- Location: Huntsville, AL
- Launch vehicle: Full-Scale
- Mission success criteria:
 - Successful flight and recovery of vehicle
 - Reusability of vehicle
 - Payload retention
 - Payload deployment and upright landing
 - Data collection - avionics and payload
 - Antenna extension
 - RF transmission

Launch Day Documentation

- ☐ 1: Avionics Checklist
- ☐ 2: Payload Checklist
- ☐ 3: Recovery Checklist
- ☐ 4: Motor Checklist
- ☐ 5: General Vehicle Checklist
- ☐ 6: Launch Rail Checklist
- ☐ 7: Launch Procedure
- ☐ 8: Igniter Installation Checklist
- ☐ 9: Safety Checklist
- ☐ 10: Post-Flight Checklist
- ☐ 11: Launch Day Responsibilities
- ☐ 12: Troubleshooting

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VADL Competition Flight 5/3/2025

Master Checklist

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Required Materials

- | | |
|--|---|
| ☐ Tarps x2 | ☐ Sand paper |
| ☐ Chairs x8 | ☐ File |
| ☐ Tables x3 | ☐ Pliers x2 |
| ☐ Garbage bags | ☐ Magloc boxes x2 |
| ☐ 5 minute epoxy | ☐ Battery Chargers x3 |
| ☐ Popsicle sticks | ☐ Snacks + drinks |
| ☐ Latex gloves | ☐ Calipers |
| ☐ Mylar | ☐ Tape measure x2 |
| ☐ Wire cutters x2 | ☐ Power drill |
| ☐ Scissors x2 | ☐ Drill set |
| ☐ Wire stripper x2 | ☐ Hammer |
| ☐ Shears | ☐ Ball hammer |
| ☐ Box cutters | ☐ Mallet |
| ☐ Sharpies and markers | ☐ "Destroyer of Worlds" |
| ☐ Gorilla tape | ☐ Clamps |
| ☐ Masking tape | ☐ Acetone |
| ☐ Electrical tape | ☐ Wipes |
| ☐ X-frames | ☐ Paper towels |
| ☐ Various sizes of zipties | ☐ Magnetizer |
| ☐ Tweezers | ☐ Adjustable wrench |
| ☐ Bungee cords | ☐ Saw |
| ☐ Hot glue gun | ☐ Tree tools |
| ☐ Hot glue | ☐ LiPo bag and LiPos |
| ☐ Battery powered solder iron | ☐ Screwdrivers |
| ☐ Hanging Scale | ☐ Nut drivers |
| ☐ Cyanoacrylate | ☐ Allen keys |
| ☐ LSM Box | ☐ Avionics Box |
| ☐ Multimeter | ☐ Recovery Hardware Box |
| ☐ Safety equipment | ☐ Launch Rail Hardware Box |
| ☐ First aid kit | ☐ Jackery x2 |
| ☐ Safety glasses | ☐ Superglue |
| ☐ Face shield | |
| ☐ Gloves | |
| ☐ Fire extinguisher | |



VADL Competition Flight
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Master Checklist

0

Signatures

Name	Signature
Ian Morgan	
Rais Nurhidajat	
Zachary Stutsman	



VADL Competition Flight 5/3/2025

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Avionics Checklist

Required Materials

- ☐ Avionics assembly
 - ☐ Avionics housing
 - ☐ Altus Metrum EasyMini altimeter (x2)
 - ☐ Screw terminal (x2)
 - ☐ 9v battery (x3)
- ☐ Avionics fore bulk plate
- ☐ Avionics aft bulk plate
- ☐ MAGloc
 - ☐ Droque primary blast charge (x3)
 - ☐ Droque backup blast charge (x3)
 - ☐ Main primary blast charge (x3)
 - ☐ Main backup blast charge (x3)
- ☐ Blast putty
- ☐ Blast caps
- ☐ Safety glasses
- ☐ Face shield
- ☐ Hardware
 - ☐ M3x20 screws x4
 - ☐ M3x16 screws x4
 - ☐ M3 nuts x8
 - ☐ 3/8-16 eyebolt (x4)
 - ☐ 10-32 radial bolts and washers (x4)
 - ☐ 10-32 screws x4
 - ☐ 10-32 nuts x4
 - ☐ Zip Ties
- ☐ Tools
 - ☐ M3 hex key
 - ☐ 1/4-20 nut driver
 - ☐ 10-32 screw driver
 - ☐ Altimeter screwdriver
 - ☐ Screw terminal screwdriver
 - ☐ Wire cutter
 - ☐ Pliers
 - ☐ Avionics Box

Required Personnel

Name	Role
Santino Clemente	Avionics Engineer
Ian Morgan	President
Zachary Stutsman	Student Safety Officer
Chris Dondanville	Rocketry Mentor

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VADL Competition Flight 5/3/2025

1

Avionics Checklist

Pre-Launch Day Procedure

PPE: Face shield and gloves.

- ☐ Verify supply of charged 9V batteries, EasyMini altimeters, and screw terminals.
- ☐ Wire altimeters to screw terminals and batteries.
- ☐ Configure altimeters for deployment at apogee and apogee + 1s for primary and backup, respectively.
- ☐ Verify altimeter configuration through pressure testing.
- ☐ Assemble **FRESH** 9v batteries into avionics housing with zip ties.

WARNING: Neglecting to complete previous steps may cause failed deployment and destruction of rocket and payload.

- ☐ Assemble altimeters and screw terminals into avionics housing with M3 screws and nuts.
- ☐ Measure and store calculated blast charge amounts for primary and backup deployment.

WARNING: Failure to store blast charge quantities will result in failed deployment.

Launch Day Procedure

PPE: Face shield and gloves.

- ☐ Arm altimeters and verify deployment settings with rocketry mentor.
- ☐ Disarm altimeters.
- ☐ Place blast charges into blast caps.

WARNING: Ensure static discharge by touching ground, else risk detonation of blast charge.

- ☐ Wire charges to altimeter.

WARNING: Ensure the altimeter is not powered and is not facing into wind, as doing so may cause inadvertent apogee detection.

- ☐ Verify connections of primary and backup wires to their respective blast charges.
- ☐ Assemble avionics bulk plates and housing with $\frac{3}{8}$ eye bolts and nuts.
- ☐ Ensure correct alignment of airframe for access to screw terminal holes.
- ☐ Place blast putty over remaining openings in the bulk plate.

Inter-Launch Procedure

PPE: Face shield and gloves.

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1

Avionics Checklist

- ☐ Disarm altimeters.

WARNING: Failure to complete this step risks inadvertent blast charge detonation.

- ☐ Verify deployment of both blast charges.
- ☐ Unscrew avionics bulk plate and remove assembly from rocket.
- ☐ Repeat Launch Day Procedure above

Post-Launch Procedure

- ☐ Repeat steps 1-3 of Inter-Launch Procedure
- ☐ Safely store all avionics equipment

WARNING: Failure to complete this step may result in damage or destruction of avionics equipment.

Signatures

Name	Signature
Santino Clemente	_____
Ian Morgan	_____
Zachary Stutsman	_____
Chris Dondanville	_____

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VADL Competition Flight 5/3/2025

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Payload Checklist

Required Materials

- ☐ Ground Station Electronics
 - ☐ Computer + Charger
 - ☐ USB-C adapter
 - ☐ SD Card reader
 - ☐ USB-MicroUSB connector
 - ☐ Diamond X-50A antenna
 - ☐ Stand for antenna
 - ☐ Attachment sleeve for antenna stand
 - ☐ USB-C to Ethernet Connector
 - ☐ Samsung Type C Thumb Drive with Linux installed
 - ☐ Pat's Laptop
 - ☐ Laptop Charger
 - ☐ Router, with charger
 - ☐ Ethernet cable
 - ☐ RTL-SDR
 - ☐ Hand-held radio x 2
 - ☐ Coaxial Cable for RTL-SDR
- ☐ Assembled Electronics Housing
 - ☐ SD card
 - ☐ VectorNav IMU
 - ☐ Battery (3S LiPO) x4
 - ☐ Raspberry Pi + PCB assembly
 - ☐ Coax cable with male SO239 connector
 - ☐ 3 way servo cable splitter
 - ☐ BigRedBee GPS module
- ☐ AEM Assembly
 - ☐ 18" tape spring x6
 - ☐ M3x10 bolts x6
 - ☐ M3 nuts x6
 - ☐ 8-32 bolts x6
 - ☐ 8-32 nuts x6
 - ☐ Jumper wires (w/ crimped
- ☐ Hardware
 - ☐ Aft aluminum bulk plate
 - ☐ PPLA+FG fore bulk plate
 - ☐ 5/18 smooth rods 24" long x2
 - ☐ 5/16 aluminum shaft collars x 20
 - ☐ 8-32 radial bolts x3 (fore bulk plate)
 - ☐ M2x16 screws x4 (PCB)
 - ☐ M3x14 screws + washers x5 (IMU + nose cone)
 - ☐ M3 nuts x2 (IMU)
 - ☐ Electrical tape
 - ☐ Multimeter
 - ☐ Zip ties
- ☐ LDM Assembly
 - ☐ Jumper wires w/ crimped ring connectors x4
 - ☐ Aluminum leg x4
 - ☐ Onyx bracket x4
 - ☐ 1/4-20 bolts x6
 - ☐ 1/4-20 capped nut x6
 - ☐ Torsional spring x6
 - ☐ LDM guide plate x2
- ☐ LSM Assembly
 - ☐ Southco latch x2
 - ☐ Aluminum latch plate
 - ☐ Parachute tube
 - ☐ Electronics bay
 - ☐ BigRedBee GPS module
 - ☐ VectorNav IMU
 - ☐ LSM LiPO battery
 - ☐ 1/4-20 threaded rods x2
 - ☐ 1/4-20 aluminum nuts x12
 - ☐ 6-32 screws x6
 - ☐ 6-32 nuts & lock washers x6

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2

Payload Checklist

- ☐ ring connectors) x3
- ☐ Female SO239 connector with crimped and soldered ring jumper
- ☐ AEM bulk plate x1
- ☐ 3x 9g servo
- ☐ Power
 - ☐ Jackery battery (wall voltage supply) x2
 - ☐ LiPo charger (including wall connector/adaptor and lipo cable) x 2
- ☐ 8-32 screws x6
- ☐ 8-32 nuts & lock washers x6
- ☐ Ballast (steel washers & nuts)
- ☐ PDM
 - ☐ Southco latch x1
 - ☐ PDM bulk plate x1
 - ☐ 8-32 bolts x4
 - ☐ 8-32 nuts x4
- ☐ Tools
 - ☐ 3mm phillips head screwdriver (Teensy/SD Mount)
 - ☐ 1.8mm allen key (Vector Nav)
 - ☐ 1.4mm allen key (Pi Mount)
 - ☐ 12mm wrench
 - ☐ 7/64 in allen wrench (shaft collars)
 - ☐ Driver set
 - ☐ Needle nose pliers
 - ☐ 7/16 wrench (LDM bolts)
 - ☐ 7/16 nut driver
 - ☐ Wire cutter
 - ☐ Wire crimper
 - ☐ Pins for Molex connectors
 - ☐ Replacement Molex connector

Required Personnel

Name	Role
Rais Nurhidajat	Vice President
Zachary Stutsman	Payload Engineer
Patrick Chen	Electronics Lead

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VADL Competition Flight 5/3/2025

2

Payload Checklist

Pre-Launch Day Procedure

- ☐ Validate LiPo Batteries and backup are charged to 12.65 V
- ☐ Verify functionality of Raspberry Pi and electronics
- ☐ Individually test AEM deployment
- ☐ Stake connectors (VN, Latches), verify software and electromechanical actuation
- ☐ Verify launch parameters
 - ☐ Launch Detection threshold
 - ☐ Timing for apogee + landing
 - ☐ Buffer size
- ☐ Unplug all electronics until morning
- ☐ Attach electronics housing to frame forward of AEMs
- ☐ Attach grounding wires to legs via hooks crimped to wires
- ☐ Verify all holes (radial and AEM slots) line up properly
- ☐ Verify electrical connection between APRS board and antennas

Launch Day Procedure

- ☐ Verify all battery charges with multimeter to 12.65 V
WARNING: Uncharged batteries will result in payload failure.
- ☐ Dry run of software initialization for launch
 - ☐ Adjust the ground-level parameter
 - ☐ Verify threshold parameters (multiple people)
 - ☐ Verify IMU connectivity (check quaternion changes)
 - ☐ Verify servo and latch actuation**WARNING: Software malfunction will lead to data corruption and payload failure.**
- ☐ Set up RTL-SDR decoding software
- ☐ Stake remaining connectors and wires
 - ☐ Latch connectors on board
 - ☐ SD card
 - ☐ Rx antennas
 - ☐ Zip tie long latch wires for stress removal
- ☐ Proceed with mechanical connections + relatching
- ☐ Confirm vector nav data stream with code
- ☐ Attach housing to skeleton and bulk plate
- ☐ Attach skeleton to payload bay and nose cone. Tighten top radial bolts
- ☐ Pull antennas into slots
- ☐ Connect Recovery Components
- ☐ Attach nose cone

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2

Payload Checklist

- ☐ Connect to main vehicle body
 - WARNING:** Failure to complete previous steps will result in payload section separation and destruction
- ☐ **Final Software Check**
 - ☐ Check I2C connectivity
 - ☐ Double-check threshold values
 - ☐ Arm software in the background
 - ☐ Check process running
 - ☐ Check I2C connectivity
 - ☐ Check RF not transmitting

Re-Launch Procedure

- ☐ Remove recovery components
- ☐ Remove nose cone
- ☐ **Recover flight data**
 - ☐ Stop the software process
 - ☐ Download flight data
 - ☐ Plot data using Jupyter Notebook script
 - ☐ Give permission to power off safely
 - ☐ Document special events during the test
- ☐ Disconnect battery
- ☐ Respool antennas
- ☐ Verify all fasteners and structure
- ☐ Change the battery (if needed, unlikely) and verify charge using a multimeter/onboard ADC
- ☐ **Power on electronics when ready for re-launch**
 - ☐ Adjust the ground-level parameter
 - ☐ Verify threshold parameters (multiple people)
 - ☐ Verify IMU connectivity (check quaternion changes)
 - ☐ Verify servo and latch actuation
- ☐ Clear APRS decoder output
- ☐ Attach skeleton to fore bulk plate
- ☐ Pull antennas into slots
- ☐ Connect recovery components
- ☐ Reattach nose cone
- ☐ Connect to main vehicle body

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2

Payload Checklist

WARNING: Failure to complete previous steps will result in payload failure or complete destruction

- ☐ **Final Software Check**
 - ☐ Check I2C connectivity
 - ☐ Double-check threshold values
 - ☐ Arm software in the background
 - ☐ Check process running
 - ☐ Check I2C connectivity
 - ☐ Check RF not transmitting

Signatures

Name	Signature
Rais Nurhidajat	_____
Zachary Stutsman	_____
Patrick Chen	_____

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Recovery Checklist

Required Materials

Drogue:

- ☐ 18" Classic Elliptical parachute
- ☐ Fore shock cord (30 ft)
- ☐ Aft shock cord (15 ft)
- ☐ Fireballs x2
- ☐ Small fire blanket

Main:

- ☐ 96" Iris Ultra parachute
- ☐ Fore shock cord (20 ft)
- ☐ Aft shock cord (10 ft)
- ☐ Fireballs x2
- ☐ Large fire blanket

General:

- ☐ Dog Barf recovery wadding
- ☐ Baby powder
- ☐ Hardware
 - ☐ 2-56 shear pins x16
 - ☐ 1200 lbs quick links x11
 - ☐ Zip Ties (plastic and metal)
 - ☐ Painters tape
 - ☐ Box cutter knife

Required Personnel

Name	Role
Ian Morgan	President & Recovery Lead
Zachary Stutsman	SSO
Chris Dondanville	Rocketry Mentor

Pre-Launch Day Procedure

- ☐ Examine the parachutes, shock cords, fireballs, fire blanket, quick links, and swivel links for any damage.
- ☐ Z-fold the shock cords and secure them with a single layer of painter's tape, ensuring not to crease or damage the tape.
- ☐ Practice assembling and storing recovery equipment in the recovery bays of the vehicle, ensuring sufficient packing volume.

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VADL Competition Flight 5/3/2025

3

Recovery Checklist

Launch Day Procedure

Main:

- ☐ Lay out and examine all recovery equipment for any damage.
- ☐ Attach quick links to outer ends of the shock cords.
- ☐ Proceed with parachute packing procedure:
 - ☐ Untangle shroud lines and separate into respective three groups
 - ☐ Unfurl parachute and lay out in its natural form, with center tucked in slightly to conform with toroidal shape.
 - ☐ Beginning with the labeled gore, fold each gore on the left side of the parachute toward the center.
 - ☐ Once complete, place weights atop folded half and repeat the process for the right side.
 - ☐ Fold halves lengthwise.
 - ☐ Lay shroud lines atop folded parachute in a snaking pattern.
 - ☐ Z fold the parachute from top to bottom and roll closed.
- ☐ Attach inner loops of the shock cords to the parachute quick link.
- ☐ Attach the fire blanket to the parachute quick link.
- ☐ Wrap the fire blanket around the parachute and tape closed until final assembly.
- ☐ Score the taped shock cords.
- ☐ Attach the 20 ft fore shock cord to the fore fireball aft quick link
- ☐ Attach the 10 ft aft shock cord to the aft fireball fore quick link
- ☐ **Ensure the avionics bay is fully assembled prior to continuing**
- ☐ Fasten the main bay to the aft avionics bulk plate
- ☐ Add dog barf to the main bay
 - WARNING:** Failure to complete this step may result in burning of the fireballs and potential zippering of the airframe
- ☐ **Ensure the tail section is fully assembled prior to continuing**
- ☐ Add dog barf to the tail section
 - WARNING:** Failure to complete this step may result in burning of the fireballs and potential zippering of the airframe
- ☐ Attach the fore fireball fore quick links to the aft avionics bulk plate eye bolts

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3

Recovery Checklist

- ☐ Coat the fire blanket with baby powder.
WARNING: Failure to complete previous steps will result in parachute deployment malfunction and subsequent destruction of vehicle and payload.
- ☐ Attach the aft fire ball aft quick link to the tail bulk plate eye bolt
- ☐ **Remove red tape from the fire blanket.**
- ☐ Load shock cords, parachute, and fireballs in the drogue recovery bay, ensuring that the parachute quick link is facing away from the blast (toward the fore section).
WARNING: quick link placed towards aft section may cause parachute tangling during deployment and prevent full unfurling.
- ☐ Adjust fireball location on shock cords such that they reside within the airframe and zip tie in place.
- ☐ Assemble vehicle sections together and fasten with shear pins.
WARNING: Failure to place fire balls may result in zippering. Failure to place shear pins will result in premature deployment of main parachute.

Drogue:

- ☐ Lay out and examine all recovery equipment for any damage.
- ☐ Attach quick links to outer ends of the shock cords.
- ☐ Proceed with parachute packing procedure:
 - ☐ Untangle shroud lines and separate into respective three groups
 - ☐ Unfurl parachute and lay out in its natural form, with center tucked in slightly to conform with toroidal shape.
 - ☐ Beginning with the labeled gore, fold each gore on the left side of the parachute toward the center.
 - ☐ Once complete, place weights atop folded half and repeat the process for the right side.
 - ☐ Fold halves lengthwise.
 - ☐ Lay shroud lines atop folded parachute in a snaking pattern.
 - ☐ Z fold the parachute from top to bottom and roll closed.
- ☐ Attach inner loops of the shock cords to the parachute quick link.
- ☐ Attach the fire blanket to the parachute quick link.
- ☐ Wrap the fire blanket around the parachute and tape closed until final assembly.

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Recovery Checklist

- ☐ Score the taped shock cords.
- ☐ Attach the 30 ft fore shock cord to the fore fireball aft quick link
- ☐ Attach the 15 ft aft shock cord to the aft fireball fore quick link
- ☐ **Ensure the avionics bay is fully assembled prior to continuing**
- ☐ Fasten the drogue bay to the fore avionics bulk plate
- ☐ Add dog barf to the drogue bay
 - WARNING:** Failure to complete this step may result in burning of the fireballs and potential zippering of the airframe
- ☐ **Ensure the payload and LSM bay are fully assembled prior to continuing**
- ☐ Add bog barf to the LSM bay
 - WARNING:** Failure to complete this step may result in burning of the fireballs and potential zippering of the airframe
- ☐ Attach the aft fire ball aft quick links to the fore avionics bulk plate eye bolts
- ☐ Coat the fire blanket with baby powder.
 - WARNING:** Failure to complete previous steps will result in parachute deployment malfunction and subsequent destruction of vehicle and payload.
- ☐ Attach the fore fire ball fore quick links to the LSM bulk plate eye bolts
- ☐ **Remove red tape from the fire blanket.**
- ☐ Load shock cords, parachute, and fireballs in the drogue recovery bay, ensuring that the parachute quick link is facing away from the blast (toward the fore section).
 - WARNING:** quick link placed towards aft section may cause parachute tangling during deployment and prevent full unfurling.
- ☐ Adjust fireball location on shock cords such that they reside within the airframe and zip tie in place.
- ☐ Assemble vehicle sections together and fasten with shear pins.

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3

Recovery Checklist

Signatures

Name	Signature
Ian Morgan	
Zachary Stutsman	
Chris Dondanville	

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Motor Checklist

Required Materials

- | | |
|---|--|
| ☐ Cesaroni L1720 reload kit x2 | ☐ Ignition system |
| ☐ Liner tube | ☐ Lithium Grease |
| ☐ Propellant grains | ☐ L1720 motor hardware x2 |
| ☐ Spacer O-rings | ☐ Motor Casing |
| ☐ Smoke Grain | ☐ Forward and Aft Closure |
| ☐ PPE | ☐ O-rings |
| ☐ Latex gloves | ☐ Igniter x2 |
| ☐ Safety glasses/face shield | ☐ Backup Retaining Ring Cap |

Required Personnel

Name	Role
Ian Morgan	President & Vehicle Engineer
Zachary Stutsman	SSO
Chris Dondanville	Rocketry Mentor

Pre-Launch Day Procedure

- ☐ Inspect motor hardware and reload kit for defects.
WARNING: Defects in motor hardware will cause rocket flight irregularities and vehicle damage. Cracked grains can result in CATO
- ☐ Store reload kit in MAGLoc for transport.

Launch Day Procedure

Reference attached L1720 motor assembly instructions for procedure. Motor assembly should be overseen by the Rocketry Mentor, the SSO, and one other team member to

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Motor Checklist

ensure that all procedures are carefully followed. These three members are designated above, and their approval of motor assembly quality is located below.

Ignition charge is not to be inserted until the rocket is on launch rail and all non-essential personnel are cleared. Rocketry Mentor is the ONLY person permitted to insert ignition charge and wire to the ignition system. Refer to the Igniter Installation Checklist and the L1720 motor assembly instructions for more details.

WARNING: Failure to clear area or handling of charge by anyone except for Rocketry Mentor may result in severe injury or death, and/or damage or destruction of the vehicle.

Signatures

Name	Signature
Ian Morgan	_____
Zachary Stutsman	_____
Chris Dondanville	_____

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General Vehicle Checklist

Required Materials

- | | |
|---|---|
| ☐ Nose cone | ☐ Hardware |
| ☐ Payload bay | ☐ 16x 10-32 phillips head screws |
| ☐ LSM bay | ☐ 16x 10-32 neoprene washer |
| ☐ Drogue bay | ☐ 20x 2-56 nylon shear pins |
| ☐ Avionics bay | ☐ 6 x 4-40 shear pins |
| ☐ Main bay | ☐ Tools |
| ☐ Tail section | ☐ Small hammer |
| ☐ Recovery equipment | ☐ Wrench |
| ☐ Cesaroni L1720 motor (x2) | ☐ Phillips head screwdriver |
| ☐ Pro-75 motor hardware (x2) | ☐ Needle nose pliers |
| ☐ Hanging scale | ☐ Flat head screwdriver |
| ☐ Silver and black sharpies | ☐ Mallet |
| ☐ Boat tail | ☐ Gloves |

Required Personnel

Name	Role
Ian Morgan	President & Vehicle Engineer
Zachary Stutsman	SSO
Saksham Sharma	Vehicle Engineer
Rais Nurhidajat	Payload Engineer

Pre-Launch Day Procedure

- ☐ Fully assemble vehicle and payload as closely to launch day conditions as possible.
- ☐ Find the dry mass of the vehicle via hanging scale.
- ☐ Find and mark CG of vehicle without ballast.
- ☐ Determine optimal ballast and place ballast at appropriate location.
- ☐ Find and mark CG of vehicle with ballast.

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General Vehicle Checklist

- ☐ Ensure rail button alignment by mounting to launch rail.

Dry mass (ballasted)	CG	CP	Dry SSM

Wet mass (ballasted)	CG	CP	Wet SSM

Ensure SSM is between 2 and 3 for both wet and dry configurations.

Launch Day Procedure

- ☐ Attach boat tail to tail section via shear pins
- ☐ Secure tail bulk plate to tail section via 10-32 bolts
- ☐ Secure aft avionics bulk plate to main bay via 10-32 bolts
- ☐ Secure fore avionics bulk plate to drogue bay via 10-32 bolts
- ☐ Ensure payload is fully assembled with nose cone
- ☐ Secure LSM to payload via EM latches
- ☐ Secure LSM bulk plate to LSM bay via 10-32 bolts
- ☐ Secure main shock cords to corresponding bulk plates via quick links
- ☐ Couple tail section and main bay
- ☐ Insert four 2-56 shear pins into shear pin holes with the small hammer
- ☐ Couple LSM bay and main bay
- ☐ Insert four 2-56 shear pins into shear pin holes with the small hammer
- ☐ ***To be completed only when ready to move to launch rail.***
 - ☐ Insert fully assembled motor into motor tube.
 - ☐ **Find wet CG and confirm adequate SSM.**

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General Vehicle Checklist

WARNING: Failure to complete any previous steps will result in improper vehicle separation and destruction of vehicle

Signatures

Name	Signature
Ian Morgan	
Zachary Stutsman	
Saksham Sharma	
Rais Nurhidajat	

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VADL Competition Flight 5/3/2025

6

Launch Rail Checklist

Required Materials

- | | |
|---|--|
| ☐ Launch rail base | ☐ Tools |
| ☐ Launch rail extension | ☐ Wrench |
| ☐ WD-40 | ☐ Hex key |
| ☐ Sand Bags | ☐ Anemometer |
| ☐ Launch Rail Hardware Box (stakes, bolts) | ☐ Level |
| ☐ Blast deflection plate | ☐ Socket wrench |
| | ☐ Crescent Wrench |
| | ☐ Mallet |
| | ☐ Angle Finder |

Required Personnel

Name	Role
Zachary Stutsman	SSO
Saksham Sharma	Vehicle Engineer
Samantha Civil	Vehicle Engineer
Sitan Huang	Simulations Lead
Chris Dondanville	Rocketry Mentor

Launch Day Procedure

- ☐ A level area must be chosen that is approved by the RSO onsite and meets all NAR and FAA regulations for high powered rocketry
Warning: Failure to follow risks damage to rocket or injury to spectators.
- ☐ Unfold launch rail legs and turn the apparatus such that it is facing into the wind. Wind speed should be determined using the anemometer.
- ☐ Ensure launch rail base is level within 2 degrees using level.
Warning: Failure to follow risks improper rocket launch conditions and potentially unstable flight
- ☐ Secure legs using stakes or sand bags.

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Launch Rail Checklist

- ☐ Slide launch rail extension onto main launch rail, ensuring that all bolt heads are guided into the slot of the X-frame. Ensure that the rail surface is flush, then tighten all bolts to secure rail.

Warning: Non-flush rail surface risks improper launch rail exit, and damage to personnel and/or rocket equipment.

- ☐ Adjust the launch rail stop so that it is a minimum of 18 inches from the ground. Ensure that the stop is tightened.
- ☐ Place blast plate on the ground beneath launch rail stop. Hold in place with a sandbag or with clamps attached to launch frame legs
- ☐ Apply WD-40 to launch rail.
- ☐ Inspect the assembled launch rail, double check tightness of all bolts, and ensure that rail is sufficiently lubricated with WD-40.
- ☐ Decide on launch rail angle based on wind conditions. Launch rail angle must be approved by the Rocketry Mentor, Chris Dondaville, the Student Safety Officer, Zachary Stutsman, and Simulations Engineer, Sitan Huang. The launch rail angle is checked using the angle finder.

***Note: The RSO will be deferred to on the matter of launch rail angle if applicable

Warning: Improper angle and launch direction may cause excessive windcocking and increase drift.



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Launch Rail Checklist

Signatures

Selected Rail Angle:	
Chris Dondanville	
Zachary Stutsman	
Sitan Huang	
Name	Signature
Zachary Stutsman	
Saksham Sharma	
Samantha Civil	



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Launch Procedure

Required Materials

- ☐ Fully assembled vehicle
- ☐ Assembled launch rail
- ☐ Ignition system

Required Personnel

Name	Role
Ian Morgan	President
Zachary Stutsman	Student Safety Officer
Santino Clemente	Avionics Engineer
Chris Dondanville	Rocketry Mentor

Pre-Launch Day Procedure

- ☐ Launch requirements from **Full-Scale Safety Checklist** are to be committed to memory by the predesignated launcher and SSO.

Launch Day Procedure

- ☐ Confirm the vehicle is fully assembled and ready for launch.
WARNING: Improper vehicle assembly risks unstable flight and/or loss of vehicle.
- ☐ Carry vehicle to launch rail location.
- ☐ Ensure that launch rail has been properly assembled and lubricated
- ☐ Lower launch rail as needed and slide the vehicle on via rail buttons.
- ☐ Raise launch rail to predetermined launch rail angle.
WARNING: Incorrect launch rail angle risks excessive windcocking, increased total drift, and/or unsafe landing location
- ☐ Ensure a minimum ground clearance of 18 inches from the bottom of the vehicle.

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Launch Procedure

PPE: Facial covering, gloves

- ☐ Arm altimeters.
 - ☐ Confirm altimeters are armed and configured by listening to series of beeps (battery voltage in decivolts, followed by “dit dah dah dit”).
 - ☐ Confirm continuity between avionics and e-matches by listening to subsequent series of beeps (“dit dah dah dit”).
 - ☐ Clear all non-essential personnel to a minimum distance of 300 ft from the launch rail.
 - ☐ Listen for continuous altimeter beeps to confirm launch has not been prematurely detected.
 - ☐ Install igniter as specified in the Igniter Installation Checklist
Must be done by team Rocketry Mentor, not team members
 - ☐ Clear all personnel from the launch rail to at least the minimum distance specified in the NAR High Powered Code. Generally, a safe area will be specified by the RSO
 - ☐ Perform launch countdown procedure:
 - ☐ Confirm all personnel are clear of the launch pad within a minimum radius of 300 ft.
 - ☐ Confirm airspace is clear.
 - ☐ Confirm all other launch minimums are met.
 - ☐ Launcher yells “CLEAR... 5... 4... 3... 2... 1...”.
 - ☐ Launch vehicle.
- WARNING: Improper clearance of area and launch callout may injure personnel and/or risk mission success.**
- ☐ In the case of failure to launch:
 - ☐ Attempt launch countdown procedure again as necessary.
 - ☐ Vehicle may not launch if any criteria are not met, per the **Safety Checklist**.
 - ☐ If launch fails again, the launcher will be disconnected from power and the team will wait at least 60 seconds and gain RSO approval before approaching the rocket



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Launch Procedure

Signatures

Name	Signature
Ian Morgan	
Zachary Stutsman	
Chris Dondanville	



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Igniter Installation Checklist

Required Materials

- | | |
|---|---|
| ☐ Igniter from motor kit | ☐ Wooden dowel |
| ☐ Painters tape | ☐ Plastic nozzle cap |

Required Personnel

Name	Role
Zachary Stutsman	Student Safety Officer
Chris Dondanville	Rocketry Mentor

Pre-Launch Day Procedure

- ☐ Ensure that igniter is present in the motor reload kit
- ☐ Ensure that Pro75 Motor instruction sheets are packed

Launch Day Procedure

WARNING: Do not proceed until the vehicle is on the launch rail and altimeters are armed. All personnel should be clear except for the rocketry mentor and SSO. **Ignition charge installation must be performed by the rocketry mentor.**

- ☐ Double check that the vehicle is properly seated on the launch rail
- ☐ Listen to ensure that the altimeters are armed (beep sequence will indicate arming)
- ☐ Scan the area to ensure that it is clear of all personnel except for the SSO and team rocketry mentor.

WARNING: Personnel too close to the launch rail risk injury in the event of accidental ignition

- ☐ Fasten the supplied igniter to a wooden dowel using painters tape

WARNING: Use only the igniter supplied by the manufacturer. Using other igniters could result in misfires or CATO.

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Igniter Installation Checklist

- ☐ Insert igniter fully into the motor, ensuring that it is in contact with the forward closure.
- ☐ Securely retain the igniter to the motor nozzle either with tape or with the provided plastic cap, routing the wires through one of the vent holes.
WARNING: Lack of vent holes risks premature igniter ejection and incomplete motor burn.
- ☐ Fasten the wires securely to the remote launch box lead wires. Place the wires on the launch platform clear of the blast area to keep them off of the ground and organized.
- ☐ Clear the launch pad and return to the safe viewing area designated by the RSO.

Signatures

Name	Signature
Zachary Stutsman	_____
Chris Dondanville	_____



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Safety Checklist

Required Materials

- ☐ Safety Binder
- ☐ Appropriate PPE (specific items listed in subsystem checklists)
- ☐ Fire extinguisher
- ☐ First aid kits (x3)
- ☐ Water and food
- ☐ Appropriate clothing

WARNING: Missing items jeopardizes team and vehicle safety.

Identification of Safety Team

Name	Role
Ian Morgan	President
Zack Stutsman	Student Safety Officer
Chris Dondanville	Rocketry Mentor

Launch Criteria

If not all criteria are met, launch must be delayed or cancelled.

Airspace

- ☐ Flight must comply with any and all applicable Federal Aviation Authority regulations.
- ☐ Airspace must be clear of all traffic.
- ☐ The predicted apogee must be below the flight ceiling for the launch

WARNING: Unclear airspace will result in hazards to rocket and/or air traffic.

Environment

- ☐ No precipitation may be falling at the time of launch.

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Safety Checklist

- ☐ Ground level wind must be below 20 mph.
WARNING: Launching in improper conditions may affect mission performance, personnel safety and vehicle health.
- ☐ All personnel must be at a distance of at least 300 ft from the launch rail at the time of launch, per NAR regulations.
WARNING: Improper clearance of personnel beyond listed minimum safety distance may result in personnel injury and vehicle damage.

Payload

- ☐ Antenna Extension Mechanism (AEM) must be wound within the outer confines of the airframe prior to launch.
- ☐ Payload must be securely attached to the LSM. This means that both EM latches are closed and locked.
WARNING: Failure to properly attach payload may result in premature deployment and mission failure.

Vehicle Readiness

- ☐ The vehicle must be deemed fit to fly by rocketry mentor Chris Dondaville.
- ☐ The recovery system and motor must likewise be deemed suitable for the launch vehicle and appropriately retained.
- ☐ Student Safety Officer and team president must also approve all subsystems and safety measures prior to launch.

The safety team and/or rocket mentor may cancel or postpone the flight for any reason, regardless of its presence on the checklist above.

Delay Procedure

- ☐ If the launch criteria have not been met, but it is expected that they may be met within three hours (maximum launch rail time), the launch may be delayed until the criteria are met.
 - ☐ If the delay-causing criteria has now been met, all criteria must be reevaluated before launch attempt.

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Safety Checklist

- ☐ If not, then refer to cancellation procedures.

WARNING: Failure to meet launch criteria may jeopardize mission success and/or launch spectator safety.

Cancellation Procedures

PPE: Face shield, gloves

- ☐ Disconnect ignition
- ☐ Disarm avionics and other electronics
- ☐ Remove vehicle from the launch rail
- ☐ Remove motor

WARNING: Failure to follow previous steps may result in accidental discharge of ignition or activation of deployment mechanisms, damaging vehicle and personnel.

- ☐ Complete post flight procedures

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Safety Checklist

Flight on:

By signing below, I acknowledge that the launch procedure was properly followed, and to the best of my knowledge, the vehicle is safe to fly.

Name

Signature

Ian Morgan

Zachary Stutsman

Chris Dondanville

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Post-Flight Checklist

Required Personnel

Name	Role
Santino Clemente	Avionics Engineer
Ian Morgan	President
Zachary Stutsman	Student Safety Officer
Chris Dondanville	Rocketry Mentor

Procedure

PPE: Face shield, gloves, close-toed shoes.

- ☐ Disarm altimeters by unscrewing screw terminals.

WARNING: Failure to complete this step risks inadvertent blast charge detonation.

- ☐ Verify deployment of both blast charges.

WARNING: Some or all blast charges may not have detonated. Therefore, no personnel can enter a 100 ft radius of the vehicle within 60 seconds of landing.

- ☐ If any blast charges have not detonated, remove carefully while wearing the PPE listed above.
- ☐ Properly dispose of black powder charges (refer to MSDS)
WARNING: ANY improper handling risks personnel and vehicle damage.
- ☐ Recover all sections of rocket from landing site and return to tent.
- ☐ Disconnect parachute from rocket.
- ☐ Unscrew avionics bulk plate and remove assembly from rocket.
- ☐ Inspect lander and vehicle for structural damage.
- ☐ Record apogee data from altimeters.
- ☐ Safely store all equipment for transportation.

WARNING: Failure to properly store equipment may result in structural or electronic damage.

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Post-Flight Checklist

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Signatures

Name	Signature
Santino Clemente	_____
Ian Morgan	_____
Zachary Stutsman	_____
Chris Dondanville	_____

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Launch Day Responsibilities

Ian

Primary:

- Oversee all launch prep activities
- Assemble vehicle

Secondary:

- Recovery assembly
- LSM assembly
- Oversee avionics and motor loading

Zack

Primary:

- Payload prep and assembly
- LSM assembly

Secondary

- Oversee avionics and motor loading
- Vehicle assembly

Rais

Primary:

- Payload prep and assembly

Secondary:

- Electronics prep
- Vehicle assembly

Saksham

Primary:

- Launch rail assembly
- Recovery assembly

Secondary:

- Vehicle assembly

Sitan

Primary:

- Record conditions and parameters for sims

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Launch Day Responsibilities

Secondary:

- Assist with data processing as needed

Santino

Primary:

- Test and assemble avionics

Secondary:

- Vehicle assembly
- Flight video recording

Sam

Primary:

- Recovery assembly
- Launch rail assembly

Secondary:

- Flight video recording

Patrick

Primary:

- Payload electronics

Secondary:

- Assist with GPS/LSM electronics as needed

Adam

Primary:

- GPS/LSM electronics

Secondary:

- Assist with payload electronics as needed

Jack

Primary:

- Keep track of all tools that are needed
- Photograph assembly process

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Launch Day Responsibilities

Secondary:

- Flight video recording

Ashley

Primary:

- Assist Patrick and Adam with electronics needs
- Photograph assembly process

Secondary:

- Flight video recording



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Troubleshooting

Pre-Launch Day

- ☐ All team members are responsible for keeping detailed notes on their subsystems
- ☐ Payload, electrical, and vehicle components will be assembled and tested prior to launch day to ensure familiarity with the full system

Launch Day Procedure

If a problem arises, it will be calmly approached from a systems engineering perspective. The steps are as follows:

- ☐ Consult the system lead for the offending subsystem
- ☐ Quickly develop a list of possible causes of the failure/issue
- ☐ Assess the probability of each of these causes
- ☐ Determine which causes can be dealt with given the resources at hand. Time constraints related to the launch day conditions should be considered here.

WARNING: Some fixes may risk damaging other components of the system. No action should be taken that is likely to fix one issue at the expense of creating others. This is likely to lead to total system failure rather than partial system failure.

- ☐ Once a root cause has been identified, take action with the appropriate steps to fix if they are feasible in the field.
- ☐ If a problem arises on the launch pad, the motor igniter is to be removed first if it has been installed. Then the altimeters are to be disarmed. After these two steps, the launch rail can be lowered and the rocket removed. This process will be cleared by the RSO before being done.

WARNING: If a critical issue cannot be fixed given the resources at hand, then launch is to be aborted.

7 Project Plan

7.1 Testing

VADL ensures design and fabrication meets NASA and team imposed standards through a rigorous testing campaign. It is not enough for the launch vehicle and payload to merely perform the Student Launch mission once, but it must be capable of re-flight within 2 hours of recovery in accordance with NASA Requirements 2.3 and 2.5. VADL continuously tests all significant changes made to the launch vehicle and payload to ensure compliance with all applicable NASA and VADL imposed requirements.

7.1.1 Required Testing

The team has defined measurable and strict testing goals to ensure each test serves to verify NASA and team-imposed requirements. Certain tests are only conducted as required due to their destructive nature. Such tests include airframe structural load tests, where specimens are loaded to failure. Tests that are partially destructive in nature but validate key parameters for mission success are repeatedly done. For example, the lander contains breakable shoes that deform and or snap on impact to lower the landing load and help stabilize the landing configuration. These parts are treated as replaceable items and their destruction does not impede repeated testing.

Tests are broadly grouped into vehicle and payload tests, with additional subcategories for further classification. Each test has a clear and specific objective, with an actionable plan for verification. The tested variable and pertinent requirement(s) are outlined, and a definitive success criteria is described. The status of each test is then listed; as of the FRR milestone, all tests are either complete. Testing outlined in the following section is defined as any tests needed to validate components, mechanisms, or subsystems for full-scale testings. The following testing does not include the Hexacopter drop tests or the launch vehicle flights. The results of full-scale testing for the payload are described in detail in Sections 4.1 and 5.4. The results from the rocket launches are in Section 5. The vehicle and payload testing tables are provided on the following pages.

Launch Vehicle		Objective	Verification Plan	Variable Tested	Pertinent Technical Requirement(s)	Success Criteria	Status
Test Code	Verification Method						
1.1 Structural Testing							
1.1.1	Drop Test	Verify that the airframe can withstand any forces sustained during a landing impact	The team will drop the each rocket section from a height that simulates the terminal velocity of the rocket body after main parachute deployment; the team will also add masses to each rocket section to reach a representative landing energy	Strength of G-12 fiberglass	NASA 2.3 VADL 2.2.1.3, 2.2.1.4	The airframe does not sustain any damage in any of the simulated impacts	Complete
1.1.2	Load Test	Determine the critical hoop stress of the fiberglass for a geometry that is representative of the rocket body	The team will place a representative geometry of the rocket body into the load frame and apply compressive stress until it fails	Critical hoop stress of fiberglass	NASA 2.3, VADL 2.2.2	The critical hoop stress is greater than the maximum loads expected to be sustained by the rocket body throughout the mission sequence by a safety factor of 2.5	Complete
1.1.3	Load Test	Determine the critical bolt tear out stress of the fiberglass for a geometry that is representative of the rocket body	The team will place a section of the rocket body into the load frame and apply tensile stress to two bulkplates attached to airframe with bolts until failure	Critical bolt bearing stress of fiberglass	NASA 2.3, VADL 2.2.2	The critical tear out stress is greater than the maximum expected load sustained by the rocket body throughout the mission sequence by a safety factor of 2.5.	Complete
1.1.4	Load Test	Determine the modulus of elasticity of fiberglass airframe material	The team will place a section of the rocket body into the load frame and apply axial compressive loading until failure.	Modulus of elasticity of fiberglass	NASA 2.3, VADL 2.2.2	Ensure that the loads on the vehicle during flight are within the elastic range of fiberglass deformation, and ensure the material is fit for use as the vehicle airframe.	Complete
1.1.5	Impact Test	Ensure the fiberglass material can withstand the expected impact loads during landing.	The team will place a section of the rocket body into the drop tower and drop a weight of 25 lbm from a height of 36 in to simulate a 75 lb-ft impact	Impact strength of fiberglass	NASA 2.3	Ensure the vehicle section can withstand a 75 lb-ft impact without permanent deformation.	Complete
1.1.6	Load Test	Determine the failure load of recovery swivel-link hardware	The team will place a quick-link and swivel-link into the load frame and apply tensile loading until failure.	Ultimate tensile strength of swivel-link	NASA 2.3, VADL 2.2.2	Ensure that the failure load on the quick-link and swivel-link are at least 2.5 times greater than the maximum expected force during recovery deployment.	Complete
1.2 Recovery Testing							
1.2.1	Battery Test	Confirm that the battery life for recovery electronics is sufficient to remain on the launch pad for three hours before launch	Team members will connect the recovery electronics to their batteries and leave them on for three hours	Battery Life	NASA 2.6.3.5 VADL 3.12	The recovery electronics still functioning after being turned on for three hours	Complete
1.2.2	Altimeter Test	Verify that the altimeter can detect apogee and trigger the parachute blast charges	The team will place the altimeter with its blast charge igniters in a vacuum chamber. The vacuum will lower the pressure in the chamber until the pressure corresponding to projected apogee is reached. Team members will listen for the sound of the ignition trigger which would deploy the drogue parachute. NOTE: There will be no gun powder present during this test.	Proper altitude detection	NASA 3.1.2	The team hears the ignition trigger at the correct pressure readings corresponding to parachute deployment	Complete
1.2.3	Deployment Test	Verify that the calculated sizes of the blast charges can separate the launch vehicle for parachute deployments	The team will detonate the blast charges inside the rocket body and see if they effectively separate the rocket sections. This test will be performed under the supervision of the SSO and the team mentor to ensure safety	Blast charge mass	NASA 3.2	The rocket body sections are separated after detonation	Complete
1.2.4	Radio Interference Test	Verify that RF interference will not have an adverse effect on the avionics electronics.	A team member shall place the avionics within a vacuum chamber alongside a RF transmitter constantly transmitting and compare the altimeter data to a simulated launch without a RF transmitter inside.	Effect of RF interference on avionics	NASA 3.13, VADL 3.5	The altimeter performs nominally with no change to the deployment timings or altimeter readings.	In-Progress

PAYLOAD		Objective	Verification Plan	Variable Tested	Pertinent Technical Requirement(s)	Success Criteria	Status
Test Code	Verification Method						
2.1 Lander Separation Mechanism							
2.1.1	Command Response Time Test	Ensure that the LSM disengages from the rocket within 1 second of command	Perform ground-based timed tests, activating the LSM through command and measuring the separation time	Separation time	VADL 4.2.1	The LSM disengages within 1 second of the command with no significant delay	Complete
2.1.2	RF Transmission Test	Verify that the LSM is capable of receiving RF transmissions	Team members shall perform RF transmission tests to ensure that the LSM successfully receives commands without interference from surrounding materials	Signal strength and reliability	VADL 4.2.2	The LSM reliably receives RF transmissions without any significant signal interference	Complete
2.1.3	Altitude Operation Test	Ensure that the LSM functions reliably at altitudes between 200-400 ft AGL	A team member shall raise the LSM to an altitude between 200-400 ft AGL, and verify the separation event occurs as expected	Altitude at activation	VADL 4.2.3	The LSM successfully disengages during the specified altitude range in multiple tests	Complete
2.1.4	Vibration and Shock Load Test	Ensure the LSM withstands vibration and shock loads experienced during rocket ascent and separation	A tem member shall connect the LSM to a shock cord and drop off the side of a stairwell to simulate the jerk of parachute opening forces.	Vibration and shock resistance	VADL 4.2.4	The LSM remains fully operational and intact after exposure to vibration and shock loads simulating rocket flight	Complete
2.2 Leg Deployment Mechanism							
2.2.1	Buckling Load Test	Ensure that the legs do not buckle due to the force caused by impact in case there is a direct impact on a leg where buckling can occur	Using a load frame, team members will apply a load upon the vertical axis of the leg to determine the critical buckling load	Critical Buckling Load, Strength of aluminum legs	VADL 4.3.1, VADL 2.2.2	The critical buckling load is above the expected load the legs are expected to experience by a safety factor of at least 2.5	Complete
2.2.2	Dynamic Loading/ Drop Tests	Ensure that the legs do not break under dynamic loading similar to that experienced during landing. This includes moments and multidirectional forces	The primary verification for this will be from multiple drop tests. The initial drop tests will be from 4 ft AGL, without parachute, to simulate the forces exerted during an actual landing. These will include a quiescent drop from 80 ft AGL, as well as angled drops and drops with initial horizontal velocity. This will prove out both the strength of the legs and the righting ability of the legs. Final step is drop from drone at 300 ft AGL under parachute to precisely emulate rocket deployment conditions.	Righting ability, strength of leg/linkage system, strength of spring/damper necessary	VADL 4.3.1, VADL 2.2.2	The legs can survive a drop, in multiple orientations, with landing speeds that are equal or greater than those expected during competition. The legs should also enable the lander to land upright under these various conditions.	Complete
2.2.3	LDM Deployment Test	Confirm that the legs will successfully deploy upon release from rocket body	Test leg opening ability in lab conditions and under simulated drop conditions	Hardware necessary for deployment (springs needed or is gravity sufficient?)	VADL 4.3.2	Legs are able to deploy and lock under forces greater than what is expected during rocket flight.	Complete
2.3 Parachute Detachment Mechanism							
2.3.1	Release Time Test	Verify that parachute detachment occurs in the proper time frame, not too early such that the landing energy increases drastically nor too late such that the parachute destabilizes the payload	The team will run PDM in the lab as if it were during flight and make sure it functions without IMU input within a proper time frame; PDM will run based on input from microcontroller timer	PDM run time/proper microcontroller algorithm	VADL 4.4.1	PDM functions within 10s of input from microcontroller, so the parachute detaches in a timely manner	Complete
2.3.2	FEA	Confirm that the PDM latch survives parachute opening forces.	The team will create a finite element analysis model of the load the mechanism will have to withstand due to the main parachute	PDM structural integrity	VADL 4.4.2	PDM can endure the load from the parachute with a safety factor of 2.5 applied	Complete
2.3.3	PDM Landing Detection/Deployment Test	Ensure that PDM safely detects landing prior to its deployment in order to prevent unacceptable landing energies	The team will place the IMU with its connections to a microcontroller and PDM inside a pressure chamber and decrease and increase the pressure inside to simulate the mission sequence. (This IMU contains a barometer.) The team then will watch to see that PDM only deploys once the pressure chamber increases back to ground pressure	IMU Performance, Microcontroller to PDM communication	VADL 4.4.1	The team observes that PDM only deploys at the proper pressure and does so within 1 second	Complete
2.3.4	PDM Locking Test	Verify that accelerations or vibrations will not cause PDM to prematurely deploy	A team member shall attach the PDM to a mass simulating the payload mass, attach the system to a shock cord, and drop the system off a 10 ft height. This will simulate the forces experienced during payload parachute opening	Robustness of PDM against accelerations and vibrations	VADL 4.4.2	PDM does not prematurely deploy under any accelerations or vibrations	Complete
2.4 Antenna Extension Mechanism							
2.4.1	Extension test	Verify that the AEM is able to fully extend to 18 in from within the airframe	A team shall repeated wind up and deploy the AEM to ensure that it does not bind internally. Antenna must also extend through the airframe without getting stuck.	Tape extension, servo unlocking	VADL 4.5.1	The AEM successfully deploys to full length 20 out of 20 times repeatedly	Complete

PAYLOAD		Objective	Verification Plan	Variable Tested	Pertinent Technical Requirement(s)	Success Criteria	Status
Test Code	Verification Method						
2.4.2	Locking Mechanism Test	Verify that the cylinder lock will not open unless actuated	A team member shall repeatedly drop the AEM and see if the locking mechanism remains locked	Locking mechanism robustness against vibrations and impact	VADL 4.5.4	The locking mechanism withstands 20 drops from a height of 4 feet AGL.	Complete
2.5 RF Transmission System							
2.5.1	Reliability Test	Ensure RF system transmits without issue for long periods of time	Send out APRS continuously for 1 hour.	APRS reliable operation duration	VADL 4.6.1	RF system transmits for at least 1 hour	Complete
2.5.2	Distance Test	Verify Transmission over long range	Send APRS over 1 mile distance with varying environmental conditions	RF system effective range	VAD: 4.6.2	RF system should be error-free when there is ground clearance, and should also function with minimal degradation in more challenging environment	Complete
2.5.3	Transmission with damaged antenna	Verify Transmission when antenna fails to deploy, or is damaged during flight	Have different sets of antenna, some optimal, some broken in different ways, and transmit signals and measure signal strength.	RF system failure resilience	VADL 4.6.2	RF system should be able to broadcast in the range half of the optimal range	Complete
2.6 Data Collection Sensors							
2.6.1	Static Bench Testing	Verify that sensors samples data at 160 Hz. 100 Hz is used for software to reduce load.	Place IMU on flat, stable surface and run for > 3 hrs	Accelerometer, gyroscope, altimeters calibration	VADL 4.7.1	Check whether accelerometers read ~9.8 m/s ² on the Z-axis, gyro reads zero, and altimeter reads stable value.	Complete
2.6.2	Drop Test and Rotation	Verify how IMU responds to changes in motion, orientation, and free fall	Drop the IMU from a known height to simulate rapid acceleration	Accelerometer	VADL 4.7.7	Ensure accelerometer detects proper acceleration changes	Complete
2.6.3	Altitude and Pressure Test	validate barometer / altimeter under varying pressure conditions to mimic altitude changes	place the sensor at different height and potentially pressure chamber	Barometer	VADL 4.7.4	Report correct altitude reading as expected	Complete
2.6.4	Apogee Detection	Confirm sensor can properly detect apogee	Test for rapid pressure drop and stabilization	Apogee	VADL 4.7.5	Report correct apogee as expected	Complete
2.6.5	Drop testing	Test complete system to ensure all sensors and detection events trigger as expected	Perform drop test using the entire payload	Landing and jettison detection	VADL 4.7.2, 4.7.3	System detects key flight events accurately	Complete
2.7 Payload Structure							
2.7.1	Load Test	Verify that the shaft collars can withstand expected loads during flight, recovery, and landing	A team member shall perform a load test on a shaft and shaft collar to determine the force required to move the collar.	Shaft collar strength	NASA 2.3, VADL 2.2.2	Shaft collars withstand a force at least 2.5 times greater than the maximum expected force	Complete
2.7.2	Drop Test	Verify the structural integrity of the internal scaffolding of the payload section by itself without the integrated payload airframe	The team will drop the internal skeleton without the outer airframe shell from a height of 5 feet repeatedly	Structural integrity of payload skeleton	NASA 2.3	Visual and physical inspection determines that the structural integrity was not compromised and no component came loose or out of place	Complete

7.1.2 Completed Testing

The team has developed a methodical approach to testing documentation. For each test the objective, test variables, success criteria, and responsible team members are outlined. The procedure is then described, and key safety considerations are outlined. Photos are included to visualize the experimental setup. Upon completion of the test, key takeaways and design implications are described. The following sheets outline the testing completed thus far. As the design evolves and full-scale vehicle fabrication begins, additional testing will be drafted and done as required to ensure no aspects of vehicle launch, recovery, or landing are overlooked. VADL remains vigilant in ensuring that the NASA Student Launch product is rigorously tested and analyzed for all possible failure modes.

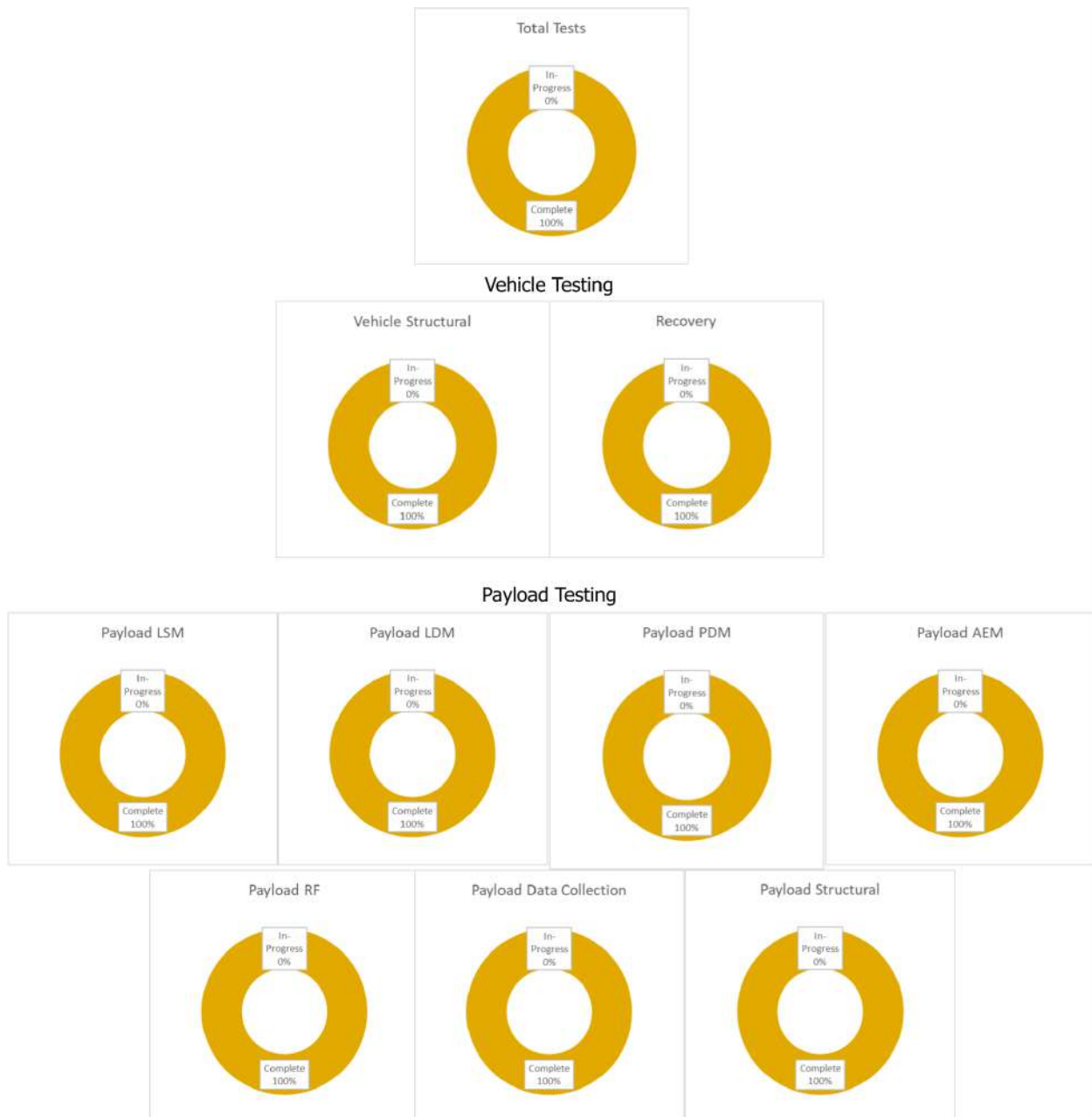


Figure 121: Status of testing as of FRR cycle. Key tests shall be repeated leading up to the PDF launch and Student Launch Competition itself.



VADL Testing Protocol

Airframe Drop Test (COMPLETED)

1.1.1

Overview:

Objective / Purpose	Verify that the airframe can withstand any forces sustained during a landing impact.
Testing Variable(s)	Strength of G-12 fiberglass.
Technical Requirement(s)	NASA 2.3, VADL 2.2.1.3, 2.2.1.4
Success Criteria	Airframe sustains no damage in any simulated impacts
Responsible Team Members	Saksham Sharma, Ian Morgan

Methodology:

Procedure

1. Load airframe to maximum section mass using ballast
2. Raise the section to a height corresponding to the maximum landing velocity for the non-jettison case
3. Drop the airframe onto a earthen surface
4. Inspect the airframe for visible damage such as hairline fracture and holes

Key Safety Considerations

- Wear safety glasses to protect against fracturing debris
- Mind sharp edges on fiberglass airframe

Experimental Setup Photos



Results

Key Takeaways

- The airframe sustained no visible damage during the simulated drops

Design Implications

The airframe material is approved for rocket flight and will not fail on impact during nominal flight. Throughout all further drop testing and rocket flight the fiberglass airframe has not failed.



VADL Testing Protocol

Airframe Hoop Load Test (COMPLETED)

1.1.2

Overview:

Objective / Purpose	To determine the structural integrity of the fiberglass airframe in hoop loading to inform vehicle design decisions and ensure adequate safety factor in flight.
Testing Variable(s)	Critical hoop stress of fiberglass airframe
Technical Requirement(s)	NASA 2.3, VADL 2.2.2
Success Criteria	Airframe returns to normal shape after a critical hoop stress at least 2.5 times greater than the maximum hoop stress expected during vehicle flight and landing.
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

Methodology:

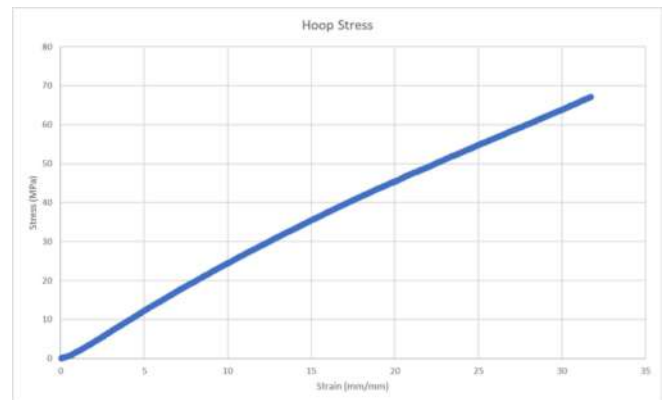
Procedure

5. Position airframe section beneath load transfer plate and base of load frame
6. Place 2 weights on either side of airframe to prevent sliding during load test
7. Close fracture shield and
8. Don safety glasses
9. Manually preload airframe section to 100 lbf
10. Initiate automated loading sequence
11. Load test is complete upon airframe section failure (cracking, delamination) abort load test, or wait for automatic cutoff once limit switches are reached
12. Reset machine and compile data to excel file

Key Safety Considerations

- Wear safety glasses and ensure the fracture shield is in front of the load frame to prevent injury due to flying debris.
- Keep clear of the load frame during operation and tie back loose hair to prevent pinch or crush injuries.
- One person must be in close proximity to the emergency stop switch at all times

Experimental Setup Photos



Results

Key Takeaways:

- Load test automatically cutoff at 22.2 kN (67.2 MPa)
- Maximum expected hoop load during recovery deployment and landing is 1.3 MPa
- Safety factor of 51 is much greater than the desired 2.5.

Design Implications:

It is unlikely that the airframe will fail in hoop loading. Even so, this will be mitigated through the use of tight-fitting bulk plates and internal skeleton.



VADL Testing Protocol

Airframe Bolt Tear Out Load Test (COMPLETED)

1.1.3

Overview:

Objective / Purpose	To determine the structural integrity of the fiberglass airframe for bolt tear out and inform vehicle design decisions and ensure adequate safety factor in flight.
Testing Variable(s)	Critical bolt tear out stress of fiberglass airframe
Technical Requirement(s)	NASA 2.3, VADL 2.2.2
Success Criteria	Critical bolt tear out stress must be 2.5 times greater than the expected maximum bolt tear out stress expected during vehicle flight and landing for the avionics bay.
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

Methodology:

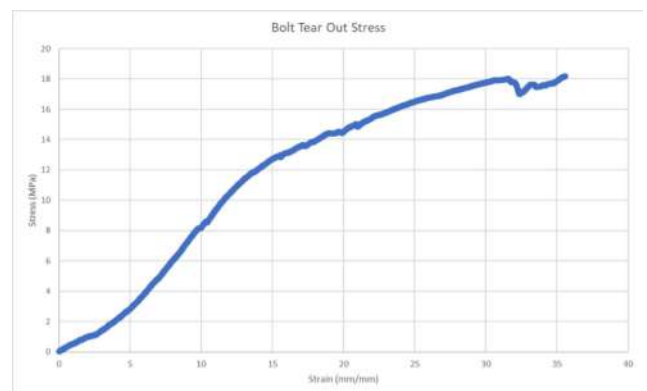
Procedure

1. Position airframe section between the 2 load frame hooks and ensure quick links are secure
2. Close fracture shield and don safety glasses
3. Manually preload airframe section to 100 lbf
4. Initiate automated loading sequence
5. Critical bolt tear out is when bolt tear out of 5mm occurs
6. Load test is complete upon bolt tear out of 1 inch or load frame limit switches are tripped
7. Reset machine and compile data to excel file

Key Safety Considerations

- Wear safety glasses and ensure the fracture shield is in front of the load frame to prevent injury due to flying debris.
- Keep clear of the load frame during operation and tie back loose hair to prevent pinch or crush injuries.
- One person must be in close proximity to the emergency stop switch at all times

Experimental Setup Photos



Results

Key Takeaways:

- Critical bolt tear out stress of 8 MPa
- Calculated maximum bolt tear out stress of 1.8 MPa, which provides a safety factor of 4.4.
- Bolts deformed severely at the end of the load test once the material was taken to complete failure (1 inch of bolt tear out)

Design Implications:

Both bolt tear out and bolt deformation will be mitigated by properly securing the bulk plates to the internal skeleton.



VADL Testing Protocol

Airframe Axial Compression Test (COMPLETED)

1.1.4

Overview:

Objective / Purpose	To determine the structural integrity of the fiberglass airframe in axial compression and inform vehicle design decisions and ensure adequate safety factor in flight.
Testing Variable(s)	Critical axial compressive stress
Technical Requirement(s)	NASA 2.3, VADL 2.2.2
Success Criteria	Critical axial compressive stress is at least 2.5 times greater than the maximum expected compressive stress experienced during vehicle flight and landing.
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

Methodology:

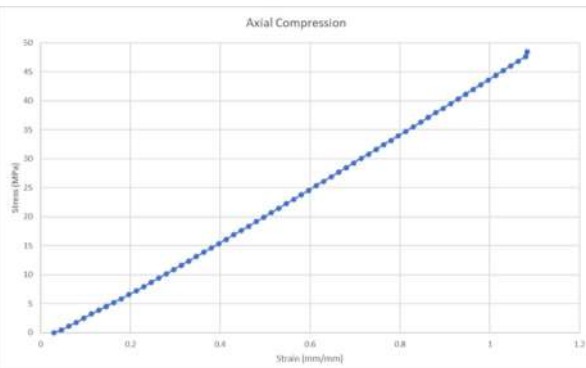
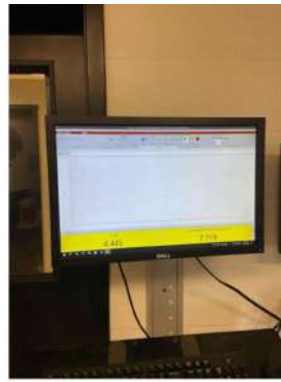
Procedure

1. Position airframe section between load transfer plate and center of load frame bed
2. Close fracture shield and don safety glasses
3. Manually preload airframe section to 100 lbf
4. Initiate automated loading sequence
5. Load test is complete upon airframe section failure (cracking, buckling) or load frame exceeds safety cutoff
6. Reset machine and compile data to excel file

Key Safety Considerations

- Wear safety glasses and ensure the fracture shield is in front of the load frame to prevent injury due to flying debris.
- Keep clear of the load frame during operation and tie back loose hair to prevent pinch or crush injuries.
- One person must be in close proximity to the emergency stop switch at all times

Experimental Setup Photos



Results

Key Takeaways

- Load frame automatically cutoff at 22.6 kN (49.5 MPa)
- Maximum expected compressive stress is 1.89 MPa
- Safety factor of 26 is much greater than the desired 2.5

Design Implications:

It is unlikely that the airframe shall fail in compressive loading during launch. To increase structural stability the motor retention rings also serve to distribute the load at the base of the rocket



VADL Testing Protocol

Airframe Impact Test (COMPLETED)

1.1.5

Overview:

Objective / Purpose	Ensure the fiberglass material can withstand the expected impact loads during landing.
Testing Variable(s)	Impact strength of fiberglass
Technical Requirement(s)	NASA 2.3
Success Criteria	Ensure the vehicle section can withstand a 75 lb-ft impact without permanent deformation.
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

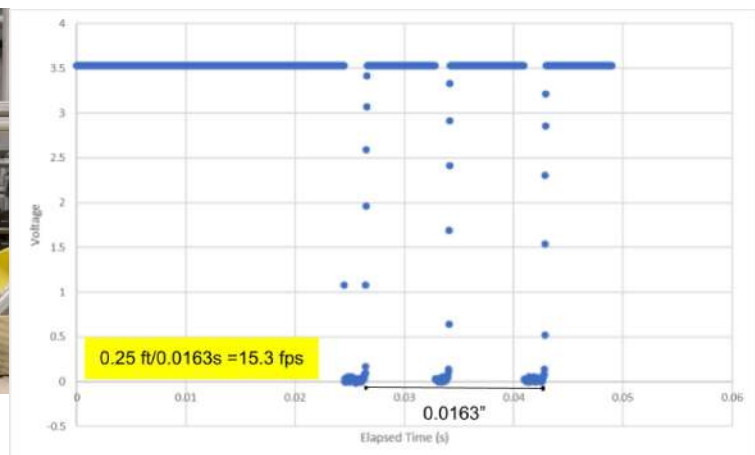
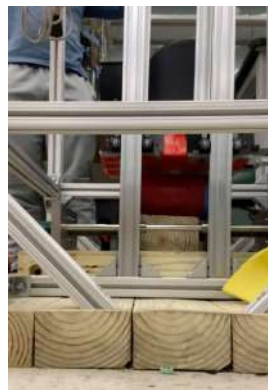
Methodology:

Procedure

1. Place the airframe section at the bottom of the drop tower and secure it in place.
2. Place weights on the drop plate and secure in place.
3. Raise drop plate to desired height.
4. Ensure that all personnel and appendages are clear of the drop zone and release the drop plate.
5. Compile data to Excel file.

Key Safety Considerations

- Wear safety glasses to prevent injury due to flying debris.
- Keep all appendages clear of the drop tower during operation to prevent crush injuries.



Experimental Setup Photos

Results

Key Takeaways

- Voltage vs. time data confirms the velocity and subsequent L.K.E. of 75 lb-ft was achieved
- Airframe withstood impact with no structural damage incurred

Design Implications:

It is unlikely that fiberglass will experience structural damage upon landing. As such, fiberglass is verified to be a suitable airframe material selection for the full-scale flight.



VADL Testing Protocol

Swivel Link Load Test (COMPLETED)

1.1.6

Overview:

Objective / Purpose	Determine the failure load of recovery swivel-link hardware
Testing Variable(s)	Ultimate tensile strength of swivel-link
Technical Requirement(s)	NASA 2.3, VADL 2.2.2
Success Criteria	Ensure that the failure load on the swivel-link is at least 2.5 times greater than the maximum expected force during recovery deployment.
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

Methodology:

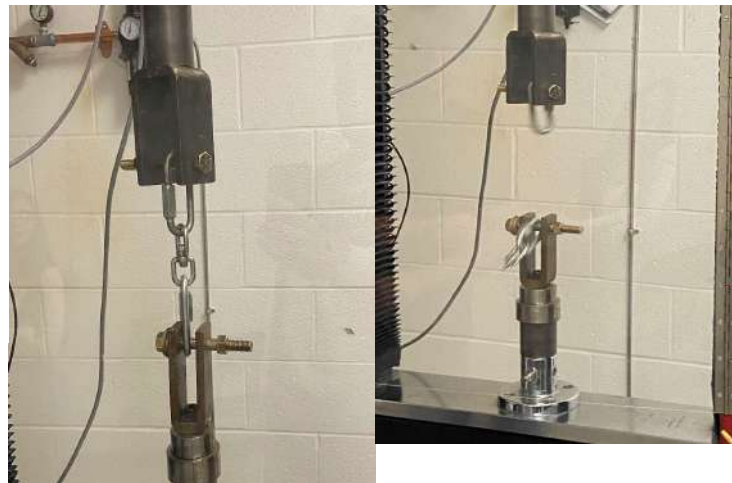
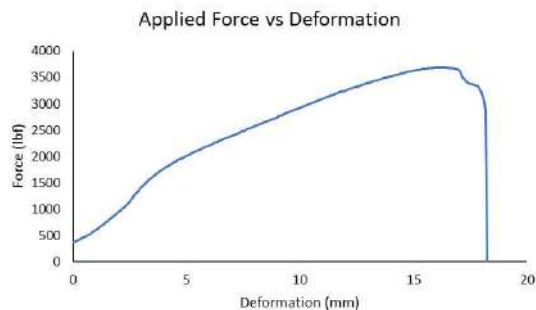
Procedure

1. Place link between load frame hooks
2. Close fracture shield and don safety glasses
1. Manually preload link to 100 lbf
2. Initiate automated loading sequence
3. Load test is complete upon link section failure (cracking, stretching) or load frame exceeds safety cutoff
4. Reset machine and compile data to excel file

Key Safety Considerations

- Wear safety glasses and ensure the fracture shield is in front of the load frame to prevent injury due to flying debris.
- Keep clear of the load frame during operation and tie back loose hair to prevent pinch or crush injuries.
- One person must be in close proximity to the emergency stop switch at all times

Experimental Setup Photos



Results

Key Takeaways

- Swivel-Link can withstand 3692 lbf before failing
- The expected loading on the link is 1204 lbf, yielding a safety factor of 3.06

Design Implications

The safety factor provided by the swivel-link is 3.06, which is well above the VADL minimum of 2.5 for recovery components, not including the shock cords. These swivel-links will be kept for full scale flight. The swivel-links are the weakest metal component in the recovery system.



VADL Testing Protocol

Recovery Avionics Battery Test (COMPLETED)

1.2.1

Overview:

Objective / Purpose	Confirm that the battery life for recovery electronics is sufficient to remain on the launch pad for three hours before launch
Testing Variable(s)	Avionics battery life
Technical Requirement(s)	NASA 2.6.3.5, VADL 3.12
Success Criteria	Avionics still functional after being turned on for over 3 hours
Responsible Team Members	Saksham Sharma, Santino Clemente

Methodology:

Procedure

1. Ensure battery voltage is above 9V
2. Connect battery to altimeter and arm
3. Every 10 minutes, validate altimeter is still live by listening to altimeter buzzer
4. Test is complete after 3 hours have elapsed

Key Safety Considerations

- Ensure battery polarity is correct before arming altimeter to prevent electrical fires or damage to altimeter electronics

Experimental Setup Photos



Results

Key Takeaways:

- 9V battery lasted well after 3 hours, and only died after 12 hours

Design Implications:

A single 9V battery is sufficient to power one of the avionics boards throughout the rocket sitting on the pad and through launch.



VADL Testing Protocol

Recovery Avionics Altimeter Test (COMPLETED)

1.2.2

Overview:

Objective / Purpose	Verify that the altimeter can detect apogee and trigger the parachute blast charges
Testing Variable(s)	Proper altitude detection
Technical Requirements(s)	NASA 3.1.2
Success Criteria	The team hears the ignition trigger at the correct pressure readings corresponding to parachute deployment
Responsible Team Members	Saksham Sharma, Santino Clemente

Methodology:

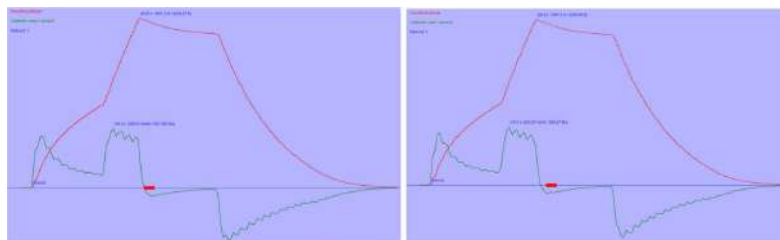
Procedure

1. Set altimeters to deploy at apogee with backup at apogee +1 second
2. Don safety glasses
3. Verify battery terminal orientation on altimeter, and connect 9V battery, and e-match
4. Arm altimeter using screw switch
5. Don safety glasses
6. Turn shock tube electronics on and open vacuum chamberPlace altimeters within the vacuum chamber and secure the clamps
7. Lower the pressure within the shock tube to correspond to 5000 ft altitude
8. Open the release valve and allow the pressure to slowly return to ground pressure within the chamber, listening for e-match ignition
9. Connect to altimeter user interface and verify deployments occurred at apogee

Key Safety Considerations

- Ensure the altimeter is properly armed and setup to prevent premature e-match ignition
- Wear safety glasses to protect eyes in the event of a premature e-match ignition

Experimental Setup Photos



Main altimeter shown on left, backup shown on right. The red line represents when deployment occurred

Results

Key Takeaways

- Altimeter results were within 15 feet, with detections matching to the tenth of a second. Deployments occurred at apogee and apogee +1 second.

Design Implications

Altimeter requires sufficiently sized port holes in order to function properly. Altimeters have slightly different calibrations, which necessitates the use of a backup altimeter to ensure deployments occur properly.



VADL Testing Protocol

Recovery Deployment Test - Subscale (COMPLETE)

1.2.3

Overview:

Objective / Purpose	Verify that the calculated sizes of the blast charges can separate the launch vehicle for parachute deployments
Testing Variable(s)	Blast charge mass
Technical Requirements(s)	NASA 3.2
Success Criteria	The rocket body sections are separated after detonation, the parachute is completely outside of the bay
Responsible Team Members	Saksham Sharma, Santino Clemente

Methodology:

Procedure

1. Don safety glasses
2. Attach recovery hardware to eyebolts with the fire blanket and fire retardant inside the parachute bay
3. Secure the rocket together with nylon shear pins
4. Attach rocket to x-frame atop tarp and foam to prevent damage to the surroundings and rocket
5. Wire e-matches to deployment box.
6. Trigger e-match and observe deployment. Verify recovery hardware is completely separated from the rocket itself

Key Safety Considerations

- Verify terminal polarity to prevent electrical fires or sparks that may cause premature detonation of the blast charge
- Wear safety glasses to shield eyes from debris
- Stand well clear of the test setup and away from the fore section of the rocket
- Keep a fire extinguisher on standby in the event the deployment causes a fire

Experimental Setup Photos



Results

Key Takeaways

- Blast charge calculations verified
- Recovery hardware properly deployed, parachute experienced no burns

Design Implications

Sufficient fire retardant is required to prevent parachute burn damage. Bulk plates must be properly designed to prevent blast charge gasses from leaking through to the avionics or other section



VADL Testing Protocol

Recovery Deployment Test - Full scale (COMPLETE)

1.2.3

Overview:

Objective / Purpose	Verify that the calculated sizes of the blast charges can separate the launch vehicle for parachute deployments
Testing Variable(s)	Blast charge mass
Technical Requirement(s)	NASA 3.2
Success Criteria	The rocket body sections are separated after detonation, the parachute is completely outside of the bay
Responsible Team Members	Saksham Sharma, Santino Clemente

Methodology:

Procedure

1. Don safety glasses
2. Attach recovery hardware to eyebolts with the fire blanket and fire retardant inside the parachute bay
3. Secure the rocket together with nylon shear pins
4. Attach rocket to x-frame atop tarp and foam to prevent damage to the surroundings and rocket
5. Wire e-matches to deployment box.
6. Trigger e-match and observe deployment.
7. Verify recovery hardware is completely separated from the rocket itself

Key Safety Considerations

- Verify terminal polarity to prevent electrical fires or sparks that may cause premature detonation of the blast charge
- Wear safety glasses to shield eyes from debris
- Stand well clear of the test setup and away from the fore section of the rocket
- Keep a fire extinguisher on standby in the event the deployment causes a fire

Experimental Setup Photos



Results

Key Takeaways

- Blast charge calculations verified
- Recovery hardware properly deployed, parachute experienced no burns

Design Implications

Sufficient fire retardant is required to prevent parachute burn damage. Bulk plates must be properly designed to prevent blast charge gasses from leaking through to the avionics or other section



VADL Testing Protocol

Radio Interference Test - Full scale (COMPLETE)

1.2.4

Overview:

Objective / Purpose	Verify that RF interference will not have an adverse effect on the avionics electronics.
Testing Variable(s)	Effect of RF interference on avionics
Technical Requirement(s)	NASA 3-13, VADL 3-5
Success Criteria	The altimeter performs nominally with no change to the deployment timings or altimeter readings.
Responsible Team Members	Santino Clemente, Ian Morgan

Methodology:

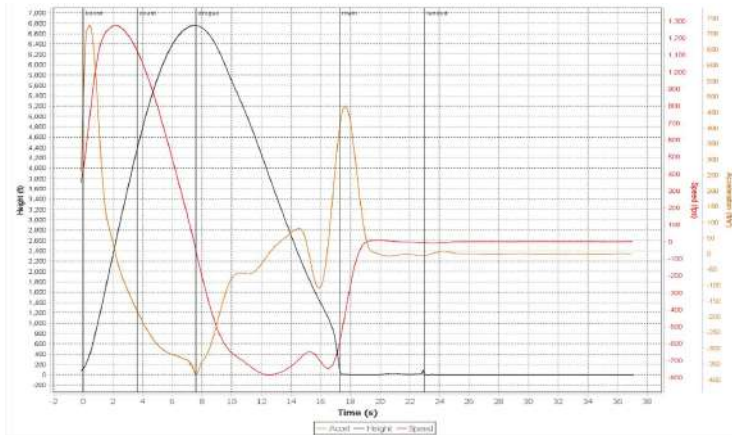
Procedure

1. Set up a Baofeng radio with the PTT (Push-To-Talk) button held down using a rubber band to ensure continuous transmission.
2. Place the radio and avionics bay inside a pressure cooker to create a controlled environment for RF exposure.
3. Simulate launch conditions by initiating altimeters and creating vacuum to simulate rocket flight.
4. After the test, retrieve and analyze the altimeter data to check for any anomalies, missing data, or interference effects.

Key Safety Considerations

1. Wear safety glasses to protect eyes around pressure cooker

Experimental Setup Photos



Results

Key Takeaways

- RF interference from mission-critical communication devices poses no risk to altimeter performance. The simulated launch data matched previous normal recordings, confirming that continuous RF exposure does not disrupt altimeter functionality.

Design Implications

RF interference from mission-critical communication devices (e.g., Baofeng radios) poses no risk to altimeter performance. No modifications or shielding are required for altimeter placement within the vehicle.



VADL Testing Protocol

LSM Command Response Test (COMPLETE)

2.1.1

Overview:

Objective / Purpose	Ensure that the LSM disengages from the rocket within 1 second of command.
Testing Variable(s)	Separation time
Technical Requirement(s)	VADL 4.2.1
Success Criteria	The LSM disengages within 1 second of the command with no significant delay
Responsible Team Members	Rais Nurhidajat, Santino Clemente

Methodology:

Procedure

1. Prepare LSM system according to setup instructions.
2. Connect LSM to a simulated payload to test release mechanism functionality.
3. Arm the system and perform pre-test checks.
4. Send command signal to initiate LSM separation.
5. Measure time from command to LSM disengagement using a high-speed camera or timer.
6. Repeat the test three times to ensure consistency.

Key Safety Considerations

- Verify electrical connections to prevent unintended activation of the LSM.

Experimental Setup Photos



Results

Key Takeaways

- The LSM successfully disengaged within 1 second of command in all three trials, meeting the success criteria.
- The separation mechanism operated without delay or malfunction, confirming system reliability.

Design Implications

No immediate changes required for LSM design. System is confirmed to be reliable and effective for mission deployment.



VADL Testing Protocol

LSM RF Transmission Test (COMPLETE)

2.1.2

Overview:

Objective / Purpose	Verify RF transmission to LSM electronics from long distances
Testing Variable(s)	RF signal strength and communication reliability over distance.
Technical Requirement(s)	VADL 4.2.2
Success Criteria	Successful transmission and response from the LSM electronics at a distance of approximately 1400 feet without signal loss or interference.
Responsible Team Members	Rais Nurhidajat, Santino Clemente, Patrick Chen

Methodology:

Procedure

1. Prepare LSM electronics with RF receiver powered and armed for testing.
2. Ensure all electrical components are safely connected and insulated.
3. Position the RF transmitter while the receiver is stationed approx. 1400 ft away.
4. Initiate RF transmissions using pre-configured command signals.
5. Monitor LSM receiver for response to each transmission and verify successful commands.
6. Record any delays, signal degradation, or failed transmissions.
7. Repeat tests multiple times to ensure consistency and reliability.

Key Safety Considerations

- Verify all electronic connections to prevent RF interference with other devices in the area.
- Maintain clear communication between testing locations for safety and efficiency.

Experimental Setup Photos



Results

Key Takeaways

- RF transmissions successfully reached the LSM receiver at 1400 feet with no signal loss or delays.

Design Implications

The RF system in the LSM is capable of maintaining communication within the required range for mission operations.



VADL Testing Protocol

LSM Altitude Operation Test (COMPLETE)

2.1.3

Overview:

Objective / Purpose	Verify the LSM operates effectively at deployment altitudes between 200 and 400 feet.
Testing Variable(s)	Deployment altitude.
Technical Requirement(s)	VADL 4.2.3
Success Criteria	The LSM successfully disengages at the specified altitude range with consistent performance and no malfunction.
Responsible Team Members	Rais Nurhidajat, Santino Clemente, Patrick Chen

Methodology:

Procedure

1. Attach the LSM to the Aurelia drone.
2. Perform pre-flight safety checks on the drone and LSM, ensuring secure attachment and proper functionality.
3. Fly the drone to the specified altitude between 300 and 400 feet.
4. Issue the release command to the LSM while maintaining altitude.
5. Observe and record the LSM's response time and deployment behavior.
6. Repeat the process at varying altitudes within the range to ensure consistent performance.

Key Safety Considerations

- Ensure the drop area is clear of personnel and obstructions.
- Verify secure attachment of the LSM to the drone to prevent accidental releases.
- Follow all FAA and drone operation guidelines for altitude and area restrictions.

Experimental Setup Photos



Results

Key Takeaways

- The LSM successfully disengaged upon command at altitudes ranging from 200 to 400 feet in all trials.
- No delays or malfunctions were observed during deployment, and the mechanism demonstrated consistent reliability.

Design Implications

The LSM is confirmed to be functional and reliable at mission-required altitude



VADL Testing Protocol

LSM Vibration and Shock Load Test (Complete)

2.1.4

Overview:

Objective / Purpose	Ensure the LSM withstands vibration and shock loads experienced during rocket ascent and separation
Testing Variable(s)	Vibration and shock resistance
Technical Requirement(s)	VADL 4.2.4
Success Criteria	The LSM remains fully operational and intact after exposure to vibration and shock loads simulating rocket flight
Responsible Team Members	Rais Nurhidajat, Santino Clemente

Methodology:

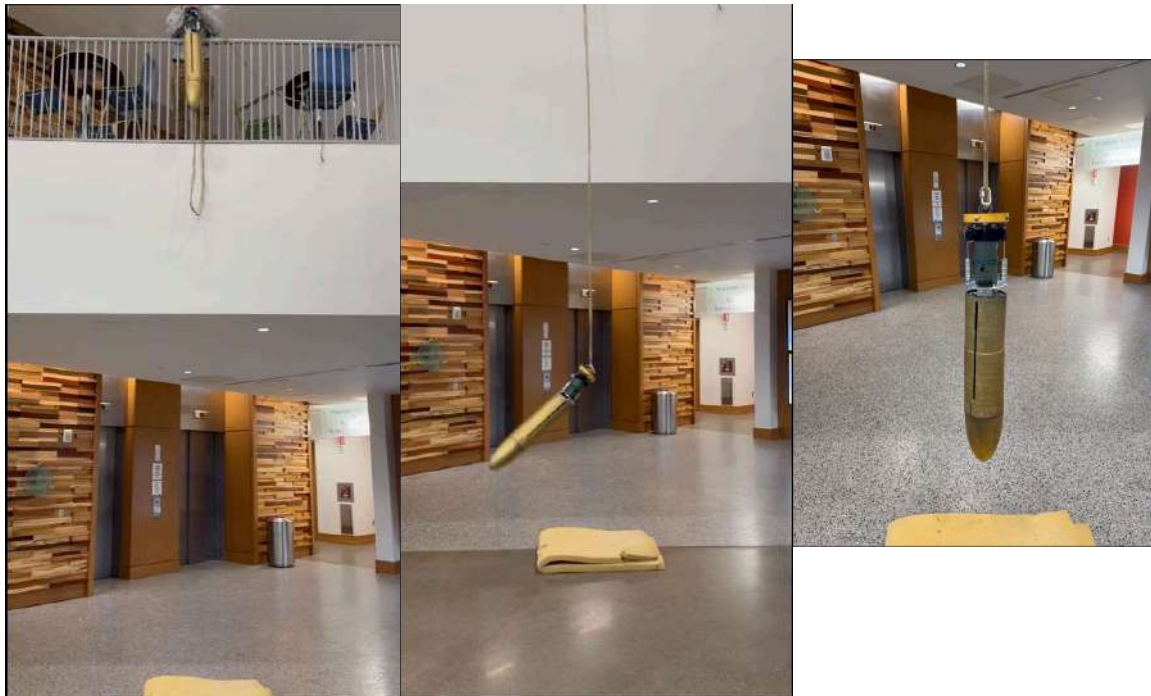
Procedure

1. Attach 10 ft shock cord to stairwell railing by looping one end of the cord through itself
2. Attach the shock cord to the LSM and payload assembly
3. Drop the assembly off the railing and observe results

Key Safety Considerations

1. Mind surroundings and ensure base of stairwell is clear of foot traffic
2. Wear eye protection to protect eyes from fracturing debris

Experimental Setup Photos



Results

Key Takeaways

- LSM successfully retained payload and sustained no damage to any components
- Both latches remained locked throughout the entire duration of the experiment

Design Implications

The LSM design performed satisfactorily and is ready for integration into the launch vehicle.



VADL Testing Protocol

LDM Dynamic Loading/ Drop Tests (Complete)

2.2.2

Overview:

Objective / Purpose	Ensure LDM can withstand landing forces without breaking
Testing Variable(s)	Loading capacity of leg bulk plate, leg bracket, and leg
Technical Requirement(s)	VADL 4.3.1, VADL 2.2.2
Success Criteria	No components break or deform past functionality
Responsible Team Members	Zachary Stutsman, Rais Nurhidajat

Methodology:

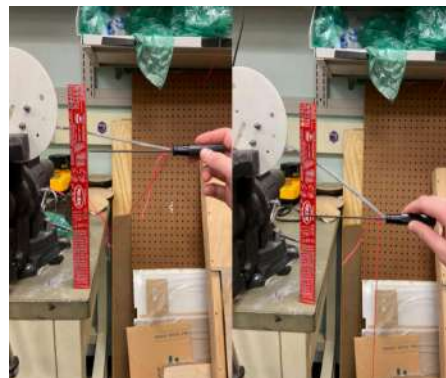
Procedure

1. Assemble one leg and bracket onto leg bulk plate
2. Clamp bulk plate into table vise so that leg is parallel to ground
3. Run string through hole at end of leg
4. Attach 1 lb weight to the end of the string and gently lower. Increase weight until 8 lb is reached
5. Once 8 lb static loading is reached, proceed to dynamic testing
6. With leg still clamped and 8 lb of mass still attached, raise weight 6 in from neutral leg position and drop
7. Continue dropping greater distances until a drop of 27 in is completed

Key Safety Considerations

- Safety glasses and closed toed shoes will be worn at all times during testing
- Team members will stand clear of dropping weights
- Incremental increases are instrumental in avoiding unexpected failures

Experimental Setup Photos



Results

Key Takeaways

- Leg and bracket withstood static testing with bending but no permanent deformation
- Dynamic testing at maximum drop height created some deformation but did not impact functioning

Design Implications

Leg bracket and bulk plate experienced no breakage or permanent deformation. Leg experienced bending without deformation during static loading and some deformation during the 27 in drop. This deformation could be undone relatively easily and was not great enough to prevent the proper functioning of the LDM.

Further testing after CDR has indicated that deformation is desired during landing in order to dissipate some of the landing energy. Based on this testing, the 2024 legs are too stiff to deform substantially under the forces experienced during landing. This guided the decision of the team to move away from 2024 aluminum for the legs and instead use 6063. All other leg components remain the same, lending confidence in the leg brackets and bulk plates to withstand landing forces.



VADL Testing Protocol

LDM Deployment Test (Complete)

2.2.3

Overview:

Objective / Purpose	Prove that legs can be deployed from airframe upon lander separation
Testing Variable(s)	Ability of legs to deploy
Technical Requirement(s)	VADL 4.3.2
Success Criteria	Consistent deployment
Responsible Team Members	Zachary Stutsman

Methodology:

Procedure

1. Fold legs of constructed payload
2. Place payload into airframe
3. Slide airframe into rocket coupler section
4. Release payload so that it can fall freely. Make sure to hold shock cord when dropping
5. Visually check to confirm that all four legs fully deploy
6. Repeat until convinced deployment is consistent

Key Safety Considerations

- Wear closed toe shoes and safety glasses
- Team members who are not dropping payload will stand clear of drop area

Experimental Setup Photos



LDM deployment re-verified in Full-scale drop test from 300 ft AGL

Results

Key Takeaways

- Legs can successfully deploy from within airframe
- Deployment is consistent and happens immediately upon separation and shock cord jerk

Design Implications

The current, torsional spring driven deployment of the legs is sufficient for successful deployment during flight. This conclusion is supported by field payload testing.



VADL Testing Protocol

PDM Release Time Test (COMPLETE)

2.3.1

Overview:

Objective / Purpose	Verify that parachute detachment occurs in the proper time frame, not too early such that the landing energy increases drastically nor too late such that the parachute destabilizes the payload
Testing Variable(s)	PDM run time/proper microcontroller algorithm
Technical Requirement(s)	VADL 4.4.1
Success Criteria	PDM functions within 10s of input from microcontroller, so the parachute detaches in a timely manner
Responsible Team Members	Rais Nurhidajat, Patrick Chen, Timothy Shaheen

Methodology:

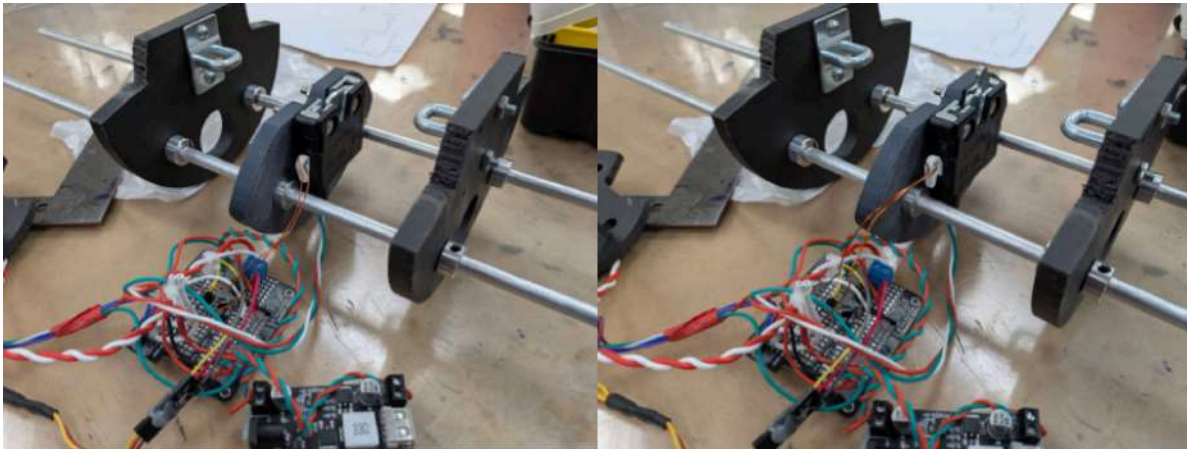
Procedure

1. Connect the PDM latch to the payload electronics and Raspberry Pi
2. Within the main launch code, adjust landing acceleration check such that it can be triggered by tapping the IMU on a hard surface.
3. Run main launch code.
4. Impose some acceleration to the IMU to simulate landing impact and observe PDM release.

Key Safety Considerations

- Tape up bare electrical wires and observe proper polarity to prevent electrical shocks and fires

Experimental Setup Photos



Results

Key Takeaways

- The PDM latch opened within 10 seconds of IMU acceleration spike for all 10 tests conducted.
- The payload electronics properly actuate the rotary latch

Design Implications

Testing and tuning will be required to properly set the acceleration threshold for landing detection.



VADL Testing Protocol

PDM Load Analysis (COMPLETE)

2.3.2

Overview:

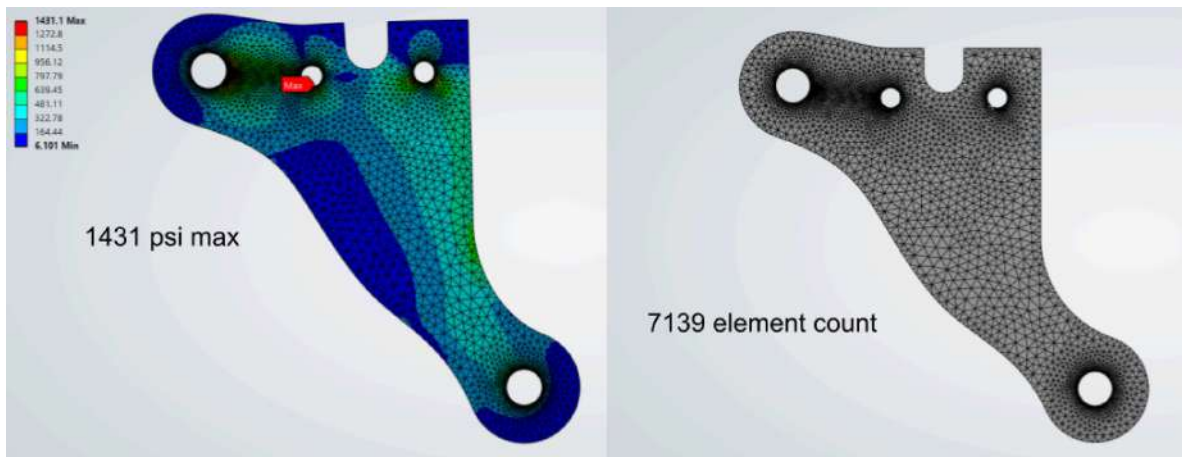
Objective / Purpose	Confirm that the PDM latch survives parachute opening forces.
Testing Variable(s)	PDM structural integrity
Technical Requirement(s)	VADL 4.4.2,2.2.2
Success Criteria	PDM can endure the load from the parachute with a safety factor of 2.5 applied
Responsible Team Members	Rais Nurhidajat

Methodology:

Procedure

1. PDM bulk plate solidworks file was imported into Ansys as a 2D surface, then thickened to 0.4 in.
2. The two rod holes in the part were fixed and a 106 lbf load was applied to the 2 latch holes
3. The FEA was run several times to ensure convergence, adjusting the edge refinement

Experimental Setup Photos



Results

Key Takeaways

- Maximum stress was calculated to be 1431 psi or 9.87 MPa.
- Onyx has a yield strength of 40 MPa, giving a safety factor of 4.1.

Design Implications

The PDM bulk plate is made with Onyx + carbon fiber inlay, meaning that the yield strength is significantly higher than that of Onyx alone. This means that the safety factor is above the calculated 4.1 one. Markforged does not give specific material properties of Onyx with carbon fiber inlay likely due to the non-isotropic nature of the material, so a specific yield strength is unknown.



VADL Testing Protocol

PDM Landing Detection Test (COMPLETE)

2.3.3

Overview:

Objective / Purpose	Ensure that PDM safely detects landing prior to its deployment in order to prevent unacceptable landing energies
Testing Variable(s)	IMU Performance, Microcontroller to PDM communication
Technical Requirement(s)	VADL 4.4.1
Success Criteria	The team observes that PDM only deploys at the proper pressure and does so within 1 second
Responsible Team Members	Rais Nurhidajat, Patrick Chen

Methodology:

Procedure

1. Set acceleration magnitude threshold to just above 10 m/s^2
2. Set altitude threshold checks to proper values
3. Power on electronics and place within the pressure cooker.
4. Depressurize and then pressurize the cooker to simulate rocket flight and landing
5. Listen for PDM actuation and while watching pressure gage

Key Safety Considerations

1. Ensure electronics are properly isolated from the pressure cooker to prevent electronic shorts

Experimental Setup Photos



Results

Key Takeaways

- The PDM actuation logic successfully once below the proper pressure altitude threshold
- The PDM did not actuate on the simulated ascent, only the descent

Design Implications

The PDM/landing logic performs satisfactorily and is ready for rocket flight.



VADL Testing Protocol

PDM Locking Test (COMPLETE)

2.3.4

Overview:

Objective / Purpose	Verify that accelerations or vibrations will not cause PDM to prematurely deploy
Testing Variable(s)	Robustness of PDM against accelerations and vibrations
Success Criteria	PDM does not prematurely deploy under any accelerations or vibrations
Responsible Team Members	Rais Nurhidajat

Methodology:

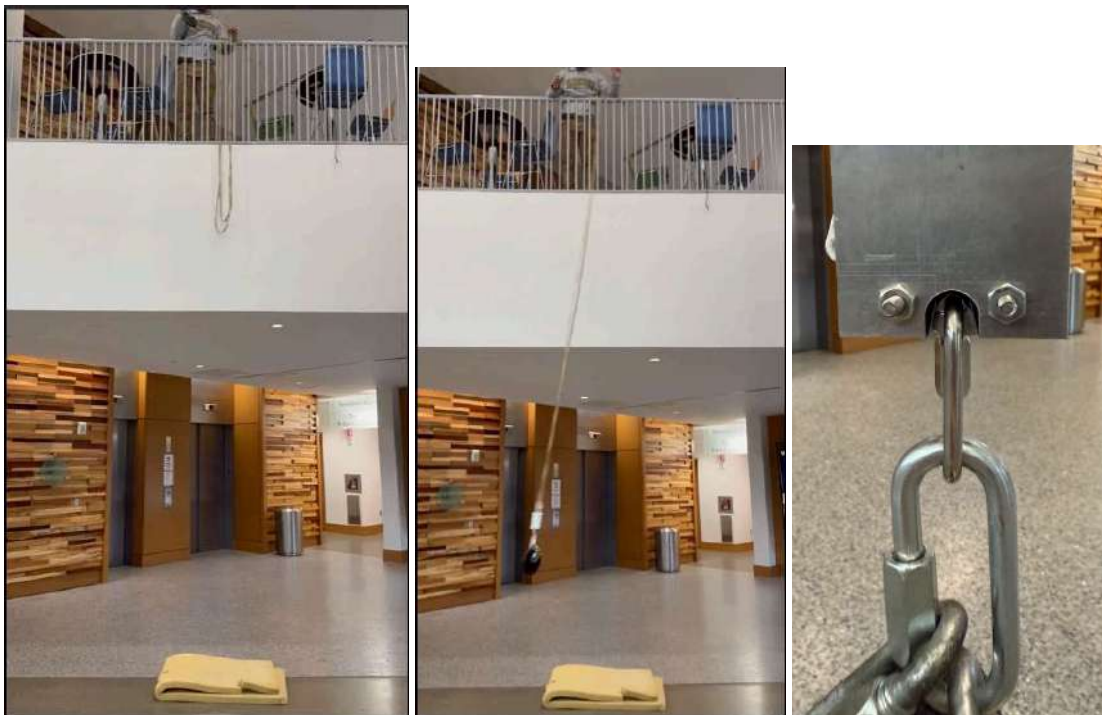
Procedure

1. Attach a 6 lb mass to the PDM latch, then attach the PDM latch to a 10 ft shock cord
2. Tie off the shock cord to a stairwell railing, drop, and observe the results

Key Safety Considerations

1. Mind surroundings and ensure base of stairwell is clear of foot traffic
2. Wear eye protection to protect eyes from fracturing debris

Experimental Setup Photos



Results

Key Takeaways

- The PDM latch did not open, as expected from the manufacturer specifications which state that the latch can withstand a force of 1061 lbf

Design Implications

The PDM latch is capable of withstanding the rigors of parachute opening and is ready for integration into the lander and launch vehicle.



VADL Testing Protocol

AEM Extension Test (COMPLETE)

2.4.1

Overview:

Objective / Purpose	Verify that the AEM is able to fully extend to 18 in from within the airframe
Testing Variable(s)	Tape extension, servo unlocking
Success Criteria	The AEM successfully deploys to full length 20 out of 20 times repeatedly
Responsible Team Members	Rais Nurhidajat

Methodology:

Procedure

1. Wind AEM and set servo lock
2. Turn on payload electronics and SSH into raspberry pi
3. Place AEM and bulkplate within airframe and pull antenna into slot cover
4. Run servo release command
5. Verify full extension length of 18 in

Key Safety Considerations

- Electronics shall be inspected prior to use to prevent shock and/or electrical fires

Experimental Setup Photos



Results

Key Takeaways

- Antenna fully extended to 18 in 20 out of 20

Design Implications

Teflon tape attached to fiberglass airframe is crucial to reduce friction and allow antennas to unfurl freely.



VADL Testing Protocol

AEM Locking Mechanism Test (COMPLETE)

2.4.3

Overview:

Objective / Purpose	Verify that the cylinder lock will not open unless actuated
Testing Variable(s)	Locking mechanism robustness against vibrations and impact
Success Criteria	The locking mechanism withstands 20 drops from a height of 4 feet AGL
Responsible Team Members	Rais Nurhidajat

Methodology:

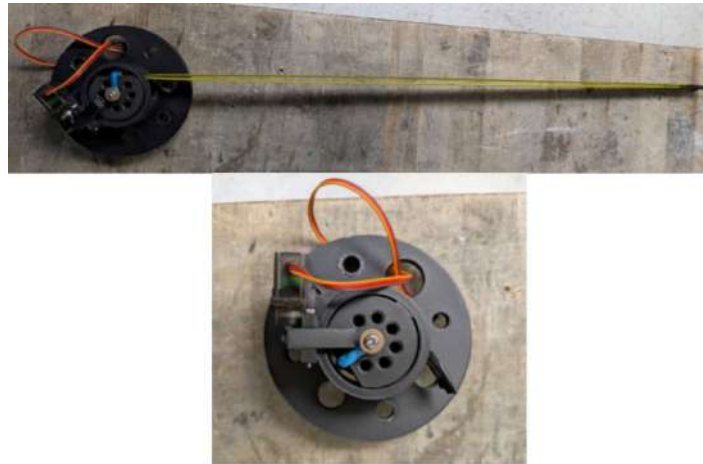
Procedure

1. Set cylinder lock by moving servo arm
2. Drop AEM repeatedly from a height of 4 feet and verify the cylinder does not unlock after repeated drops

Key Safety Considerations

- Wear safety glasses to protect eyes from potential debris breaking off AEM

Experimental Setup Photos



Results

Key Takeaways

- AEM cylinder did not unlock after over 20 drops
- Antenna can be jostled leading to slight expansion within the AEM case. When this was observed, the cylinder was unlocked to ensure tape extension. In these trials the tape did not extend as smoothly.

Design Implications

The antenna must be wrapped as tightly around the cylinder as possible prior to each launch. Teflon tape will be added to the tip of the antenna to lower friction between the antenna and case as much as possible.



VADL Testing Protocol

RF Range Test with unfurled antennas (**COMPLETE**)

2.5.2

Overview:

Objective / Purpose	Verify RF Transmission at 1 mile distance
Testing Variable(s)	RF system effective range
Technical Requirement(s)	VADL 4.6.2
Success Criteria	RF system should be error-free when there is ground clearance, and all APRS packages should be received with no break between any packets.
Responsible Team Members	Patrick Chen, Rais Nurhidajat

Methodology:

Procedure

1. Set up the ground station and antenna at the top of love circle
2. Place the lander on the ground at a 1 mile distance from love circle ensuring that there are no buildings in the line of sight between the receiver and the lander
3. Power on lander electronics and begin RF transmission

Key Safety Considerations

1. Be cautious when handling electronics to prevent damage or electrical shorts
2. When in public areas keep aware of pedestrian or vehicular traffic

Experimental Setup Photos



Results

Key Takeaways

- The RF system was able to successfully transmit all APRS packets at a distance of 1 mile
- Telemetry was compared to expected values (temperature, “landing” acceleration, etc.) to ensure that

Design Implications

The RF system is capable of fulfilling the Student Launch mission beyond the expected ranges.



VADL Testing Protocol

RF Range Test with collapsed antennas (COMPLETE)

2.5.3

Overview:

Objective / Purpose	Verify Transmission to 1 mile when antenna fails to deploy, or is damaged during flight
Testing Variable(s)	RF system effective range
Technical Requirement(s)	VADL 4.6.2
Success Criteria	RF system should be error-free when there is ground clearance, and all APRS packages should be received with no break between any packets.
Responsible Team Member	Patrick Chen, Rais Nurhidajat

Methodology:

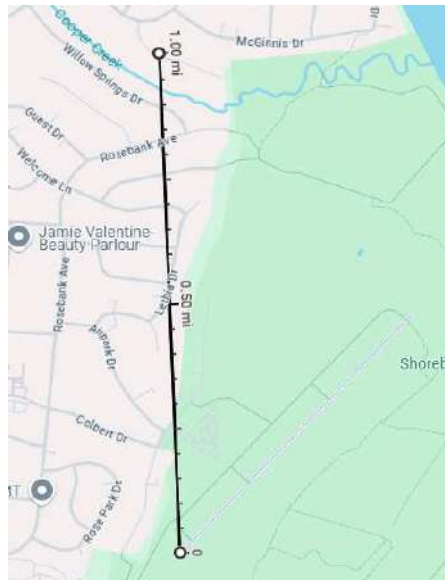
Procedure

1. Set up the ground station and antenna 1 mile away from the lander position
2. Power on electronics and begin transmission
3. Ensure that packets arrive properly

Key Safety Considerations

1. Be cautious when handling electronics to prevent damage or electrical shorts
2. When in public areas keep aware of pedestrian or vehicular traffic

Experimental Setup Photos



Results

Key Takeaways

- Transmission was achieved when the lander was not placed behind any obstacles - the lander needed a certain height
- When LOS is achieved the collapsed antennas are able to transmit all APRS packages reliably

Design Implications

While unlikely, in the case that all antennas cannot come out the lander can still transmit to the ground station receive



VADL Testing Protocol

Sensor Drop Test (COMPLETE)

2.6.2

Overview:

Objective / Purpose	Ensure IMU can detect dramatic changes in the state of the payload. Verify its reliability in free-fall conditions.
Testing Variable(s)	Acceleration and altitude data changes during the test.
Technical Requirement(s)	VADL 4.7.7
Success Criteria	Confirm the accelerometer detects proper acceleration changes, including rapid deceleration upon impact.
Responsible Team Members	Patrick Chen

Methodology:

Procedure

1. Prepare the IMU for testing by securely mounting it with the system on a prototype payload with a parachute.
2. Drop the IMU from the selected height.
3. Collect accelerometer data during the drop, ensuring the IMU records acceleration changes in all axes.
4. Analyze the data to ensure it reflects expected behavior.

Key Safety Considerations

- Conduct the test in a clear and designated area free from bystanders, obstacles, and hazardous materials.
- Ensure the landing surface is aligned with the drop point to avoid unexpected payload trajectory.
- Test the parachute system beforehand to confirm proper deployment and functionality.
- Maintain a safe distance from the drop zone during testing.

Experimental Setup Photos



Results

Key Takeaways

- Acceleration spike is correctly recorded and plotted.

Design Implications

The original plan to use the landing acceleration spike for detection has been successfully validated. This confirms the feasibility and effectiveness of the approach.



VADL Testing Protocol

Payload Altimeter Test (COMPLETE)

2.6.3

Overview:

Objective / Purpose	Ensure barometer data is changing with respect to the change of altitude.
Testing Variable(s)	Pressure and altitude readings from the barometer/altimeter.
Technical Requirement(s)	VADL 4.7.4
Success Criteria	The sensor must report correct altitude readings consistent with the expected pressure changes.
Responsible Team Members	Patrick Chen

Methodology:

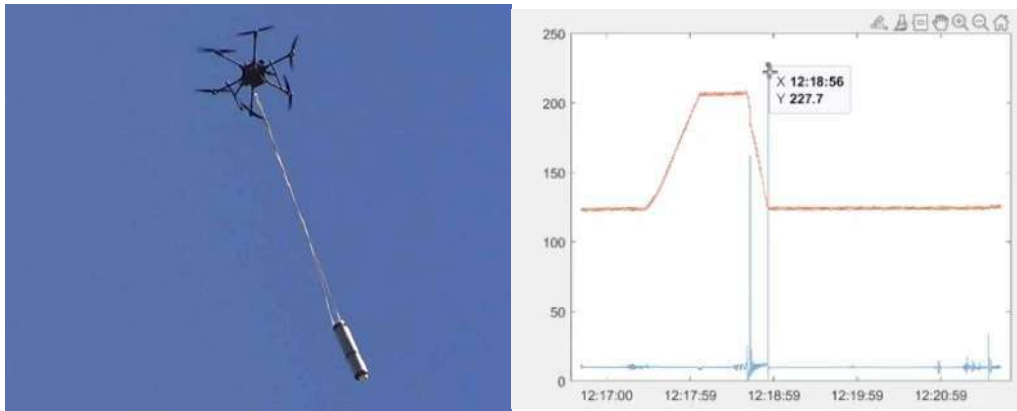
Procedure

1. Attach the payload system, including the barometer/altimeter, securely to a drone.
2. Verify the drone's flight controls and payload detachment mechanism are functioning properly.
3. Fly the drone to the desired altitude for testing.
4. At the designated height, detach the payload and deploy the parachute to ensure a controlled descent.
5. Retrieve the payload after landing.
6. Plot the data to visualize altitude changes over time and validate sensor performance.

Key Safety Considerations

- Conduct the test in a designated safe area, clear of bystanders, to minimize risks.
- Ensure the payload and parachute system are securely attached and tested beforehand to avoid malfunctions.
- Maintain visual control of the drone throughout the flight and descent.

Experimental Setup Photos



Results

Key Takeaways

- Both acceleration and altimeter data are confirmed to be working.

Design Implications

The core components of the landing detection algorithm have been successfully validated. As a result, the team will proceed with the originally proposed algorithm, which utilizes altitude changes and acceleration spikes to reliably detect landing events.



VADL Testing Protocol

Payload Apogee Detection Test (COMPLETE)

2.6.4

Overview:

Objective / Purpose	Ensure the apogee of the flight can be detected and recorded by the payload system.
Testing Variable(s)	Barometric pressure readings from the altimeter to identify changes indicative of apogee
Technical Requirement(s)	VADL 4.7.5
Success Criteria	The sensor correctly detects apogee by identifying the rapid pressure drop. The result is cross-referenced against simulation data.
Responsible Team Members	Patrick Chen, Sitan Huang

Methodology:

Procedure

1. Attach the payload system to a drone, ensuring the altimeter sensor is calibrated and securely mounted.
2. Fly the drone to a predetermined altitude simulating apogee conditions.
3. Collect barometric pressure readings during the ascent, at apogee (when the drone stabilizes at maximum altitude), and during descent.
4. Analyze the data to detect the pressure drop at apogee and confirm stabilization, comparing the results against simulation data.

Key Safety Considerations

- Conduct the test in a designated flight zone free of bystanders, obstacles, and restricted airspace.
- Ensure the drone's flight controls and payload attachment mechanisms are thoroughly inspected before the test.
- Maintain a safe distance and clear communication among team members during the flight.

Experimental Setup Photos



Results

Key Takeaways

- The apogee was correctly recorded and was within $\pm 5\%$ of the commercial altimeter on the drone (plus the length of the shock cord).
- The apogee data was also successfully transmitted via APRS.

Design Implications

This functionality was verified and requires no changes to the system.



VADL Testing Protocol

Landing Detection Test (COMPLETE)

2.6.5

Overview:

Objective / Purpose	Landing detection must be verified. It triggers all post-landing procedures.
Testing Variable(s)	Accelerometer, Barometer data. Confirm the activation of key detection events.
Technical Requirement(s)	VADL 4.7.2,4.7.3
Success Criteria	The latch needs to detach, the AEM needs to deploy, and RF transmission needs to begin immediately after landing.
Responsible Team Members	Patrick Chen

Methodology:

Procedure

1. Prepare the payload system, ensuring all sensors are calibrated and detection mechanisms are operational.
2. Attach the payload to a controlled drop mechanism (drone) to simulate the descent and landing phases.
3. Drop the payload from a predetermined height to mimic real flight conditions.
4. Validate that the landing and jettison events are detected accurately and match the expected behavior.

Key Safety Considerations

- Perform the test in a secure area away from bystanders and obstacles.
- Use protective materials to minimize potential damage to the payload upon landing.
- Maintain a safe distance for all team members during the drop.

Experimental Setup Photos



Results

Key Takeaways

- Landing was detected in 4 out of 5 tests.
- The only time of not detecting landing was caused by human error in calibrating the threshold data.

Design Implications

The checklist was updated to avoid human error. Now the software system requires three people to cross-check to ensure reliability in calibration processes.



VADL Testing Protocol

Shaft Collar Load Test (COMPLETE)

2.7.1

Overview:

Objective / Purpose	Verify that the shaft collars can withstand expected loads during flight, recovery, and landing
Testing Variable(s)	Shaft collar strength
Technical Requirement(s)	VADL 2.2.2
Success Criteria	Shaft collars withstand a force at least 2.5 times greater than the maximum expected force
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

Methodology:

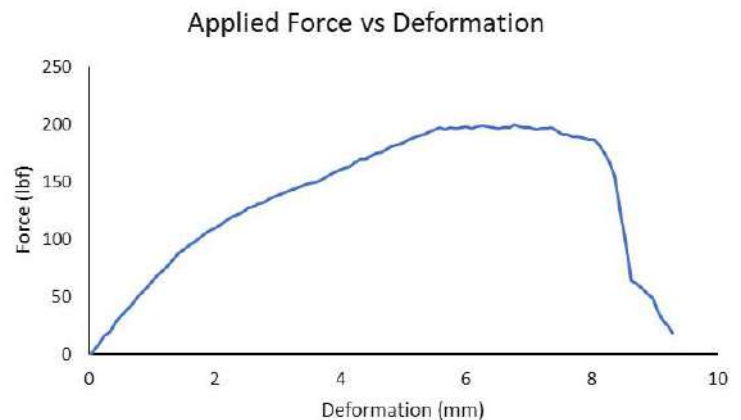
Procedure

1. Place shaft collar and aluminum rod assembly between load frame arms
2. Close fracture shield and don safety glasses
3. Initiate automated loading sequence
4. Load test is complete upon shaft collar tear out
5. Reset machine and compile data to excel file

Key Safety Considerations

- Wear safety glasses and ensure the fracture shield is in front of the load frame to prevent injury due to flying debris.
- Keep clear of the load frame during operation and tie back loose hair to prevent pinch or crush injuries.
- One person must be in close proximity to the emergency stop switch at all times

Experimental Setup Photos



Results

Key Takeaways

- A single shaft collar can withstand a force of 199.8 lbs
- Assuming that the entire payload mass is being held by a set of 2 shaft collars, the maximum force expected would be 110 lbf. (6.6 lbf x 32 G acceleration / 2 rods)
- This yields a safety factor of 1.82 for each shaft collar, which is insufficient for VADL requirement 2.2.2 which mandates a safety factor of 2.5

Design Implications

The shaft collars are not strong enough to retain the entire payload with only 2 rods. Nuts will be used instead to retain the rods to the aft bulk plate that connects to the LSM and launch vehicle. The shaft collars are strong enough to hold all other components such as the LDM guides, electronics bay, and battery bay



VADL Testing Protocol

Payload Skeleton Drop Test (COMPLETE)

2.7.2

Overview:

Objective / Purpose	Verify the structural integrity of the internal scaffolding of the payload section by itself without the integrated payload airframe
Testing Variable(s)	Structural integrity of payload skeleton
Technical Requirement(s)	NASA 2.3
Success Criteria	Visual and physical inspection determines that the structural integrity was not compromised and no component came loose or moved out of place
Responsible Team Members	Rais Nurhidajat

Methodology:

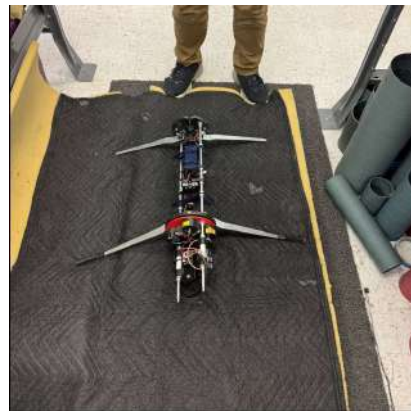
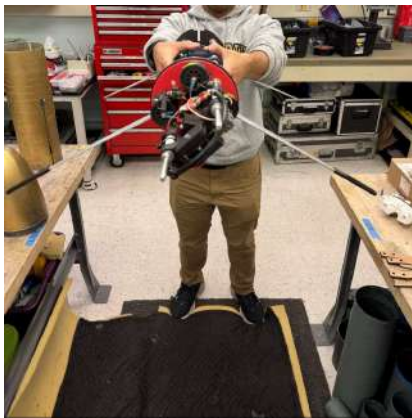
Procedure

1. Prepare lander skeleton with all electronic components and mechanical subsystems
2. Raise the lander to a height of 5 feet above the ground and drop
3. Inspect the skeleton for damage
4. Repeat test at least 20 times

Key Safety Considerations

1. Don safety glasses to protect eyes from flying debris that may come off skeleton
2. Ensure electronics batteries are not punctured at all during drop tests
3. Mind electronics to not accidentally short them, and to prevent electric shocks

Experimental Setup Photos



Results

Key Takeaways

- The lander skeleton is capable of withstanding the loads of landing without the airframe supporting it.
- All internal components such as electronics remained properly fixed to the skeleton

Design Implications

The lander skeleton is ready for rocket flight and continued testing.

7.2 Requirements Compliance

VADL has further advanced its requirements compliance with the completion of the full-scale vehicle demonstration flights (Section 5), additional payload drop tests (Section 4.1.3), and other vehicle and payload testing (Section 7.1). The following sections provide an update on the verification plans for both NASA and team-derived requirements.

7.2.1 NASA Requirements

VADL has updated its verification plans for the NASA requirements. Consistent with previous reports, VADL utilizes Analysis (A), Demonstration (D), Inspection (I), and Test (T) verification methods, as reflected in the following tables. Further, each requirement is designated as Verified (V), Not Verified (NV), or In Progress (IP). The updated requirements verification table is provided on the following pages. With the exception of requirements pertaining to the Payload Demonstration flight and Final Flight, both of which have yet to occur, all requirements have been successfully verified.

NASA General Requirements		Design Requirements	Method	Verification Details	Team Member(s) - Initials	Status
Req #	Description					
1.1.	Students on the team will do 100% of the project, including design, construction, written reports, presentations, and flight preparation with the exception of assembling the motors and handling black powder or any variant of ejection charges, or preparing and installing electric matches (to be done by the team's mentor). Teams will submit new work. Excessive use of past work will merit penalties.	The team shall complete 100% of USLI project work.	I	The team has maintained accountability through self-inspection to verify only students work on the project, with the exception of motors and energetics handled by the team mentor.	All	V
1.2.	The team will provide and maintain a project plan to include, but not limited to the following items: project milestones, budget and community support, checklists, personnel assignments, STEM engagement events, and risks and mitigations.	The team shall regularly update key project management items, to be presented in key NASA deliverables.	I	The team has met weekly outside of design meetings to discuss these aspects of the project; the President, Student Safety Officer, and Outreach Lead has managed these items to ensure compliance.	All	V
1.3.	Team members who will travel to the Huntsville Launch shall have fully completed registration in the NASA Gateway system before the roster deadline. Team members shall include:	The team shall announce all members attending launch week by CDR and ensure one mentor and no more than two educators.	I	The team lead has inspected that all members, mentors, and educators attending launch week were included on the submitted Huntsville roster.	IM	V
1.3.1	Students actively engaged in the project throughout the entire year.					
1.3.2	One mentor (see requirement 1.13).					
1.3.3	No more than two adult educators.					
1.4	Teams shall engage a minimum of 250 participants in Educational Direct Engagement STEM activities. These activities can be conducted in-person or virtually. To satisfy this requirement, all events shall occur between project acceptance and the FRR addendum due date. A template of the STEM Engagement Activity Report can be found on pages 40 – 43.	The team has planned events to remain on track with this requirement and shall continue to engage more participants.	I	The team has inspected completed engagement reports to ensure outreach milestones were met in a timely manner.	All	V
1.5	The team shall establish and maintain a social media presence to inform the public about team activities.	The team shall designate a social media manager to ensure timely and frequent social media presence.	I	The social media manager has inspected all social media presence and ensured timely, informative postings. The team has inspected these accounts as well.	SS	V
1.6.	Teams shall email all deliverables to the NASA project management team by the deadline specified in the handbook for each milestone. In the event that a deliverable is too large to attach to an email, inclusion of a link to download the file will be sufficient. Late submissions of PDR, CDR, or FRR milestone documents will NOT be accepted. Teams that fail to submit the PDR, CDR, or FRR milestone documents will be eliminated from the project.	All deliverables shall be sent to the NASA project management team in a timely manner.	I	The team lead, with assistance from other team members, has ensured that all deliverables were submitted by the deadline.	IM	V
1.7	Teams who do not satisfactorily complete each milestone review (PDR, CDR, FRR) will be provided action items to be completed following their review and will be required to address action items in a delta review session. After the delta session, the NASA management panel will meet to determine the teams' status in the program, and the teams will be notified shortly thereafter.	N/A				
1.8.	All deliverables shall be in PDF format.	The team shall submit deliverables in PDF format.	I	The team lead, with assistance from other team members, has inspected submission documents to ensure they were submitted in PDF format.	IM	V
1.9.	In every report, teams will provide a table of contents including major sections and their respective sub-sections.	A table of contents shall be included in every report.	I	The table of contents for each report has been updated as the report was constructed, as inspected by the team lead.	IM	V
1.10.	In every report, the team shall include the page number at the bottom of the page.	Page numbers shall be included on every report page.	I	The report outline for each report has automatically included page numbers, as inspected by the team lead.	IM	V
1.11.	The team shall provide any computer equipment necessary to perform a video teleconference with the review panel. This includes, but is not limited to: a computer system, video camera, speaker telephone, and a sufficient Internet connection. Cellular phones should be used for speakerphone capability only as a last resort.	Team members shall procure and test video conferencing equipment ahead of review panel meetings.	T	The team has tested all equipment prior to presentations and ensured it meets required parameters.	All	V
1.12.	All teams attending Launch Week shall be required to use the launch pads provided by Student Launch's launch services provider. No custom pads will be permitted at the NASA Launch Complex. At launch, 8-foot 1010 rails and 12-foot 1515 rails will be provided. The launch rails will be canted 5 – 10 degrees away from the crowd on Launch Day. The exact cant will depend on Launch Day wind conditions.	The launch vehicle shall be designed to integrate with provided launch pads.	I	The Vehicle Team Lead has ensured the vehicle is constructed to integrate with the provided launch rails.	SS	V
1.13.	Each team shall identify a "mentor." A mentor is defined as an adult who is included as a team member, who will be supporting the team (or multiple teams) throughout the project year, and may or may not be affiliated with the school, institution, or organization. The team mentor shall not be a student team member. The mentor shall maintain a current certification, and be in good standing, through the National Association of Rocketry (NAR) or Tripoli Rocketry Association (TRA) for the motor impulse of the launch vehicle and must have flown and successfully recovered (using electronic, staged recovery) a minimum of two flights in this or a higher impulse class, prior to PDR. The mentor is designated as the individual owner of the rocket for liability purposes and must travel with the team to Launch Week. One travel stipend will be provided per mentor regardless of the number of teams he or she supports. The stipend will only be provided if the team passes FRR and the team and mentor attend Launch Week in April.	The team shall identify their rocketry mentor in the General Information section of all reports.	I	The mentor certifications have been inspected by the team lead to ensure they are up-to-date and meet requirements.	IM	V
1.14.	Teams will track and report the number of hours spent working on each milestone.	Team members shall record their weekly hours in a common spreadsheet.	I	The team lead has inspected the hours tracking spreadsheet and ensured all members are tracking hours in a timely and accurate manner.	IM	V
NASA Vehicle Requirements		Design Requirements	Method	Verification Details	Team Member(s)	Status
Req #	Description					
2.1.	The vehicle shall deliver the payload to an apogee altitude between 4,000 and 6,000 feet above ground level (AGL). Teams flying below 3,500 feet or above 6,500 feet on their competition launch will receive zero altitude points towards their overall project score and will not be eligible for the Altitude Award.	Team simulations will utilize confidence intervals to confirm that the vehicle will deliver the payload to the desired apogee range regardless of environmental conditions.	AD	The vehicle altitude has been analyzed with redundant simulation methods, and demonstrated in a full-scale test flight to comply with the altitude requirement.	SH	V
2.2.	Teams shall declare their target altitude goal at the CDR milestone. The declared target altitude shall be used to determine the team's altitude score.	The target apogee shall be specified in VADL derived requirement 1.1.1.	AI	The target altitude has been analyzed with redundant simulation methods and reported in the CDR submission.	SH	V
2.3	The launch vehicle shall be designed to be recoverable and reusable. Reusable is defined as being able to launch again on the same day without repairs or modifications.	The launch vehicle shall be designed to be reusable, and the payload will be entirely reusable with battery recharging.	AD	Calculations, computational models, and the full-scale vehicle demonstration flight have verified the launch vehicle is recoverable and reusable.	All	V

2.4.	The launch vehicle shall have a maximum of four (4) independent sections. An independent section is defined as a section that is either tethered to the main vehicle or is recovered separately from the main vehicle using its own parachute.	The launch vehicle shall have four independent sections.	I	The launch vehicle design has been inspected to confirm there are 3 independent sections pre-jettison and 4 post-jettison.	All	V
2.4.1	Coupler/airframe shoulders which are located at in-flight separation points shall be at least two airframe diameters in length. (One body diameter of surface contact with each airframe section.)	The separation point couplers shall all be designed with 2 body diameter length.	I	The launch vehicle design has been inspected to ensure separation point couplers and shoulders are all at least 2 body diameter in length.	All	V
2.4.2	Coupler/airframe shoulders which are located at non-in-flight separation points shall be at least 1.5 airframe diameters in length. (0.75 body diameter of surface contact with each airframe section.)	There are no non-in-flight separation points in the full-scale vehicle design.				
2.4.3	Nosecone shoulders which are located at in-flight separation points shall be at least 1/2 body diameter in length.	There is no nosecone shoulder in the full-scale vehicle design.				
2.5.	The launch vehicle shall be capable of being prepared for flight at the launch site within 2 hours of the time the Federal Aviation Administration flight waiver opens.	The launch vehicle shall be constructed in a modular fashion to allow for rapid assembly.	D	The launch vehicle was demonstrated to be prepared in less than 2 hours of the launch waiver opening during the full-scale vehicle demonstration flight.	All	V
2.6.	The launch vehicle and payload shall be capable of remaining in launch-ready configuration on the pad for a minimum of 3 hours without losing the functionality of any critical on-board components, although the capability to withstand longer delays is highly encouraged.	The payload shall be equipped with sufficient battery power and cooling methods to ensure launch-ready configuration for greater than two hours.	AT	The launch vehicle and payload has been analyzed and tested to ensure its launch-ready configuration can be maintained for at least 3 hours.	All	V
2.7.	The launch vehicle shall be capable of being launched by a standard 12-volt direct current firing system. The firing system shall be provided by the NASA-designated launch services provider.	The launch vehicle will be designed to integrate with a 12-volt direct current firing system.	D	The full-scale vehicle demonstration flight has demonstrated that a launch can occur with a 12-volt direct current firing system.	All	V
2.8.	The launch vehicle shall require no external circuitry or special ground support equipment to initiate launch (other than what is provided by the launch services provider).	The launch vehicle shall be designed to operate without reliance on special ground support equipment.	ID	All design and integration plans have been inspected to ensure no external support equipment is needed, as validated by the full-scale vehicle demonstration flight.	PC, SS	V
2.9	Each team shall use commercially available ematches or igniters. Hand-dipped igniters shall not be permitted.	The team shall use MJG Firewire electric matches to ignite blast charges.	I	The team lead and student safety officer have inspected the electric match selection to ensure that they are commercially available and not hand-dipped.	IM, ZS	V
2.10.	The launch vehicle shall use a commercially available solid motor propulsion system using ammonium perchlorate composite propellant (APCP) which is approved and certified by the National Association of Rocketry (NAR), Tripoli Rocketry Association (TRA), and/or the Canadian Association of Rocketry (CAR).	The launch vehicle shall use the TRA-certified Cesaroni L1720 motor.	I	Motor selection and vehicle plans have been inspected to ensure the rocket motor meets NASA requirements.	SS, SH	V
2.10.1	Final motor choices shall be declared by the Preliminary Design Review (PDR) milestone.	The team shall verify their final motor selection by PDR.	I	The team lead has inspected that the final motor choice was included in the PDR report.	IM	V
2.10.2	Any motor change after PDR shall be approved by the NASA management team or NASA Range Safety Officer (RSO). Changes for the sole purpose of altitude adjustment shall not be approved. A scoring adjustment against the team's overall score shall be incurred when a motor change is made after the PDR milestone. The only exception is teams switching to their secondary motor choice provided the primary motor choice is unavailable due to a motor shortage.	The team did not require any motor changes.				
2.11.	The launch vehicle shall be limited to a single motor propulsion system.	The launch vehicle shall be a single stage.	I	Inspection of vehicle designs confirm the vehicle is a single stage.	All	V
2.12.	The total impulse provided by a College or University launch vehicle shall not exceed 5,120 Newton-seconds (L-class).	The motor selected will not exceed the maximum impulse limits.	I	Inspection of the motor selection confirms the motor does not exceed L-class limitations.	SS, SH	V
2.13.	Pressure vessels on the vehicle must be approved by the RSO and shall meet the following criteria:	The launch vehicle and payload shall not include any pressure vessels.				
2.13.1.	The minimum factor of safety (Burst or Ultimate pressure versus Max Expected Operating Pressure) will be 4:1 with supporting design documentation included in all milestone reviews.					
2.13.2.	Each pressure vessel shall include a pressure relief valve that sees the full pressure of the tank and is capable of withstanding the maximum pressure and flow rate of the tank.					
2.13.3.	The full pedigree of the tank shall be described, including the application for which the tank was designed and the history of the tank. This will include the number of pressure cycles put on the tank, the dates of pressurization/depressurization, and the name of the person or entity administering each pressure event.					
2.14.	The launch vehicle shall have a minimum static stability margin of 2.0 at the point of rail exit. Rail exit is defined at the point where the forward rail button loses contact with the rail.	The design of the rocket shall estimate the SSM to exceed 2.0 at the rail exit, and then verified by two methods of simulation.	A	Simulations confirm that the static stability margin exceeds 2.0 at the launch rail.	SH	V
2.15.	The launch vehicle shall have a minimum thrust to weight ratio of 5.0 : 1.0.	The motor choice shall provide a high enough thrust relative to the vehicle weight to meet the requirement.	A	Simulations confirm that the thrust to weight ratio exceeds NASA requirements.	SH	V
2.16.	Any structural protuberance on the rocket shall be located aft of the burnout center of gravity. Camera housings will be exempted, provided the team can show that the housing(s) causes minimal aerodynamic effect on the rocket's stability.	There are no structural protuberances in the full-scale vehicle design.				
2.17.	The launch vehicle shall accelerate to a minimum velocity of 52 fps at rail exit.	The simulations shall calculate velocity at rail exit.	ADT	Simulations confirm the minimum velocity is exceeded, and the full-scale vehicle demonstration flight confirmed the launch rail exit velocity exceeds 52 fps.	SH	V
2.18.	All teams shall successfully launch and recover a subscale model of their rocket. Success of the subscale is at the sole discretion of the NASA review panel. The subscale flight may be conducted at any time between proposal award and the CDR submission deadline. Subscale flight data shall be reported in the CDR report and presentation at the CDR milestone. Subscale models are required to use a minimum motor impulse class of E (Mid Power motor).	The team shall conduct a subscale launch on Dec. 7, 2024.	ID	The team conducted a subscale flight test on December 7 and ensured it was reported in the CDR report.	All	V
2.18.1.	The subscale model should resemble and perform as similarly as possible to the full-scale model; however, the full-scale will not be used as the subscale model.	The subscale shall be designed and built to represent a scaled version of the full-scale vehicle.	ID	The subscale was designed and built, with its performance validated through a successful flight.	All	V
2.18.2.	The subscale model shall carry an altimeter capable of recording the model's apogee altitude.	The subscale model shall carry an onboard altimeter.	I	Subscale inspection confirmed redundant Entacore AIM altimeters were used to capture altitude data.	SS, SC	V

2.18.3.	The subscale rocket shall be a newly constructed rocket, designed and built specifically for this year's project.	The subscale rocket shall be designed from scratch and constructed entirely separately from the full-scale vehicle.	I	The subscale was verified by the team as a newly constructed vehicle.	All	V
2.18.4.	Proof of a successful flight shall be supplied in the CDR report.	Subscale proof of flight requirements shown below.				
2.18.4.1	Altimeter flight profile graph(s) OR a quality video showing successful launch, recovery events, and landing as deemed by the NASA management panel are acceptable methods of proof. Altimeter flight profile graph(s) that are not complete (liftoff through landing) will not be accepted.	The team shall include altimeter flight profile graphs in the CDR report.	I	The team lead will inspect the CDR report to ensure that altimeter flight profile graphs were included.	IM	V
2.18.4.2	Quality pictures of the as landed configuration of all sections of the launch vehicle shall be included in the CDR report. This includes, but is not limited to: nosecone, recovery system, airframe, and booster.	The team shall include pictures of the rocket and payload sections in their landed configurations in the CDR report.	I	The team lead inspected the CDR report to ensure that the required pictures were included.	IM	V
2.18.5.	The subscale rocket shall not exceed 75% of the dimensions (length and diameter) of your designed full-scale rocket. For example, if your full-scale rocket is a 4" diameter 100" length rocket your subscale shall not exceed 3" diameter and 75" in length.	The subscale rocket shall be designed such that the length and diameter do not exceed 75% of the full-scale rocket.	I	Subscale design inspection confirmed the rocket's dimensions did not exceed 75% of the full-scale vehicle	SS, SC, IM	V
2.19.	All teams shall complete demonstration flights as outlined below.	Verification is outlined as follows.				
2.19.1.	Vehicle Demonstration Flight — All teams shall successfully launch and recover their full-scale rocket prior to FRR in its final flight configuration. The rocket flown shall be the same rocket to be flown for their competition launch. The purpose of the Vehicle Demonstration Flight is to validate the launch vehicle's stability, structural integrity, recovery systems, and the team's ability to prepare the launch vehicle for flight. A successful flight is defined as a launch in which all hardware is functioning properly (drogue chute at apogee, main chute at the intended lower altitude, functioning tracking devices, etc.). The following criteria shall be met during the full-scale demonstration flight:	The team shall launch and recover their full-scale rocket prior to FRR.	ID	The team has verified the launch vehicle is the same as the launch day vehicle, and the flight demonstration has validated its success.	All	V
2.19.1.1.	The vehicle and recovery system shall have functioned as designed.	The vehicle and recovery system data shall be analyzed to ensure proper functionality.	ID	The vehicle and recovery performance has been inspected and validated through the demonstration flight to ensure proper functionality.	All	V
2.19.1.2.	The full-scale rocket shall be a newly constructed rocket, designed and built specifically for this year's project.	The team shall design and construct an entirely new full-scale rocket.	I	The team inspected the launch vehicle to ensure it is a newly constructed rocket.	All	V
2.19.1.3.	The payload does not have to be flown during the full-scale Vehicle Demonstration Flight. The following requirements still apply:	The team shall fly its payload during the full-scale demonstration flight.	I	The team inspected the launch vehicle and payload to ensure it is the same payload being flown during the competition launch.	All	V
2.19.1.3.1	If the payload is not flown, mass simulators shall be used to simulate the payload mass.					
2.19.1.3.2	The mass simulators shall be located in the same approximate location on the rocket as the missing payload mass.					
2.19.1.4.	If the payload changes the external surfaces of the rocket (such as camera housings or external probes) or manages the total energy of the vehicle, those systems will be active during the full-scale Vehicle Demonstration Flight.	The payload will not include external surfaces.				
2.19.1.5.	Teams shall fly the competition launch motor for the Vehicle Demonstration Flight. The team may request a waiver for the use of an alternative motor in advance if the home launch field cannot support the full impulse of the competition launch motor or in other extenuating circumstances.	The team shall fly the L1720 motor for the Vehicle Demonstration Flight.	ID	The vehicle lead inspected that the launch motor was the L1720 for the Vehicle Demonstration Flight.	SS	V
2.19.1.6.	The vehicle will be flown in its fully ballasted configuration during the full-scale test flight. Fully ballasted refers to the maximum amount of ballast that will be flown during the competition launch flight. Additional ballast shall not be added without a re-flight of the full-scale launch vehicle.	The team does not intend to fly with ballast for the full-scale vehicle.				
2.19.1.7.	After successfully completing the full-scale demonstration flight, the launch vehicle or any of its components shall not be modified without the concurrence of the NASA management team or Range Safety Officer (RSO).	Any changes made after the full-scale demonstration flight will only be performed with concurrence of the NASA RSO.	I	The team has confirmed that no launch vehicle components will be modified without NASA RSO approval.	All	V
2.19.1.8.	Proof of a successful flight shall be supplied in the FRR report.	Proof of successful full-scale flight requirements shown below.				
2.19.1.8.1	Altimeter flight profile data output with accompanying altitude and velocity versus time plots is required to meet this requirement. Altimeter flight profile graph(s) that are not complete (liftoff through landing) shall not be accepted.	The team shall provide proof of a successful flight in FRR with altimeter flight profile data and required analysis.	I	The team lead inspected that the FRR flight data was included in the FRR report.	IM	V
2.19.1.8.2	Quality pictures of the as landed configuration of all sections of the launch vehicle shall be included in the FRR report. This includes, but is not limited to: nosecone, recovery system, airframe, and booster.	The team shall provide pictures of the landed rocket and payload sections in the FRR report.	I	The team lead inspected that the required pictures were included in the FRR report.	IM	V
2.19.1.8.3.	Raw altimeter data shall be submitted in .csv or .xlsx format.	The team shall submit raw altimeter data in .csv or .xlsx format.	I	The team lead inspected that the raw altimeter data was submitted in the proper format.	IM	V
2.19.1.9.	Vehicle Demonstration flights shall be completed by the FRR submission deadline. No exceptions will be made. If the Student Launch office determines that a Vehicle Demonstration Re-flight is necessary, then an extension may be granted. THIS EXTENSION IS ONLY VALID FOR RE-FLIGHTS, NOT FIRST TIME FLIGHTS. Teams completing a required re-flight shall submit an FRR Addendum by the FRR Addendum deadline.	The team shall complete the full-scale flight prior to FRR and request an extension if re-flight is deemed to be necessary.	ID	The team ensured that the Vehicle Demonstration flight was completed prior to the FRR submission deadline by having it on February 22.	All	V
2.19.2	Payload Demonstration Flight — All teams shall successfully launch and recover their full-scale rocket containing the completed payload prior to the Payload Demonstration Flight deadline. The rocket flown shall be the same rocket to be flown as their competition launch. The purpose of the Payload Demonstration Flight is to prove the launch vehicle's ability to safely retain the constructed payload during flight and to show that all aspects of the payload perform as designed. A successful flight is defined as a launch in which the rocket experiences stable ascent and the payload is fully retained until it is deployed (if applicable) as designed. The following criteria shall be met during the Payload Demonstration Flight:	The team shall fly a payload demonstration flight in Spring 2025 at a partner launch location.	ID	The completed flight vehicle and payload will be inspected to ensure it is ready for launch, and the launch demonstration will validate that the system meets all requirements.	All	NV
2.19.2.1.	The payload shall be fully retained until the intended point of deployment (if applicable), all retention mechanisms shall function as designed, and the retention mechanism shall not sustain damage requiring repair.	The payload shall be retained via the LSM until 300 ft.	ID	The team will inspect that all components are secured and that the payload is jettisoned at 300 ft, demonstrated through flight.	All	NV
2.19.2.2.	The payload flown shall be the final, active version.	The team shall complete a flight in its final configuration prior to the FRR deadline and follow proper NASA procedures for vehicle re-flights.	I	The team will inspect the payload and ensure it is ready to be flown in its final version.	All	NV

2.19.2.3.	If the above criteria are met during the original Vehicle Demonstration Flight, occurring prior to the FRR deadline and the information is included in the FRR package, the additional flight and FRR Addendum are not required.	The team shall complete a flight in its final configuration prior to the FRR deadline and follow proper NASA procedures for vehicle re-flights.	I	FRR requirements inspection will ensure a successful Vehicle Demonstration flight, and inspection of an FRR addendum will ensure a successful Payload Demonstration flight as well.	All	NV
2.19.2.4.	Payload Demonstration Flights shall be completed by the FRR Addendum deadline. NO EXTENSIONS WILL BE GRANTED.	The team shall complete a flight in its final configuration prior to the FRR deadline and follow proper NASA procedures for vehicle re-flights.	I	The team will verify that all requirements of the FRR and FRR addendum are validated.	All	NV
2.20.	An FRR Addendum shall be required for any team completing a Payload Demonstration Flight or NASA-required Vehicle Demonstration Re-flight after the submission of the FRR Report.					NV
2.20.1.	Teams required to complete a Vehicle Demonstration Re-Flight and failing to submit the FRR Addendum by the deadline will not be permitted to fly a final competition launch.					NV
2.20.2.	Teams who complete a Payload Demonstration Flight which is not fully successful may petition the NASA RSO for permission to fly the payload during launch week. Permission will not be granted if the RSO or the Review Panel have any safety concerns.					NV
2.21.	The team's name and Launch Day contact information shall be in or on the rocket airframe as well as in or on any section of the vehicle that separates during flight and is not tethered to the main airframe. This information shall be included in a manner that allows the information to be retrieved without the need to open or separate the vehicle.	The team shall provide the team name and contact information on each section of the airframe.	I	The Vehicle Lead has ensured that the required information is present on the vehicle in the manner requested.	SS	V
2.22.	All Lithium Polymer batteries shall be sufficiently protected from impact with the ground and will be brightly colored, clearly marked as a fire hazard, and easily distinguishable from other payload hardware.	The team shall protect lithium polymer batteries and provide identifiable colorings and fire hazard markings.	I	The Avionics Lead has ensured that all LiPo batteries are sufficiently protected and marked, inspected by the safety lead.	SC, ZS	V
2.23.	Vehicle Prohibitions:		I	The Vehicle Lead/Safety Lead has confirmed that the motor selection and integration does not violate any vehicle prohibitions.	SS, ZS	V
2.23.1.	The launch vehicle shall not utilize forward firing motors.					
2.23.2.	The launch vehicle shall not utilize motors that expel titanium sponges (Sparky, Skidmark, MetalStorm, etc.)					
2.23.3.	The launch vehicle shall not utilize hybrid motors.					
2.23.4.	The launch vehicle shall not utilize a cluster of motors.					
2.23.5.	The launch vehicle shall not utilize friction fitting for motors.					
2.23.6.	The launch vehicle will not exceed Mach 1 at any point during flight.		ID	The Simulations Lead has verified that the launch meets these stated requirements, and the full-scale Vehicle Demonstration flight has proven compliance.	SH	V
2.23.7.	Vehicle ballast shall not exceed 10% of the total unballasted weight of the rocket as it would sit on the pad (i.e. a rocket with an unballasted weight of 40 lbs. on the pad may contain a maximum of 4 lbs. of ballast).					
2.23.8.	Transmissions from onboard transmitters, which are active at any point prior to landing, shall not exceed 250 mW of power (per transmitter).	The transmitter shall be selected to ensure that it abides by prohibition requirements.	IT	The Payload Lead has inspected and tested relevant systems to ensure these requirements are satisfied.	RN	V
2.23.9.	Transmitters shall not create excessive interference. Teams shall utilize unique frequencies, handshake/passcode systems, or other means to mitigate interference caused to or received from other teams.	The transmitter shall be selected to ensure that it abides by prohibition requirements.	IT	The Payload Lead has inspected and tested relevant systems to ensure these requirements are satisfied.	RN	
2.23.10.	Excessive and/or dense metal shall not be utilized in the construction of the vehicle. Use of light-weight metal will be permitted but limited to the amount necessary to ensure structural integrity of the airframe under the expected operating stresses.	The launch vehicle shall be designed with a combination of aluminum and rapid prototyping materials.	I	The team has inspected the design of the vehicle to ensure proper material selection, and FEA has been performed to verify the structural integrity of load-bearing components.	All	V
NASA Recovery System Requirements		Design Requirements	Method	Verification Details	Team Member(s)	Status
Req #	Description					
3.1	The full scale launch vehicle shall stage the deployment of its recovery devices, where a drogue parachute is deployed at apogee, and a main parachute is deployed at a lower altitude. Tumble or streamer recovery from apogee to main parachute deployment is also permissible, provided that kinetic energy during drogue stage descent is reasonable, as deemed by the RSO.	The launch vehicle drogue parachute will be designed to deploy within one second of apogee, and the main parachute will be designed to deploy at 750 ft.	AD	The altimeters in the avionics bay have been configured such that the blast charges responsible for parachute deployment are triggered at apogee for drogue and 750 ft for main. Dual deployment capabilities were demonstrated in the Vehicle Demonstration flight.	SC	V
3.1.1.	The main parachute shall be deployed no lower than 500 feet.	The main parachute deployment shall occur at 750 ft.	AD	The altitude sensing performance of the altimeter has been tested prior to launch, and was demonstrated with the full-scale Vehicle Demonstration flight.	SC	V
3.1.2.	The apogee event shall contain a delay of no more than 2 seconds.	The apogee event shall occur within one second of apogee.	AD	The apogee event timing has been analyzed by altimeter data of the full-scale Vehicle Demonstration flight to verify event timing compliance.	SC	V
3.1.3.	Motor ejection is not a permissible form of primary or secondary deployment.		The mission sequence shall not involve motor ejection.			
3.2.	Each team shall perform a successful ground ejection test for all electronically initiated recovery events prior to the initial flights of the subscale and full scale vehicles.	A ground test of parachute ejection shall be performed in a safe outdoor location, with events monitored by the rocketry mentor and Student Safety Officer.	TD	The team performed successful ejection tests of the blast charges prior to the subscale flight and Vehicle Demonstration flight.	All	V
3.3.	Each independent section of the launch vehicle will have a maximum kinetic energy of 75 ft-lbf at landing. Teams whose heaviest section of their launch vehicle, as verified by vehicle demonstration flight data, stays under 65 ft-lbf will be awarded bonus points.	The maximum kinetic energy shall be calculated in simulations and design parameters shall incorporate the maximum kinetic energy limitations.	AD	Flight simulations have verified that the launch vehicle landing energy does not exceed the limits, and the Vehicle Demonstration flight verified compliance	All	V
3.4.	The recovery system shall contain redundant, commercially available barometric altimeters that are specifically designed for initiation of rocketry recovery events. The term "altimeters" includes both simple altimeters and more sophisticated flight computers.	The vehicle avionics shall contain two EasyMini altimeters for the recovery system.	I	The team inspected the design and the constructed vehicle to ensure the recovery system contains redundant altimeters.	All	V
3.5.	Each altimeter shall have a dedicated power supply, and all recovery electronics shall be powered by commercially available batteries.	Each altimeter shall be powered by independent, commercially available 9V batteries.	IT	The Avionics Lead has ensured each altimeter has independent power supplies, and has tested that they are sufficient to power the altimeters.	SC	V
3.6.	Each altimeter shall be armed by a dedicated mechanical arming switch that is accessible from the exterior of the rocket airframe when the rocket is in the launch configuration on the launch pad.	The rocket body tube shall contain external entry points for altimeter access on the launch pad.	ID	The vehicle design has been inspected to ensure the altimeters are externally accessible, and the Vehicle Demonstration flight validated successful arming systems.	SC	V

3.7.	Each arming switch shall be capable of being locked in the ON position for launch (i.e. cannot be disarmed due to flight forces).	The avionics arming switch shall lock into the ON position for launch via screw switches.	ID	Screw switches have been used in the avionics design. The Avionics Lead ensured the arming switch can be locked in the ON position during the Vehicle Demonstration flight.	SC	V
3.8.	The recovery system, GPS and altimeters, and electrical circuits shall be completely independent of any payload electrical circuits.	The recovery system electronic shall be housed in the avionics bay, separated from payload circuits.	I	Design plan inspections by the Vehicle Lead and Payload Lead have ensured that the electrical circuits are independent of payload circuits.	SS, RN	V
3.9.	Removable shear pins shall be used for both the main parachute compartment and the drogue parachute compartment.	Shear pins shall hold all three rocket sections together prior to deployment.	AD	Mathematical calculations have verified the shear pin requirements, and ejection tests confirmed they meet requirements.	All	V
3.10.	Bent eyebolts shall not be permitted in the recovery subsystem.	The team shall not include bent eyebolts in the recovery subsystem.	I	The vehicle lead has inspected and confirmed that no bent eyebolts are included in the design of the recovery subsystem.	SS	V
3.11.	The recovery area shall be limited to a 2,500 ft. radius from the launch pads.	The flight vehicle shall be designed to land within a 2,500 ft. radius of the launch pad	AD	Redundant flight simulations have confirmed that the vehicle and payload drift do not exceed 2,500 ft, and the Vehicle Demonstration flight verified compliance.	SH	V
3.12.	Descent time of the launch vehicle shall be limited to 90 seconds (apogee to touch down). Teams whose launch vehicle descent, as verified by vehicle demonstration flight data, stays under 80 seconds will be awarded bonus points.	The descent time shall be calculated in simulations and design parameters shall change to accommodate this requirement if needed .	AD	Redundant flight simulations have confirmed vehicle and payload descent time do not exceed 90 seconds, and the Vehicle Demonstration flight validated the results.	SH	V
3.13.	An electronic GPS tracking device shall be installed in the launch vehicle and will transmit the position of the tethered vehicle or any independent section to a ground receiver.	A GPS device shall be installed in the launch vehicle.	TD	The GPS tracking device has been tested utilizing ground tests, and the Vehicle Demonstration flight validated the device's inclusion.	AD	V
3.13.1.	Any rocket section or payload component, which lands untethered to the launch vehicle, shall contain an active electronic GPS tracking device.	The payload shall contain a GPS tracking device.	TD	The GPS tracking device has been tested utilizing ground tests, and the Vehicle Demonstration flight validated the device's inclusion.	AD	V
3.13.2.	The electronic GPS tracking device(s) shall be fully functional during the official competition launch.	The GPS device shall communicate with the base station on launch day.	ID	The GPS device has been successfully demonstrated in the Vehicle Demonstration flight, and it will be inspected on competition day to ensure it is fully functional.	AD	IP
3.14.	The recovery system electronics shall not be adversely affected by any other on-board electronic devices during flight (from launch until landing).	The Recovery system components shall all be housed in an independent avionics bay and shielded from all other onboard electronics to prevent interference.	IT	The vehicle designs have been inspected to ensure recovery electronics are sufficiently isolated from other electronics, and testing has validated the reliability of the designs.	All	V
3.14.1.	The recovery system altimeters shall be physically located in a separate compartment within the vehicle from any other radio frequency transmitting device and/or magnetic wave producing device.	The Recovery system components shall all be housed in an independent avionics bay and shielded from all other onboard electronics to prevent interference.	IT	The vehicle designs have been inspected to confirm the altimeters are in a separate compartment, and testing has validated they are shielded from wave producing devices.	SS, SC	V
3.14.2.	The recovery system electronics will be shielded from all onboard transmitting devices to avoid inadvertent excitation of the recovery system electronics.	The Recovery system components shall all be housed in an independent avionics bay and shielded from all other onboard electronics to prevent interference.	ID	The vehicle designs have been inspected to ensure proper isolation and shielding, and the Vehicle Demonstration flight validated reliability.	All	V
3.14.3.	The recovery system electronics shall be shielded from all onboard devices which may generate magnetic waves (such as generators, solenoid valves, and Tesla coils) to avoid inadvertent excitation of the recovery system.	The Recovery system components shall all be housed in an independent avionics bay and shielded from all other onboard electronics to prevent interference.				
3.14.4.	The recovery system electronics shall be shielded from any other onboard devices which may adversely affect the proper operation of the recovery system electronics.	The Recovery system components shall all be housed in an independent avionics bay and shielded from all other onboard electronics to prevent interference.				
NASA Payload Experiment Requirements		Design Requirements	Method	Verification Details	Team Member(s)	Status
Req #	Description					
4.1.	Teams are tasked with designing, building, and flying a STEMnaut flight capsule capable of safely retaining four STEMnauts and transmitting, via radio frequency, relevant rocket and STEMnaut landing site data to a NASA-owned receiver located at the launch site. STEMnauts are physical representations of the crew on-board the rocket. The method(s)/design(s) utilized to complete the payload mission shall be at the team's discretion and will be permitted so long as the designs are deemed safe, obey FAA and legal requirements, and adhere to the intent of the challenge. NASA reserves the right to require modifications to a proposed payload.	The team shall design, build, and fly a STEMnaut flight capsule to complete the 2025 NASA Student Launch Competition challenge.	ADIT	Analysis, demonstration, inspection, and testing have all been undertaken in order to satisfy the core NASA mission requirements. Further, the payload was demonstrated to complete the NASA mission during the Vehicle Demonstration flight.	All	V
4.2.	STEMCRaFT Mission Requirements:					
	Mission requirements are met as follows.					
4.2.1.	Teams shall choose a minimum of 3 pieces of data from the below list to a maximum of 8 to transmit to the NASA receiver. Temperature of landing site • Apogee reached • Battery check/power status • Orientation of on-board STEMnauts • Time of landing • Maximum velocity • Landing velocity, G-forces sustained • Calculated STEMnaut crew survivability	The team shall implement the data collection and transmission systems to collect and transmit a minimum of 3 pieces of data from the NASA-provided list.	ADIT	Analysis, demonstration, inspection, and testing has been performed to ensure completion of the data collection and RF transmission mission. Further, the payload was demonstrated to successfully collect and transmit data during the Vehicle Demonstration flight.	All	V
4.2.2.	The payload shall not have any protrusions from the vehicle prior to apogee that extends beyond a quarter inch exterior to the airframe.	The payload shall be designed such that it can be stowed entirely within the nose cone and payload bay.	I	The payload lead and vehicle lead have inspected the payload design to confirm that no protrusions exist.	RN, SS	V
4.2.3.	Payload shall transmit on the 2-M band. A specific frequency shall be given to the teams later. NASA shall use the FTM-300DR transceiver.	The RF transmission system shall be designed such that it is capable of transmitting on the designated band.	DT	The RF transmission system has been thoroughly tested on the ground and demonstrated on the subscale and Vehicle Demonstration flights.	PC	V

4.2.4.	All transmissions shall start and stop with team member call sign.	The transmission signals shall be configured to begin and end with the team member call sign.	DT	The transmission signals have been confirmed to contain the call sign through ground testing, payload drop testing, and the Vehicle Demonstration flight.	PC	V
4.2.5.	Teams shall submit a list of what data they will attempt to transmit by NASA receiver by March 17.	The team shall finalize the data selections no later than March 17	D	The electrical lead has finalized the data list, and the team lead confirmed the information was sent to NASA prior to March 17.	PC, IM	V
4.2.6.	Teams shall transmit with a maximum of 5W and transmissions shall not occur prior to landing.	The transmission system shall be designed such that its power usage does not exceed 5W nor begin transmitting prior to landing.	DT	Thorough testing has been conducted to verify power usage stays within the specified limit; the Vehicle Demonstration flight confirmed that transmission does not occur prior to landing.	PC	V
4.2.6.1.	Teams shall not transmit on the specified NASA frequency on launch day prior to landing.	The team shall refrain from transmitting on the specified NASA frequency on launch day prior to the payload mission occurring.	D	The team will verify this requirement by not transmitting on the NASA frequency at any time on launch day prior to landing.	All	NV
4.3.	General Payload Requirements:					
4.3.1.	Black Powder and/or similar energetics are only permitted for deployment of in-flight recovery systems. Energetics will not be permitted for any surface operations.	Energetics shall only be used for deployment of launch vehicle recovery systems.	I	The team has inspected system designs to ensure energetics are used in the proper manner.	All	V
4.3.2.	Teams shall abide by all FAA and NAR rules and regulations.	The team shall follow all FAA and NAR rules and regulations, overseen by the Student Safety Officer.	I	The team has continuously self-inspected procedures to confirm they abide by regulations.	All	V
4.3.3.	Any experiment element that is jettisoned during the recovery phase shall receive real-time RSO permission prior to initiating the jettison event, unless exempted from the requirement at the CDR milestone by NASA.	The lander separation mechanism shall be designed such that it receives a manual release signal following RSO approval.	DT	The team has demonstrated through drop tests that the lander separation mechanism executes following manual receipt of the separation signal. The team will confirm separation from the rocket during the Payload Demonstration flight.	All	IP
4.3.4.	Unmanned aircraft system (UAS) payloads, if designed to be deployed during descent, shall be tethered to the vehicle with a remotely controlled release mechanism until the RSO has given permission to release the UAS.	The payload shall be structurally retained to the vehicle.	I	The team has inspected the payload to ensure it is retained to the vehicle prior to separation and has proven retention through the Vehicle Demonstration flight.	All	V
4.3.5.	Teams flying UASs will abide by all applicable FAA. regulations, including the FAA's Special Rule for Model Aircraft (Public Law 112-95 Section 336; see https://www.faa.gov/uas/faqs).	The team shall follow all FAA and NAR rules and regulations, overseen by the Student Safety Officer.	I	The team has continuously self-inspected procedures to confirm they abide by regulations.	All	V
4.3.6.	Any UAS weighing more than .55 lbs. shall be registered with the FAA and the registration number marked on the vehicle.	The team shall register the lander with the FAA and mark the registration number on the vehicle.	I	The team has inspected that the payload is registered and the number is marked.	All	V
NASA Safety Requirements						
Req #	Description	Design Requirements	Method	Verification Details	Team Member(s)	Status
5.1.	Each team will use a launch and safety checklist. The final checklists will be included in the FRR report and used during the Launch Readiness Review (LRR) and any Launch Day operations.	The team shall develop a launch and safety checklist for FRR and launch.	I	The team has inspected all launch and safety procedures and ensured proper checklists are developed for FRR. The team lead will ensure these checklists are used for LRR.	IM	IP
5.2.	Each team shall identify a student safety officer who will be responsible for all items in section 5.3.	The team shall identify a student safety officer.	I	The team has designated Zachary Stutsman as the Student Safety Officer.	IM	V
5.3.	The role and responsibilities of the safety officer will include, but are not limited to:	Verifications are documented for each requirement below.				
5.3.1.	Monitor team activities with an emphasis on safety during:					
5.3.1.1.	Design of vehicle and payload	The Student Safety Officer shall ensure designs comply with NASA, local, state, NAR, and FAA rules and regulations.	I	The team has self-inspected its practices to ensure all rules and regulations are followed, as enforced by the SSO and team lead.	ZS, IM	V
5.3.1.2.	Construction of vehicle and payload components	The Student Safety Officer shall ensure construction protocols are written and followed during construction.	I	The team self-inspected its practices to ensure all construction protocols were followed, as enforced by the SSO and team lead.	ZS, IM	V
5.3.1.3.	Assembly of vehicle and payload	The Student Safety Officer shall ensure assembly protocols are written and followed during assembly.	I	The team has self-inspected its practices to ensure all assembly protocols are followed, as demonstrated during the Vehicle Demonstration flight.	ZS, IM	V
5.3.1.4.	Ground testing of vehicle and payload	The Student Safety Officer shall ensure that ground testing procedures are written and followed for all ground testing.	I	The team has self-inspected its practices to ensure all ground testing protocols are followed.	ZS, IM	V
5.3.1.5.	Subscale launch test(s)	The Student Safety Officer shall produce subscale launch checklists and ensure procedures are followed.	I	The team has self-inspected its practices to ensure all subscale testing protocols were followed, with checklists verified for launches.	ZS, IM	V
5.3.1.6.	Full-scale launch test(s)	The Student Safety Officer shall produce full-scale launch checklists and ensure procedures are followed.	I	The team self-inspect its practices to ensure all full-scale testing protocols were followed for the Vehicle Demonstration flight.	ZS, IM	V
5.3.1.7.	Competition Launch	The Student Safety Officer shall produce and compile all launch day checklists and ensure procedures are followed during launch.	I	The team will self-inspect its practices to ensure all competition testing protocols are followed, with checklists verified for launches.	ZS, IM	NV
5.3.1.8.	Recovery activities	The Student Safety Officer shall develop a recovery checklist and ensure procedures are followed during launch.	I	The team self-inspected its practices to ensure all recovery protocols were followed for the subscale and Vehicle Demonstration flights.	ZS, IM	V

5.3.1.9.	STEM Engagement Activities	The Student Safety Officer shall communicate with educational outreach members to ensure STEM engagement activities follow proper safety protocol.	I	Safety procedures have been inspected prior to all engagement activities.	ZS, IM	V		
5.3.2.	Implement procedures developed by the team for construction, assembly, launch, and recovery activities.	The Student Safety Officer shall develop a comprehensive set of operational protocols to ensure safety in regards to personal protective equipment, operational checklists, and testing overview.	I	The safety procedures have been followed by the team and enforced and updated by the Student Safety Officer.	ZS, IM	V		
5.3.3.	Manage and maintain current revisions of the team’s hazard analyses, failure modes analyses, procedures, and MSDS/chemical inventory data.	The Student Safety Officer shall maintain a safety binder in the lab and digitally containing all required safety documentation.	I	The safety binder has been inspected regularly to ensure all information is present and relevant.	ZS, IM	V		
5.3.4.	Assist in the writing and development of the team’s hazard analyses, failure modes analyses, and procedures.	The Student Safety Officer shall lead the development of the team’s hazard analyses, failure modes analyses, and procedures.	I	Hazard and failure analyses and procedures have been inspected by the team and lead by the Student Safety Officer.	ZS, IM	V		
5.4.	During test flights, teams will abide by the rules and guidance of the local rocketry club’s RSO. The allowance of certain vehicle configurations and/or payloads at the NASA Student Launch does not give explicit or implicit authority for teams to fly those vehicle configurations and/or payloads at other club launches. Teams should communicate their intentions to the local club’s President or Prefect and RSO before attending any NAR or TRA launch.	The Student Safety Officer shall communicate with the local NAR organization and rocketry mentor to ensure compliance with pertinent rules and regulations.	I	Procedural and system-wide inspections have taken place to ensure the team abides by the rules and regulations of NAR and TRA launches.	ZS, IM	V		
5.5.	Teams will abide by all rules set forth by the FAA.	The Student Safety Officer shall ensure compliance with FAA rules are regulations.	I	Inspections have taken place to confirm that procedures and mission profile abide by FAA rules and regulations.	ZS, IM	V		
NASA Final Flight Requirements		Design Requirements	Method	Verification Details	Team Member(s)	Status		
Req #	Description							
6.1.	NASA Launch Complex							
6.1.1.	Teams are not permitted to show up at the NASA Launch Complex outside of launch day without permission from the NASA management team.	The team shall abide by all NASA Launch Complex rules and regulations.	I	The team lead will continuously inspect activities throughout the Launch Week to verify compliance with rules and regulations.	IM	NV		
6.1.2.	Teams shall complete and pass the Launch Readiness Review conducted during Launch Week.							
6.1.3.	The team mentor shall be present and oversee rocket preparation and launch activities.			The team lead will continuously inspect activities throughout the Launch Week to verify compliance with rules and regulations.				
6.1.4.	The scoring altimeter shall be presented to the NASA scoring official upon recovery. The scoring altimeter shall be one of the altimeters used for recovery events.							
6.1.5.	Teams may launch only once. Any launch attempt resulting in the rocket exiting the launch pad, regardless of the success of the flight, will be considered a launch. Additional flights beyond the initial launch, will not be scored and will not be considered for awards.	The team shall abide by all NASA Launch Complex rules and regulations.						
6.2.	Commercial Spaceport Launch Site							
6.2.1.	The launch shall occur at a NAR or TRA sanctioned and insured club launch. Exceptions may be approved for launch clubs who are not affiliated with NAR or TRA but provide their own insurance, such as the Friends of Amateur Rocketry. Approval for such exceptions shall be granted by NASA prior to the launch.	The team shall comply with all Commercial Spaceport Launch Site guidelines.	I	The team lead will actively monitor team activities during Launch Week to ensure adherence to all guidelines.	IM	NV		
6.2.2.	Teams shall submit their rocket and payload to the launch site Range Safety Officer (RSO) prior to flying the rocket. The RSO shall inspect the rocket and payload for flight worthiness and determine if the project is approved for flight. The local RSO shall have final authority on whether the team’s rocket and payload may be flown.							
6.2.3.	The team mentor shall be present and oversee rocket preparation and launch activities.							
6.2.4.	BOTH the team mentor and the Launch Control Officer shall observe the flight and report any off- nominal events during ascent or recovery on the Launch Certification and Observations Report.							
6.2.5.	The scoring altimeter shall be presented to BOTH the team’s mentor and the Range Safety Officer. The scoring altimeter shall be one of the altimeters used for recovery events.	The team shall comply with all Commercial Spaceport Launch Site guidelines.	I	The team lead will actively monitor team activities during Launch Week to ensure adherence to all guidelines.	IM			
6.2.6.	The mentor, the Range Safety Officer, and the Launch Control Officer must be three separate individuals who must ALL complete the applicable sections of the Launch Certification and Observations Report. The Launch Certification and Observations Report document will be provided by NASA upon completion of the FRR milestone and shall be returned to NASA by the team mentor upon completion of the launch.							
6.2.7.	The Range Safety Officer and Launch Control Officer certifying the team’s flight shall be impartial observers and shall not be affiliated with the team, individual team members, or the team’s academic institution.							
6.2.8.	Teams may launch only once. Any launch attempt resulting in the rocket exiting the launch pad, regardless of the success of the flight, will be considered a launch. Additional flights beyond the initial launch will not be scored and will not be considered for awards.							

7.2.2 VADL Derived Requirements

As with the NASA requirements, VADL has also updated the verification plans and statuses for the team-derived requirements. Provided in the first table is a description and justification for each VADL requirement. The second table contains the resulting design requirement, relevant report section, verification method(s), verification details, responsible team member(s), and status. Important to note is that at this point, all team-derived requirements have been verified, with the exception of VADL 4.1.9 which mandates that the payload be capable of completing the NASA mission in both jettisoned and non-jettisoned configurations. While this has been proven between payload drop tests and the full-scale Vehicle Demonstration flight for jettisoned and non-jettisoned cases, respectively, the Payload Demonstration flight will verify the full jettisoned sequence for rocket flight.

1. VADL General Requirements		
Req #	Description	Justification
1.1 Budgetary Requirements		
1.1.1	The team shall maintain an itemized list of all expenditures	Having this updated list of expenses will keep the team within budget
1.1.2	The president and treasurer shall review the budget and all expenses	Review by multiple people will prevent any budget mishaps
1.1.3	The team shall keep 10% of the budget in reserve for emergency expenses	Maintaining a healthy amount of contingency funding is necessary for unforeseen expenditures, such as an expensive component failing unexpectedly.
1.1.4	All purchases shall be reviewed and approved by team leadership	Allows for time to review lab inventory before purchases, minimize the risk of unnecessary purchases
1.2 Social Media Requirements		
1.2.1	The team shall post to the VADL instagram at least once a week	A social media presence is necessary for proper engagement with the greater outreach community
1.2.2	Key news and major updates shall be cross-posted to the VADL twitter account	Twitter is a key resource used by the University, NASA, and Academia in general, so important updates must be viewable by those audiences
1.3 STEM Engagement Requirements		
1.3.1	VADL shall conduct STEM outreach and engagement events both in the Nashville community and across the country	Engaging with the community, both in Nashville and elsewhere, is a key tenet of VADL's outreach program
1.3.2	VADL shall reach a minimum of 500 participants	Outreach is of the utmost importance to VADL, so the team wishes to engage at least twice the total designated by NASA requirement 1.4
1.3.3	VADL shall have at least 8 STEM engagement events over the lifetime of this project cycle	Consistent STEM engagement over the project lifecycle is key to any outreach program
1.3.4	Every team member shall participate in at least one STEM outreach and engagement event	STEM engagement is a responsibility for all team members, so that outreach participants will learn about all aspects of the team's project
1.3.5	All team members shall lead one engagement event in their hometown over Thanksgiving or Winter break	VADL would like to impact not just the surrounding TN communities, but give back to the teachers and educators who helped the team along their educational career
1.4 Quality Control Requirements		
1.4.1	Any technical writing shall be proofread by at least two other team members before submission	Proofreading will maintain quality of any written material
1.4.2	All requirement verification that demands testing shall be done according to a rigorous testing protocol that clearly defines success parameters	Rigorous testing is the only way to ensure the verification of the requirements that VADL has imposed on itself
2. VADL Vehicle Requirements		
Req #	Description	Justification
2.1 System-Level Requirements		
2.1.1	The vehicle shall reach an apogee within 5% of the predicted apogee of 4200 ft.	The NASA-specified apogee range is between 4000 and 6000 ft, and flight simulations indicate a projected apogee of 4200 ft.
2.1.2	The launch vehicle shall have a static stability margin between 2.0 and 2.5 calibers on the launch pad.	The launch vehicle static stability margin must be large enough to ensure a stable launch, but not too large so that it causes overstable flight and excessive pitch of the vehicle into the wind.
2.1.3	Both the full scale and subscale launch vehicles shall have an average Thrust-to-Weight ratio of 5.5 minimum.	A higher thrust to weight ratio ensures that the rocket leaves the launch rail with an aerodynamically stable velocity.
2.1.4	The launch vehicle shall be designed and fabricated to be recoverable and reusable, meaning that it can be re-launched within 2 hours after recovery.	The team wants to take sustainability into account; this would aid especially for large scale applications where a reusable vehicle would save costs and materials.
2.2 Structural Requirements		
2.2.1.1	The vehicle shall be manufactured with a lightweight and durable material.	The vehicle must withstand launch and landing loads and in order to do so must be manufactured with durable material.
2.2.1.2	The nose cone shall withstand the maximum landing kinetic energy expected for the lander.	The nose cone must protect the STEMnauts and the rest of the payload components during landing.
2.2.1.3	The fins shall avoid fluttering.	Fin flutter must be prevented, as it can cause structural failures.
2.2.1.4	LiPo batteries shall be properly stored and protected from excessive damage and heat.	LiPo batteries can catch fire if they are damaged/punctured, so it's important to properly protect them from potential hazards.
2.2.1.5	The motor shall provide sufficient thrust to achieve the required apogee designated in Req. 2.1.1	Achieving the projected apogee is an important element of mission success.
2.2.1.6	All structural components shall withstand expected temperatures on the launch pad.	Structural components must be designed to withstand launch pad temperatures so that they do not deform and cause structural damage.
2.2.2	All structural components shall have a safety factor of at least 2.5.	Structural components must be designed with large safety factors in order to mitigate any unpredicted errors that may occur.
2.3 Flight Analysis Requirements		
2.3.1	Flight Simulations shall be conducted that shall predict the flight trajectory at winds from omph to 20mph and launch angles from 0 to 10 degrees	Full flight simulations verify that all flight requirements will be met regardless of environmental conditions
2.3.2	Simulations shall benchmark the performance of various motors	These motor simulations will allow the team to choose the best motor for flight that also adheres to NASA's launch and flight requirements
2.3.3	Analysis from commercial software shall verify the vehicle's SSM	Redundancy with regards to the SSM will ensure flight safety. Several softwares such as RockSim and OpenRocket are valuable alternatives to MATLAB program
2.3.4	Descent time and drift calculations will be calculated for varying wind speeds and deployment heights	Necessary to determine feasible range of parachute sizes and deployment times
2.3.5	Flight simulations in the case of denied jettison shall be conducted.	Jettison is not guaranteed; simulations are necessary to ensure compliance against NASA requirements in this context.
2.3.6	Analysis performed in MATLAB shall track and simulate pitch angle throughout the duration of flight	Pitch angle is a difficult flight parameter to predict, and a data-informed prediction will improve accuracy of the secondary payload algorithm
2.3.7	The launch thrust-to-weight ratio shall be estimated with simulation data and verified after construction of the vehicle	Determining a value for the thrust-to-weight ratio will confirm the team has selected the right motor and that the vehicle will achieve the necessary velocity for launch rail exit and the necessary acceleration to reach the estimated apogee
2.3.8	Massing estimates shall be continually updated as designs progress, and verified once component masses are known	Accurate massing is essential to achieve accurate SSM and flight trajectory analysis
2.4 Aerodynamic Analysis Requirements		
2.4.1	CFD shall be conducted to verify the drag coefficient values yielded by OpenRocket	The drag coefficient dictates the descent time and landing energy, and it should be well understood
2.4.2	Thorough CFD analysis shall be performed at various flight speeds and at different angles of attack	The axial and normal coefficients of drag play a major role in determining flight dynamics and altitude performance
2.4.3	OpenRocket analysis shall be used to optimize design of the boat tail.	Proper boat tail design is essential for the aerodynamic performance of the rocket.
2.4.4	OpenRocket analysis shall be used to optimize design of the fins.	Proper fin design is necessary for ensuring the stability of the vehicle.
3. VADL Recovery Requirements		
Req #	Description	Justification
3.1	There shall exist a primary and backup charge for both main and drogue deployment	A larger backup charge will be used to ensure that recovery hardware deploys to avoid catastrophic recovery failure
3.2	In the event of failed jettison, the descent velocity of the vehicle shall not exceed 20 fps upon landing.	Parachute selections should be designed to ensure soft landing even in case of jettison failure.
3.3	Shock cords shall be sown, as opposed to being tied in knots	Knots create unnecessary stress concentration and additional points of failure
3.4	Parachutes shall be wrapped in fireproof fabric such that they can easily be dislodged and inflated	Fire and heat can create holes or melt parts together given the synthetic construction material. Either poses a massive hazard to recovery system performance
3.5	All recovery electronics shall be shielded from RF interference in an independent section	RF interference can impede the proper functionality of altimeters and cause premature / delayed deployment. Extension of NASA 3.8
3.6	The shock cords shall be configured such that no independent sections have to roll over and reconfigure during main deployment	Unnecessary reorienting and maneuvering increases the chance of collision between sections and dimensions

3.7	Independent altimeters shall be responsible for the primary and backup charges	In case of failure, a redundant altimeter be able to activate charges in both the main and drogue bays
3.8	A recoverable, electrically independent 'black box' shall be included that is intended to record all flight data, as well as transmit GPS data	An independent, redundant system is useful to guarantee that flight data is recovered regardless of the performance of the primary payload
3.9	Recovery system shall utilize firebails to prevent zippering of the rocket body due to tensioning of the shock cord	To comply with NASA requirement 2.3, the vehicle should not incur severe / irreparable damage during deployment
3.10	Batteries used for recovery electronics shall be tested prior to launch day, and backups brought to the field	Mission success is required at all launches, therefore thorough power supply testing and options will be necessary prior to any rocket ignition
3.11	The terminal descent velocity under drogue parachute of the rocket shall not exceed 115 fps.	Substantially high drogue terminal velocity places dangerous stress to rocket structures and shock cords during main parachute deployment.
4. VADL Payload Requirements		
Req #	Description	Justification
4.1 General Payload Requirements		
4.1.1	The payload shall provide a retention method for four STEMnauts.	To comply with NASA requirement 4.1, it is necessary to provide a retention method for the STEMnauts
4.1.2	The payload shall land with a velocity no greater than 24 fps.	To meet survivability requirements for the STEMnauts, it is necessary to limit the landing velocity
4.1.3	The payload shall have an antenna capable of transmitting flight data via RF	To transmit data via RF, as imposed by NASA requirement 4.1, an antenna is needed
4.1.4	The payload shall be capable of being prepared in the field in less than 45 min	Flight preparation within a two hour window necessitates an efficient payload preparation in order to ensure enough time is available for full vehicle assembly
4.1.5	The team shall develop software unit tests for all software integrated systems	Integration of software into the payload is both mission critical and involves many potential failure points, so rigorous testing is required
4.1.6	The Dynamic Response Index (DRI) for the STEMnauts shall be no greater than 17.7 in the vertical axis.	To comply with NASA requirement 4.1, it is necessary to provide a metric for determining whether or not the STEMnauts survive flight.
4.1.7	The payload shall land in an upright orientation.	To maximize the chances for RF transmission, orientation of the transmission antenna requires a known payload landing orientation
4.1.8	The payload electronics shall be able to withstand launch acceleration and landing impact	Electronics must be able to withstand these forces for data collection and transmission to be achieved
4.1.9	The payload shall be capable of completing the NASA mission in both jettisoned and non-jettisoned configurations	The team must ensure that the mission can be completed even if RSO approval to jettison is not granted
4.2 Lander Separation Mechanism		
4.2.1	The LSM shall disengage the lander from the main rocket body within 1 second of reaching the target altitude and receiving command signal.	Timely separation is critical to ensure that the lander can safely deploy its parachute and legs
4.2.2	The LSM shall be able to receive RF transmission	Approval for payload release requires the ability to communicate with the mechanism in flight, necessitating RF capability.
4.2.3	The mechanism shall be capable of operating at altitudes between 200 and 400 ft AGL	Variable altitude capabilities are necessary to determine the optimal release point of the payload
4.2.4	The mechanism shall be capable of withstanding maximum expected shock loads of 32G.	The mechanism must survive launch and flight conditions to ensure successful operation.
4.3 Leg Deployment Mechanism		
4.3.1	The legs shall withstand maximum anticipated landing loads.	Legs must be able to withstand the maximum possible landing force in order to ensure mission success.
4.3.2	The legs shall deploy passively upon lander separation.	Passive deployment is preferable to an active mechanism to limit mechanical complexity and remove potential failure points.
4.3.3	The legs shall be packaged within the body of the payload.	Packaging the legs within the payload body decreases risk of entanglement with parachute.
4.4 Parachute Detachment Mechanism		
4.4.1	The parachute shall detach within 1 second of detected landing	Parachute must be able to detach quickly in order to prevent dragging or toppling of payload.
4.4.2	The detachment latch shall withstand the maximum acceleration experienced during flight.	The latch must stay locked throughout the entire flight to prevent premature deployment
4.5 Antenna Extension Mechanism		
4.5.1	The AEM shall extend the antenna 18"	The antenna requires a 0.5m/18" transmission length
4.5.2	The AEM shall be capable of extending even if partially obstructed	The antenna needs to achieve the required clearance regardless of obstructions in its path
4.5.3	The AEM shall not interfere with the RF transmission	The antenna must transmit collected data across the field to NASA ground station. Any interference in the RF system from the use of non-transmitting metal within the RF system must be minimized.
4.5.4	The AEM locking mechanism shall withstand maximum expected loading during flight	The AEM must not extend until the payload has landed
4.5.5	The AEM shall be capable of extending regardless of payload jettison, and have at least 1 antenna extending away from the ground regardless of landing orientation	RF transmission is required regardless of whether or not the payload is jettisoned from the launch vehicle. Without jettison and subsequent leg deployment, orientation is not guaranteed.
4.6 RF Transmission System		
4.6.1	Transmission shall occur given any landing orientation.	Data transmission is the primary mission objective and must occur regardless of landing orientation
4.6.2	The antenna system shall be able to reliably transmit up to a distance of 1 mile.	The maximum predicted drift distance is 2500ft. Applying a safety factor of 2 yields approximately 1 mile as the maximum transmission distance
4.6.3	RF transmission shall cease 5 minutes after landing	NASA has imposed a time limit of 5 minutes of RF transmission after landing
4.7 Data Collection Sensors		
4.7.1	Data sampling frequency shall be 160 Hz (TBR) throughout data collection.	Data sampling frequency must be high enough to collect all relevant data and is required for calculations
4.7.2	The sensors shall be able to detect landing	Provides data to be transmitted and prevents transmission from occurring before landing
4.7.3	The sensors shall be able to determine altitude	Altitude is needed for transmission of apogee value, as well as for the landing detection sequence
4.7.4	The sensors shall be able to detect apogee	Required to send data selected by VADL
4.7.5	The sensors shall be able to detect orientation	
4.7.6	The sensors shall be able to detect acceleration	
4.7.7	The sensors shall be able to detect battery voltage	
4.7.8	The software shall be able to calculate velocity	Software velocity calculation is preferable to sensor reading because it reduces the number of instruments required on the payload
4.7.9	The software shall be able to calculate survivability	Verifying STEMnaut survival is required for successful mission completion
5. VADL Safety Requirements		
Req #	Description	Justification
5.1 General Safety Requirements		
5.1.1	Team members shall wear proper PPE as indicated in VADL safety protocols.	Team safety depends on the proper use of PPE
5.1.2	Team members shall clean up after themselves and engage in collective lab cleanings when deemed necessary by the Student Safety Officer.	A clean lab promotes the safety and good organization bolsters team efficiency.
5.1.3	Team members shall participate in mandatory weekly safety briefings delivered by the Student Safety Officer.	Safety briefings help maintain safe practices throughout the year and are an important part of the VADL commitment to safety.
5.1.4	Energetic material shall be stored in appropriate explosives magazines and shall be kept locked at all times.	The safety of team members and general community is reliant on proper storage of energetic material.
5.1.5	Team members shall only use equipment that they are trained to operate.	Proper equipment training helps to reduce preventable accidents.

VADL General Requirements		Design Requirements	Section	Verification Method	Verification Details	Team Member(s)	Status
Req #	Description						
1.1 Budgetary Requirements							
1.1.1	The team shall maintain an itemized list of all expenditures	The president and treasurer shall track all expenses incurred by the team in a spreadsheet	7.3.3	I	The expense spreadsheet has been regularly inspected to ensure it is up to date and compliant with the budget	IM, TS	V
1.1.2	The president and treasurer shall review the budget and all expenses	The president and treasurer shall discuss the budget and the expenses that must be made at various points during the project	7.3	I	The president and treasurer have regularly compared the expenses and budget to ensure that the team remains within the budget	IM, TS	V
1.1.3	The team shall keep 10% of the budget in reserve for emergency expenses	The team shall not spend all of its budget and shall maintain an emergency reserve of 10% of the total budget	7.3	I	The president and treasurer inspected the expenses on a weekly basis and inspected funding sources to ensure that adequate funding is available to maintain a 10% contingency margin	IM, TS	V
1.1.4	All purchases shall be reviewed and approved by team leadership	All team purchases shall go through the president and treasurer	7.3.3	I	All purchases have been inspected to confirm that they were signed off by both the president and the treasurer	IM, TS	V
1.2 Social Media Requirements							
1.2.1	The team shall post to the VADL instagram at least once a week	The outreach lead shall post to the VADL instagram at least three times a month	VADL Instagram	I	Team members with instagram accounts have followed the VADL instagram to ensure compliance with this requirement	SS	V
1.2.2	Key news and major updates shall be cross-posted to the VADL twitter account	The outreach lead shall post noteworthy news, updates, and accomplishments on the VADL twitter account	VADL Twitter	I	Team members with twitter accounts have followed the VADL twitter account to ensure compliance with this requirement	SS	V
1.3 STEM Engagement Requirements							
1.3.1	VADL shall conduct STEM outreach and engagement events both in the Nashville community and across the country	All team members shall participate in STEM outreach events in Nashville and in their hometowns. VADL as a whole shall reach at least 500 participants	N/A	I	The outreach lead has contacted organizations and maintained a regular schedule of events through a shared events spreadsheet	SS	V
1.3.2	VADL shall reach a minimum of 500 participants	The outreach lead shall track participation at all events to ensure that a minimum of 500 participants have been engaged	N/A	I	The outreach lead continuously inspected the number of participants reached to ensure that the minimum requirement was achieved	SS	V
1.3.3	VADL shall have at least 8 STEM engagement events over the lifetime of this project cycle	The outreach lead with the help of graduate student mentors shall schedule at least 8 STEM engagement events over the length of the project cycle	N/A	I	Team members kept track of STEM engagement events to ensure VADL had at least 8 STEM engagement events	All	V
1.3.4	Every team member shall participate in at least one STEM outreach and engagement event	Team members shall sign up for events via the event spreadsheet to ensure they have met the minimum requirement for participation	N/A	I	The outreach lead tracked who went to each event to confirm that everyone attended at least one event	SS	V
1.3.5	All team members shall lead one engagement event in their hometown over Thanksgiving or Winter break	Team members shall make the necessary arrangements to organize and lead the event	N/A	I	The outreach lead inspected that all team members conducted outreach events in their hometowns	SS	V
1.4 Quality Control Requirements							
1.4.1	Any technical writing shall be proofread by at least two other team members before submission	Team members shall proofread material that they did not write such that all writing is proofread at least twice	N/A	I	A chart was used for each report to track which sections were proofread and by whom they were proofread	All	V
1.4.2	All requirement verification that demands testing shall be done according to a rigorous testing protocol that clearly defines success parameters	The team shall create a testing plan that defines each test and its success criteria. Tests and verification shall be performed according to NASA Technical Standard 5001-B	7.1	I	The testing plan was updated and proper documentation of the tests were kept to ensure compliance with this requirement	All	V
VADL Vehicle Requirements		Design Requirements	Section	Verification Method	Verification Details	Team Member(s)	Status
Req #	Description						
2.1 System-Level Requirements							
2.1.1	The vehicle shall reach an apogee within 5% of the predicted apogee of 4200 ft.	Team members shall model the vehicle and its flight in MATLAB, OpenRocket, and RockSim simulations to verify the apogee	3.3	AD	As the design of the rocket changed, new simulations were performed to verify its flight performance. Compliance with this requirement was also verified through the full-scale Vehicle Demonstration flight.	SS, SH	V
2.1.2	The launch vehicle shall have a static stability margin between 2.0 and 2.5 calibers on the launch pad.	Vehicle mass distribution and fin design shall be completed with the focus of maintaining a SSM in this range.	3.1	AD	Simulations were performed under various constraints to optimize the design and achieve a reasonable SSM within the expected range. Compliance with this requirement was also verified through the full-scale Vehicle Demonstration flight.	SS, SH	V
2.1.3	Both the full scale and subscale launch vehicles shall have an average Thrust-to-Weight ratio of 5.5 minimum.	Team members shall use OpenRocket simulations to conduct trade studies for selected motors.	3.1	AD	Multiple simulation platforms and trade studies were used to determine the best motor selection to optimize thrust to weight ratio. Compliance with this requirement was verified through the full-scale Vehicle Demonstration flight.	SS, SH	V
2.1.4	The launch vehicle shall be designed and fabricated to be recoverable and reusable, meaning that it can be re-launched within 2 hours after recovery.	Team members shall ensure the material durability and reusability and analyze them in datasheets and simulations.	3.2	AD	Simulations were performed under different wind conditions to ensure that the vehicle complied with this requirement under any flyable conditions (winds up to 20 MPH). Compliance with this requirement was verified through the Vehicle Demonstration flight.	SS, SH	V
2.2 Structural Component Requirements							
2.2.1.1	The vehicle shall be manufactured with a lightweight and durable material.	The vehicle airframe shall be manufactured using fiberglass.	3.1.3	AT	Massing analysis has been performed to ensure the vehicle is sufficiently lightweight, and load testing has been completed to ensure the airframe can withstand expected loading.	SS, SC	V
2.2.1.2	The nose cone shall withstand the maximum landing kinetic energy expected for the lander.	The nose cone shall be manufactured using fiberglass cloth laid up on a 3D printed mold.	3.1.3	T	After fabrication, drop tests were conducted to ensure durability with added mass to mimic the landing energy.	SS, SC	V
2.2.1.3	The fins shall avoid fluttering.	The fins shall be sized such that fin flutter is avoided.	3.1.4	AD	Calculations were performed to ensure fin flutter is avoided. After fabrication, the team conducted a Vehicle Demonstration flight to verify the reliability and structural integrity of the fins.	SS, SC	V
2.2.1.4	LiPo batteries shall be properly stored and protected from excessive damage and heat.	Secure, protective structures for LiPo batteries shall be designed.	3.2.4	IT	Casings and hardware were thoroughly inspected, and tests were conducted to ensure durability.	SS, PC	V

2.2.1.5	The motor shall provide sufficient thrust to achieve the required apogee designated in Req. 2.1.1	The Cesaroni L1720 shall be used as the competition motor.	3.3	AD	MATLAB simulations were conducted to confirm motor performance. Compliance with this requirement was verified through full-scale Vehicle Demonstration flight.	SS, SH	V
2.2.1.6	All structural components shall withstand expected temperatures on the launch pad.	Structural components shall be made with aluminum or 3D printed materials that can withstand temperatures up to at least 100F.	3.1	AT	Material thermal properties were analyzed, and components were tested in a thermal environment at temperatures exceeding that expected on launch day.	SS, SC	V
2.2.2	All structural components shall have a safety factor of at least 2.5.	Structural components shall be designed to fail at least 2.5x the maximum load experienced during main parachute deployment.	3.1	AD	FEA was conducted to analytically verify the safety factor of structural components. Performance was demonstrated through the full-scale Vehicle Demonstration flight.	SH	V
2.3 Flight Analysis Requirements							
2.3.1	Flight Simulations shall be conducted that shall predict the flight trajectory at winds from 0mph to 20mph and launch angles from 0 to 10 degrees	Members shall simulate flight under various conditions using MATLAB	3.3	A	MATLAB simulations have verified the flight trajectory will meet all flight requirements under any flyable condition	SH	V
2.3.2	Simulations shall benchmark the performance of various motors	Members shall conduct a trade study and simulate the performance of different motors with the full-scale rocket vehicle	3.3	A	MATLAB simulations have verified that the motor choice will satisfy flight requirements	SH	V
2.3.3	Analysis from commercial software shall verify the vehicle's SSM	The team shall analyze the vehicle with software to determine its SSM	3.3	A	Outside software and MATLAB simulations have verified the calculated SSM to ensure flight stability	SH	V
2.3.4	Descent time and drift calculations will be calculated for varying wind speeds and deployment heights	The team shall analyze the flight trajectory under different flight scenarios for different mass distributions	3.3	A	MATLAB simulations have verified maximum drift requirement of 1750 ft is met	SH	V
2.3.5	Flight simulations in the case of denied jettison shall be conducted.	Members shall simulate the flight scenario where payload jettison is denied	3.3	A	MATLAB simulations have determined flight performance in the case that payload jettison is denied	SH	V
2.3.6	Analysis performed in MATLAB shall track and simulate pitch angle throughout the duration of flight	The team shall estimate pitch angles to inform apogee prediction	3.3	A	MATLAB simulations of various pitch angles have verified the reliability of the team's predicted apogee	SH	V
2.3.7	The launch thrust-to-weight ratio shall be estimated with simulation data and verified after construction of the vehicle	MATLAB simulations shall predict thrust-to-weight ratio before verification during construction	3.3	AI	MATLAB simulations used vehicle mass and motor data to determine thrust-to-weight ratio; proper massing during construction validated results	SH	V
2.3.8	Massing estimates shall be continually updated as designs progress, and verified once component masses are known	The team shall keep up-to-date massing estimates that shall be used in all simulations	N/A	I	Proper documentation ensured the most accurate data was used during any calculations	SS, SC	V
2.4 Aerodynamic Analysis Requirements							
2.4.1	CFD shall be conducted to verify the drag coefficient values yielded by OpenRocket	The team shall create a CFD model of the entire vehicle	3.3.7	A	A CFD analysis has been performed and results tabulated and compared with expected values from OpenRocket	JL	V
2.4.2	Thorough CFD analysis shall be performed at various flight speeds and at different angles of attack	Members shall perform CFD analysis for various flight scenarios	3.3.7	A	A thorough CFD model has been analyzed to validate flight performance under any flyable condition	JL	V
2.4.3	OpenRocket analysis shall be used to optimize design of the boat tail.	The team shall utilize the OpenRocket optimizer tool to facilitate design of the boat tail.	3.1	A	The OpenRocket analysis has been thoroughly examined to ensure adequate boat tail design	SS, SC	V
2.4.4	OpenRocket analysis shall be used to optimize design of the fins.	The team shall utilize the OpenRocket optimizer tool to facilitate design of the fins.	3.1	A	The OpenRocket analysis has been thoroughly examined to ensure adequate fin design	SS, SC	V
VADL Recovery Requirements							
Req #	Description	Design Requirements	Section	Verification Method	Verification Details	Team Member(s)	Status
3.1	There shall exist a primary and backup charge for both main and drogue deployment	The team shall place independent blast charges inside the vehicle for separate parachute deployment events	3.2.3	IA	Inspection during vehicle construction ensured that separate charges were placed in their respective chambers; analysis prior to launch determined the size of the charges	SC	V
3.2	In the event of failed jettison, the descent velocity of the vehicle shall not exceed 20 fps upon landing.	The team shall select a drogue parachute such that the descent velocity is greater than 70 fps	3.3	AD	Simulations in MATLAB have determined descent velocity for various drogue parachute sizes; the subscale launch provided altimeter data to verify the simulation	SS, SH	V
3.3	Shock cords shall be sown, as opposed to being tied in knots	Team members shall sew all shock cords used in the vehicle	N/A	I	Inspection during vehicle construction verified that all shock cords are sown rather than tied in knots	SS	V
3.4	Parachutes shall be wrapped in fireproof fabric such that they can easily be dislodged and inflated	Team members shall use fireproof fabric when wrapping the parachutes during vehicle construction	N/A	I	Inspection of the recovery hardware ensured that fireproof fabric was used to wrap the parachutes	SS	V
3.5	All recovery electronics shall be shielded from RF interference in an independent section	The team shall shield the avionics bay with RF-opaque aluminum bulk plates	3.1.5	I	Inspection of the fabricated avionics bay confirmed the use of aluminum bulk plates to shield the avionics bay from the RF transmitters.	SS, SC	V
3.6	The shock cords shall be configured such that no independent sections have to roll over and reconfigure during main deployment	The team shall determine proper shock cord lengths and configurations such that main parachute deployment runs smoothly	N/A	I	Inspection of the assembled recovery hardware confirmed the proper selection of cord lengths.	SS	V
3.7	Independent altimeters shall be responsible for the primary and backup charges	The vehicle shall contain two independent altimeters for primary and backup blast charges	3.2.3	I	Inspection of the assembled avionics ensured the use of two independent altimeters.	SC	V
3.8	A recoverable, electrically independent 'black box' shall be included that is intended to record all flight data, as well as transmit GPS data	The team shall create an independent electrical system that shall record all flight data and transmit GPS data	3.2.5	I	Inspection of the assembled vehicle ensured that the GPS is electrically isolated from other systems	PC	V
3.9	Recovery system shall utilize fireballs to prevent zippering of the rocket body due to tensioning of the shock cord	Team members shall include fireballs on the shockcords	N/A	I	The assembled recovery hardware was inspected prior to flight to ensure the use and proper placement of fireballs	SS	V
3.10	Batteries used for recovery electronics shall be tested prior to launch day, and backups brought to the field	The team shall thoroughly test the batteries for any electronic equipment	7.1	T	Battery life testing was performed to verify the lifetimes of the batteries used on launch day	PC	V
3.11	The terminal descent velocity under drogue parachute of the rocket shall not exceed 115 fps.	The drogue parachute shall be selected such that the rocket descent velocity does not exceed 115 fps	3.3	AD	Flight simulations were conducted to verify the descent velocity, and data from the Vehicle Demonstration flight was reviewed to ensure compliance with this requirement	SH	V
VADL Payload Requirements							
Req #	Description	Design Requirements	Section	Verification Method	Verification Details	Team Member(s)	Status
4.1 General Payload Requirements							

4.1.1	The payload shall provide a retention method for four STEMnauts.	The payload shall contain a STEMnaut crew module to house the four STEMnauts	4.2.5	D	The requirement has been verified by demonstrating that the crew module has been built and integrated into the lander	BR	V
4.1.2	The payload shall land with a velocity no greater than 24 fps.	The payload shall descend under parachute, with sizing informed by analysis	4.1.3	AT	Descent simulations have been run to verify that the landing velocity does not exceed the imposed limit, and drop tests have been performed to empirically confirm landing velocity	SH	V
4.1.3	The payload shall have an antenna capable of transmitting flight data via RF	A quarter-wave ground plane antenna shall be designed and integrated into the lander to perform RF transmission	4.3.3	D	The requirement has been verified by demonstrating the payload's ability to transmit RF data through drop tests	PC, AK	V
4.1.4	The payload shall be capable of being prepared in the field in less than 45 min	In order to complete assembly of the rocket within two hours, the team shall be able to assemble the payload in the field in less than 45 min	N/A	D	The team has practiced preparing the payload and timed how long it takes to prepare in the field during full-scale payload drop tests and the Vehicle Demonstration flight	All	V
4.1.5	The team shall develop software unit tests for all software integrated systems	The team shall create robust software by developing software unit tests	4.3.1	D	Software unit tests have been performed to verify that all electronics and software are integrated properly and are functioning nominally	PC	V
4.1.6	The Dynamic Response Index (DRI) for the STEMnauts shall be no greater than 17.7 in the vertical axis.	The payload recovery system and leg damping shall be designed such that the DRI value is below the specified threshold	4.3.7	DA	Simulation and flight data has been analyzed to ensure that the STEMnauts do not experience a DRI greater than 17.7	SH, PC	V
4.1.7	The payload shall land in an upright orientation.	The lander shall be designed with a low CG and wide legs.	4.1.2, 4.2.2	D	Drop tests under varying conditions have been conducted to demonstrate that the payload repeatedly and reliably lands upright	ZS, RN	V
4.1.8	The payload electronics shall be able to withstand launch acceleration and landing impact	The electronics system shall be thoroughly ruggedized to ensure survival throughout flight	4.3.5	DT	Electronic connections have been checked in the lab as well as in the field post-launch. Payload operation and transmission of data have been analyzed to indicate sufficient ruggedization of electronics	PC	V
4.1.9	The payload shall be capable of completing the NASA mission in both jettisoned and non-jettisoned configurations	The team must ensure that the mission can be completed even if RSO approval to jettison is not granted	5.2.2	DT	The Vehicle Demonstration flight verified that the payload can complete the NASA mission in the case of non-jettison, and the Payload Demonstration flight will verify that the mission can be completed in the jettison case	RN, ZS, PC	V
4.2 Lander Separation Mechanism							
4.2.1	The LSM shall disengage the lander from the main rocket body within 1 second of reaching the target altitude and receiving command signal.	Proper sensor selection and friction reduction shall be implemented for the LSM.	4.2.1	T	The requirement has been verified by conducting timed separation tests during ground-based trials, ensuring that the lander consistently separates from the main rocket body within 1 second of command. Cameras and timing devices were used to measure separation time.	SC	V
4.2.2	The LSM shall be able to receive RF transmission	No materials that potentially interfere with RF signals shall be used around the LSM	4.2.1	T	RF transmission tests have been conducted by transmitting signals from ground to the LSM to ensure there is no signal degradation. The team verified that the chosen materials do not obstruct RF signals.	SC	V
4.2.3	The mechanism shall be capable of operating at altitudes between 200 and 400 ft AGL	The team shall test the capability of the LSM to reliably function at these altitudes	4.2.1	T	The requirement was verified by flight testing the LSM at different altitudes between 200 and 400 ft AGL.	SC	V
4.2.4	The mechanism shall be capable of withstanding maximum expected shock loads of 32G.	The team shall use durable materials and perform shock load tests on the LSM simulating launch conditions	4.2.1	DT	The requirement was verified through vibration and shock load testing in a controlled environment. The Vehicle Demonstration flight further showed that the mechanism can withstand the maximum loading during flight.	SC	V
4.3 Leg Deployment Mechanism							
4.3.1	The legs shall withstand maximum anticipated landing loads.	Legs shall be constructed out of material which will not break under the forces experienced at landing	4.2.2	DT	Extensive testing under different loading conditions, dynamic load testing with integrated leg system, and drop testing from drone have been performed to verify leg strength.	ZS	V
4.3.2	The legs shall deploy passively upon lander separation.	Legs shall be spring loaded and released by separation from main body	4.2.2	DT	Extensive testing under different initial conditions, static testing of opening mechanism in lab, testing when being released under tether, and testing when being dropped under parachute have been performed to ensure full deployment under any scenario.	ZS	V
4.3.3	The legs shall be packaged within the body of the payload.	The lander separation bay airframe shall retain the legs within the payload bay airframe until jettison	4.2.2	ID	Inspection of the integrated vehicle and payload, along with the Vehicle Demonstration flight and payload drop tests, ensured that the legs are packaged within the vehicle prior to jettison.	ZS	V
4.4 Parachute Detachment Mechanism							
4.4.1	The parachute shall detach within 1 second of detected landing	The PDM shall utilize an EM latch	4.2.3	T	The PDM has been thoroughly tested through drop tests, and the data was reviewed to confirm that detachment occurs within 1 second of landing detection.	TS	V
4.4.2	The detachment latch shall withstand the maximum acceleration experienced during flight.	The latch shall be secured in place using sufficiently robust materials. The latch will be capable of withstanding the maximum expected loadings	4.2.3	T	The PDM has been thoroughly tested through drop tests and demonstrated to withstand parachute jerk	TS	V
4.5 Antenna Extension Mechanism							
4.5.1	The AEM shall extend the antenna 18"	The AEM spring tape shall be 18" long	4.2.4	T	Testing has ensured that the AEM is capable of extending even after being subjected to high G loads and impacts	RN	V
4.5.2	The AEM shall be capable of extending even if partially obstructed	Back to back spring tapes shall be rolled together to strengthen their extension force	4.2.4	T	The extension force of the AEM has been measured and verified to ensure that light debris does not prevent the AEM from extending to full length	RN	V
4.5.3	The AEM shall not interfere with the RF transmission	The AEM tape spring serves as the antenna itself	4.2.4	T	Testing has been conducted to ensure that transmission is not degraded by the use of a metal spring tape in the AEM	RN	V

4-5.4	The AEM locking mechanism shall withstand maximum expected loading during flight	The AEM cylinder lock shall be designed to be robust and able to withstand high loads and accelerations	4.2.4	T	The AEM has been dropped from a height that simulates landing velocity to ensure that the mechanism remains locked	RN	V
4-5.5	The AEM shall be capable of extending regardless of payload jettison, and have at least 1 antenna extending away from the ground regardless of landing orientation	There shall be three AEMs equally spaced radially to ensure one is always facing upward	4.2.4	T	The payload bay has been dropped repeatedly to ensure that one AEM is facing upward in all landed configurations	RN	V
4.6 RF Transmission System							
4.6.1	Transmission shall occur given any landing orientation.	The signal lost relative to optimal orientation should be less than 10 percent.	4.3.3	T	Signal strength in dBm has been measured at different landing orientations to ensure sufficient strength for compliance with this requirement.	NB	V
4.6.2	The antenna system shall be able to reliably transmit up to a distance of 1 mile.	Need to have reliable transmission within 1 mi in clear air space.	4.3.3	T	Testing in different environments has been performed to ensure reliable transmission for distances up to 1 mi.	NB, PC	V
4.6.3	RF transmission shall cease 5 minutes after landing	RF transmission must be cut-off after 5 minutes of landing	4.3.3	T	The timer for the RF transmission logic has been tested several times in launches and on its own to ensure compliance with this requirement	NB, PC	V
4.7 Data Collection Sensors							
4.7.1	Data sampling frequency shall be 160 Hz (TBR) throughout data collection.	Electronics shall be configured such that they sample at 160 Hz	4.3.1	T	The sampling frequency was tested by leaving the system running for >1hr, which ensured that data points are consistent with minimal noise and without other nonmeaningful values.	AF, PC, JH	V
4.7.2	The sensors shall be able to detect landing	The thresholds for landing values shall be calibrated to avoid premature parachute detachment. Backup plan is a timer.	4.3.1	T	Landing detection logic has been thoroughly tested with debugging techniques such as blinky and different scenarios, such as at different locations.	AF, PC, JH	V
4.7.3	The sensors shall be able to determine altitude	Pressure readings shall be used to calculate altitude	4.3.1	D	Drone drop test data has been used to verify the functionality of the altimeter.	AF, PC, JH	V
4.7.4	The sensors shall be able to detect apogee	Changes in acceleration shall be utilized to determine when apogee is reached	4.3.1	D	Drone drop test data will be used to verify the functionality of the accelerometer.	AF, PC, JH	V
4.7.5	The sensors shall be able to detect orientation	The selected IMU shall be capable of determining orientation	4.3.1	DT	Orientation values collected during drop testing were reviewed to ensure they match the actual orientation of the payload. Additional testing of orientation readings were conducted in the lab to verify reliability	AF, PC, JH	V
4.7.6	The sensors shall be able to detect acceleration	The selected IMU shall be capable of determining acceleration	4.3.1	DT	IMU has been tested in the field to ensure that it can read acceleration in this environment	AF, PC, JH	V
4.7.7	The sensors shall be able to detect battery voltage	Battery voltage shall be detected through Raspberry Pi.	4.3.1	T	Battery has been monitored during field testing to verify that this capability functions as expected	AF, PC, JH	V
4.7.8	The software shall be able to estimate velocity	Altitude data shall be used to estimate velocity via a first order filter.	4.3.1	T	Velocity readings have been compared to velocity that can be calculated from drop videos and other sensors in order to verify that it is accurate	AF, PC, JH	V
4.7.9	The software shall be able to estimate survivability	Acceleration data shall enable the calculation of DRI, which allows survivability to be estimated.	4.3.1	DT	Survivability was calculated during drop testing to ensure that code logic and sensors are working properly	AF, PC, JH	V
VADL Safety Requirements							
Req #	Description	Design Requirements	Section	Verification Method	Verification Details	Team Member(s)	Status
5.1 Safety Requirements							
5.1.1	Team members shall wear proper PPE as indicated in VADL safety protocols.	All team members shall follow VADL safety protocols with respect to PPE.	6.1	I	The Student Safety Officer has regularly inspected lab activities to ensure that team members are properly wearing the required PPE	ZS	V
5.1.2	Team members shall clean up after themselves and engage in collective lab cleanings when deemed necessary by the Student Safety Officer.	Team leadership shall enforce guidelines regarding the cleanliness of the lab.	6.1	I	The Student Safety Officer has regularly inspected the lab to ensure the cleanliness of the lab is up to lab standards	ZS	V
5.1.3	Team members shall participate in mandatory weekly safety briefings delivered by the Student Safety Officer.	The Student Safety Officer and Team Lead shall enforce attendance at weekly safety briefings.	6	I	The Student Officer and Team Lead have recorded attendance and inspected that all team members are present at each meeting.	ZS, IM	V
5.1.4	Energetic material shall be stored in appropriate explosives magazines and shall be kept locked at all times.	The Student Safety Officer shall store energetic material in a locked MAGloc case.	6	I	The Student Safety Officer has regularly inspected that energetic material is stored in the MAGloc case and that it remains locked at all times.	ZS	V
5.1.5	Team members shall only use equipment that they are trained to operate.	The Student Safety Officer shall maintain a list of personnel trained on each piece of equipment.	6	I	All team members have inspected the training certification list to ensure that nobody uses equipment they are not certified to operate.	ZS	V

7.3 Budgeting

As the project has matured, so too have the team’s budgeting practices. The following sections provide information on VADL’s funding, budget, and expenses. This documentation has enabled the team to develop a full-scale, competition-ready vehicle and payload while remaining financially responsible.

7.3.1 Funding Plan

The funding plan for the 2025 competition year remains unchanged from the CDR milestone. The total funding of \$16,000 is derived from several sources: lab benefactor Mark Dalton has contributed \$7,000, \$5,000 comes from a grant awarded by the Vanderbilt University School of Engineering, and a combination of surplus funds and alumni contributions make up the remaining \$4,000. Figure 122 provides a visual representation of this funding breakdown.

Total Funding: \$16,000

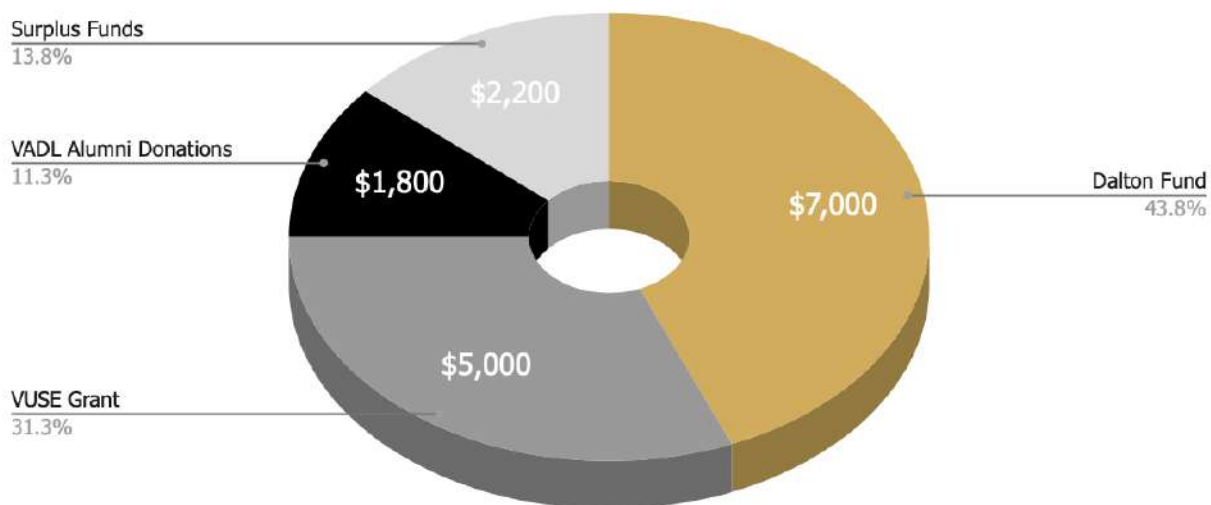


Figure 122: Funding breakdown for the 2025 competition year.

7.3.2 Budget Summary

The team has updated its budget from the CDR milestone. The budget has increased slightly, from \$14,836 at CDR to \$14,946 at FRR. This increase is primarily due to multiple unanticipated expenses, namely the new 60” Fruity Chutes Iris Ultralight parachute and the X50-A antenna. A graphical breakdown of the budget is provided in Figure 123, and an itemized table is included in Table 77. The budget is broken down into vehicle, payload, and travel categories, with “other” encompassing other expenses that do not fall into one of these three areas. Each category is further decomposed into subcategories to indicate the specific aspect of the project that is being detailed. As always, a contingency margin of 10% is applied to ensure that unanticipated expenses do not result in the team exceeding its allocated funding of \$16,000.

Budget: \$14,946

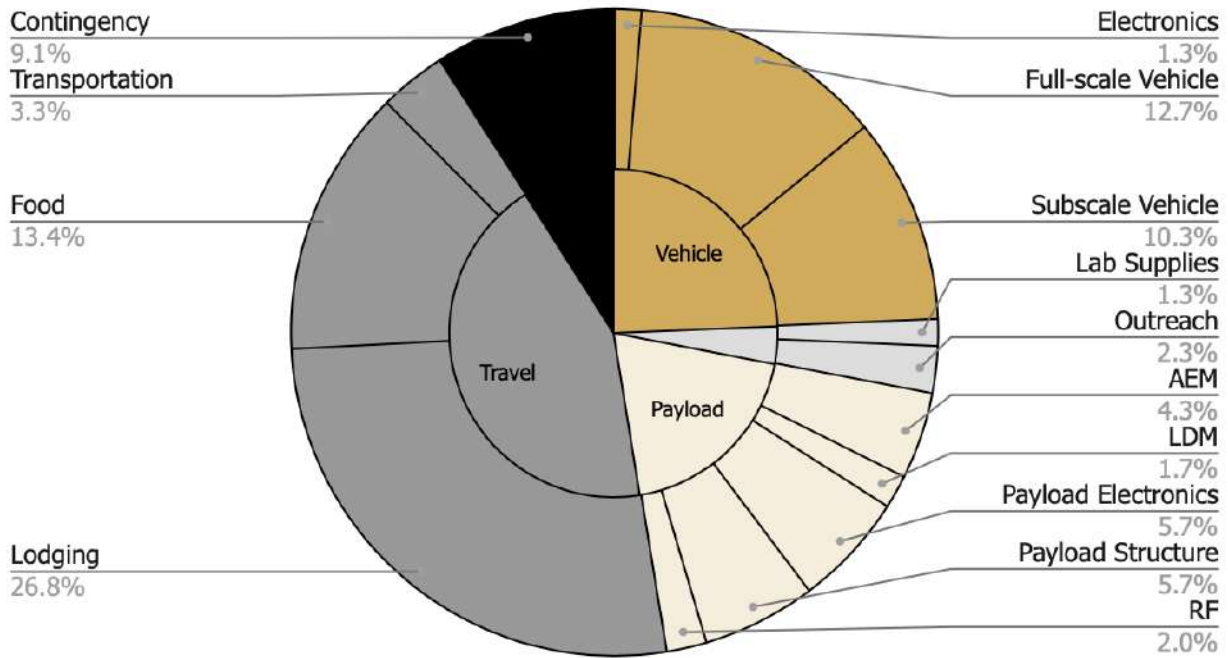


Figure 123: Visual breakdown of VADL's budget for the 2025 competition year.

Table 77: Itemized budget list for the 2025 competition year.

Category	Subcategory	Cost
Vehicle		\$3,637
	Full-scale Vehicle	\$1,900
	Subscale Vehicle	\$1,537
	Electronics	\$200
Payload		\$2,900
	AEM	\$650
	LDM	\$250
	Payload Electronics	\$850
	Payload Structure	\$850
	RF	\$300
Travel		\$6,500
	Lodging	\$4,000
	Food	\$2,000
	Transportation	\$500
Other		\$550
	Lab Supplies	\$200
	Outreach	\$300
Contingency		\$1,359
Total		\$14,946

7.3.3 To-Date Spending

VADL regularly tracks expenses to ensure spending does not exceed the team's funding. A visual summary of to-date spending is provided in Figure 124, and a full breakdown of all items procured is listed in the following table. Total spending has increased from \$6,073.96 at CDR to \$6,779.51 at FRR.

Total Spending: \$6,779.51

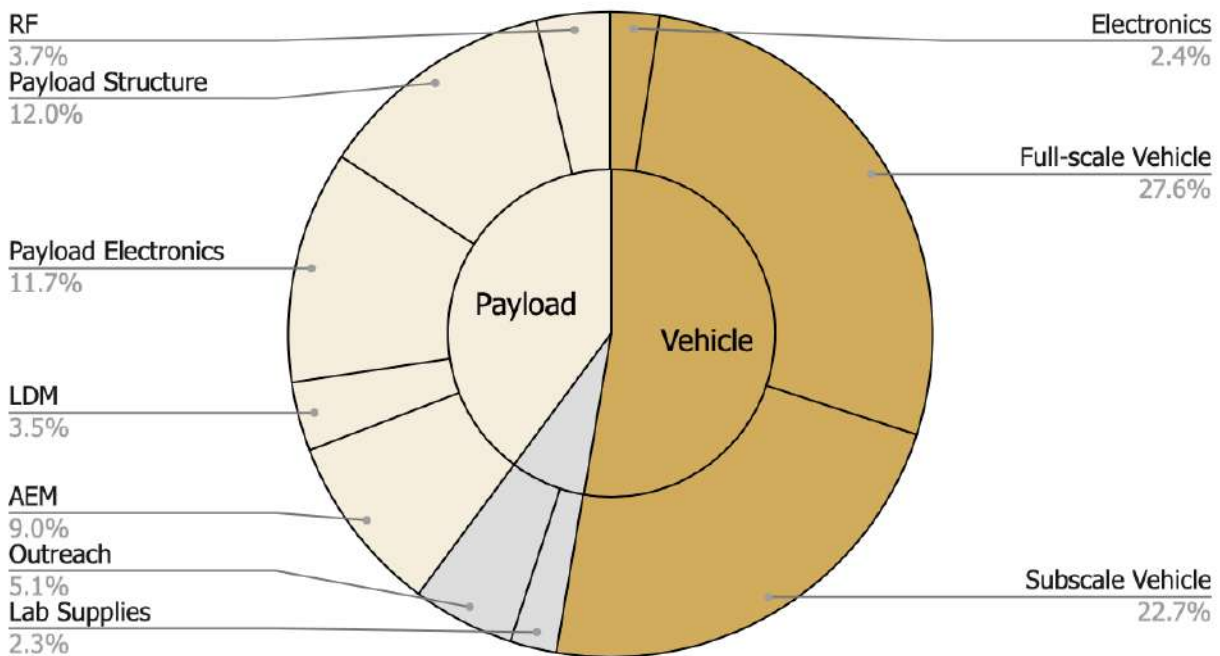


Figure 124: Total spending, as of the FRR milestone.

Project Segment	Date Ordered	Vendor	Quantity	Cost	Total Cost	Description
Category: AEM					\$610.58	
PDR	9/12/24	Amazon	1	\$13.67	\$13.67	2 count 1/4" x 10' tape measure
PDR	9/12/24	Amazon	1	\$7.59	\$7.59	1/2" x 12' tape measure
PDR	9/12/24	Amazon	1	\$14.99	\$14.99	2 count mini solenoid
PDR	9/14/24	McMaster-Carr	2	\$11.40	\$22.80	1-3/8" ID needle roller bearing, 1/2" wide
PDR	9/14/24	McMaster-Carr	1	\$13.05	\$13.05	1/8" diameter flush bearing, 50 ct
PDR	9/24/24	McMaster-Carr	5	\$11.40	\$57.00	1-3/8" ID needle roller bearing, 1/2" wide
PDR	9/24/24	Amazon	3	\$7.59	\$22.77	1/2" x 12' tape measure
PDR	9/26/24	McMaster-Carr	5	\$11.97	\$59.85	Compression Spring 0.375" Long, 0.180" OD, 0.132" ID
PDR	9/26/24	McMaster-Carr	2	\$33.23	\$66.46	Rotary Speed Limiter Gearless, 360 Degree Dual Direction, 21 in.-oz. Torque
PDR	9/26/24	McMaster-Carr	1	\$18.86	\$18.86	Powder Lubricant 1 lbs. Can
PDR	9/26/24	McMaster-Carr	1	\$16.43	\$16.43	Captive Pins 18-8 Stainless Steel, 1/8" Diameter, 1/2" Long, Tps-125-8v
PDR	9/26/24	McMaster-Carr	1	\$12.73	\$12.73	Dowel Pin Aluminum, 1/8" Diameter, 3/4" Long
PDR	9/26/24	McMaster-Carr	10	\$1.51	\$15.10	Dry-Running MDS-Filled Nylon Sleeve Bearing Flanged, for 1/8" Shaft Diameter and 1/4" Housing ID, 1/4" Long
PDR	10/17/24	Amazon	1	\$15.99	\$15.99	Seymour 16-o83 Good Grip Slip Resistant Coating Spray, Black 12 Ounce (Pack of 1)
PDR	10/17/24	Amazon	1	\$14.75	\$14.75	Teflon Glass Cloth Tape, (2-inch x 33 Feet)
PDR	10/17/24	Amazon	3	\$19.09	\$57.27	Rotary Speed Limiter Geared, 360 Degree Clockwise, 2.88 in.-oz. Torque
CDR	11/6/24		2	\$7.59	\$15.18	1/2" x 12' tape measure
CDR	11/6/24	Amazon	1	\$7.99	\$7.99	4 Way Servo Wire Extension Cord
CDR	11/6/24	Amazon	1	\$27.99	\$27.99	10 ct 9g servo
CDR	11/6/24	McMaster-Carr	3	\$6.31	\$18.93	Torsion Spring 90 Degree Left-Hand Wound, 0.16" OD
CDR	11/6/24	Amazon	1	\$12.99	\$12.99	Dip And Grip (Black) Rubberized Plastic Coating Cables Resists Moisture Chemical Abrasion And Corrosion Slip Resistance Air Dry (Black) 8 Fl. Oz
CDR	11/19/24	McMaster-Carr	3	\$19.09	\$57.27	Rotary Speed Limiter Geared, 360 Degree Clockwise, 2.88 in.-oz. Torque
CDR	11/19/24	McMaster-Carr	6	\$3.42	\$20.52	Ultra-Low-Friction Oil-Embedded Sleeve Bearing for 3 mm Shaft Diameter, for 6 mm Housing ID, 4 mm Long
CDR	11/19/24	McMaster-Carr	6	\$3.40	\$20.40	302 Stainless Steel Torsion Spring 180 Degree Right-Hand Wound, 0.217" OD
CDR	11/19/24	McMaster-Carr	1	\$6.72	\$6.72	Compression Spring 0.25" Long, 0.25" OD, 0.21" ID
Category: Electronics					\$166.08	
PDR	9/14/24	Amazon	2	\$6.50	\$13.00	SMA female to UHF male PL-259
PDR	9/14/24	Amazon	2	\$8.88	\$17.76	UHF PL259 Cable RG58 3.3ft(1M) PL259 SO239 Male to Female Extension Cable Low
PDR	9/14/24	Amazon	2	\$6.87	\$13.74	SMA Coax Connector Cable Gender Changers,SMA Male/Female

PDR	9/24/24	Digikey	3	\$12.95	\$38.85	MPU6050 IMU
CDR	11/6/24	Amazon	1	\$30.88	\$30.88	Router
FRR	1/9/25	Amazon	1	\$38.90	\$38.90	SMD 0603, Resistors, Capacitors, and Transistors
FRR	1/29/25	Pololu	1	\$12.95	\$12.95	5V, 3.4A Step-Down Voltage Regulator
Category: Full-scale Vehicle					\$1,869.15	
PDR	9/12/24	McMaster-Carr	1	\$352.47	\$352.47	6" Diameter Al Mandrel 1/2" thickness. 3 Foot length
PDR	9/24/24	Other	2	\$141.27	\$282.54	6" G-12 Fiberglass airframe
PDR	9/12/24	HomeDepot	1	\$74.66	\$74.66	100 ft. Highly Reflective Mylar
PDR	9/12/24	Fiberglast	1	\$27.95	\$27.95	PVA Release Film 13-B (Gallon)
PDR	9/12/24	Amazon	1	\$14.51	\$14.51	Clay tape / Colorimetrics Gray Putty Tape/Butyl Tape
PDR	10/3/24	Fiberglast	3	\$32.95	\$98.85	OLFA® Rotary Cutter
PDR	9/12/24	Amazon	1	\$17.99	\$17.99	PLA Filament - BLACK
PDR	9/12/24	Amazon	1	\$18.99	\$18.99	PLA Filament - WHITE
PDR	9/12/24	Amazon	1	\$23.99	\$23.99	PLA Filament - GOLD
PDR	9/24/24	Other	2	\$190.00	\$380.00	800cc onyx filament
CDR	11/12/24	Amazon	1	\$11.99	\$11.99	Squeegees for epoxy application
CDR	12/11/24	Other	4	\$124.00	\$496.00	6" G-12 Fiberglass coupler
FRR	1/9/25	Metal Supermarkets	1	\$45.75	\$45.75	Round Aluminum 6061 Bar Stock
FRR	1/14/25	Apogee Rockets	2	\$11.73	\$23.46	Rail Buttons
Category: Lab Supplies					\$158.35	
PDR	9/12/24	Amazon	1	\$20.13	\$20.13	73-Piece Wrench and Screwdriver Set
PDR	10/3/24	Amazon	2	\$7.28	\$14.56	Super glue
PDR	10/3/24	Amazon	1	\$8.11	\$8.11	Band-aids
CDR	11/6/24	Amazon	2	\$8.97	\$17.94	Left-handed scissors
CDR	11/6/24	Amazon	3	\$6.48	\$19.44	Regular use scissors
CDR	11/7/24	Amazon	1	\$12.99	\$12.99	24 Gauge Wire
CDR	11/12/24	Amazon	1	\$7.75	\$7.75	Isopropyl alcohol
CDR	11/12/24	Amazon	1	\$7.99	\$7.99	Cleaning rags
CDR	11/12/24	Amazon	2	\$8.99	\$17.98	Combination Lock
CDR	11/19/24	Amazon	1	\$10.99	\$10.99	1" by 30" sanding belt 24 pack
FRR	1/14/25	Amazon	1	\$8.48	\$8.48	Fast Orange Permatex hand cleaner
FRR	1/29/25	Amazon	1	\$11.99	\$11.99	Sandpaper, Variety Pack
Category: LDM					\$235.12	
PDR	9/24/24	Amazon	1	\$7.59	\$7.59	8mm ID Linear Bearing
PDR	9/24/24	Amazon	1	\$11.99	\$11.99	8mm Linear Shaft
PDR	10/12/24	McMaster-Carr	5	\$15.55	\$77.75	Spring Pins for Leg Locking System
PDR	10/12/24	McMaster-Carr	10	\$2.25	\$22.50	8mm Set Screw Shaft Collar
PDR	10/12/24	Amazon	1	\$7.59	\$7.59	8mm ID Linear Bearing
PDR	10/12/24	Amazon	2	\$11.99	\$23.98	8mm Linear Shaft
PDR	10/12/24	McMaster-Carr	8	\$5.79	\$46.32	8mm Clamping Shaft Collar
FRR	2/6/25	McMaster-Carr	2	\$18.70	\$37.40	Architectural 6063 Aluminum Bar, 1/8" Thick x 2" Wide
Category: Outreach					\$347.79	
PDR	9/14/24	Amazon	10	\$16.99	\$169.90	Carboard space shuttles
PDR	10/3/24	Amazon	1	\$7.99	\$7.99	Counters for outreach events
PDR	10/17/24	Amazon	10	\$16.99	\$169.90	Space Shuttles for Outreach
Category: Payload Electronics					\$790.64	

PDR	9/12/24	Amazon	4	\$6.00	\$24.00	SO-239 Chassis Connector
PDR	9/12/24	Amazon	1	\$7.59	\$7.59	16-14AWG Wire Crimp Connectors
PDR	9/24/24	Other	1	\$115.00	\$115.00	Light APRS tracker
PDR	9/26/24	Digikey	100	\$13.48	\$13.48	EM Latch 6-Pin Connectors
PDR	9/26/24	Digikey	200	\$16.04	\$16.04	Crimp
PDR	9/27/24	Spark Fun	5	\$1.05	\$5.25	Screw Terminals
PDR	9/27/24	Spark Fun	5	\$1.05	\$5.25	Screw Terminals
PDR	9/27/24	Amazon	2	\$7.99	\$15.98	Desoldering Wick
PDR	9/27/24	Amazon	1	\$16.92	\$16.92	Lipo Battery
PDR	9/27/24	Amazon	1	\$22.69	\$22.69	LiPo Battery Charger
PDR	9/27/24	Amazon	1	\$9.99	\$9.99	DC 5V Converter
PDR	9/27/24	Amazon	1	\$13.99	\$13.99	DC 5V Converter
PDR	10/10/24	Microcenter	1	\$79.99	\$79.99	Raspberry Pi 5
PDR	10/12/24	Microcenter	1	\$79.99	\$79.99	Raspberry Pi 5
PDR	10/12/24	Amazon	3	\$8.79	\$26.37	Micro HDMI to HDMI Display Cable
CDR	11/6/24	Amazon	1	\$35.95	\$35.95	Soldering Iron
CDR	11/6/24	Amazon	3	\$6.11	\$18.33	Photoresistor
CDR	11/6/24	Other	3	\$8.66	\$25.98	Prototyping Board for Raspberry Pi
CDR	11/6/24	Amazon	1	\$39.99	\$39.99	Circuit Board Holder
CDR	11/12/24	Amazon	1	\$15.99	\$15.99	DROK Buck Converter 12v to 5v Pack of 2
CDR	11/19/24	Amazon	3	\$14.99	\$44.97	Electronics-Salon Pi Screw Terminal Block Breakout Module, for Raspberry Pi.
CDR	11/19/24	Amazon	3	\$18.99	\$56.97	GeeekPi GPIO Screw Terminal Block Breakout Board HAT
CDR	11/19/24	Amazon	1	\$22.49	\$22.49	BTECH APRS-K1 Multi-Function Universal Audio Interface Cable
FRR	1/9/25	Amazon	1	\$33.29	\$33.29	Zee 3S 2200mAh Lipo Battery 11.1V
FRR	1/9/25	Amazon	1	\$5.49	\$5.49	Molex Female Micro-Fit 3.0 Crimp Terminal Pins
FRR	2/6/25	Amazon	1	\$21.57	\$21.57	Heat Shrinkable Braided Sleeving
FRR	2/14/25	Amazon	1	\$17.09	\$17.09	Wire Shielder
Category: Payload Structure					\$813.62	
PDR	9/26/24	McMaster-Carr	4	\$7.78	\$31.12	1/4"-20 Eye Nuts
PDR	10/10/24	Other	2	\$128.50	\$128.50	1/8" Kevlar shock cord
PDR	10/17/24	McMaster-Carr	1	\$10.19	\$10.19	Copper wire (.016")
PDR	10/17/24	McMaster-Carr	1	\$9.16	\$9.16	Copper Wire (.025")
PDR	10/17/24	McMaster-Carr	1	\$8.83	\$8.83	Copper Wire (.051")
PDR	10/17/24	McMaster-Carr	1	\$12.08	\$12.08	Aluminum wire (.061")
PDR	10/17/24	McMaster-Carr	1	\$12.48	\$12.48	Aluminum wire (.032")
PDR	10/17/24	Amazon	1	\$9.99	\$9.99	Solder wire (2 mm)
CDR	11/6/24	Metal Supermarkets	5	\$9.40	\$46.98	0.313" Diameter by 26" long 6061-T6511 Aluminum round bar
CDR	11/6/24	Amazon	1	\$88.00	\$88.00	Raspberry Pi 5 8GB
CDR	11/6/24	McMaster-Carr	30	\$4.37	\$131.10	5/16" Aluminum Shaft Collar
FRR	1/29/25	Fruity Chutes	1	\$325.19	\$325.19	60" Fruity Chutes Iris Ultralight
Category: RF					\$251.49	
PDR	10/10/24	Spark Fun	5	\$1.25	\$6.25	SMA Connector
PDR	10/10/24	Other	1	\$115.00	\$115.00	Light APRS tracker
PDR	10/12/24	Spark Fun	5	\$1.25	\$6.25	SMA Connector
FRR	1/29/25	Amazon	1	\$123.99	\$123.99	X50-A Diamond Original 144/440 MHz Dual Band Base Antenna
Category: Subscale Vehicle					\$1,536.69	

PDR	9/12/24	McMaster-Carr	1	\$266.88	\$266.88	4" Diameter Al Mandrel 1/2" thickness. 3 Foot length
PDR	9/26/24	Other	2	\$141.27	\$282.54	4" G-12 Fiberglass airframe
PDR	10/3/24	Fiberglast	1	\$37.50	\$37.50	4 oz Fiberglass Cloth x 50" Wide
PDR	10/3/24	Other	1	\$25.75	\$25.75	High temperature mold release
CDR	11/6/24	Madcow Rocketry	2	\$46.00	\$92.00	4" G-12 Coupler
CDR	11/6/24	DragonPlate	1	\$222.84	\$222.84	Quasi-isotropic Solid Carbon Fiber Sheet ~ 1/8" x 12" x 24"
CDR	11/6/24	Wildman Rocketry	1	\$71.39	\$71.39	G-12 Fiberglass air frame
CDR	11/6/24	Other	3	\$82.99	\$248.97	Aerotech I366R motor reload kits for subscale
CDR	11/6/24	Other	1	\$6.48	\$6.48	Regular use scissors
CDR	11/7/24	Apogee Rockets	1	\$108.97	\$108.97	RMS 38/600 Casing w/ forward seal disk
CDR	11/7/24	Apogee Rockets	1	\$51.10	\$51.10	38mm Aft Closure
CDR	11/7/24	Apogee Rockets	1	\$62.22	\$62.22	38mm Standard Forward Closure
CDR	11/7/24	Other	1	\$25.29	\$25.29	18-38mm Delay Tool
CDR	11/7/24	Apogee Rockets	2	\$121.15	\$242.30	Entacore AIM Dual Deployment Altimeters
CDR	11/7/24	Amazon	1	\$11.44	\$11.44	9V Batteries
CDR	11/19/24	McMaster-Carr	4	\$6.27	\$25.08	5/16"-18 steel threaded rods
CDR	11/19/24	McMaster-Carr	2	\$2.33	\$4.66	5/16"-18 aluminum threaded rods
CDR	11/20/24	Amazon	4	\$4.54	\$18.16	5/16"-18 aluminum threaded rods
Total:				\$6,779.51		

7.3.4 Expense Summary

An updated expense projection summary is shown below in Figure 125. Important to note is that at the CDR milestone, projected total spending for March was anticipated to be 7,071, *roughly* 300 greater than the actual total spending as of the FRR submission. This point is important to note, demonstrating that while the intentionally conservative budget projection did increase slightly, spending is actually within the projections from CDR. The team, therefore, feels confident that total spending will fall well within the total budget allocated for the competition year.

Project Spending Over Time

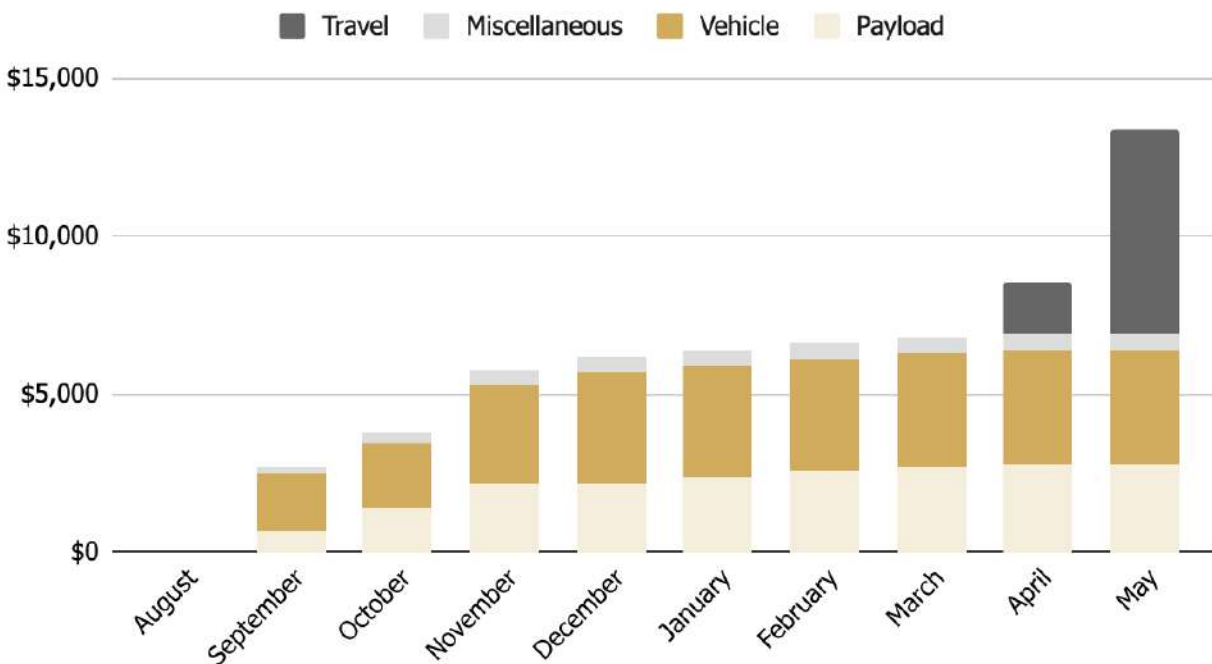


Figure 125: Project spending over time, as of FRR milestone, with projections for April and May.

A References

- [1] *1L10.50 LOCAL GRAVITY CHART*. URL: https://physics.montana.edu/demonstrations/video/1_mechanics/demos/pics/localgravitychart.jpg.
- [2] *Aerodynamic Coefficients in Cambridge Rocketry*. URL: <https://cambridgerocket.sourceforge.net/AerodynamicCoefficients.pdf>.
- [3] *Cesaroni 3660L1720-P Motor Data*. URL: <https://www.thrustcurve.org/motors/Cesaroni/3660L1720-P/>.
- [4] L. D. Duncan and R. J. Ensey. *Six degree of freedom digital simulation model for unguided fin-stabilized rockets*. 1964.
- [5] Sitan Huang. *Declarative Rocket Simulation Software*. 2024. URL: <https://github.com/SitanHuang/DRSS>.
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- [7] C. Lawrence et al. “Orion Crew Member Injury Predictions during Land and Water Landings”. In: *Earth Space 2008: Engineering, Science, Construction, and Operations in Challenging Environments* (2008). DOI: <https://ntrs.nasa.gov/api/citations/20080013357/downloads/20080013357.pdf>.
- [8] *Simulation and Testing of Rail Launchers*. URL: <https://nar.org/member-only-reports/rd/NARAM-61/N61-Simulation-Testing-Rail-Launchers-Flanigan.pdf>.

B VADL Safety Agreement

Written Safety Agreement

By signing below, I agree to abide by all laws, standards, regulations and guidelines set forth by the National Association of Rocketry (NAR), the Federal Aviation Administration (FAA), the state of Tennessee, and pertinent Material Safety Data Sheets (MSDS). I acknowledge that our rocket must undergo a range safety inspection prior to launch, and I will adhere to the results of that inspection. The ultimate authority on rocket safety lies with the Range Safety Officer (RSO), who has the right to deny our rocket's launch if any safety concerns arise. As the team mentor bears the final responsibility for ensuring the safe flight and retrieval of our rocket, we will not proceed with the launch unless the mentor has reviewed the design, evaluated the construction, and confirmed that it complies with established amateur rocketry design and safety standards. My signature confirms that I have read, understand, and will adhere to all relevant safety procedures as outlined by the Safety Officer.

Print Name

Ian Morgan	Signature: <u>[Signature]</u>	Date: <u>9/6/24</u>
Rais Nurhidajat	Signature: <u>[Signature]</u>	Date: <u>9/6/24</u>
Saksham Sharma	Signature: <u>[Signature]</u>	Date: <u>9/7/24</u>
Sitan Huang	Signature: <u>[Signature]</u>	Date: <u>9/7/24</u>
Zachary Stutsman	Signature: <u>[Signature]</u>	Date: <u>9/6/24</u>
Benjamin Racape	Signature: <u>[Signature]</u>	Date: <u>9/8/24</u>
Santino Clemente	Signature: <u>[Signature]</u>	Date: <u>9/7/24</u>
Samantha Civil	Signature: <u>[Signature]</u>	Date: <u>9/1/24</u>
Timothy Shaheen	Signature: <u>[Signature]</u>	Date: <u>09/07/24</u>
Jonathan Zak	Signature: <u>[Signature]</u>	Date: <u>09/07/24</u>
Jae-Hoon Smith	Signature: <u>[Signature]</u>	Date: <u>9/8/24</u>
Jack Liedel	Signature: <u>[Signature]</u>	Date: <u>9/10/24</u>
Alend Khoshnaw	Signature: <u>[Signature]</u>	Date: <u>9/8/24</u>
Jillian Hornby	Signature: <u>[Signature]</u>	Date: <u>09/07/24</u>
Patrick Chen	Signature: <u>[Signature]</u>	Date: <u>9/6/24</u>
Ashley Fu	Signature: <u>[Signature]</u>	Date: <u>9/8/24</u>
Neal Bagai	Signature: <u>[Signature]</u>	Date: <u>09/07/24</u>

C VUSE Shop Safety Guidelines

Vanderbilt University School of Engineering

Environmental Health and Safety Policy

1.0 Purpose

The School of Engineering is strongly committed to protecting the environment and human health in all of its operations. To fulfill this commitment, the School recognizes that pro-active efforts must be taken to ensure that sound environmental and safety planning is integrated into every level of School decision-making. The School expects that all faculty, staff, and students will share this responsibility for safety and security of themselves, fellow faculty and staff, students, and guests, and maintain reasonable care when using School and University property. It is important that all members of the School of Engineering comply with University written policies and guidelines set forth by offices including but not limited to Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>, as well as Vanderbilt University Police Department (VUPD), <http://police.vanderbilt.edu/>, and all Federal, State and local regulatory bodies, including fire and police agencies.

Effective environmental, health and safety performance is important to the School of Engineering in its relationships with students, faculty, staff, regulatory agencies and the general public. All employees are expected to participate in formulation of and to comply with safety practices relating to their work and University guidelines. To assist managers in the School of Engineering in meeting their environmental and safety responsibilities, the following policy is established.

2.0 Policy

It is the policy of the School of Engineering to:

- 2.1 ensure that all operations are conducted safely and in compliance with all Federal, State and local applicable environmental, health, and safety laws and regulations. In the absence of specific laws or regulations, good management practices shall be followed;
- 2.2 design, construct, and operate all facilities in a manner that protects human health, safety of the occupants, and the environment;
- 2.3 ensure that environmental health and safety considerations are an integral part of the School's master planning efforts;
- 2.4 work with Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>, to develop, follow, and continually improve environmental, health, and safety procedures and practices for all facilities, projects, and operations;
- 2.5 perform periodic internal environmental, health, and safety program reviews to correct deficiencies, establish goals, and identify priorities;

- 2.6 promote environmental health, and safety awareness among all faculty, staff and students and ensure that Principal Investigators (PI's) and supervisors proper training using, as appropriate, training courses and other resources provided by Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>;
- 2.7 ensure adherence to University laboratory and Departmental safety procedures;
- 2.8 ensure, as far as is reasonably practicable, maintenance of adequate first aid facilities throughout School premises;
- 2.9 ensure, as far as is reasonably practicable, provision of adequate and correct safety equipment throughout School premises;
- 2.10 ensure, as far as is reasonably practicable, provision of protective clothing and equipment as is necessary to maintain the health and safety of the School's staff, faculty, students and visitors;
- 2.11 encourage the reporting of violations of safe practices to Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>;
- 2.12 ensure the reporting of on-the-job accidents in a timely manner in accordance with OSHA and State Department of Health guidelines to the Risk Management Office (Supervisors are responsible for filing out a First Report of Work injury form as described in HR policy #HR-011, <http://www.vanderbilt.edu/HRS/policies/workaccident.html> . The form itself is available at <http://polproc.mc.vanderbilt.edu/policy/pol9-23.html>.);
- 2.13 ensure the seeking of diagnosis and possible treatment from the Vanderbilt Occupational Health Clinic, <http://www.vanderbilt.edu/HRS/wellness/occhealth.htm>, in the event of personal injury from an on-campus accident requiring immediate attention;
- 2.14 encourage personal accountability and emphasize compliance with standards and conformance with University and School policies and best practices during employee training and in performance reviews;
- 2.15 communicate our desire to continuously improve our performance and foster the expectation that every employee, student, and contractor on Engineering School premises will follow this policy and report any environmental, health, or safety concern to Engineering School management.
- 2.16 Cooperate with any environmental, health, and safety program reviews by Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>, or external regulators and ensure prompt improvement of deficiencies.

**Vanderbilt University
School of Engineering**

Environmental Health and Safety: Roles and Responsibilities

<u>Personnel</u>	<u>Roles and Responsibilities</u>
Dean's Office or Designee	• Supports University policies and training programs, as administered by Vanderbilt Environmental Health and Safety, http://www.safety.vanderbilt.edu, related to the use of appropriate environmental health and safety practices and facilities in research and teaching activities. • Supports safety programs to implement environmental health and safety policies and procedures for safety and compliance in specific departments, institutes, centers, and units. • Reviews annual reports of activities, successes, and problems in environmental health and safety in the Engineering School, and supports departmental efforts to address improvements. • Appoints Chair and members of School of Engineering Safety Committee and ensures that the committee is active.
Department Chair/Center Director	• Supports University and School policies and training programs, as administered by Vanderbilt Environmental Health and Safety, http://www.safety.vanderbilt.edu, related to the use of appropriate environmental health and safety practices and facilities in research and teaching activities in their department/center. • Establishes departmental-specific safety programs and procedures to implement environmental health and safety policies and procedures for safety and compliance in their department/center. • Assumes responsibility for ensuring that good safety practices and environmental compliance are enforced within their department/center. • Supports and establishes the authority of the departmental Safety and Environmental Compliance Officer (SECO) to carry out their duties and to function effectively within the department. • Reports non-compliance and unsafe working

**Principal Investigator
(P.I.)/Faculty**

conditions to the Vanderbilt Compliance Office,
<http://www.vanderbilt.edu/compliance/html/office.htm>

- Ensures departmental personnel receive required training that is appropriate to their work.
- Supports activities of departmental faculty that are assigned to safety committees.
-
- Has knowledge of institutional, local, state, and federal environmental health and safety policies applicable to the research being performed. P.I.'s are responsible for safety, health and compliance issues in their labs or for those experiments being done under their supervision.
- Has knowledge of potential hazardous and/or regulated materials used in research and is trained in methods to minimize the risks of the hazards and to meet institutional compliance requirements.
- Seeks appropriate approvals/guidance through the departmental SECO, School Safety Committee, and VEHS, <http://www.safety.vanderbilt.edu>, for the use of potentially hazardous or regulated materials. Installation requirements of equipment, exhaust systems, etc. involving potentially hazardous or regulated materials should be coordinated through the Vanderbilt Campus Planning and Construction Department, <http://www.cpc.vanderbilt.edu>, prior to requests for construction estimates. Depending on the requirements, engineering analysis and assessment may be required.
- At the initiation of each laboratory intensive project and at least annually during the course of the project, identifies the safety and environmental compliance requirements of that project and discusses them with the departmental SECO.
- Implements appropriate safety policies as they apply to the research being performed.
- Ensures the integrity of equipment and facilities designed to contain potential hazards by providing recommended maintenance and calibration.
- Facilitates internal and external inspections of research facilities and equipment.
- Facilitates or provides environmental health and safety training and education on a continuing basis for research staff and students.
- Takes prompt action when unsafe acts or conditions are reported or noted.

Departmental Safety and Environmental Compliance Officer (SECO)

- Follows written emergency procedures.
- Reports all on-the-job accidents, illnesses, and/or potential exposures to the Vanderbilt Occupational Health Clinic, <http://www.vanderbilt.edu/HRS/wellness/occhealth.htm> . (The P.I. is responsible for filing out a First Report of Work injury form as described in HR policy #HR-011, <http://www.vanderbilt.edu/HRS/policies/workaccident.html> . The form itself is available at <http://polproc.mc.vanderbilt.edu/policy/pol9-23.html> .);
- Requests medical assistance as required and provides information regarding hazards involved to medical personnel.
- Develops and maintains a written Chemical Hygiene Plan (for laboratories that make use of chemicals and hazardous materials) that includes a) Standard Operating Procedures for laboratory processes, b) current inventory of hazardous materials, c) Material Safety Data Sheet (MSDS) for all chemicals, d) Personal Protective Equipment (PPE) for lab personnel, e) proper storage and disposal methods for all hazardous wastes, and f) written emergency procedures.
-
- Maintains a thorough working knowledge of the Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> , web site and the resources provided by VEHS.
- Liaison with Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>, Lab Compliance Assistance Program.
- Is current in all required training appropriate for the hazards present in their department.
- At the initiation of each laboratory intensive project and at least annually during the course of the project, identify the safety and environmental compliance requirements of that project and discuss them with the principal investigator.
- Maintains documentation necessary to demonstrate compliance with applicable safety and environmental compliance requirements.
- Notify the Department Chair and Principal Investigator (P.I) when a project or laboratory is out of compliance and communicate the actions required to be

in compliance.

- Periodically monitor instructional and research labs to be sure that health, safety and environmental compliance measures are being properly observed.
- Provides assistance in directing questions from faculty, staff, or students to appropriate resources at VEHS, <http://www.safety.vanderbilt.edu> , or elsewhere.
- Provides assistance in ensuring that lab workers have participated in appropriate courses offered through VEHS, <http://www.safety.vanderbilt.edu> .
- Provides assistance in obtaining appropriate signage through VEHS, <http://www.safety.vanderbilt.edu> , for safety warning, chemical disposal, etc.
- Assists P.I.'s with Chemical Hygiene Plan development.
- Coordinates with the Department Chair and, through the department representative, with the School of Engineering Safety Committee.

**Laser Safety Officer
(LSO)**

- Monitors and enforces the control of laser hazards and effects the knowledgeable evaluation and control of laser hazards for all classes of lasers used in the School. The American National Standard for Safe Use of Lasers (ANSI Z136.1) is adopted as the standard for the safe use of lasers and laser systems. Users of lasers and their supervisors must adhere to the practices and procedures described in the ANSI Standard.
- Classifies laser systems and establishes the Nominal Hazard Zone (NHZ) for all lasers used in the School. Class 3b and 4 unclosed lasers require a well-designated NHZ around the laser within which a person can be exposed to radiation in excess of the Maximum Permissible Exposure (MPE).
- Approves Standard Operating Procedures (SOP's) for all lasers used in the School. Written SOP's are required of all class 3b and 4 lasers and are enforced by the LSO.
- Approves signs and labels.
- Audits, recommends, or approves controls used in ensuring safety.
- Prior to use of new laser systems, approves the laser installation facilities and equipment.
- Periodically audits safety features in each of the laser installations in the School.
- Assists in ensuring that adequate education and

training is obtained by all personnel. As a minimum this requires taking the Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>, training course on Laser Safety. For class 3b and 4 lasers, only authorized, trained personnel may be present in the vicinity of the laser during its operation.

- Determine personnel categories.
- Ensure that appropriate medical surveillance through the Vanderbilt Occupational Health Clinic (OHC), <http://www.vanderbilt.edu/HRS/wellness/occhealth.htm>, is obtained for all workers with class 3b or 4 lasers.
- Maintains records of laser systems in use in the School.
- Investigates accidents and ensures corrective action.

Shop Machinery Safety Officer (SMSO)

- Ensures that safe working conditions are maintained in all areas of the School where machine tools and equipment are in use.
- Provide frequent inspections of work methods and materials/equipment used. Any unsafe equipment/material shall be tagged and rendered inoperative or physically removed from its place of operation.
- Assist in ensuring that all shop users are thoroughly trained in the use of protective equipment, guards, and the safe operation of equipment, machines, and tools they use or operate. Only users who have been trained shall be allowed to use shop equipment, machines, and tools.
- Assist in ensuring that tools and equipment are properly maintained and used.
- Assist in ensuring that required guards and protective equipment are provided, used, and properly maintained.
- Take immediate action to report and correct any violation of safety rules observed or reported to the SMSO.
- Ensure that shop doors are closed after hours or when the supervisor is not present. Unauthorized or untrained people are not allowed into the shop. All shop users must be accompanied by an authorized person at all times.

School Safety Committee

- The School Safety Committee is charged with the continuing development of the overall School Safety and Compliance Plan.

- Serves as a consultative body to the departmental Safety and Environmental Compliance Officers, the School Laser Safety Officer, the Shop Machinery Safety Officer, and the department Chairs.
- Serves as a “translator” of University policy to school safety and environmental compliance needs.
- Serves to evaluate specific safety issues within the School as they arise.
- Performs periodic internal laboratory auditing.

Department of Vanderbilt Environmental Health and Safety (VEHS)

- The Vanderbilt Environmental Health and Safety Department, <http://www.safety.vanderbilt.edu>, serves as the primary technical resource for the Vanderbilt teaching and research community regarding environmental health and safety policies, procedures, and regulatory issues.
- Appoints members for and coordinates the activities of regulatory-required university environmental health and safety committees (e.g., Radiation Safety Committee, Institutional Biosafety Committee, Chemical Safety Committee, Environmental Affairs Committee, Fire Protection Committee).
- Executes enforcement actions as prescribed by institutional policy or committee decision.
- Assists P.I.'s or departments in creating and maintaining inventories of potentially hazardous and regulated materials used in research at Vanderbilt.
- Conducts routine as well as special risk assessments and provide recommendations on appropriate practices and facilities to conduct research and laboratory instruction safely and in compliance with regulatory requirements.
- Conducts periodic review of research and laboratory instructional facilities and/or procedures to ensure compliance.
- Facilitates regulatory inspections by external regulatory entities.
- Provides operational and/or technical support for research and or instructional laboratory activities that require monitoring, tracking, decontamination, or disposal of potentially hazardous and/or regulated materials.
- Provides emergency response as appropriate in handling to spills or other critical events involving potentially hazardous and/or regulated materials.
- Identifies and/or develops training resources for the

	- Vanderbilt teaching and research community. - Identifies, monitors, and interprets local, state, and federal agency actions related to environmental health and safety for implementation within the Vanderbilt community. - Develops draft policies/procedures for oversight committee review. - Provides an annual report of activities to the Environmental Health and Safety Oversight Committee.
Vanderbilt Environmental Health and Safety Committees (Radiation Safety Committee, Institutional Biosafety Committee, Chemical Safety Committee, Environmental Affairs Committee, Fire Protection Committee)	- Have knowledge of institutional, local, state, and federal environmental health and safety policies applicable to research and laboratory instruction. - Establish institutional policies related to content area of committee (e.g., radiation safety, biosafety, etc.). - Assist in notifying the Vanderbilt research and teaching community of responsibilities to environmental health and safety in the conduct of research. - Review research for compliance through assessment of facilities, procedures, practices, and the training/expertise of those conducting the research. - Conduct periodic review of research to ensure compliance. - Take action on non-compliance and unsafe working conditions, as needed. - Report problems, violations, etc. to institutional officials and/or specific regulatory agencies, as appropriate. - Provide an annual report of activities to the Environmental Health and Safety Oversight Committee. - Refer research issues to other research committees as needed.
Division of Sponsored Research (University Central), http://www.vanderbilt.edu/SponsoredResearch/homepg.html	Communicates and coordinates with Environmental Health and Safety, http://www.safety.vanderbilt.edu, regarding appropriate approvals for studies conducted with radioactive, chemical, or biological materials prior to proposal submission (or prior to approval of budget for funded project).
Office of Contract and Grant Accounting (University Central),	Communicates and coordinates with Environmental Health and Safety, http://www.safety.vanderbilt.edu, regarding appropriate approvals for studies conducted

<http://www.vanderbilt.edu/ocga>

with radioactive, chemical, or biological materials prior to approval of budget for funded project if the proper approvals are not apparent.

**Students and others
working in the lab**

- Undergo necessary training as identified by the department SECO.
- Report to the laboratory in appropriate attire.
- Follow safe practices in the laboratory.
- Report safety concerns and/or problems to the department SECO.
- Obtain approval from P.I. or SECO for any deviations from established experimental procedures.

Vanderbilt University School of Engineering

Environmental Health and Safety: General Lab Policies and Procedures

Training: Appropriate training shall be provided for all new personnel (faculty, staff, students) using Research Labs, Teaching Labs, and other Technical Facilities, including Machine Shops, in the School of Engineering. Particular attention should be given to safety and procedures for handling Regulated Hazardous Materials (RHM). Lab Directors and Managers (e.g., PI's, Shop Foreman, Teaching Lab Manager) are responsible for ensuring that this training is delivered in a timely manner and that it is appropriate to the particulars of the Lab, Shop, or Facility. At a minimum, this training should include a review of relevant University Policy and training courses available at Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>.

Annual Retraining: All personnel who use Regulated Hazardous Materials must receive annual retraining. That retraining is delivered by Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>. Training courses are provided by VEHS in several ways – online through Vandysafe, in regularly scheduled classes, or by the request of the department.

New Projects: At the initiation of each laboratory intensive project and at least annually during the course of the project, the PI shall identify the safety and environmental compliance requirements of that project and discuss them with the departmental Safety and Environmental Compliance Officer (SECO).

Installation Requirements Involving Hazardous or Regulated Materials: PI's shall seek appropriate approvals/guidance through the departmental Safety and Environmental Compliance Officer (SECO), School Safety Committee, and VEHS, <http://www.safety.vanderbilt.edu>, for the use of potentially hazardous or regulated materials. Installation requirements of equipment, exhaust systems, etc., involving potentially hazardous or regulated materials should be coordinated through the Vanderbilt Campus Planning and Construction Department, <http://www.cpc.vanderbilt.edu>, prior to requests for construction estimates. Depending on the requirements, engineering analysis and assessment may be required. Direct requests for installations/construction requirements of this type to the Vanderbilt Plant Operations – Estimating And Design Center, <http://www.vanderbilt.edu/PlantOps/request.html>, will only result in a delay.

Laboratory Closeout Guidelines: Principal Investigators at Vanderbilt are responsible for the safe operation of their laboratory or suite of laboratories. This includes leaving any and all of these facilities in a safe condition when, and if, the premises are vacated. Guidelines outlining the Principal Investigator's responsibilities in the Laboratory Close Out Process may be found on the VEHS web site, <http://www.safety.vanderbilt.edu>.

Student RHM Disposition: RHM should be labeled at all times. As a requirement for graduation, all students must properly dispose of any Regulated Hazardous Materials (RHM) that they have generated and their advisor shall provide a written affirmation to that effect and submit it to the departmental Safety and Environmental Compliance Officer (SECO) for filing.

Flammables Storage Lockers: Storage lockers containing flammable materials must be clearly labeled at all times.

Emergency Contact List: All Research Labs, Teaching Labs, Machine Shops, or other Technical Facilities shall have the Vanderbilt emergency number, 1-1911, posted in a clearly visible location. All P.I.'s must provide one emergency contact with a non-work phone number that emergency responders can contact to determine status and contents of lab. This contact information will be posted on a hazard sign on the door of each lab.

Machine Shop Training: The Shop Machinery Safety Officer (SMSO) shall provide training for all personnel who wish to use the machine shop located on the first floor of Featheringill Hall. The Shop Supervisor for the Model Shop in Olin Hall or PI's, who use machine tools in their laboratory, are responsible for ensuring that users of their equipment are properly trained. They may avail their students of the instruction offered by the Shop Machinery Safety Officer. In all cases, the Shop Machinery Safety Officer is responsible for ensuring that safe working conditions are maintained in all areas of the School where machine tools and equipment are used, and that users have been properly trained in the safe operation of equipment, machines, and tools they use or operate. The Shop Machinery Safety Officer has the final say in who may use any of the School's Machine Shop Facilities and may revoke that privilege at any time for any unsafe or inappropriate use of the facility.

Shop Self-Inspection: Shop Supervisors and/or PI's shall perform and document self-inspections at least annually as part of OSHA's requirement for an effective Injury and Illness Prevention Program (IIPP). The School of Engineering's Shop Self-Inspection Form shall be used for this purpose. This self-inspection form will help document shop safety inspections and will assist shop personnel in identifying and correcting many common unsafe practices and conditions. The results of the self-inspection shall be shared with the Shop Machinery Safety Officer (SMSO), who will maintain a file copy.

Shop Safety Handout: The School of Engineering Shop Safety Handout shall be provided to all personnel using machine tools and related equipment in the School.

"Buddy System": No one shall work with chemicals or perform other hazardous lab work, such as energizing power supplies, alone in the lab.

Laboratory Attire: The School of Engineering operates a number of different laboratories with different potential safety hazards. All personnel, working in any laboratory, shall wear clothing and safety equipment that is appropriate to the laboratory and protective with respect to the hazards that may be present. All personnel in chemical-

using labs should wear a lab coat, closed-toe shoes, and safety glasses at a minimum. More specific attire or protective equipment might be required dependent on the specific hazards in that lab. Standards for appropriate dress, required safety equipment and good laboratory practices shall be developed for each laboratory by the SECO, in concert with VEHS and the department chair, and communicated to all laboratory personnel.

Eating and Drinking in Labs: Eating and drinking are not permitted in School of Engineering Labs unless in an approved clear area.

Vanderbilt University School of Engineering

Environmental Health and Safety: Laser Safety

1. Policy

Lasers can present health and safety hazards to personnel. This policy and procedures manual on laser safety in the School of Engineering outlines methods for assessing and controlling the hazards associated with lasers for the protection of potentially exposed personnel. Installation and operation of lasers for research purposes must be in compliance with the current ANSI Z136.1 standard and the ANSI Z136.5 standard for Safe Use of Lasers in Educational Institutions. A copy of this standard is on file with the School Laser Safety Officer. All laser installations in the School of Engineering must be coordinated with and approved by the School Laser Safety Officer. All Class 3b and 4 lasers must be registered with the Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>. A comprehensive Vanderbilt University Laser Safety Program Guide is available on the VEHS web site. A number of additional sources of information on lasers and laser safety can be found on the Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>, web site at Safety Links/Laser Safety. The links include a link to the OSHA Technical Manual – Laser Safety, which is an excellent source of additional information. A training course on Laser Safety can be found at Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>. All personnel using Class 3a, 3b, or 4 lasers in their research or teaching should avail themselves of this Laser Safety training course.

2. Responsibilities

2.1 Principal Investigators are responsible for:

- The immediate supervision of lasers in the laboratory.
- Providing, implementing and enforcing the safety recommendations and requirements prescribed in this policy.
- Assuring that all laser operators are properly trained and proficient in safety procedures before operating a laser.
- Classifying and labeling all of their lasers.
- Completing a Laser Registry Form and providing it to VEHS.
- Notifying VEHS immediately in the event of any incident involving lasers.

2.2 Laser Operator s are responsible for:

- Following laboratory safety procedures and laser standard operating procedures.
- Keeping the Principal Investigator fully informed of any departure from established safety procedures. This includes notification of an exposure incident.
- Receive Laser Safety Training before working with lasers.

2.3 Vanderbilt Environmental Health & Safety and/or School of Engineering Laser Safety Officer is responsible for:

- Conducting lab inspections to ensure that safety requirements are followed.
- Providing assistance in evaluating and controlling hazards.
- Updating the Laser Safety Policy.
- Maintaining all records of lasers and laser operators.
- Coordinating laser safety training for personnel who are assigned to an area where lasers are operated.
- Participating in accident investigations involving lasers.
- Advising laser operators of the requirements of the Medical Surveillance program.

3. Laser Hazard Classification and Primary Precautions

Lasers are identified by type, e.g., Helium-Neon, wavelength(s), or Laser Hazard Classification. The Laser Hazard Classification is determined by the wavelength(s), maximum duration of exposure, and the average power output of the laser. Manufacturers of lasers are required to label all lasers with the appropriate Laser Hazard Classification “Caution” or “Danger” warnings. If an unlabeled laser is found, contact the School of Engineering Laser Safety Officer. Class 1, 2, and 3a lasers should have a yellow and black “Caution” label, while Class 3b, and 4 lasers should have a red, black, and white “Danger” label. The universal symbol for a laser hazard is a sunburst pattern of short spokes with a single long spoke radiating from the center of the sunburst. Details of the laser classes can be found in the Vanderbilt University Laser Safety Program Guide available at VEHS, <http://www.safety.vanderbilt.edu>.

4. Laser Safety Control Measures

Class 1 Lasers:

- Prior to operation, the laser shall be classified as to its hazard class, and equipped with the appropriate warning label. As previously stated, Manufacturers of lasers are required to label all lasers with the appropriate Laser Classification warning.
- Prior to maintenance, repair, or modifications, a safety plan should be developed to ensure personnel will not be exposed to health or safety hazards.
- A protective housing shall encase the laser.
- The housing shall be interlocked to prevent operation of the laser while the housing is removed.

Class 2 Lasers:

- The laser safety control measures for Class 1 lasers shall be complied with in addition to the following.
- Do not stare into the beam.

- Control all unnecessary shiny reflecting surfaces from the work area and yourself.
- A non-reflective surface should be used as a beam stop. The area along all parts of the path of the beam should be cleared of personnel and reflective material.
- Do not use at eye height.
- A protective housing shall encase the laser.
- The housing shall be interlocked to prevent operation of the laser while the housing is removed.

Class 3a Lasers:

- The laser safety control measures for Class 2 lasers shall be complied with in addition to the following.
- Do not stare into the beam.
- Control all unnecessary shiny reflecting surfaces from the work area and yourself.
- A non-reflective surface should be used as a beam stop. The area along all parts of the path of the beam should be cleared of personnel and reflective material.
- Do not use at eye height.
- Post “Caution” signs alerting personnel to the use of Class 3a lasers in the work area.
- Personnel should have access to laser safety glasses for the appropriate wavelength.
- Personnel using a Class 3a laser should receive laser safety training.
- A protective housing shall encase the laser.
- The housing shall be interlocked to prevent operation of the laser while the housing is removed.

Class 3b Lasers:

- The laser safety control measures for Class 3a lasers shall be complied with in addition to the following.
- The path of the laser beam shall be enclosed, and inaccessible to personnel.
- Control all unnecessary shiny reflecting surfaces from the work area and yourself.
- A protective housing shall encase the laser.
- The housing shall be interlocked to prevent operation of the laser while the housing is removed.
- A permanent non-reflective beam stop shall be incorporated.
- Post “Danger” signs alerting personnel to the use of Class 3b lasers in the work area.
- If used with a chamber, all windows not in use shall be constructed of or covered with material that will not transmit light of the wavelength generated by the laser.

- Personnel shall have access to laser safety glasses for the appropriate wavelength. See the Vanderbilt University Laser Safety Program Guide available on the VEHS web site, <http://www.safety.vanderbilt.edu> .
- The area of the laser shall not be in personnel traffic areas.
- The user of the Class 3b laser shall provide documentation to the School Laser Safety Officer regarding safety measures taken and planned.
- Personnel using a Class 3b laser shall receive laser safety training provided by Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu> , or other appropriate laser safety training instruction.
- The School Laser Safety Officer shall be notified when alterations which may affect safety are made to equipment.

Class 4 Lasers:

- The laser safety control measures for Class 3 lasers shall be complied with in addition to the following.
- A protective housing shall encase the laser.
- The housing shall be interlocked to prevent operation of the laser while the housing is removed.
- The path of the laser beam shall be enclosed, and inaccessible to personnel. This enclosure shall be interlocked.
- A permanent non-reflective beam stop shall be incorporated.
- An alarm or revolving light shall be in use during operation of the laser.
- If used with a chamber, all windows not in use shall be constructed of or covered with material that will not transmit light of the wavelength generated by the laser.
- Personnel shall have access to laser safety glasses for the appropriate wavelength. See the Vanderbilt University Laser Safety Program Guide available on the VEHS web site, <http://www.safety.vanderbilt.edu> .
- The area of the laser shall not be in personnel traffic areas, and is considered a Laser Controlled Area.
- The user of the Class 4 laser shall provide documentation to the School Laser Safety Officer regarding safety measures taken and planned.
- Personnel using a Class 4 laser shall receive laser safety training provided by Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu> , or other appropriate laser safety training instruction.
- The School Laser Safety Officer shall be notified when alterations which may affect safety are made to equipment.
- Post “Danger” signs alerting personnel to the use of Class 4 lasers in the work area.

5. Procedure for Using a Laser or Laser System

To safely use a laser or laser system in the School of Engineering, and to protect personnel in the area from laser hazards, a Laser Standard Operating Procedure must be

utilized. A Laser Standard Operating Procedure Template is available on the VEHS web site, <http://www.safety.vanderbilt.edu>. The Laser Standard Operating Procedure for Class 1 and 2 lasers is satisfied by following the established School of Engineering Safety Control Measures as described in section 3 of this manual. No written documentation is required for Class 1 or 2 lasers. More involved Laser Standard Operating Procedures are required for Class 3a, 3b, and 4 lasers or laser systems. Listed below are the components required in the written Laser Standard Operating Procedure submitted to the School Laser Safety Officer before work may begin with the applicable class laser.

Class 1 Lasers:

See Laser Safety Control Measures (section 3) for Class 1 lasers. No written Laser Standard Operating Procedure is required. Ensure a “Caution” label is attached to the laser housing, as required of the manufacturer.

Class 2 Lasers:

See Laser Safety Control Measures (section 3) for Class 2 lasers. No written Laser Standard Operating Procedure is required. Ensure a “Caution” sign appropriate for Class II lasers is posted in the work area.

Class 3a Lasers:

See Laser Safety Control Measures (section 3) for Class 3a lasers. Ensure a “Caution” sign appropriate for Class 3a lasers is posted in the work area. Documentation should be provided to the School Laser Safety Officer, which should include a diagram of the intended setup, evidence of laser safety training for involved personnel, a description of the Class 3a laser, and evidence of laser safety protective eyewear for involved personnel.

Class 3b Lasers:

A written Laser Standard Operating Procedure shall be submitted to the School Laser Safety Officer prior to operation of a Class 3b laser or laser system. The written plan shall contain a diagram of the intended setup, documented laser safety training of involved personnel and an outline of the laser safety training program, documentation describing the intended experiment and use of the Class 3b laser or laser system (including a description of the Class 3b laser or laser system), a checklist indicating that all steps required by the Laser Safety Control Measures (see section 3) for Class 3b lasers have been completed, and documentation outlining the engineering controls, personal protective equipment, and administrative procedures implemented to safeguard users of the laser and personnel in the area. No experiments with or testing of a Class 3b laser or laser system shall begin without a written plan received and approved by the School Laser Safety Officer. Ensure a “Danger” sign appropriate for Class 3b lasers is posted in the work area.

Class 4 Lasers:

A written Laser Standard Operating Procedure shall be submitted to the School Laser Safety Officer prior to operation of a Class 4 laser or laser system. The written plan shall contain a diagram of the intended setup, documented laser safety training of involved personnel and an outline of the laser safety training program, documentation describing

the intended experiment and use of the Class 4 laser or laser system (including a description of the Class 4 laser or laser system), a checklist indicating that all steps required by the Laser Safety Control Measures (see section 3) for Class 4 lasers have been completed, and documentation outlining the engineering controls, personal protective equipment, and administrative procedures implemented to safeguard users of the laser and personnel in the area. No experiments with or testing of a Class 3b or 4 laser or laser system shall begin without a written plan received and approved by the School Laser Safety Officer. Ensure a “Danger” sign appropriate for Class 4 lasers is posted in the work area.

6. Medical Surveillance

A Medical evaluation is required for all Class 3b and 4 laser users in the university and medical center. This evaluation may be scheduled in the Vanderbilt Occupational Health Clinic, <http://www.vanderbilt.edu/HRS/wellness/occhealth.htm> , and must be completed prior to laser use, and upon termination of employment or transfer to non-laser related duty.

The initial medical evaluation establishes the faculty/staff member's baseline, so that changes in the eye or skin resulting from an acute laser exposure may be accurately and efficiently diagnosed and treated. The exam also documents pre-existing eye or skin conditions which could become worse with exposure to lasers. The exam is also an opportunity for the faculty/staff member to discuss laser related health issues with the medical provider.

7. Recordkeeping

The School of Engineering Laser Safety Officer will maintain all records pertaining to laser operator training, laboratory Standard Operating Procedures (SOP's), laser safety inspections, laboratory incidents, School of Engineering Laser Policy, VEHS Laser Safety Guide, and regulatory standards.

The laboratory PI will maintain copies of records pertaining to laser operator training, laboratory SOP's, laser safety inspections, School of Engineering Laser Policy, and VEHS Laser Safety Guide.

Vanderbilt University School of Engineering

Environmental Health and Safety: Shop Safety Handout

Shop Safety

The School of Engineering's principal Machine (Model) Shop is located on the first floor of Featheringill Hall in the west wing. A smaller Model Shop is located in Olin Hall on the first level. Additional machine tools and associated equipment are located in some of the research labs in the School. The single most important issue in the School in using any of these Shop facilities is safety. The primary objective of all people that use the Machine Shops must be to uphold safety. There is nothing that can be designed or built using the Machine Shops that is worth trading for a permanent disability. This handout is intended to give you some guidelines to working safely in the Machine Shops as well as familiarize you with some of the hazards.

The Machine Shop is a more hazardous environment than people are used to. All personnel must be properly trained before using the equipment. When you enter the shop, you should make a conscious effort to adjust mentally and physically to the increased hazards around you. Consult the first five safety rules below to adjust yourself to existence in a shop environment. Scrupulously follow safe procedures. Be careful to think through each operation before you execute it. Use your common sense. Do not do things that seem unsafe.

The shop supervisor or faculty PI or trained assistant is there to help you use the shop. However, they cannot be everywhere at once. Users of the shop must assume responsibility for the safe use of the facility. If you are unsure of the proper method to do something, wait until the shop supervisor or other trained person can give you guidance. If you notice an unsafe condition in the shop, fix it or bring it to the attention of the shop supervisor. If you think someone is using the equipment unsafely, suggest a safer alternative or bring that activity to the attention of the shop supervisor.

General Safety Rules

The rules listed below apply to all people doing work in the School's Machine (Model) Shops. They are not all encompassing. Most shop areas and machines have specific safety rules that relate to their use, and these should be consulted. The following rules are a good general guide to safe use of the facility.

1) **Never work alone.**

When you are working in the shops you must have a second person present who is capable of rendering assistance in case of an accident. Accidents do happen. You need to have a second person present so that if you become unable to help yourself,

the second person can render aid and contact outside help. At Vanderbilt, the emergency number to call for assistance is 1-1911.

- 2) **Never work when you are impaired. This includes times when you are too tired, stressed, or otherwise inhibited from exercising appropriate caution in the shop.** Do not enter the shop when you are under the influence of any intoxicants or medications that might make you drowsy or alter your ability to be alert to reality.

Do not use the shop when you are too frantic to think clearly and carefully. Try to avoid last minute shop work in favor of a consistent weekly effort. Deadline-driven haste can lead to ruined projects and/or serious injuries.

Do not use the shop if you are wearing a cast or bandage that limits your mobility. In an emergency you must be able to react quickly to avoid injury.

- 3) **Wear all necessary protective gear and clothing. This always includes safety glasses and shoes that completely enclose your feet. Some shop activities will require additional safety gear.**

All persons, including visitors, entering the Machine Shop shall wear eye protection appropriate to the hazard. All such eye protection must conform to ANSI standard Z87.1. Safety glasses that conform to the standard are marked with a “Z87”.

If you wear glasses already you should be aware that most eyeglasses purchased in the United States conform to ANSI Z87.1 for frontal impact. You may wear such glasses for work in the shop. To protect your eyes from side impact you can add side shields to your existing glasses or purchase prescription safety glasses for use in the shop.

Do not wear contact lenses in the welding shop. The intense light from the arc welding torches can cause contact lenses to damage your eyes. Additional information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Persons exposed to foot injury hazards shall wear foot protection appropriate to the hazard. Leather shoes are necessary to protect your feet from stray droplets of molten metal in the welding area. Additional information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Certain dusts and fumes require respiratory protection. In the event that a respirator is necessary, the Shop Supervisor may require you to obtain and wear one for operations that expose you to airborne hazards. Additional information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health. Any employee or personnel wearing any respirator other than a dust mask must have medical surveillance through the Vanderbilt Occupational Health Clinic, <http://www.vanderbilt.edu/HRS/wellness/occhealth.htm>, and be fit-tested by VEHS.

When it is not practical to reduce sound levels below ninety decibels, then persons exposed to such sound levels shall wear hearing protection in the form of foam ear inserts.

Leather gloves are available in the welding areas for protection from the heat and radiation of welding. Additional information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Welding requires the use of welding masks with appropriate eye-protective lenses and natural fiber clothes that completely cover your skin. The electric arc in particular is an intense source of ultraviolet and infrared radiation. Even a brief exposure of your unprotected eyes can cause severe damage to your sight. Do not look at an electric without the correct eye protection. If you expose bare skin to the light of the arc, the radiation is intense enough to give you severe sunburn. Natural fiber clothes are required because they will not melt onto you in the presence of high heat, as some man made materials will. Additional information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

4) Long hair must be tied up securely.

Most of the power tools in the shop are based around a rapidly rotating shaft. In use the shaft is frequently sticky with oil. Long, loose hair can stick to such a rotating shaft and pull the owner of the hair into the tool. Long hair must be kept out of harm's way by tucking it into a cap, tying it up. Or knotting it in a way that prevents it from dangling.

5) Remove all personal accessories and loose clothing that might get caught in moving machinery. This includes rings, watches, jewelry, personal stereos, shop rags, ties, and open jackets.

Like long hair, things that dangle from your person can get caught in rotating machinery. Regardless of the fashion, it is not worth risking your health.

Loose garments must not be worn in the shop. Tuck in loose shirttails and sweat pants ties. Keep tight fitting jackets or coats closed.

Do not keep shop rags in your pockets.

Do not wear personal stereos or headphones while you are working in the shop.

6) Never leave a machine running unattended.

Some of the tools in the shop can be set to cut automatically. You must keep your attention focused on the machining operation. If you are focused on the process, you will be more likely to react appropriately in the event of an accident.

The Machine (Model) Shops in the School are used primarily for building prototypes. As a result most set-ups are not tried and tested in a production sense. It is inevitable

that some set-ups will go bad. If you are paying attention to the operation, you may be able to avert injury to yourself and the people around you.

7) **Never leave a chuck key in a chuck or a drawbar wrench on a drawbar. If the key or wrench is in use your hand must be on it.**

Chuck keys and drawbar wrenches can be accidentally launched across the shop if they are left in place. The startup power of the lathes and mills can throw the tools with enough force to puncture a body cavity. You must be careful not to leave the chuck keys and drawbar wrenches in place when they are not in use.

The same rule applies to the chuck keys used to tighten the chuck on the drill presses.

8) **Keep your hands well away from the point of contact between the work piece and the cutter. If you must hand hold the work to keep it in place, your set-up is unsafe; improve it.**

While working on the lathes and mills you should never hand hold your work. If the tool or work piece is vibrating, the chances are high for a sudden shift in the set-up. If you are hand holding the tool or work piece, you might not be able to get out of the way as the parts come together. Besides, the cutting forces are too large for you to be effective in securing the work piece or tool.

When using the drill presses, especially with large drills and tough to cut materials, you must clamp the work to the drill press table. If the drill catches an unclamped work piece, the rotating part can cut your hand.

On the band saws, table saw, router table, chop saw and any other tool that is designed to be used with a hand fed work piece, be certain to keep your hands, fingers, and other body parts out of the path of cutters and away from the point between the work piece and the cutter.

Keep all parts of yourself at least 6 inches from the point of contact between any work piece and any cutter.

Do not remove chips with your fingers. Use brushes, pliers, or compressed air.

9) **Support work pieces and cutting tools as securely as possible.**

A vibrating set-up is usually an indication that the work piece and/or tool are not held strongly enough to resist the applied cutting forces. You must take the time to secure the set-up to resist the force of cutting or use a different operation to do the job. Do not try to make do with a flimsy set-up.

Expect to spend a lot (80% or more) of your time in the shops making set-ups. It is the nature of prototyping. Unfortunately, it is also hard to visualize when you are thinking about work in the shop. This is where a well-metered approach to work in the shop will pay off. If you think things are taking longer than you expected, you can

scale down or redesign your work. At the last minute it is hard to make such large changes.

10) Have the shop supervisor or other trained person check you out the first time you use each machine or process in the shop. Do not operate any machinery with which you are unfamiliar.

Each and every tool in the shop has safe operating procedures associated with it. Do not work on any tool in the shop until the shop supervisor or other trained person has introduced you to its safe operation. You should get checked out even on tools you have been taught to operate elsewhere. The tools in the shop may have idiosyncrasies that you should know about.

Clean Up Procedures

Careful clean up is part of the safe operation of the shops. If a person cannot find a needed tool in its regular place or if a person is required to work in someone else's mess, the result will be frustration. It is difficult to keep safety in mind when you are frustrated. So it is important, especially in times of heavy use, to keep the shop clean with everything returned to its correct place at the end of its use.

1) Shut off power to the machine.

Turn off the main power switch for the machine.

Disengage all power feeds and lead screws.

2) Un-mount all cutters

Remove all end mills, lathe tools, drills, and similar tools from their tool holders.

3) Put away all measuring tools, hand tools, material scraps, and drawings.

Put away all objects that do not belong permanently with the machine. If you do not know where something goes, ask the shop supervisor. If the shop supervisor is unavailable, leave whatever it is in plain sight on a table. It is better to leave it out than to put it away in the wrong place.

4) Clean chips and excess oil from machines and chip pans.

Protect your hands from sharp chips with a shop rag.

Most machines can be wiped down completely with a shop rag. If you must use compressed air, be careful with it. Do not point compressed air guns at people. Blown chips can become lodged in eyes and the compressed air itself can do severe damage to a person's body. Use the compressed air early in the clean up process. Otherwise you will blow chips all over the areas you have already cleaned.

Do not use brooms or brushes on the machine tools. The brushes and brooms pick up abrasive dirt from the floor. If abrasive gets on the machine tools they will wear very quickly.

You should clean the equipment well enough that the next user will not be able to tell what material you were using.

- 5) **Put a light coat of way oil on the machine ways.**
Move the machine slides to one extreme position and oil the exposed ways. Next move the ways to the middle of their travel and oil the newly exposed sections. Leave the machine slides at the approximate center of travel.
- 6) **Sweep the floor in the vicinity of the machines you have used.**
Collected debris should be either recycled or thrown in a scrap container.
- 7) **Ensure all hazardous materials such as used paint, oils, solvents, degreasers, etc. are properly labeled and stored.**
Vanderbilt Environmental Health and Safety (VEHS),
<http://www.safety.vanderbilt.edu>, should be notified of anything requiring disposal.

**Vanderbilt University
School of Engineering**

Environmental Health and Safety: Shop Self-Inspection Form

The School of Engineering's principal Machine (Model) Shop is located on the first floor of Featheringill Hall in the west wing. A smaller Model Shop is located in Olin Hall on the first level. Additional machine tools and associated equipment are located in some of the research labs in the School. All of these shops are required to perform and document self-inspections at least annually as part of OSHA's requirement for an effective Injury and Illness Prevention Program (IIPP). This self-inspection form will help document shop safety inspections and will assist shop personnel in identifying and correcting many common unsafe practices and conditions.

Shop Supervisors and/or PI's shall perform and document a self-inspection at least once a year, using this form or an equivalent. Each question should be answered by checking "Yes" (satisfactory), "No" (needs correction), or "N/A" (if the question does not apply to your shop).

The results of the self-inspection shall be shared with the Shop Machinery Safety Officer (SMSO), who will maintain a file copy.

This form was designed to help ensure compliance with University/State/OSHA regulations that require documented periodic inspections of all work areas as part of an effective IIPP. However, completion of this self-inspection checklist and correction of any findings noted herein does not ensure that State/OSHA will not issue citations during an inspection. In additions, the School's Shop Machinery Safety Officer (SMSO), representatives from the School Safety Committee and/or Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu> , will periodically be verifying that self-inspections have been documented on-site.

_____ Shop Location	_____ Department
_____ Shop Supervisor's Name (printed)	_____ Shop Supervisor's Signature
_____ Inspector's Name (printed)	_____ Inspector's Signature
_____ Date of Inspection	

Administrative

1. Are training records maintained to indicate which faculty/staff/students are trained and qualified to use shop/lab equipment?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Establish a training file. Prohibit untrained personnel from using shop equipment.
Completion Date:_____.
2. Is an operator's manual or other written safe operating procedures available for each piece of equipment and machine in shop?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Obtain operator's manuals or write safe operating procedures for each piece of equipment.
Completion Date:_____.
3. Is there a lockout/tagout program in place to prevent accidental energizing of equipment or machines that are being repaired, adjusted, or undergoing tool changes?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Institute a lockout/tagout (or equivalent program).
Completion Date:_____.

Housekeeping

4. Are work areas (including equipment and machinery) kept clean and orderly, so as to prevent trip and fire hazards?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Include housekeeping as part of regular shop activities (e.g., the job is not finished until all tools and materials are put away, the work area swept and waste is disposed of.)
Completion Date:_____.
5. Have shop users been advised that they should not consume food or drinks in shop areas unless there is a designated clean area?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Prohibit consuming food or beverages in shop areas. Post signs if necessary.
Completion Date:_____.
6. Are floors and work surfaces dry and/or made slip-resistant?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Keep surfaces dry or install slip-resistant material on surfaces that can not be kept dry.
Completion Date:_____.
7. Do all work areas have adequate ventilation, particularly for hazardous operations (e.g., welding, soldering, spray coating, using solvents)?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> , if there is a concern. Provide additional exhaust ventilation for operations that create dust, fumes, mists or vapors.
Completion Date:_____.

8. Are all compressed gas cylinders adequately secured with non-combustible restraints to keep the cylinder(s) from falling? Also, are all compressed gas cylinders capped when not in use?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> , for advice on how to properly secure cylinders. Train shop users to cap compressed gas cylinders when not in use.
Completion Date:_____.

Fire/Emergency Safety

9. Is the Vanderbilt Emergency telephone number, 1-1911, clearly posted?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Post Vanderbilt Emergency telephone number, 1-1911, near the telephone(s).
Completion Date:_____.

10. Are fire alarm pull boxes clearly identifiable and unobstructed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that all fire alarm pull boxes are clearly labeled. Clear any obstructions in front of fire alarm pull boxes.
Completion Date:_____.

11. Are fire hose stations and portable extinguishers clearly identifiable and unobstructed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that all fire hoses and extinguishers are clearly identifiable. If they are not clearly marked, contact Vanderbilt Plant Operations, <http://www.vanderbilt.edu/PlantOps> . Clear any obstructions in front of fire hoses and extinguishers.
Completion Date:_____.

12. Are portable fire extinguishers tagged with current annual inspections?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact Vanderbilt Plant Operations, <http://www.vanderbilt.edu/PlantOps> , if they have not been within the last 12 months.
Completion Date:_____.

13. Are fire escapes, exit doors, stairwells, and corridors kept clear and unobstructed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Remove any stored equipment and materials from exit doors, stairwells and corridors. Prohibit storage in these areas.

Completion Date:_____.

14. Are flammable materials stored in approved safety containers or safety cabinets?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that flammable materials are stored in approved safety containers or cabinets. If there are questions, consult Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> . Prohibit storage of flammable materials outside safety containers or cabinets.

Completion Date:_____.

15. Are combustible materials kept at least 35 feet away from welding operations, or is fireproof covering provided?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Move combustible material at least 35 feet from welding operations or place fireproof covering over them. Detailed safety information regarding welding and cutting can be found at the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Completion Date:_____.

Electrical Safety

16. Are electrical panels accessible and are circuit breakers clearly labeled?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that there is a minimum of 36 inches of clearance in front of all electrical panels. If circuit breakers are not labeled, notify Vanderbilt Plant Operations, <http://www.vanderbilt.edu/PlantOps> .

Completion Date:_____.

17. Is all electrical shop equipment properly grounded? (Ensure that the grounding pin has not been removed, and that 3-pin to 2-pin adapters are not used.)

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Inspect all electrical connections for grounding plugs or wires.

Completion Date:_____.

18. Is there a lockout/tagout program in place to prevent accidental energizing of equipment or machines that are being repaired, adjusted, or undergoing tool changes?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Institute a lockout/tagout (or equivalent program).

Completion Date:_____.

19. Are extension cords and power strips in good condition (e.g., no breaks or exposed wiring), used only as temporary wiring (less than 30 days), and not connected in series?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Remove all extension cords connected in series, permanent use power strips, and permanent use extension cords. Contact Vanderbilt Plant Operations, <http://www.vanderbilt.edu/PlantOps> , to arrange for installation of permanent wiring. Dispose of or repair all electrical cords that are not in good condition

Completion Date:_____.

Personal Protective Equipment

20. Are shop users provided with eye and face protection such as safety glasses and face shields where needed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Provide and ensure the use of approved eye protection for all shop users and visitors upon entry to shop work areas. Ensure that face protection is used when flying materials could cause injury to the face. For welding and cutting operations, consult the American Welding Society, <http://www.aws.org> , web site under Safety & Health.

Completion Date:_____.

Portable Power-Operated Tools

21. Are power tools grounded or double insulated?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that power tools are inspected prior to each use. Repair or replace power tools that are not grounded or identified as double insulated.

Completion Date:_____.

22. Are rotating or moving parts of equipment or tools guarded (e.g., hand-held grinder wheel)?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that all guards are in place and used during tool operation.

Completion Date:_____.

Shop Equipment and Machines

23. Are abrasive wheel grinders equipped with a work rest adjusted to within 1/8 inch of the wheel, and side guards that cover the spindle, nut, flange and 75% of the wheel diameter?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Prohibit the use of wheel grinders until work rests and proper side guards are installed and/or properly adjusted.

Completion Date:_____.

24. Are radial arm saws provided with a spreader and automatic return?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Prohibit the use of radial arm saws until spreaders and automatic returns are installed.

Completion Date:_____.

25. Do table saws have a hand guard, spreader, and anti-kickback device for use when applicable?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Prohibit the use of table saws until anti-kickback devices are installed.

Completion Date:_____.

26. Are all pieces of equipment and machinery provided with adequate guarding? (Rotating parts, nip points and v-belts must be guarded.)

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Prohibit the use of equipment and machinery until adequate guards are installed.

Completion Date:_____.

27. Are pieces of equipment or machinery securely anchored to the floor or a bench?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure equipment or machinery that can move from its operation or vibration or the vibration of nearby equipment or machinery is securely anchored.

Completion Date:_____.

Welding and Cutting

28. Is adequate ventilation provided in areas where welding and/or cutting occurs?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>, to have ventilation evaluated if there is a concern. Additional information can be obtained at the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Completion Date:_____.

29. Are welders provided with appropriate personal protection equipment (e.g., eye protection, gloves, flame resistant clothing or leathers)?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that welders are provided with and use the proper personal protective equipment. Detailed information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Completion Date:_____.

30. Are welders provided with a fire extinguisher within ten feet of their working area?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that there is a fire extinguisher with a current inspection tag within 10 feet of welding work areas. Additional information can

be obtained from the American Welding Society, <http://www.aws.org> , web site under Safety & Health.

Completion Date:_____.

Hazardous Materials

31. Is there a Material Safety Data Sheet (MSDS) for each chemical (solvent, degreaser, etc.) stored in the shop?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Information regarding MSDS's as well as specific MSDS's can be obtained at Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> .

Completion Date:_____.

32. Are all containers of chemicals labeled with the contents and primary hazards?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Properly label all chemical containers, including stored and temporary containers

Completion Date:_____.

33. Has the shop submitted an updated chemical inventory to Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> , within the last 12 months?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Complete and submit a chemical inventory to Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu> .

Completion Date:_____.

34. Are all paint, oil, degreasers, solvents, and other chemicals labeled and stored properly?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Properly label all paint, oil, degreasers, solvents and other chemicals and store in a proper location.

Completion Date:_____.

35. Has Vanderbilt Environmental Health and Safety (VEHS),

<http://www.safety.vanderbilt.edu>, been notified of any unwanted or waste hazardous materials requiring disposal?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Notify VEHS of any unwanted or waste hazardous materials requiring disposal.

Completion Date:_____.

D Material Safety Data Sheets

Composite Fibers:

- Fibre Glast Part #262 4 oz. Fiberglass Fabric:
<https://www.fibreglast.com/downloads/PDCT-MSDS-00056.pdf>
- Fibre Glast Part #1069 - 5.7oz Graphite 3K 2 x 2 Twill Carbon Fiber:
<https://www.fibreglast.com/downloads/PDCT-MSDS-00111-C.pdf>

Epoxy:

- Devcon 2 Ton Epoxy [1:1]: 5-Minute Epoxy Resin, 2-Ton Hardener:
<https://itwperformancepolymers.com/media/15407/14260-devcon-2-ton-epoxy.pdf>
- Devcon 5 Minute Epoxy amber [1:1]: 5-Minute Epoxy Resin, 5-Minute Epoxy Hardener:
<https://itwperformancepolymers.com/wp-content/uploads/umb/15412/14310-devcon-2-ton-epoxy.pdf>
- Fibre Glast Part #2000 System 2000 Epoxy Resin:
<https://www.fibreglast.com/downloads/PDCT-MSDS-00130.pdf>
- Fibre Glast Part #2120 Epoxy Hardener:
<https://www.fibreglast.com/downloads/PDCT-MSDS-00133.pdf>
- J-B Weld Steel Reinforced Epoxy Resin and Hardener - Twin Tubes:
https://cdn.shopify.com/s/files/1/0411/5921/files/Steel_Reinforced_Epoxy_Twin_Tubes.pdf?785811878289892783
- Loctite Fast Cure Epoxy:
<https://www.grainger.com/ec/pdf/Loctite-Fixmaster-Fast-Cure-Data-Sheet.pdf>

Explosives:

- Aerotech L940X Motor:
<https://www.apogeerockets.com/downloads/MSDS/Aerotech/Motors.pdf>
- Cesaroni L1350 Motor:
http://www.pro38.com/MSDS/SDS--ProX--ver4-03-EN_FR.pdf
- Owen Compliance Black Powder:
<http://aiaacrocketry.org/wp-content/uploads/2010/11/black-powder-4f.pdf>

Glues/Adhesives:

- Ethyl 2-Cyanoacrylate:
<http://www.sigmaaldrich.com/MSDS/MSDS/DisplayMSDSPage.do?country=US&language=en&productNumber=E1505&brand=ALDRICH&PageToGoToURL=http%3A%2F%2Fwww.sigmaaldrich.com%2Fcatalog%2Fsearch%3Fterm%3DEthyl%2B2-cyanoacrylate%26interface%3DProduct%2520Name%26N%3D0%2B%26mode%3Dmode%2520matchpartialmax%26lang%3Den%26region%3DUS%26focus%3DproductN%3D0%2520220003048%252019853286%252019853269>
- Gorilla Hot Glue:
<https://www.gorillatough.com/wp-content/uploads/Gorilla-Hot-Glue.pdf>
- 3M Super 77 Adhesive, Bulk:
https://multimedia.3m.com/mws/mediawebserver?mwsId=SSSSSuUn_zu8l00x482xN8mvNv70k17zHvu9lxtD7SSSSSS--

Liquid Foams:

- 3M Polyurethane Foam Insulation Adhesive CR-20, Parts A and B:
<https://tapecasecom.blob.core.windows.net/product/download/3m%20cr-20-c14bbee6-4b5d-47b5-bb43-eba14796649b.pdf>

Mold Compounds:

- Fibre Glast Part #1153 - FibRelease:
<https://s3.amazonaws.com/cdn.fibreglast.com/downloads/1153.pdf>
- Meguiar's M08, Mirror Glaze Maximum Mold Release Wax (23-135A): M0811:
<https://www.fibreglast.com/downloads/PDCT-MSDS-00030.pdf>

Solvents:

- Ethyl Alcohol, 70%:
<https://ehss.syr.edu/wp-content/uploads/2020/05/ThermoFisher-70-Ethanol-SDS.pdf>
- Klean-Strip Acetone:
<https://www1.mscdirect.com/MSDS/MSDS00052/70250444-20190911.PDF>

E FAA Regulations

§ 101.1 Applicability.

- (a) This part prescribes rules governing the operation in the United States, of the following:
- (1) Except as provided for in § 101.7, any balloon that is moored to the surface of the earth or an object thereon and that has a diameter of more than 6 feet or a gas capacity of more than 115 cubic feet.
 - (2) Except as provided for in § 101.7, any kite that weighs more than 5 pounds and is intended to be flown at the end of a rope or cable.
 - (3) Any amateur rocket except aerial firework displays.
 - (4) Except as provided for in § 101.7, any unmanned free balloon that -
 - (i) Carries a payload package that weighs more than four pounds and has a weight/size ratio of more than three ounces per square inch on any surface of the package, determined by dividing the total weight in ounces of the payload package by the area in square inches of its smallest surface;
 - (ii) Carries a payload package that weighs more than six pounds;
 - (iii) Carries a payload, of two or more packages, that weighs more than 12 pounds; or
 - (iv) Uses a rope or other device for suspension of the payload that requires an impact force of more than 50 pounds to separate the suspended payload from the balloon.
- (b) For the purposes of this part, a *gyroglider* attached to a vehicle on the surface of the earth is considered to be a kite.

[Doc. No. 1580, 28 FR 6721, June 29, 1963, as amended by Amdt. 101-1, 29 FR 46, Jan. 3, 1964; Amdt. 101-3, 35 FR 8213, May 26, 1970; Amdt. 101-8, 73 FR 73781, Dec. 4, 2008; 74 FR 38092, July 31, 2009; Amdt. 101-9, 81 FR 42208, June 28, 2016; Amdt. Nos. 101-10, 85 FR 79826, Dec. 11, 2020]

§ 101.3 Waivers.

No person may conduct operations that require a deviation from this part except under a certificate of waiver issued by the Administrator.

[Doc. No. 1580, 28 FR 6721, June 29, 1963]

§ 101.5 Operations in prohibited or restricted areas.

No person may operate a moored balloon, kite, amateur rocket, or unmanned free balloon in a prohibited or restricted area unless he has permission from the using or controlling agency, as appropriate.

[Doc. No. 1457, 29 FR 46, Jan. 3, 1964, as amended at 74 FR 38092, July 31, 2009]

§ 101.7 Hazardous operations.

(a) No person may operate any moored balloon, kite, amateur rocket, or unmanned free balloon in a manner that creates a hazard to other persons, or their property.

(b) No person operating any moored balloon, kite, amateur rocket, or unmanned free balloon may allow an object to be dropped therefrom, if such action creates a hazard to other persons or their property.

(Sec. 6(c), Department of Transportation Act (49 U.S.C. 1655(c)))

[Doc. No. 12800, 39 FR 22252, June 21, 1974, as amended at 74 FR 38092, July 31, 2009]

Subpart B - Moored Balloons and Kites

Source: Docket No. 1580, 28 FR 6722, June 29, 1963, unless otherwise noted.

§ 101.11 Applicability.

This subpart applies to the operation of moored balloons and kites. However, a person operating a moored balloon or kite within a restricted area must comply only with § 101.19 and with additional limitations imposed by the using or controlling agency, as appropriate.

§ 101.13 Operating limitations.

(a) Except as provided in paragraph (b) of this section, no person may operate a moored balloon or kite -

- (1) Less than 500 feet from the base of any cloud;
- (2) More than 500 feet above the surface of the earth;
- (3) From an area where the ground visibility is less than three miles; or
- (4) Within five miles of the boundary of any airport.

(b) Paragraph (a) of this section does not apply to the operation of a balloon or kite below the top of any structure and within 250 feet of it, if that shielded operation does not obscure any lighting on the structure.

§ 101.15 Notice requirements.

No person may operate an unshielded moored balloon or kite more than 150 feet above the surface of the earth unless, at least 24 hours before beginning the operation, he gives the following information to the FAA ATC facility that is nearest to the place of intended operation:

- (a) The names and addresses of the owners and operators.
- (b) The size of the balloon or the size and weight of the kite.
- (c) The location of the operation.
- (d) The height above the surface of the earth at which the balloon or kite is to be operated.
- (e) The date, time, and duration of the operation.

§ 101.17 Lighting and marking requirements.

- (a) No person may operate a moored balloon or kite, between sunset and sunrise unless the balloon or kite, and its mooring lines, are lighted so as to give a visual warning equal to that required for obstructions to air navigation in the FAA publication “Obstruction Marking and Lighting”.
- (b) No person may operate a moored balloon or kite between sunrise and sunset unless its mooring lines have colored pennants or streamers attached at not more than 50 foot intervals beginning at 150 feet above the surface of the earth and visible for at least one mile.

(Sec. 6(c), Department of Transportation Act (49 U.S.C. 1655(c)))

[Doc. No. 1580, 28 FR 6722, June 29, 1963, as amended by Amdt. 101-4, 39 FR 22252, June 21, 1974]

§ 101.19 Rapid deflation device.

No person may operate a moored balloon unless it has a device that will automatically and rapidly deflate the balloon if it escapes from its moorings. If the device does not function properly, the operator shall immediately notify the nearest ATC facility of the location and time of the escape and the estimated flight path of the balloon.

Subpart C - Amateur Rockets

§ 101.21 Applicability.

- (a) This subpart applies to operating unmanned rockets. However, a person operating an unmanned rocket within a restricted area must comply with § 101.25(b)(7)(ii) and with any additional limitations imposed by the using or controlling agency.

(b) A person operating an unmanned rocket other than an amateur rocket as defined in § 1.1 of this chapter must comply with 14 CFR Chapter III.

[Doc. No. FAA-2007-27390, 73 FR 73781, Dec. 4, 2008]

§ 101.22 Definitions.

The following definitions apply to this subpart:

(a) ***Class 1 - Model Rocket*** means an amateur rocket that:

- (1) Uses no more than 125 grams (4.4 ounces) of propellant;
- (2) Uses a slow-burning propellant;
- (3) Is made of paper, wood, or breakable plastic;
- (4) Contains no substantial metal parts; and
- (5) Weighs no more than 1,500 grams (53 ounces), including the propellant.

(b) ***Class 2 - High-Power Rocket*** means an amateur rocket other than a model rocket that is propelled by a motor or motors having a combined total impulse of 40,960 Newton-seconds (9,208 pound-seconds) or less.

(c) ***Class 3 - Advanced High-Power Rocket*** means an amateur rocket other than a model rocket or high-power rocket.

[Doc. No. FAA-2007-27390, 73 FR 73781, Dec. 4, 2008]

§ 101.23 General operating limitations.

(a) You must operate an amateur rocket in such a manner that it:

- (1) Is launched on a suborbital trajectory;
- (2) When launched, must not cross into the territory of a foreign country unless an agreement is in place between the United States and the country of concern;
- (3) Is unmanned; and
- (4) Does not create a hazard to persons, property, or other aircraft.

(b) The FAA may specify additional operating limitations necessary to ensure that air traffic is not adversely affected, and public safety is not jeopardized.

[Doc. No. FAA-2007-27390, 73 FR 73781, Dec. 4, 2008]

§ 101.25 Operating limitations for Class 2-High Power Rockets and Class 3-Advanced High Power Rockets.

When operating *Class 2-High Power Rockets* or *Class 3-Advanced High Power Rockets*, you must comply with the General Operating Limitations of § 101.23. In addition, you must not operate *Class 2-High Power Rockets* or *Class 3-Advanced High Power Rockets* -

- (a) At any altitude where clouds or obscuring phenomena of more than five-tenths coverage prevails;
- (b) At any altitude where the horizontal visibility is less than five miles;
- (c) Into any cloud;
- (d) Between sunset and sunrise without prior authorization from the FAA;
- (e) Within 9.26 kilometers (5 nautical miles) of any airport boundary without prior authorization from the FAA;
- (f) In controlled airspace without prior authorization from the FAA;
- (g) Unless you observe the greater of the following separation distances from any person or property that is not associated with the operations:
 - (1) Not less than one-quarter the maximum expected altitude;
 - (2) 457 meters (1,500 ft.);
- (h) Unless a person at least eighteen years old is present, is charged with ensuring the safety of the operation, and has final approval authority for initiating high-power rocket flight; and
- (i) Unless reasonable precautions are provided to report and control a fire caused by rocket activities.

[74 FR 38092, July 31, 2009, as amended by Amdt. 101-8, 74 FR 47435, Sept. 16, 2009]

§ 101.27 ATC notification for all launches.

No person may operate an unmanned rocket other than a Class 1 - Model Rocket unless that person gives the following information to the FAA ATC facility nearest to the place of intended operation no less than 24 hours before and no more than three days before beginning the operation:

- (a) The name and address of the operator; except when there are multiple participants at a single event, the name and address of the person so designated as the event launch coordinator,

whose duties include coordination of the required launch data estimates and coordinating the launch event;

- (b) Date and time the activity will begin;
- (c) Radius of the affected area on the ground in nautical miles;
- (d) Location of the center of the affected area in latitude and longitude coordinates;
- (e) Highest affected altitude;
- (f) Duration of the activity;
- (g) Any other pertinent information requested by the ATC facility.

[Doc. No. FAA-2007-27390, 73 FR 73781, Dec. 4, 2008, as amended at Doc. No. FAA-2007-27390, 74 FR 31843, July 6, 2009]

§ 101.29 Information requirements.

(a) ***Class 2 - High-Power Rockets.*** When a Class 2 - High-Power Rocket requires a certificate of waiver or authorization, the person planning the operation must provide the information below on each type of rocket to the FAA at least 45 days before the proposed operation. The FAA may request additional information if necessary to ensure the proposed operations can be safely conducted. The information shall include for each type of Class 2 rocket expected to be flown:

- (1) Estimated number of rockets,
- (2) Type of propulsion (liquid or solid), fuel(s) and oxidizer(s),
- (3) Description of the launcher(s) planned to be used, including any airborne platform(s),
- (4) Description of recovery system,
- (5) Highest altitude, above ground level, expected to be reached,
- (6) Launch site latitude, longitude, and elevation, and
- (7) Any additional safety procedures that will be followed.

(b) ***Class 3 - Advanced High-Power Rockets.*** When a Class 3 - Advanced High-Power Rocket requires a certificate of waiver or authorization the person planning the operation must provide the information below for each type of rocket to the FAA at least 45 days before the proposed operation. The FAA may request additional information if necessary to ensure the proposed operations can be safely conducted. The information shall include for each type of Class 3 rocket expected to be flown:

- (1) The information requirements of paragraph (a) of this section,
- (2) Maximum possible range,
- (3) The dynamic stability characteristics for the entire flight profile,
- (4) A description of all major rocket systems, including structural, pneumatic, propellant, propulsion, ignition, electrical, avionics, recovery, wind-weighting, flight control, and tracking,
- (5) A description of other support equipment necessary for a safe operation,
- (6) The planned flight profile and sequence of events,
- (7) All nominal impact areas, including those for any spent motors and other discarded hardware, within three standard deviations of the mean impact point,
- (8) Launch commit criteria,
- (9) Countdown procedures, and
- (10) Mishap procedures.

[Doc. No. FAA-2007-27390, 73 FR 73781, Dec. 4, 2008, as amended at Doc. No. FAA-2007-27390, 74 FR 31843, July 6, 2009]

Subpart D - Unmanned Free Balloons

Source: Docket No. 1457, 29 FR 47, Jan. 3, 1964, unless otherwise noted.

§ 101.31 Applicability.

This subpart applies to the operation of unmanned free balloons. However, a person operating an unmanned free balloon within a restricted area must comply only with § 101.33 (d) and (e) and with any additional limitations that are imposed by the using or controlling agency, as appropriate.

§ 101.33 Operating limitations.

No person may operate an unmanned free balloon -

- (a) Unless otherwise authorized by ATC, below 2,000 feet above the surface within the lateral boundaries of the surface areas of Class B, Class C, Class D, or Class E airspace designated for an airport;
- (b) At any altitude where there are clouds or obscuring phenomena of more than five-tenths coverage;

- (c) At any altitude below 60,000 feet standard pressure altitude where the horizontal visibility is less than five miles;
- (d) During the first 1,000 feet of ascent, over a congested area of a city, town, or settlement or an open-air assembly of persons not associated with the operation; or
- (e) In such a manner that impact of the balloon, or part thereof including its payload, with the surface creates a hazard to persons or property not associated with the operation.

[Doc. No. 1457, 29 FR 47, Jan. 3, 1964, as amended by Amdt. 101-5, 56 FR 65662, Dec. 17, 1991]

§ 101.35 Equipment and marking requirements.

(a) No person may operate an unmanned free balloon unless -

- (1) It is equipped with at least two payload cut-down systems or devices that operate independently of each other;
- (2) At least two methods, systems, devices, or combinations thereof, that function independently of each other, are employed for terminating the flight of the balloon envelope; and
- (3) The balloon envelope is equipped with a radar reflective device(s) or material that will present an echo to surface radar operating in the 200 MHz to 2700 MHz frequency range.

The operator shall activate the appropriate devices required by paragraphs (a) (1) and (2) of this section when weather conditions are less than those prescribed for operation under this subpart, or if a malfunction or any other reason makes the further operation hazardous to other air traffic or to persons and property on the surface.

- (b) No person may operate an unmanned free balloon below 60,000 feet standard pressure altitude between sunset and sunrise (as corrected to the altitude of operation) unless the balloon and its attachments and payload, whether or not they become separated during the operation, are equipped with lights that are visible for at least 5 miles and have a flash frequency of at least 40, and not more than 100, cycles per minute.
- (c) No person may operate an unmanned free balloon that is equipped with a trailing antenna that requires an impact force of more than 50 pounds to break it at any point, unless the antenna has colored pennants or streamers that are attached at not more than 50 foot intervals and that are visible for at least one mile.
- (d) No person may operate between sunrise and sunset an unmanned free balloon that is equipped with a suspension device (other than a highly conspicuously colored open parachute) more than 50 feet along, unless the suspension device is colored in alternate bands of high

conspicuity colors or has colored pennants or streamers attached which are visible for at least one mile.

(Sec. 6(c), Department of Transportation Act (49 U.S.C. 1655(c)))

[Doc. No. 1457, 29 FR 47, Jan. 3, 1964, as amended by Amdt. 101-2, 32 FR 5254, Mar. 29, 1967; Amdt. 101-4, 39 FR 22252, June 21, 1974]

§ 101.37 Notice requirements.

(a) **Prelaunch notice:** Except as provided in paragraph (b) of this section, no person may operate an unmanned free balloon unless, within 6 to 24 hours before beginning the operation, he gives the following information to the FAA ATC facility that is nearest to the place of intended operation:

- (1) The balloon identification.
- (2) The estimated date and time of launching, amended as necessary to remain within plus or minus 30 minutes.
- (3) The location of the launching site.
- (4) The cruising altitude.
- (5) The forecast trajectory and estimated time to cruising altitude or 60,000 feet standard pressure altitude, whichever is lower.
- (6) The length and diameter of the balloon, length of the suspension device, weight of the payload, and length of the trailing antenna.
- (7) The duration of flight.
- (8) The forecast time and location of impact with the surface of the earth.

(b) For solar or cosmic disturbance investigations involving a critical time element, the information in paragraph (a) of this section shall be given within 30 minutes to 24 hours before beginning the operation.

(c) **Cancellation notice:** If the operation is canceled, the person who intended to conduct the operation shall immediately notify the nearest FAA ATC facility.

(d) **Launch notice:** Each person operating an unmanned free balloon shall notify the nearest FAA or military ATC facility of the launch time immediately after the balloon is launched.

§ 101.39 Balloon position reports.

(a) Each person operating an unmanned free balloon shall:

- (1) Unless ATC requires otherwise, monitor the course of the balloon and record its position at least every two hours; and
 - (2) Forward any balloon position reports requested by ATC.
- (b) One hour before beginning descent, each person operating an unmanned free balloon shall forward to the nearest FAA ATC facility the following information regarding the balloon:
- (1) The current geographical position.
 - (2) The altitude.
 - (3) The forecast time of penetration of 60,000 feet standard pressure altitude (if applicable).
 - (4) The forecast trajectory for the balance of the flight.
 - (5) The forecast time and location of impact with the surface of the earth.
- (c) If a balloon position report is not recorded for any two-hour period of flight, the person operating an unmanned free balloon shall immediately notify the nearest FAA ATC facility. The notice shall include the last recorded position and any revision of the forecast trajectory. The nearest FAA ATC facility shall be notified immediately when tracking of the balloon is re-established.
- (d) Each person operating an unmanned free balloon shall notify the nearest FAA ATC facility when the operation is ended.

F NAR High Power Safety Code

NAR High Power Rocket Safety Code

1. **Certification.** I will only fly high power rockets or possess high power rocket motors that are within the scope of my user certification and required licensing.
2. **Materials.** I will use only lightweight materials such as paper, wood, rubber, plastic, fiberglass, or when necessary ductile metal, for the construction of my rocket.
3. **Motors.** I will use only certified, commercially made rocket motors, and will not tamper with these motors or use them for any purposes except those recommended by the manufacturer. I will keep smoking, open flames, and heat sources at least 25 feet away from these motors.
4. **Ignition System.** I will launch my rockets with an electrical launch system, and with electrical motor igniters that are installed in the motor only after my rocket is at the launching or prepping area. My launch system will have a safety interlock that is in series with the launch switch that is not installed until my rocket is ready for launch, and will use a launch switch that returns to the "off" position when released. The function of onboard energetics and firing circuits will be inhibited except when my rocket is in the launching position.
5. **Misfires.** If my rocket does not launch when I press the button of my electrical launch system, I will remove the launcher's safety interlock or disconnect its battery, and will wait 60 seconds after the last launch attempt before allowing anyone to approach the rocket.
6. **Launch Safety.** I will use a 5-second countdown before launch. I will ensure that a means is available to warn participants and spectators in the event of a problem. I will ensure that no person is closer to the launch pad than allowed by the accompanying Minimum Distance Table. When arming onboard energetics and firing circuits I will ensure that no person is at the pad except safety personnel and those required for arming and disarming operations. I will check the stability of my rocket before flight and will not fly it if it cannot be determined to be stable. When conducting a simultaneous launch of more than one high power rocket I will observe the additional requirements of NFPA 1127.
7. **Launcher.** I will launch my rocket from a stable device that provides rigid guidance until the rocket has attained a speed that ensures a stable flight, and that is pointed to within 20 degrees of the vertical. If the wind speed exceeds 5 miles per hour I will use a launcher length that permits the rocket to attain a safe velocity before separation from the launcher. I will use a blast deflector to prevent the motor's exhaust from hitting the ground. I will ensure that there is no dry grass within a clear distance of each launch pad determined by the accompanying Minimum Distance table, and will increase this distance by a factor of 1.5 and clear that area of all combustible material if the rocket motor being launched uses titanium sponge in the propellant.
8. **Size.** My rocket will not contain any combination of motors that total more than 40,960 N-sec (9208 pound-seconds) of total impulse. My rocket will not weigh more at liftoff than one-third of the certified average thrust of the high power rocket motor(s) intended to be ignited at launch.
9. **Flight Safety.** I will not launch my rocket at targets, into clouds, near airplanes, or on trajectories that take it directly over the heads of spectators or beyond the boundaries of the launch site, and will not put any flammable or explosive payload in my rocket. I will not launch my rockets if wind speeds exceed 20

miles per hour. I will comply with Federal Aviation Administration airspace regulations when flying, and will ensure that my rocket will not exceed any applicable altitude limit in effect at that launch site.

10.Launch Site. I will launch my rocket outdoors, in an open area where trees, power lines, occupied buildings, and persons not involved in the launch do not present a hazard, and that is at least as large on its smallest dimension as one-half of the maximum altitude to which rockets are allowed to be flown at that site or 1500 feet, whichever is greater, or 1000 feet for rockets with a combined total impulse of less than 160 N-sec, a total liftoff weight of less than 1500 grams, and a maximum expected altitude of less than 610 meters (2000 feet).

11.Launcher Location. My launcher will be at least one half the minimum launch site dimension, or 1500 feet (whichever is greater) from any occupied building, or from any public highway on which traffic flow exceeds 10 vehicles per hour, not including traffic flow related to the launch. It will also be no closer than the appropriate Minimum Personnel Distance from the accompanying table from any boundary of the launch site.

12.Recovery System. I will use a recovery system such as a parachute in my rocket so that all parts of my rocket return safely and undamaged and can be flown again, and I will use only flame-resistant or fireproof recovery system wadding in my rocket.

13.Recovery Safety. I will not attempt to recover my rocket from power lines, tall trees, or other dangerous places, fly it under conditions where it may recover in spectator areas or outside the launch site, or attempt to catch it as it approaches the ground.

MINIMUM DISTANCE TABLE					
Installed Total Impulse (N-sec)	Equivalent Motor Type	Minimum Clear Distance (ft.)		Minimum Personnel Distance (ft.)	Minimum Personnel Distance (Complex Rocket) (ft.)
0 - 320.00	H or smaller	50		100	200
320.01 - 640.00	I	50		100	200
640.01 - 1280.00	J	50		100	200
1280.01 - 2560.00	K	75		200	300

2560.01 - 5120.00	L	100		300	500
5120.01 - 10,240.00	M	125		500	1000
10,240.01 - 20,480.00	N	125		1000	1500
20,480.01 - 40,960.00	O	125		1500	2000

Note: A complex rocket is one that is multi-staged or that is propelled by two or more rocket motors.

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